

Overlapping Generations

Methods, Models, and Morphology

By

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To our wives, Ann and Sara

Who put up with all the time that went into this work

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Introduction

In the early days of personal computing, there were two competing computer operating systems vying for consumers' dollars. One was Microsoft's MS-DOS system, and the other was Apple's MacIntosh system.

Microsoft's system was a clunky command line system, based on the original IBM disk operating system. Apple's system used a windowed graphical user interface based on Xerox's original X-windows interface for Unix. Of these two systems, the MacIntosh was by far the easier to use. Switching between applications was as simple as mousing the pointer from one window to another and clicking. On a Microsoft machine, switching involved quitting out of one application, entering commands to start a new application, and then settling down to work in the new application, hopefully without needing to go back to the first one!

It is well-known by now that of these two competing computer operating systems, Microsoft's became dominant, despite its obvious inferiority to the MacIntosh. The reason for this is the stuff of an Industrial Organization course case study, but the factors leading to Microsoft's dominance include the state of technology, political and economic differences of opinion, and the lock-in effects generated by the network of applications that ran on the two different platforms. That the MacIntosh was the better technology, though, is clearly apparent from the fact that MS-DOS steadily evolved from its initial command line structure into a system centered on Xerox's windowed graphical user interface, which Microsoft now calls Windows.

This particular example of what we call the standardization trap is not new. Careful study of the history of the automobile clearly shows that electric cars developed alongside steam- and gasoline-powered cars, with electric vehicles being the better technology in the early phases

of development. The market for automobiles tipped to internal combustion only after Henry Ford's improvement on Ransom Olds' patented assembly line process allowed him to mass produce cheap Model T's that ran on gasoline. The superiority of electric vehicles is only now becoming apparent, with the advent of serious repercussions due to global warming, and with the development of high-capacity batteries that allow electric cars to travel long distances. But even before these recent developments, diesel-electric hybrid technology replaced steam as the optimal power source for heavy freight trains, and this technology has recently found its way into automobiles as well.

In the history of electrification, competing systems developed by Thomas Edison – direct current – and Nicola Tesla – alternating current – existed side by side, with alternating current finally winning out for the almost trivial reason that it is easier to change voltages with AC. Today, DC adapters routinely make voltage adjustments, and, as a result, all modern electronics run on DC current. As locally generated electricity from renewable sources starts to come online, the electricity market might well tip back to Edison's favored DC technology.

Standardization traps aren't limited to new technological innovations. They occur in the realm of ideas as well. In fact, the scientific method operates by allowing competing hypotheses to be tested against empirical facts, with the hypothesis best able to explain the data enjoying at least temporary status as king of the hill. How long different hypotheses can stay on top depends on the quality of the empirical data available for testing, and on the sharpness of predictions the reigning hypothesis generates.

Science characterizes reigning hypotheses as paradigms, although one should note that the original meaning of the word paradigm from the Greek was one of side-by-side comparison.

Thomas Kuhn originated the notion of a paradigm shift for the replacement or selection of one theory or hypothesis over others as the premier object of study.

In contemporary physics, we see this process at work in the parallel development of different theories for unifying general relativity's characterization of gravity with the underlying structure of quantum mechanics. As yet, there has been no compelling reason for one approach to supplant others, so no standardization trap or paradigm shift has occurred.

Historically, the debate over whether the universe was static or dynamic allowed both perspectives equal validity, to the point that Einstein made his famous blunder in setting the cosmological constant in the field equations of general relativity so that expansive and contractive forces would balance out, leaving the universe static. This despite the fact that in 1922, Alexander Friedmann had shown that the general relativity equations could accommodate different cosmological shapes and dynamics. Only in 1929, after Edwin Hubble discovered the uniform red shift of galaxies in all directions (with the implication that space was itself expanding), did Einstein correct his blunder. With this correction, a paradigm shift occurred, and the Big Bang became the standardized theory of cosmology. Whether a new theory will ultimately replace the Big Bang (and hence whether this was a trap) remains to be seen.

In the social sciences, a much slower research cycle, a relative paucity of empirical data, and limited possibilities for conducting experiments that generate new data, leads to the possibility of much longer coexistence between competing theories, and greater opportunities for incomplete or incorrect theories to get locked in as the dominant paradigm, with all the attendant problems this raises for correcting the science when resources flow elsewhere.

We will argue here in this book that this is precisely the situation that macroeconomics and general equilibrium theory currently face. The problem arises out of the fact that two competing dynamic, stochastic, general equilibrium (DSGE) models have evolved as workhorse models for macroeconomic analysis, which we define broadly to include monetary economics, business cycle theory, economic growth, public finance/optimal taxation, and fiscal policy analysis.

To indicate the conundrum macro faces with these models, let us start with the crudest possible characterization of them.

The first model consists of a single, so-called “representative” agent who lives forever. Time enters the model as a sequence of discrete periods. The agent has access to technology for producing goods and services. The agent values these goods and services according to a utility function defined over consumption in each period, and the agent discounts utility in each period using a constant discount factor $0 < \beta < 1$. With these features in place, the agent’s economic problem is to choose a series of investments in capital (or technology more generally) to maximize intertemporal utility subject to the constraint that invested resources can’t be consumed.

The second model consists of a sequence of agents born at different times (with time also entering discretely) who overlap with and can trade with agents born earlier. Agents in this model live finite lifetimes. As with the first model, agents have access to technology for producing goods and services, and value consumption of goods and services via a utility function defined over consumption in each period of life. Agents in this model may or may not discount future utility. With these features in place, agents in this second model choose lifetime

consumption and saving to maximize lifetime intertemporal utility, with savings becoming available for investment in technology.

By way of simplifying nomenclature, we will follow the conventional economic practice of referring to the first model as the Infinite Lived Agent model (ILA), and the second model as the Overlapping Generations model (OLG).

The ILA model was originally developed by Frank Ramsey in 1928 as a way of modeling economic growth. Ramsey viewed the optimization problem posited in the model as the goal of a central planner, with the objective function of the problem (the representative agent's intertemporal utility) viewed as a social welfare function that put specified welfare weights on future generations' utility. The model was adopted and elaborated on by John Von Neumann (1937) and saw parallel development in the 1940's and '50's by Maurice Allais (1947) and Edmond Malinvaud (1953), and by Robert Solow and Paul Samuelson (1956).

Work by David Cass in the mid-1960's showed how to completely characterize the optimal trajectories in this infinite-horizon general equilibrium model. Since Cass's work coincided with work by the then more senior and better known Tjalling Koopmans, the resulting model is now universally known within the economics profession as the Ramsey-Cass-Koopmans (RCK) model, though it is sometime also referred to as the neoclassical growth model. For details on these developments, see Spear and Young (2014).

The OLG model first appeared in 1947 in the technical appendix of Maurice Allais' book *Economie et Interet*. Because the model appeared only in the appendix and the work in which it appeared was published in French, Allais' application of the model received very little attention in the English-speaking sector of the economics profession.

The OLG model was reinvented (or rediscovered) a decade later by Paul Samuelson (1958). Samuelson's model generated intense interest among general equilibrium theorists and monetary economists. General equilibrium theorists were interested in the fact that Samuelson's model could admit competitive equilibria which were not Pareto optimal, a contradiction of the First Welfare theorem for static general equilibrium models. Monetary economists focused on the fact that a role for money appeared naturally in Samuelson's model as a means of saving for retirement. A few years later, Peter Diamond (1965) reworked the OLG approach by introducing production in the model. This made possible a more realistic way of dealing with government debt, and extended OLG into the area of public finance, fiscal policy, and social security.

At roughly the same time that Samuelson was working on his early version of the OLG model, Gerard Debreu (1954) published work showing that Ramseyan-type models satisfied the second welfare theorem, allowing him to explicitly find prices that would support the Pareto optimal allocations obtained by the central planner as competitive equilibrium allocations for a large number of identical agents whose preferences were given by the central planner's social welfare function. This result was generalized by Takashi Negishi (1960) who showed how to extend Debreu's result to allow for any finite number of infinite lived agents. This appears to be the first instance in which the central planner was reinterpreted as an actual economic actor.

Now, on its face, a model in which actual economic agents live forever is problematic, and it isn't surprising that while there was some work being done on Samuelson's model in the twenty years after his seminal paper, there was no work (per a search of EconLit for "representative agent") being done on ILA models (as distinct from the planning version of the RCK model) during the comparable period.

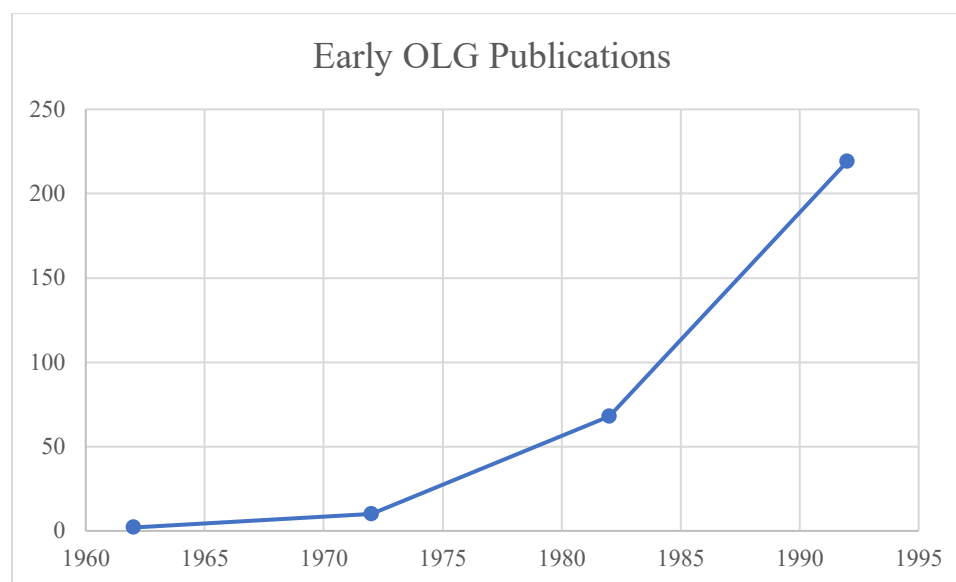
It is also not surprising that perhaps the most important paper in modern macroeconomics, Robert Lucas' 1972 paper "Expectations and the Neutrality of Money," was based on a very sophisticated stochastic extension of the OLG model. The OLG framework allowed (as we noted above) the introduction of money, and individual agents (as opposed to a planner), together with random shocks to both the demand for goods and the supply of money, which then allowed Lucas to present a simple model of how a signal extraction problem could generate a short-run positive relationship between inflation and real output. He also introduced macroeconomists and general equilibrium theorists to the rational expectations paradigm along the way.

Shortly after Lucas published his path breaking '72 paper, Robert Barro (1974) took Samuelson's OLG framework and showed how it could be transformed into the representative agent ILA model of Debreu and Negishi. Barro's insight was that if one interpreted the finite-lived agents of the OLG model as families, and families propagated lineally, and each parent family left positive bequests to their immediate offspring such that the resulting intergenerational allocation maximized the discounted infinite sum of utilities of all generations, then the OLG model with bequests was identical to the representative agent ILA model. As an added benefit, Barro's construction showed how forward dynamic programming techniques introduced in economics a decade earlier by Roy Radner (1967) could be applied to indicate the generational separation between agents in the ILA model, and to solve the infinite horizon optimization without transversality conditions.

Barro's reinterpretation of OLG came at roughly the same time that William Brock and Leonard Mirman were showing how to extend the RCK model to stochastic environments, a development which then fed directly into the work of Finn Kydland and Edward Prescott which

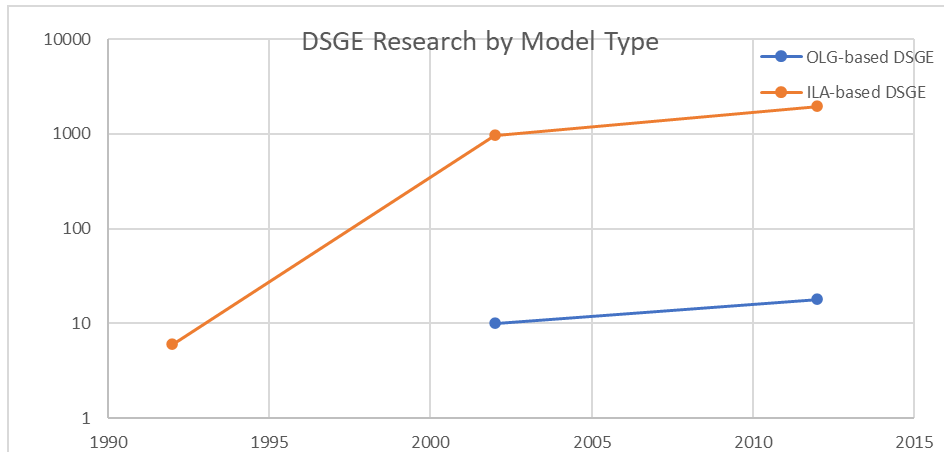
launched real business cycle theory – based on the stochastic representative agent ILA model – in the mid-1980's. For details on this history, see Young (2014).

The chart below, based on a search of the American Economics Association's EconLit database shows the growth of publications based on the OLG model from its introduction through the early 1990's.



Compiled by authors from Econlit database, 1960-1995

Viewed in isolation, this seems like impressive growth in interest in and work on OLG models. However, when we do an EconLit comparison of the DSGE research (that is, using the search term “dynamic stochastic general equilibrium”) using models based on the OLG framework with those based on the ILA framework, the result is stunning. Prior to the early 1990's, there is no research of any type on DSGE. The topic then explodes for ILA-based models, with some growth in the early 2000's for OLG-based models, as illustrated in the chart below.

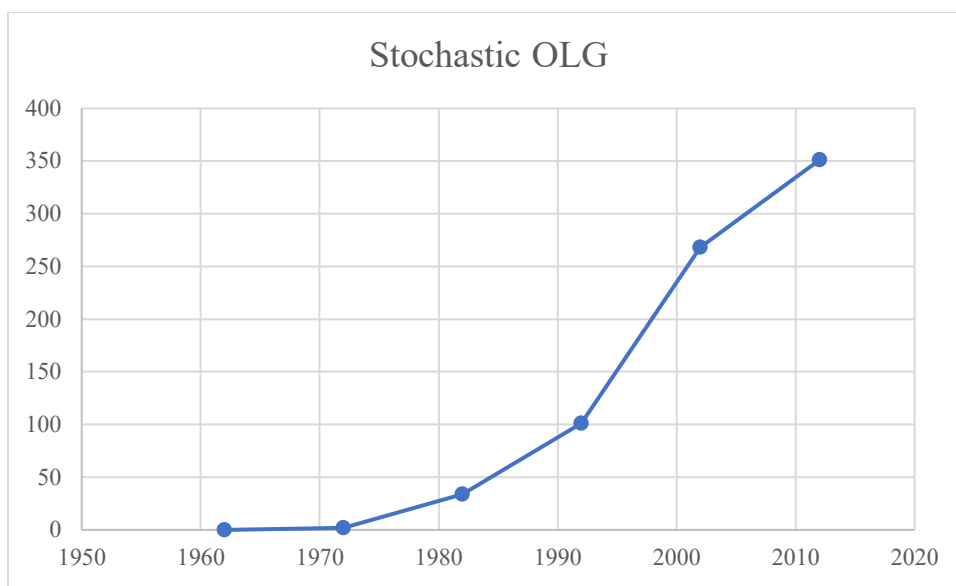


Compiled by authors from Econlit database, 1990-2015

We use a log scale for this chart because the growth in ILA-based DSGE papers has been exponentially greater than that for OLG-based models.

While macroeconomists routinely describe OLG and ILA models as workhorses, it is clear that the ILA model is pulling a great deal more weight than OLG. At the same time, a search on “stochastic” and “overlapping generations”¹ generates the following:

¹ Strictly speaking we search on “generations” since early use of the model did not refer to overlapping generations at the time.



Compiled by authors from Econlit database, 1950-2020

So, work on stochastic OLG models did continue apace – much of it stimulated by the publication of Lucas’s ’72 paper – and in fact grew, but at nowhere near the rate at which applications of the ILA model in macro were growing. Indeed, as we will see, even Lucas, who pioneered the use of stochastic OLG models changed workhorses in midstream.

An additional piece of data that supports this contention comes from a study by Kim et al. (2006). Based on a list of papers appearing in major economics journals over the period 1970-2005 with at least 500 citations in the ISI Web of Science/Social Science Citation Index, the authors concluded that there had been a significant shift in emphasis placed on theory as against empirical work, and on the various subfields that attracted the attention of researchers in economics, with a shift from both micro and macroeconomics to growth and development during the 1990s.

What is relevant for our purpose is the placement of four watershed papers – three based upon the dynastic/infinite-lived agent model, as against a single paper based on the overlapping

generations model – in their listing of articles receiving more than 500 citations over the period. The three ILA papers were those by Barro (1974), Kydland and Prescott (1982), and Lucas (1988). The OLG-based paper is Lucas (1972). The 1988 *Journal of Monetary Economics* paper by Lucas is listed at number 15, with 1772 citations. The 1974 *Journal of Political Economy* paper by Barro is listed at number 29, with some 1209 citations. The 1982 *Econometrica* paper by Kydland and Prescott had 814 citations, coming in at 61. The OLG paper by Lucas, published a decade earlier, in the *Journal of Economic Theory* in 1972, had 838 citations, and comes in only three places before Kydland and Prescott, at number 58. Although published a decade after Lucas's OLG-based paper, the ILA-based Kydland and Prescott paper had caught up in terms of citation impact to that of Lucas. Lucas' own 1988 paper "On the Mechanics of Economic Development," which uses the ILA methodology, had also clearly eclipsed his earlier OLG paper.

This illustrates the twilight of the old OLG-based approach as proxied by Lucas (1972) and the rise of the new ILA-based approach as proxied by Kydland and Prescott (1982) and Lucas' own adoption of the ILA framework as his basic research agenda, albeit retaining OLG as an analytical device to be applied in specific cases; see e.g., Lucas (1986)². But more is at stake here. The Kim et al. piece clearly shows the shift away from theory toward empiricism, and the high placement of Barro's 1974 paper likely reflects the way macro people often started by positing an OLG model, but then allowed bequests and invoked Barro to justify working with the more convenient, recursive methodology provided by the ILA framework.

² We could also add Lucas' other seminal 1978 paper, "Asset Prices in an Exchange Economy," to this list, since it had 772 citations and came in at number 70 in the Kim et. al. paper.

This is exactly the process we are trying to describe. The key question is why did macroeconomists feel the need to do this? Computational efficacy? Mathematical tractability? Ideological predisposition? If so, it illustrates the process of a particular paradigm coming to "rule the roost." And the fact that – with the exception of the 1972 paper by Lucas – there was nothing on the Kim et. al. list that reflects OLG, shows exactly what occurred. ILA came to rule the roost, not perhaps because it was the “best” or most realistic model, but rather because it was the easiest to work with.

Our final data showing the relative ascendance of the ILA model over OLG comes from the ProQuest database of dissertations awarded between 1960 and 2020. When we search by various model-related keywords – “overlapping generations,” “consumption loans,” “Ramsey-Cass-Koopmans,” “real business cycles,” and “growth” – we get the following numbers:

	Total Dissertations Awarded				Stochastic Growth
	OLG	CL	RCK	RBC	
1960-1979	2	8			84
1980-1999	522	128	12	407	491
2000-2020	1254	133	49	1394	639
Totals	1778	269	61	1801	1214
OLG Total	2047				
ILA Total	3076				

Compiled by authors from ProQuest Dissertations Database

As with the data on publications based on each model, while there has been growth in the number of doctoral students trained in both models, ILA outpaces OLG significantly.

The ILA applications in macro were not without critics and substantive critiques. Among the early critiques was a paper by Kyle Bagwell and B. Douglas Bernheim which made the

obvious point that families don't propagate lineally. Rather, given standard taboos against incest, families propagate as trees. Bernheim and Bagwell (1988) showed that if family structures were modeled realistically, and all families were interconnected via positive transfers (as in Barro's model), then everything was neutral. Markets played no role in the Bernheim-Bagwell model for the simple reason that internal transfer adjustments between families could restore optimality of allocations without the need for market interventions following any perturbations to the original optimal equilibrium. A more recent study by Engineer and Welling (2004) argues persuasively that the overlapping generations structure and the economic consequences generated by age heterogeneity conform closely to anthropological structures delineated in Stewart (1977) known as graded age-group sets. As Engineer and Welling argue, their results

establish that OLG models can accurately represent a large subset of actual age-group societies called graded age-set societies. Thus, an OLG model bears a close resemblance to reality. (Engineer and Welling [2004] p. 454)

Together with data showing that most families do not leave substantial bequests to their children – investing instead in human capital bequests via K-12 education and beyond – it is hard not to come away from a study of these two workhorse models without the feeling that one of them is a PC and the other is a Mac.

Our purpose here, then, is to elucidate the technical and intellectual forces that have generated the current situation in which macroeconomics seems to be caught in a standardization trap based on a problematic choice of models.

Chapter 1: Origins of the Workhorse Models

Section 1.1 Infinite Lived Agents and The Neoclassical Growth Model

Of the two workhorse models, ILA is somewhat older, having its origin in a remarkable paper by Frank Ramsey (1928). Ramsey's specification of the model was consistent with contemporary specifications, with the exception of his assumption that society eventually became satiated in consumption. Ramsey's objective function for society was a standard sum over time of instantaneous utility payoffs (possibly discounted), which he interpreted explicitly as the valuation of a utilitarian social planner. This explicit normative interpretation would persist until the mid-1970's. Ramsey needed the satiation assumption to force the dynamics of the model to converge over time to a steady state at the social bliss point. This assumption was later dispensed with when the mathematics for analyzing the more general case of non-satiated utility became available.

Ramsey's model received scant attention during the 1930's and WWII years, although the model was adopted and elaborated on by John Von Neumann (1937). An additional citation to the paper comes in Paul Samuelson (1943). There, Samuelson terms Ramsey's approach as "brilliant" (p. 67), though he elided the section of his 1943 paper that contained his summary and citation of Ramsey from his *Foundations of Economic Analysis* (Samuelson [1947]).

During the immediate post-war period (1945-1955), Ramsey's paper is noted by the major growth theorists, including Koopmans, Malinvaud, Samuelson, and Tinbergen. As noted above, these authors followed Ramsey's lead in interpreting the social object function

as that of the central planner. Samuelson (1943) discussed this interpretation explicitly (p.62):

Let us assume, therefore, a condition of perfect certainty and an economy consisting of one or more individuals. We further assume, since otherwise the discussion can end before it starts, that there is no intrinsic rate of time preference. We need not speculate as to whether or not this implies infinite life expectations for the individual, for his family, etc., etc., since in any case we are not concerned here with the realism or the usefulness of the argument. For our purposes it is convenient to adopt the quite arbitrary assumption that utility is a given function of income in each period; more specifically, that it is the same function at each instant of time and that the individual acts so as to maximize the sum of utilities thus defined over all future time.

The culmination of what we might call the Ramsey project comes in the early 1960's with the work of Cass, Koopmans and Malinvaud. These authors complete the correct mathematical characterization of the solution to the planner's problem without the assumption of satiated preferences. Malinvaud's key contribution is his demonstration of the importance of capital value minimization in preventing inoptimal overaccumulation of capital, which he characterized mathematically in what we now call a transversality condition. Cass's key contribution is the application of mathematical techniques from Pontryagin et. al. (1960) to derive the transversality condition, which insured convergence of any optimal trajectory to the optimal long-run steady state (and which implies Malinvaud's capital value minimization condition). For a detailed history of these developments, see Spear and Young (2014) on the Cass-Malinvaud-Koopmans nexus.

The positivist development of growth starts after WWII with the work of John von Neumann and was then picked up and moved forward by Paul Samuelson, Robert Solow and their coauthors. The positive approach is partial equilibrium in nature, focusing as it does on the role of production and capital accumulation as drivers of economic growth. Specifically, this literature takes a distinctly Keynesian view of the consumption side of the economy, using the

empirical relationship consumption and income to calibrate the savings rate in the various models. Even in treatments that do posit a central planner, most of the action in these models is driven by production, as noted explicitly by Malinvaud (1985):

In the previous sections we have seen how theories of interest, capital and growth may prove interesting properties resulting only from the fact that most productive operations can be described as taking place within one period or several successive periods. For, apart from a reference to consumer preference in the last example, we have so far considered only production. (p. 307)

Among the important results developed in this literature are the delineation of turnpike properties of balanced growth, the remarkable no-arbitrage relationship between the value of capital as an output of production and its value as an input in subsequent periods, and the Solovian convergence results. All of these results arise from the optimality calculations for general multi-sector capital models of the type first considered by von Neumann.

At roughly the same time that the two distinct strands of the neoclassical growth literature were developing in the post-war period, Gerard Debreu put forward an abstract, theoretical analysis of the same model. Debreu (1954) extends the finite-dimensional Arrow-Debreu-McKenzie model to allow for infinitely-many commodities. He explicitly cites Malinvaud's work on the capital model as the justification for considering an infinite-dimensional commodity space. He then establishes that the model satisfies the two standard welfare theorems and shows how to decentralize the Pareto optimum found by the Ramseyan central planner as a competitive equilibrium. While Debreu does not concern himself with the question of whether the people who inhabit his model can live forever, he is explicit in interpreting the agents in the model as individuals or households.

Malinvaud (1985) notes the key problem with this abstract, but still rather cavalier specification of the consumption side of the economy:

We see immediately that analysis by periods is appropriate for such problems. We must certainly take account of the fact that an individual lives through several successive periods; but we must also note that no individual lives indefinitely. We should also recognise that generations are renewed from one period to another. (p. 308)

The question of how to interpret the infinite-lived representative agent – is it a planner or an actual participant in the underlying economy – remains much as Samuelson originally described in his 1943 paper: a convenient abstraction from reality. Ironically, it is Samuelson's development of the overlapping generations model in 1958, plus about a decade's worth of work on the model by others, that leads ultimately to the dynastic family interpretation of the ILA model at the hands of Robert Barro.

Section 1.2 The Overlapping Generations Model: Origins and Dissemination

The overlapping generations model is distinctly different from the infinite-lived representative agent or household in the Ramsey-Cass-Koopmans model. OLG agents live finite lifetimes, with new agents being born into the economy over time, so that the economy goes on forever with older and younger generations overlapping. Unlike the ILA model examined by Debreu (1954), the OLG model admits competitive equilibria which violate the welfare theorems, a fact that has led to a number of surprising discoveries and corresponding contentious issues in the on-going debate over the appropriate role of government in the economy.

The genesis of the idea that the human lifecycle and the structure of human families ought to play an important role in economic activity is probably lost to history, since the idea

isn't particularly deep and certainly not radical. The first written consideration of these factors in economics, however, can be traced back to Irving Fisher (1907) and Fisher (1930). While Fisher has no formal overlapping generations-like model in his 1930 book *The Theory of Interest, as Determined by Impatience to Spend Income and Opportunity to Invest It*, he does make repeated references to intergenerational considerations as a determinant of interest rates. The key passage in his 1930 book is from section 9.5:

But whereas the shortness and uncertainty of life tend to increase impatience, their effect is greatly mitigated by the fifth circumstance, solicitude for the welfare of one's heirs. Probably the most powerful cause tending to reduce the rate of interest is the love of one's children and the desire to provide for their good. Wherever these sentiments decay, as they did at the time of the decline and fall of the Roman empire, and it becomes the fashion to exhaust wealth in self-indulgence and leave little or nothing to offspring, impatience and the rate of interest will tend to be high. At such times the motto, "after us the deluge," indicates the feverish desire to squander in the present, at whatever cost to the future.

On the other hand, in a country like the United States, where parents regard their lives as continuing after death in the lives of their children, there exists a high appreciation of the needs of the future. This tends to produce a low degree of impatience. For persons with children, the prospect of loss of earnings through death only spurs them all the more to lay up for that rainy day in the family. For them the risk of loss of income through death is not very different from the risk of cessation of income from any ordinary investment; in such a case the risk of cessation of future income through death tends to lower their impatience for income. This act supplies the motive for life insurance. A man with a wife and children is willing to pay a high insurance premium in order that they may continue to enjoy an income after his death. This is partly responsible for the enormous extension of life insurance. At present in the United States the insurance on lives amounts to over \$100,000,000,000. This represents, for the most part, an investment of the present generation for the next.

An unmarried man, on the other hand, or a man who cares only for self-indulgence and does not care for posterity, man, in short, who wishes to "make the day and the journey alike," will not try thus to continue the income after his death. In such a case uncertainty of life is especially calculated to produce a high rate of time preference. Sailors, especially unmarried sailors, offer the classic example. They are natural spendthrifts, and when they have money use it lavishly. The risk of shipwreck is always before them, and their motto is, "a short life and a merry one." The same is even more true of the unmarried soldier. For such people the risk of cessation of life increases their impatience, since there is little future to be patient for.

Not only does regard for one's offspring lower impatience, but the increase of offspring has in part the same effect. So far as it adds to future needs rather than to immediate needs, it operates, like a descending income stream, to diminish impatience. Parents with growing families often feel the importance of providing for future years far more than parents in similar circumstances but with small families. They try harder to save and to take out life insurance; in other words, they are less impatient. Consequently, an increase of the average size of family would, other things being equal, reduce the rate of interest.

This proposition does not, of course, conflict with the converse proposition that the same prudent regard for the future which is created by the responsibilities of parenthood itself tends to diminish the number of offspring. Hence it is that the thrifty Frenchman and Dutchman have small families. (pp. 85-87)

The last two paragraphs could have come from Richard Easterlin's work on the demographic transition, and indicate clearly that Fisher viewed the internal dynamics of the family, and the resulting intergenerational dynamics of the economy, as important factors in determining intertemporal economic activity.

The first formal overlapping generations models were introduced to the economics profession in two waves. In the first, Maurice Allais wrote down a version of the model in an appendix to his book (published only in French) *Economie et Interet* (Allais [1947]). In the second wave, coming eleven years after the publication of Allais' book, Paul Samuelson published his 1958 article. This article introduced English speaking economists to the overlapping generations model, and, for many years, Samuelson's paper was cited as the original source for the model. Indeed, until Malinvaud published papers in 1986 and 1987 pointing up Allais' priority, the economics profession uniformly credited Samuelson with the introduction of model. (See Grandmont [1989] for a detailed explication of Allais' development of the OLG model.) Even today, Allais' contribution is frequently overlooked. Weil (2008) starts his paper celebrating the golden jubilee of the OLG model by writing: "Paul Samuelson's (1958) overlapping generations model has turned 50." It is only in a footnote two pages into the article

that Weil notes that Maurice Allais derived the same model in 1947. While Weil is quite gracious in noting Allais' priority, the misassignment of priority to Samuelson for the OLG model is quite common. The answer to the question of why Allais' contribution was not noted until well after Samuelson's is likely a matter of a) the work being published only in French; and b) as Malinvaud (1987) notes: "The treatment in the appendix is analytically rather complex, leading to consideration of many cases and to introduction of long formulas." (p. 103)

Malinvaud (1953) used Allais' OLG structure, though as he himself notes in his 1987 *Journal of Economic Literature* communication (Malinvaud [1987]), the OLG structure didn't play a major role there. Malinvaud used the OLG structure in a manner similar to that used thirty years later by Balasko and Shell in their well-known papers on the overlapping generations model. Specifically, he used the fact that with the OLG demographic structure, only finitely many agents would be alive at a given time. This allowed him to write down individual allocation vectors as infinite-dimensional vectors (given the infinite horizon of the model), having only finitely-many non-zero entries. It also allowed him to add up allocations over the full economy with all aggregate entries being finite. Finally, he used a time truncation to produce a sequence of economies of increasingly longer (but finite) time horizons that he could apply the standard finite-dimension separating hyperplane theorem to, together with a nested set argument to show existence of support prices for characterizing the optimal accumulation sequence.

Samuelson's OLG development in his 1958 paper worked with a pure exchange economy and emphasized both positive and normative elements. On the positive side, he developed the initial model assuming that agents lived three period lives, so that there was no so-called "Samuelson-Gale" dichotomy in the original model. The three period life cycle allowed

Samuelson to capture the essential empirical fact of the hump-shaped age-income relationship. By assuming that agents in their third period of life had very low endowments, Samuelson built in the need for retirement saving that was an obvious fact of the life cycle by the middle of the 20th century. His assumption that goods were completely perishable, and the absence of a social credit system, captured the harsh reality of penury in old age that Samuelson had likely observed growing up during the depression, but which was also an historical fact of life throughout the ages when elderly parents had to count on the love of their children to keep them alive. Indeed, Shakespeare based one of his best-known plays, *King Lear*, on this theme. Finally, Samuelson showed how a social credit system – which he characterized as the “social contrivance of money” could alleviate this hardship and co-dependence within families by allowing for independent asset accumulation.

The normative elements of Samuelson’s model included his demonstration of the existence of competitive equilibrium which were not Pareto optimal, and his demonstration, via the introduction of a fiat money asset, of a role for government in improving the economic functions of society via management of the money supply or provision of Social Security. Finally, in looking at the optimal solution with money, Samuelson deduced that the equilibrium rate of interest would be precisely equal to the biological rate of growth of population in the economy. If we take the population growth rate to be zero, this yields an equilibrium interest rate of zero, since in equilibrium, whatever savers don’t consume, dis-savers consume (given the complete perishability of goods). This result sparked a good deal of debate, with two formal criticisms of the model published in the *Journal of Political Economy (JPE)* by Abba Lerner and William Meckling, which we examine in detail below. But we should emphasize that Samuelson’s only actual use of the two period lived agents version of his model was in the

section where he tries to starkly illuminate the nature of the social welfare trade-off implicit in the model.

It is also of considerable interest that Samuelson had a strong interest in population dynamics in biology. This work is documented in Ronald Lee (2019), where Lee notes

Who now knows that Samuelson published two papers on the two-sex problem in demography? Or two papers on predator-prey models? A paper on the reversibility of time in population processes? Five papers generalizing and critiquing Fisher's concept of reproductive value, a concept that plays a key role in evolutionary theory? Analyses of evolutionary theories of altruism and kin selection? (p. 474)

For our purposes, the most interesting aspect of this work is that in the biological study of animal population dynamics, there are two approaches to looking at how animal populations evolve. One assumes a lineal process, typically based on looking at the females of a species, which then unfolds over time under the influence of environmental and genetic factors that bear on survival and reproduction. The second type of model assumes that generations of organisms overlap, with some resource consideration required of parent generations to ensure the survival of offspring. Given Samuelson's interest in and work on population dynamics, he would surely have been aware of the explicit use of the term "overlapping generations" from the biology literature, and yet in his own economic application of the core idea, he opted not to refer to overlapping generations.

This then raises the question of how the economics profession came to call the model not by Samuelson's "consumption loans" moniker, but by the OLG terminology that is now standard. A search of the EconLit database for the phrase "overlapping generations" restricted to the time period 1940-1970 returns no hits, because Samuelson routinely referred to his model as one of "consumption loans," and all of the early follow-on papers followed Samuelson's lead on

this. A similar search on Google scholar returns a large number of hits, but all of this work is in biology, specifically population dynamics studies. As we noted above, it is ironic that Samuelson would have been so well-versed in this field, but not think to (or actively resist thinking to) call his model one of human overlapping generations. In any event, the first use we have found of the phrase – albeit in the context of a comment on Diamond (1965) – appears in Bierwag *et al.* (1969), while in the same year Phelps and Shell (1969) used the term “population overlap” in the context of their interpretation and extension of Samuelson’s 1958 model. Barro (1974) also uses the term in reference Samuelson (1958). The next references found using the phrase are to be found in Buchanan (1976) commenting on Barro’s paper, in Shell (1977), and in Wallace (1977).

Since that time, the use of “consumption loans” has died out in favor of the standard overlapping generations terminology. So, going forward, we will refer to the model as overlapping generations or OLG rather than the earlier Samuelsonian terminology.

Chapter 2: OLG – The Next Generations, 1960-1970

Samuelson's model generated significant interest among the up-coming young generation of both general equilibrium theorists and capital theorists. Demographers generally refer to the people born between 1921 and 1940, which includes this group, as the silent generation. As we will see, the economists among them were anything but silent. This group included David Cass, Peter Diamond, David Gale, Karl Shell, David Starrett, and Menahem Yaari. Their work in following up on the issues and conundrums raised by Samuelson's article set the research agenda on the OLG model for the next thirty years.

To understand how this group of theorists shaped this agenda, though, we need to first examine the economics profession's immediate response to the publication of Samuelson's paper. This was captured in a series of communications published in the *JPE* between Samuelson and Abba Lerner in 1959, and Samuelson and William H. Meckling in 1960. So, we turn first to these.

Section 2.1 The Lerner-Samuelson Debate

The first shot in the Samuelson-Lerner debate was fired by Lerner (1959). Lerner's objection to Samuelson's analysis turns on his definition of optimality, which he bases (as Samuelson notes in his reply) on a utilitarian social welfare function which weights generations according to their sizes. Given this, he and Samuelson agree that the optimal gross interest rate should be one when population is not growing. Lerner asserts, however, that in the case where population is growing, the optimal interest rate should remain zero. In his reply to Lerner, Samuelson (1959) shows that this would be true if the social planner's optimization problem was

based on the utilitarian objective function. Crucially, though, Samuelson notes that his analysis was not normative, but positive in asking what kind of equilibrium would be supported if there were some mechanism for allowing intergenerational transfers to occur. This mechanism is Samuelson's "social contrivance of money," though in Lerner's telling it becomes the Social Security system and Lerner's chief objection is the governmental taking required for a pay-as-you-go system to operate.

Samuelson's reply to Lerner does make the point that for normative analyses, the choice of social welfare function is important, though this shouldn't detract from the use of positive economic models for analyzing social institutions such as money. The focus in this discussion on social welfare also echoes the earlier concerns with the question of how to weight different generations, going back to Ramsey (1928), the concerns of the capital theorists about how to handle growth models which didn't discount the future, the interest expressed by the Vatican in the subject – manifested in their hosting of a scientific conference on development and growth in 1963 – and continuing into contemporary debates over global warming.

A second, important feature of Samuelson's reply to Lerner is that it works exclusively in the two period lived agents setting, in contrast to his own use of the three-period model in the paper, and to Lerner's casting of his own (non-mathematical) arguments in the three-period setting. This setting highlights the various issues involving intergenerational transfers in a particularly stark fashion, and likely gave impetus to future work which focused solely on the two-period model. Samuelson closes his reply to Lerner with the droll statement:

In defending the logic of my own exposition, I hope I shall not be construed as belittling Lerner's contribution. Who knows, if I read his paper a third time, I might become convinced of the errors of my way. (p.522)

Section 2.2 The Meckling-Samuelson Debate

In the aftermath of Samuelson's debate with Lerner, Meckling (1960) published a comment also taking exception to Samuelson's results, though using more formal mathematical reasoning, which forced Samuelson, in his reply, to clarify a number of points from the original paper in a way that now looks like a blueprint for the OLG research program that was subsequently carried out by the next generation.

Like Lerner, Meckling insists that Samuelson's zero interest rate equilibrium can only arise if one takes the view of a central planner who equally weights all generations currently living in any given time period. He adds that the biological interest rate also seems to fly in the face of Irving Fisher's observation that resources and preferences jointly determine interest rates, noting that in Samuelson's model, any preferences (not giving rise to corner solutions) give the same equilibrium interest rate.

Meckling goes on to insist, however, that absent a benevolent planner, the arrangement of loans and their repayment must be a matter of bilateral negotiations. In the context of Samuelson's original three-period model, this means that only the young and middle-aged can borrow and lend to each other, with the old only capable of taking repayment of loans negotiated earlier in life. This leads him to conclude that the only possible equilibrium is one involving intergenerational autarky. Finally, Meckling also makes the interesting observation that the Samuelsonian equilibrium is likely to be unstable in the dynamics of the model around the zero interest rate steady state and suggests that such instability makes observing such an equilibrium as a positive economic outcome unlikely.

Meckling concludes his critique by allowing for the possibility of the “social contrivance of money” or related social contracts as a way of supporting the Samuelsonian equilibrium, but in a footnote (fn. 9 on p.75) to this section, he relates these mechanisms to what we have come to call Ponzi schemes.

Samuelson (1960), replying to Meckling is a *tour de force*.

His analysis of the equilibrium equations governing the model begins by noting that in looking at steady states, he is interested only in the roots of the equations given by the market-clearing condition for the model and the budget constraints (i.e., Walras’ law). These equations are worth noting, and take the form

$$\begin{aligned} S_1(R, R) + S_2(R, R) + S_3(R, R) &= 0 \\ S_1(R, R) + RS_2(R, R) + R^2S_3(R, R) &= 0 \end{aligned}$$

Here, R is the gross interest rate, and $S_i(R, R)$ is the savings of young, middle-aged and retired agents respectively. Multiplying the first equation (market-clearing) by R and subtracting it from the second yields

$$(1 - R)S_1(R, R) + (1 - R)RS_3(R, R) = 0.$$

If $R < 1$, then the common term can be cancelled, leaving Meckling’s basic equilibrium equation (corresponding to intergenerational autarky). But, as Samuelson notes, if $R = 1$, there are other equilibrium savings schemes (which implement the Samuelsonian outcome) that also satisfy the equilibrium equations. This feature shows up again later (in the work of Balasko and Shell, and of Kehoe and Levine for pure exchange models) in the guise of an equilibrium condition of the form

$$(1 - R)m = 0$$

which states that either the equilibrium to the model involves no intergenerational resource transfers (in which case $R < 1$) or the equilibrium will have agents holding valued quantities of money, $m > 0$.

On the Fisher critique, Samuelson replies that in looking at the possible steady state equilibria of the model (which he refers to as “statical” equilibria), it is the interaction of consumer demand (determined by budget constraints and preferences) together with the resource constraints that determine the equilibrium prices and allocations, just as in Fisher or any other general equilibrium model. While this is a relatively minor point in both Lerner’s and Meckling’s critiques, it foreshadows coming work showing how differences in the distribution of resources, or the introduction of productive assets would affect the equilibrium interest rates in the model.

On the dynamic properties of the model, Meckling observes correctly – and Samuelson gives him credit for this -- that for a general three period model, the perfect foresight equilibrium dynamic system would involve a third-order difference equation, and that the Samuelsonian equilibrium for this model would not be forward stable. Samuelson acknowledges this and points to the fact that he demonstrated the same result in a two-period model with log linear preferences. He notes, though, that this was done in the context of showing that *laissez-faire* market forces would lead to a demonstrably suboptimal equilibrium.

For the general dynamic system, Samuelson then makes the observation that it is in fact indeterminate, since it consists of a third-order difference equation but only two initial conditions. He notes that for it to be possible for the system to recursively determine a sequence

of (non-stationary) equilibrium interest rates, a terminal condition would need to be appended to the model. In the absence of a central planner able to impose a condition such as Malinvaud's capital value minimization condition, there is no natural terminal condition for the OLG dynamic system. This observation also foreshadows a large body of work undertaken by subsequent generations of OLG theorists on the role of the dynamics in determining the optimality properties of the model.

As a final note on the dynamic systems analysis in Samuelson's reply to Meckling, the issue of whether or not a steady state was stable would merge with Samuelson's indeterminacy observation in what became a long-running (indeed, continuing) debate over the determinacy of equilibrium in OLG economies. At issue here is the distinction between whether or not unstable behavior can ever be observed. Meckling argues (in his quaint parable of Martian versus Terran economics) that the stable equilibrium is the only relevant one. Samuelson's response to this points out that stability depends on the boundary conditions imposed on the difference equation system, and that Meckling's example imposes a particular knife-edge condition which then gives rise to Meckling's stability result.

Curiously, Samuelson doesn't bring his knowledge of saddle-path stability in the capital model into the discussion, by way of suggesting that rational optimization can lead to otherwise unstable equilibria being determinate, in the modern sense. Indeed, in Samuelson-Solow (1956) the authors state that:

Our economic problem is like the unconventional case of a point of maximum rather than minimum potential energy - as e.g., a stiff pendulum with its bob upright. Touch it with a molecule and it departs ever further from the unstable equilibrium point.

For any point near the equilibrium there will be one special set of initial velocities such that [...] the resulting motion will asymptotically approach the equilibrium level. This

special motion - and only it - will truly minimize the Hamilton-Lagrange integral over all time. Such a motion would be unstable with respect to any exogenous shocks or displacements, but if an intelligence bent on truly minimizing Hamilton's integral guided the pendulum, it could re-aim the pendulum so as to seek a true minimum. This re-aiming is, so to speak, what an optimizing society is constantly doing. (p.548)

This process of “re-aiming” would be grist for the OLG theory mill for years to come.

On the question of whether the Samuelsonian equilibrium could be sustained, raised by both Lerner and Meckling, Samuelson makes the observation that while one can demonstrate the Pareto optimality of the competitive equilibrium for any finite truncation of the infinite OLG economy – which Balasko and Shell later dubbed Weak Pareto Optimality – determining full optimality for the infinite economy requires (as Samuelson puts it in footnote 6 of his reply to Meckling) a reformulation of the rules for optimality. Lerner and Meckling’s argument against the rationality of the Samuelsonian equilibrium is that if exponential population growth were ever to halt, the last young generation to pay into the social Ponzi scheme would not receive full compensation because of the slowdown in growth. This argument has been manifested in other forms, such as the fact that each new young generation always has an incentive to undertake a currency reform which eliminates the existing old generation’s claim to resources, while re-starting the Ponzi scheme in their own favor. Samuelson’s reply is that:

Laissez faire competitive pricing will, teleologically speaking, condemn a society to a negative interest-rate configuration that leaves, say, thousands of generations at a sub-Bentham optimum in order that the single terminal generation's property rights not be infringed. (p. 78)

This argument hardly settles the issue, as the contemporary debate over whether U.S. Social Security can survive the baby boomers’ retirement shows. What the argument does do is show that later work by David Gale, David Starrett, and Peter Diamond tried to address questions already raised by the earlier Lerner-Meckling-Samuelson debate, albeit with a different

focus, as will be seen below. This has not been noted until now. Moreover, the Lerner-Meckling-Samuelson debate can be seen to foreshadow later work on the core of an OLG economy.

We conclude our discussion of the Lerner-Meckling-Samuelson debate with a final quote from Samuelson's reply to Meckling, ostensibly made in the context of whether societies can adjust to changed economic conditions, but perhaps indicative of Samuelson's attitude toward the flak he was getting on the new model:

Now, aside from the resulting interpersonal equity problem between the generations, there may well be the noteworthy problem of "cultural lag," in which the modes of behavior of one epoch are painfully clung to even after the time for them is past. (p.77)

After the publication of Samuelson's seminal paper and the debate between Samuelson and the old guard about his new model's implications, it fell to a new generation of economic theorists to further the research agenda on overlapping generations models. These theorists – Peter Diamond (MIT), David Cass and Menahem Yaari (Yale), Karl Shell (MIT/Penn), David Starrett (Stanford) – were all trained in the then-new general equilibrium theory of Arrow and Debreu, and did their thesis work on applications of general equilibrium to growth. To this group we add a member of the old guard, David Gale, whose training in mathematics and interest in economic models led him to address some of the early OLG conundrums. We consider the key works of each of these pioneers in order of their publication.

But before this, we have to examine related work by Samuelson between 1968 and 1976. For example, in Samuelson (1975a), Samuelson wrote that in his earlier 1968 paper "The Two Part Golden Rule Deduced as the Asymptotic Turnpike of Catenary Motions" (Samuelson [1968]), both he and Diamond (1965) "combined the Solow model...with its pivotal role for capital, with the 1958 model of overlapping generations" (1975a, 533). In the same issue of

International Economic Review, Samuelson published “Optimum Social Security in a Life-Cycle Growth Model” (Samuelson [1975b]), in which he introduced “An exact model of steady state, exponential-growth social security...into the life-cycle model of Modigliani [1954, 1974], Diamond [1965], Cass and Yaari [1967] and Samuelson [1958]” (1975b, p. 539). Using a combination of his two-generation 1958 approach and Solow’s neoclassical model (1956), he dealt with the cases of no social security, as against the case where it exists, and then derived the characteristics of what he called “optimum social security” (1975b, pp. 541-543). He later acknowledged that this paper “was designed as a companion piece...and accordingly they were published back to back” (1976, p. 519).

However, it seems that Samuelson did not check the second order conditions in his optimum growth rate for population paper, which lead to a somewhat overlooked critique by Deardorff (1976), and subsequent admission of error by Samuelson (1976). While tangential to the main thrust of this volume, the exchange revolved around Samuelson’s extension of his 1958 model to include production, and thus deserves a brief mention here. Deardorff wrote:

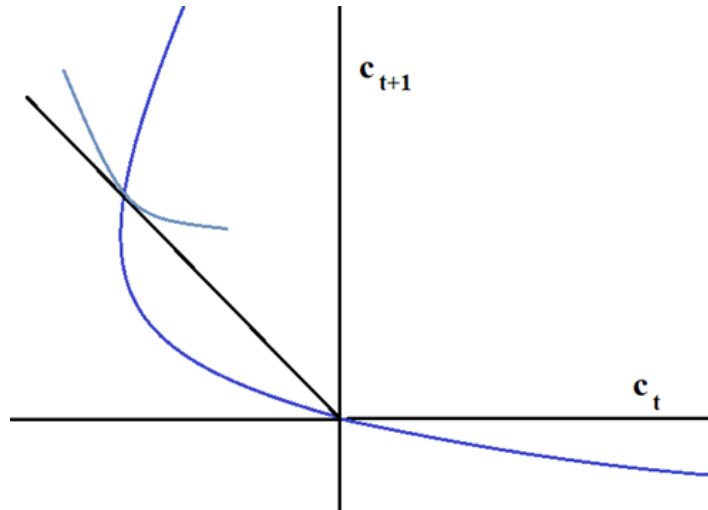
“In a recent contribution to this *Review*, Paul Samuelson...adapted his earlier consumption-loan model...to include production, and attempted to derive a rate of population growth that would maximize utility in steady state growth. His solution derived from necessary conditions for optimality, and he acknowledged the need for, but did not state or evaluate, the corresponding sufficiency conditions that would assure that his solution was actually optimal...Samuelson's solution is in fact *not* optimal in general. We concentrate primarily on the special case in which both utility and production functions are Cobb-Douglas. In that case we show that Samuelson's solution, for those parameter values for which it exists, provides a *global minimum* of steady state utility, not a maximum.” [italics in original] (p. 310)

Samuelson’s reply was “*Mea culpa*” (1976, p. 516), albeit he attempted to ease the severity of his oversight by posing further questions and tentatively pleading “*nolo contendere?*” (1976, p. 52).

Section 2.3 The Diamond Model

Diamond (1965) took the logical next step of introducing production into the model, extending Samuelson's pure exchange framework. One immediate and important effect this had was the demonstration that the introduction of productive assets would lead to equilibria in which interest rates were positive, as opposed to the case Samuelson highlighted when population was not growing, where the optimal interest rate was zero. Diamond developed his model under the assumption that capital never depreciates, and we can illustrate the nature of his results in a very simple setting by introducing a durable productive asset like land or a Lucas tree.

To illustrate the features of this equilibrium, we work with a two-period model and make use of the standard offer-curve diagram. In Figure 1, we show the simplest case of no population growth (or other exogenously driven productivity growth), with only money available for transferring wealth between periods. We center the offer curve on the (assumed stationary) endowment point for agents, under the assumption that young agents have more endowment than old, i.e., the case Samuelson initially examined. The dark blue curve is the representative agent's offer curve between consumption at time t and $t+1$. The 45-degree line in the negative quadrant shows the zero interest budget constraint, which is also the economy's resource constraint, since all saving gets transferred from young to old one for one. Where the budget/resource constraint crosses the offer curve, we have a competitive equilibrium. The light blue indifference curve shows the representative agent's optimal choice for this case.

Figure 1³

Standard offer curve diagram for two period OLG

We can illustrate Samuelson's biological rate of interest result in the same diagram (reproduced below as Figure 2) by introducing population growth, which makes more consumption available in period $t+1$ because there are more young agents in the economy than there were in period t .

³ Unless otherwise indicated, figures were produced by the authors.

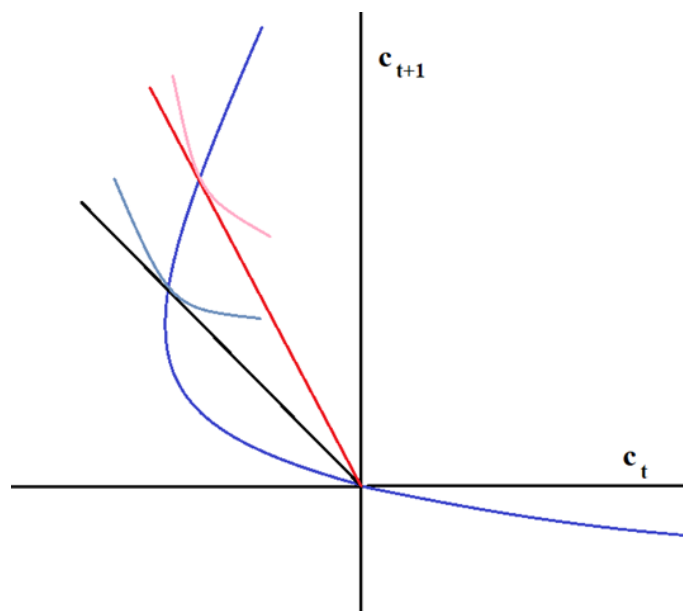


Figure 2

Offer curve with population growth

In this diagram, the red line (having slope equal to $1+n$, where n is the rate of population growth) is the economy's resource/budget line, and where this intersects the offer curve determines the equilibrium. Again, we put in the pink indifference curve to illustrate the representative agent's optimal choice.

If we now introduce a productive asset, *a la* Diamond, this changes the total resources available in all periods in the model. Specifically, if we have a Lucas tree which drops a dividend in the amount of $D > 0$, this will shift the economy's resource constraint out along both the period t and $t+1$ axes. We illustrate this in Figure 3 below.

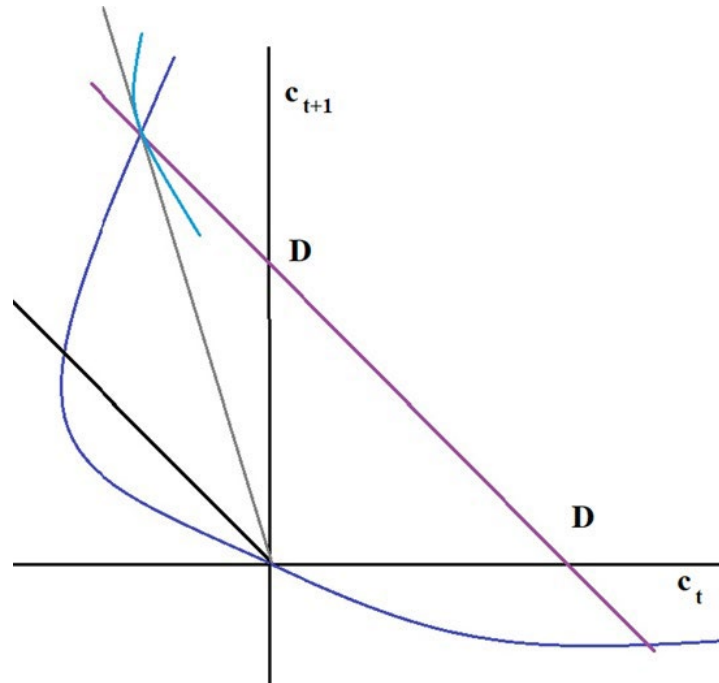


Figure 3

Offer curve with a productive asset

In this diagram, the offer curve is as before, while the purple line indicates the total resource constraint due to the productive dividend generated by the asset. The equilibrium for the economy is determined by the intersection of the resource constraint with the offer curve. The representative agent's budget constraint in this case, though, is different. It must pass through the origin (which, recall, represents the agent's endowment point) because the agent must use the sale of current endowments to finance the purchase of equity shares in the asset. The equilibrium price for the asset (in terms of consumption) is determined by the intersection of the agent's budget constraint with the offer curve at the point where the resource constraint crosses the offer curve. Since the slope of this line is greater than one in absolute value, the interest rate is positive. For reference, the dark line shows the old resource/budget constraint

absent a productive asset. As before, we indicate in light blue the representative agent's indifference curve at the optimal consumption bundle.

We can also use this diagram to illustrate what we will call the quasi-inefficiency of the asset equilibrium generated by over saving. Specifically, with no growth, the golden rule for balanced growth (see, e.g., Phelps [1965]) requires the interest rate to be zero. From the diagram above, it is clear that with a productive asset, the interest rate must be positive. The diagram below shows a blow up of Figure 3, indicating that optimal golden rule allocation, occurring where the representative agent's indifference curve is just tangent to the resource constraint:

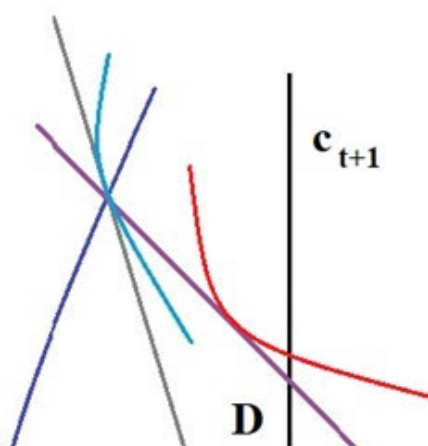


Figure 4

Offer curves showing competitive equilibrium and golden rule

We call this a quasi-inefficiency because, as should be clear from the diagram, in order to get to the golden rule allocation, a central planner would need to transfer resources from some initial group of old agents to the first group of young agents, and then maintain this transfer scheme in perpetuity. This redistribution makes every generation after the initial one better off,

but it makes the initial old worse off, and hence, cannot be characterized as a Pareto improvement. We will see below that this point was first made formally by David Gale in 1972.

Returning to Diamond's model, we can use our diagrammatic set up to illustrate the inefficiency identified by Diamond and detailed in the example in section 10 of the paper, using log linear utility functions and Cobb-Douglas production. Figure 5 below shows the equilibrium for a productive economy (again using the Lucas tree set up), where the competitive equilibrium generates an allocation and associated interest rate which is below the rate of growth of population. In the diagram, the purple line indicates the economy's resource constraint, the black line is the competitive equilibrium budget constraint for the representative young agent, and the light grey line is the budget constraint that would occur if the interest rate were equal to the rate of population growth. The same diagram can illustrate an efficient asset equilibrium by shifting the purple resource constraint out until it passes through or lies above the grey population growth line. The corresponding shift in the black budget constraint line will then show that the resulting equilibrium interest rate is greater than or equal to the population growth rate.

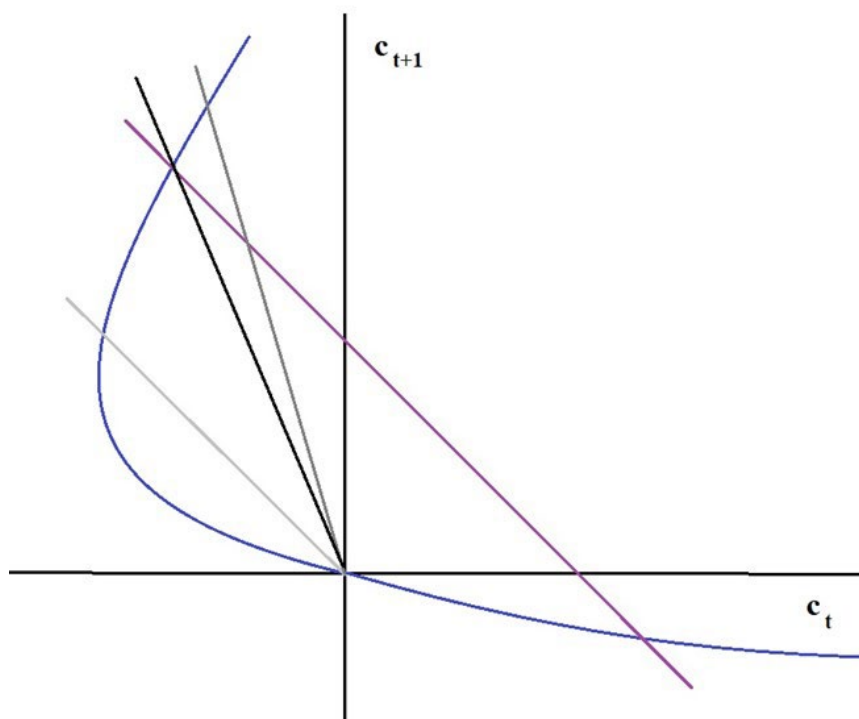


Figure 5

Offer curves for inefficiency with productive asset

Diamond then looks at the effect of introducing different types of government debt and shows that if the economy was originally operating efficiently (i.e., where interest rates are greater than or equal to the population growth rate), debt reduces welfare. When the economy is operating in an inefficient region, debt may improve or decrease overall welfare.

We can also interpret the diagrams we've presented here for the case where productive capital must be accumulated (as in Diamond's one-sector setting), by interpreting the dividend in the Lucas tree case as the amount of output net of undepreciated capital plus new investment along the steady state balanced growth path for the economy.

Before leaving Diamond's 1965 approach, we look at the development of his OLG approach over the period 1964-1973. After graduation from Yale as a mathematics major and a

period as research assistant to Tjalling Koopmans at Cowles (see Koopmans, Diamond and Richard Williamson, [1962], [1964]), he entered MIT as a graduate student in mathematics, but transferred to economics. (KDW [1964]). His MIT Ph.D. thesis, supervised by Solow, consisted of three chapters. The first chapter – his job market paper – was based upon his work with Koopmans on the infinite choice problem. After reading the thesis on growth of Srinivasan – his friend and colleague at Yale and Cowles – Diamond wrote a second thesis chapter extending Srinivasan's model (see Diamond [2014]). Then, following up on a suggestion by Solow, Diamond converted Salter's ideas about technical change, productivity, and cost reductions (Salter [1960]) into a growth model approach in a third thesis chapter entitled “Technical change and economic growth” (Diamond [1963]). What is interesting to note is that in the acknowledgments in his dissertation, Diamond thanked the members of his thesis committee, Samuelson and Fisher, and “the members of the intertemporal seminar – both in seminar meetings and outside.”

Diamond left MIT and took up a position at Berkeley, which included teaching a class in public finance, and a lecture he gave turned into a 1964 University of California at Berkeley Working Paper, which afterwards became his classic 1965 *American Economic Review* (*AER*) paper. As he recalled (email November 6, 2018):

In both 1963-4 and 1964-5, I taught a long year public finance course with a prerequisite in microeconomics, so it was for a small number of very good students. Having a whole year, I could go systematically through parts of the literature. When I was lecturing on topics where I did not follow an existing article closely, I prepared a handout. That lecture was the start of my use of OLG on the debt. I used OLG on taxation in two papers after I left Berkeley.

The 1964 working paper was issued by the Berkeley Institute of Business and Economic Research. The 1965 *AER* version contains additions in both text and footnotes, including

discussion of the effects of external and internal debt, the absence of bequests, Walrasian versus Marshallian stability, and changes in wording, albeit there were more substantial differences, to which we now turn. After submission of his working paper to the *AER*, Diamond received referee's comments (provided as an attachment to the email communication) and based on them changed and corrected text and equations, and re-drew a diagram. For example, the referee suggested generalizing the utility function in the example of the Cobb-Douglas production and utility functions that appeared in the working paper, and Diamond did this accordingly. Finally, the referee called Diamond's attention to the fact that Modigliani did in fact deal “with the feedback effect of the national debt on the stock of capital via reduced income” (referee's report, paragraph 4). In his final footnote in the 1965 *AER* version, Diamond added this accordingly.

Regarding the 1964 working paper and 1965 *AER* paper, and his interaction with David Gale while at Berkeley, Diamond recalled (email November 5, 2018) that he had:

[N]o memory of the exact version used at that time (1964), but I presented the model at the joint Berkeley-Stanford seminar at Stanford (as was the custom of those who presented) ... at the time, there was a good turnout at the joint seminars from both departments. I did note that Ken Arrow was there and seemed to find it interesting.... I did interact with David Gale while I was at Berkeley ... no memory of interacting with Gale on my paper – remember talking with him about his. My only memory of discussions with Gale is my suggesting to him that in a real model, which his was, it was wrong to interpret results as being about inflation.

Regarding his use of OLG in “two papers” after he left Berkeley, let us now turn to Diamond's 1970 *JET* and 1973 International Economics Association (IEA) Conference Proceedings papers. In an email dated November 6, 2018, Diamond recalled that he spent 1965-66 at Churchill College, Cambridge, where he wrote his paper “Incidence of an interest income tax” (Diamond [1967, 1970], subsequently published in *JET* in 1970). After he had returned to MIT in September 1966, he said that he “talked economics a lot” with Karl Shell, who had since

joined the department as an assistant professor. In the published version of the paper, Diamond wrote that he was indebted to Shell for “helpful discussion.” Diamond used a “two-period” OLG model for analysis (1967, pp. 1-2; 1970, pp. 211-212).

The second of his two post-1965 OLG papers was his contribution to the 1970 IEA conference that appeared in the Conference Proceedings (Mirlees and Stern [1973]). In this paper, entitled “Taxation and public production in a growth setting,” Diamond presented a welfare function as “a discounted sum of individual utilities where individuals are assumed to live for two periods” (1973, p. 215) and noted that “the presence of overlapping generations and the optimal growth model introduces two concepts of the intertemporal social marginal rate of substitution (1973, p. 216). He maintained that “in the first, one considers transferring the consumption of a single consumer over time; In the second, commodities are transferred between members of successive generations” (1973, p. 236). The advantage of this approach, according to Diamond, was that “the latter concept fitted with the discount rates that arise in standard optimal growth models, while the former is the concept which we can observe by watching individuals lend or borrow” (1973, Comments by Diamond, 236-237).

Section 2.4 Cass and Yaari

David Cass and Menahem Yaari also introduced a version of Samuelson’s consumption loans model with production, done in a continuous time framework, in 1965, though this was after Diamond’s paper was accepted for publication, since it is cited as forthcoming in the Cass-Yaari (1965) text. Like Diamond, they found that the competitive equilibrium generally did not

correspond to the golden rule equilibrium. This paper was published in *Essays in the Theory of Optimal Economic Growth* (K. Shell, ed.), MIT, 1967.

Cass and Yaari published a second paper on Samuelson's model, Cass-Yaari (1966a) the following year focused on what Samuelson had called the "infinity paradox," the fact that the overlapping generations framework could give rise to inefficient competitive equilibria, and that these inefficiencies carried over to models with productive capital as well.

To quote from the Introduction to their paper:

Professor Samuelson's article (1958) on an exact consumption-loan model of interest led to an interesting controversy. At issue were the determination and properties of interest rates in a dynamic economic system with no capital. One might have thought that these exchanges, between Samuelson (1959) and Lerner (1959) on the one hand, and between Samuelson (1960) and Meckling (1960) on the other, would result, if not in a complete resolution of the disagreements, at least in the emergence of a clear picture of what the issues are and how they might be treated. Unfortunately, such was not the case. The Samuelson-Lerner and the Samuelson-Meckling dialogues leave the reader rather perplexed, as though he had just watched a New Wave film -- executed with brilliance, enjoyable while in progress, but not quite clear as to what is happening and never giving one a sense of resolution. (*Ibid*, p. 353)

Working with the pure exchange, two-period model of Samuelson's paper, the Cass-Yaari analysis first notes that the only economic activity possible in such a setting is the distribution of total resources between members of the two overlapping generations. Their model assumes that young agents are endowed with one unit of the consumption good, while old agents have nothing. They then proceed to define the rate of interest for the economy as simply the return to consumption between an agent's two periods of life. Following their notation, we let c_t^a be the consumption of an agent born at time t in age a =young, old (or, for simplicity, young=1 and old=2). The interest rate is then given by

$$1 + r_t = \frac{c_{t-1}^2}{1 - c_{t-1}^1}.$$

With population growing at rate n , they then show that the economy's resource feasibility constraint is given by

$$c_t^1 + \frac{c_{t-1}^2}{1 + n} = 1.$$

After some simple algebra, they are then able to show that as long as $c_t^a \neq 1$ then it has to be the case that $r_t = n$. What this construct shows is that the rate of interest is simply (and, as Cass-Yaari note) mechanically determined by the resource constraints, without any appeal to preferences or biology. Quoting from the paper:

The main cause for suspicion seems to be the fact that a rate of interest has been determined without any reference to impatience and time preference or, more generally, to the utility function U . Somehow, the fact that r is a completely mechanistic construct, having no reference to markets, seems to have become blurred [in the Samuelson-Lerner and Samuelson-Meckling debates]. (*Ibid*, p. 355)

The second important clarification that Cass and Yaari make to the understanding of the OLG inefficiency is the demonstration that technology, *per se*, doesn't make the potential inefficiency go away. Specifically, they drop the assumption that goods are completely perishable and introduce a simple storage technology which allows young agents to transfer resources one-for-one between youth and old age. The analysis then shows that young agents would optimally choose some positive level of storage (as long as preferences are such that consumption smoothing is desired) and make the point (see Figure 2 in the paper) that this choice will lie strictly below the resource constraint generated by taking population growth into account.

One can also see from this analysis that even if population were not growing, the storage equilibrium is not efficient, since it requires that a positive quantity of resources sit idle rather than being used for consumption. Cass and Yaari note that, as in Samuelson's paper, trade in this situation is not possible, since no young agent will be able to find someone willing to borrow from him now in order to receive the (population growth augmented) consumption when he is old. So, once again, there needs to be some form of intermediation between the generations in order for the economy to move away from the inefficient autarky allocation. They show, however, that in conventional accounting terms, no private intermediary would ever willingly undertake the job since their net worth would always be negative. Hence, this task is one that government (either directly or through the exercise of its monetary authority) would need to undertake.

The middle section of the paper examines a three-period model in order to show that there can be a role for competitive internal credit markets, though, in the absence of a social institution for promoting intergenerational trade, the competitive equilibrium would remain one of autarky.

The last part of the paper is, in many ways, the most interesting, since it begins to delve in detail into the question of what causes the OLG inefficiency. In the section titled "Efficiency and Infinity," Cass and Yaari begin by noting:

The possible inefficiency (or non-optimality) of the competitive mechanism, as demonstrated by Samuelson and Diamond, has given rise to a certain amount of speculation, mostly on an informal basis. Many people (including Samuelson [for example, 1958, p. 474; 1959, p. 522] and Diamond [1965, p. 1134]) seem to feel that this phenomenon has something to do with infinity. What apparently leads one to point an accusing finger at "infinity" is the fact that for the standard general equilibrium model (which is finite) we have theorems which tell us that the competitive mechanism always leads to an optimum (and, a fortiori, to efficiency). Nevertheless, the role played by

"infinity" in leading the competitive mechanism astray has remained, at best, rather vague. (*Ibid*, p. 364)

Cass and Yaari then set up a finite model that (in its steady state) looks like an OLG model by truncating time and identifying the last period with the first, so that the model becomes circular⁴. In this setting, the OLG structure is manifested in the fact that agents have preferences over goods they don't produce and must, therefore, seek out mutually beneficial trades. By setting up this model with more than two periods, Cass and Yaari are able to show that there can be no double coincidence of wants, and that barter is therefore not tenable without further intermediation. In his later work with Balasko and Shell, Cass would refer to this friction as one of restricted participation on markets. In the OLG setting, the fact that agents live finite lives restricts participation on markets open when the agents aren't alive. In the circular model, this is captured via restrictions on preferences, via the observation that finite-lived agents can be viewed as infinite-lived agents who only care about consumption in finitely many periods. While the restricted participation argument is valid on its face, the circular structure Cass and Yaari impose on their example does permit the infinite shifting of resources among agents in the model, in the same way the normal OLG timeline does, so it can be argued that Cass-Yaari haven't really dispensed with infinity.

Cass and Yaari had a third OLG-related paper from 1966 (Cass-Yaari [1966b]) written after the "Re-examination" paper, though never published. In this paper Cass and Yaari elaborated on the relationship between inefficient competitive equilibrium without money, and the existence of monetary equilibrium. They show, in particular, that there can exist a valued fiat money equilibrium which is not Pareto optimal in a model where young agents have access to a

⁴ In private communication with us, Yaari recalls that they ended up considering this version of the model after being asked about the possibility by Herb Scarf at a Cowles Foundation seminar presentation of the paper.

storage technology (as in the “Re-examination” paper) if the price of money is low enough that agents wish to save via inventory. The paper also introduces the idea of a commodity money, which enters the first period utility function and shows by this example that non-monetary equilibrium (intermediated by commodity money) may be Pareto optimal. Interestingly, these results are a precursor to the results Cass later developed in his paper with Masahiro Okuno and Itzhak Zilcha, which we discuss below. It is also of note that Cass and Yaari anticipated Gale’s results (also discussed below) in footnote 8 of this paper, where they note that if the representative agent’s intertemporal marginal rate of substitution is bigger than one in absolute value, the associated allocation will be optimal.

Yaari (1965) was a solo authored paper, which dealt not with overlapping generations, but with consumer decision making when consumers faced uncertain lifetimes. In the paper, Yaari analyzed the effects of various considerations a consumer might face, including bequest motives, the need for life insurance, and the possibility of using annuities to finance retirement consumption. This work was later picked up by Blanchard (1985) who used Yaari’s result to develop what is now referred to as a “perpetual youth” model (when not referred to simply as the Blanchard-Yaari model). In this model, young agents are born every period (or instance, in continuous time models), and have a fixed (age-independent) probability of dying. These agents purchase annuities (among other assets) which closes the model in the sense that when agents die, their terminal wealth accrues to the issuer of the annuity. This formulation generates a type of OLG model, though because of the assumption that the probability of dying doesn’t vary with age, the model lacks much of the age-related heterogeneity found in the traditional OLG setting. We will discuss this model and variants of it later.

Section 2.5 Shell

The next paper we consider is Shell (1971), “Notes on the Economics of Infinity.” As the paper’s title indicates, Shell is also tackling the question of what drives the OLG inefficiency result, and his analysis is somewhat at odds with that of Cass and Yaari. Shell attributes the inefficiency primarily to the doubly infinite structure of the OLG model – infinitely many agents and infinitely many commodities (consumption bundles in each time period) – relative to that of the neoclassical growth model, where the number of agents (or agent types) is restricted to be finite.

Shell’s paper starts by demonstrating the inefficiency of competitive equilibrium in a particularly simple setting. He examines a two-period pure exchange model with one period overlap of generations, in which agents are endowed with and can consume a single good (which Shell calls chocolate). Preferences are linear, so that only the total amount of chocolate consumed over an agent’s lifetime matters. Endowments are initially set so that each agent owns one unit of chocolate in each period of life. To close the model, there is an initial agent who lives through period zero who only values chocolate consumed in that period and is endowed with one unit in that period.

With this set up, Shell then shows that there is a simple competitive equilibrium in which the price of chocolate in every period is one, and every agent consumes their endowment. (In general equilibrium terms, this would be a no-trade equilibrium.) Shell then shows that it is possible to Pareto improve on this equilibrium by having each young agent at time t give their endowment of chocolate to the old agent at time t . Given the linearity of preferences, no agent born after period 0 is made worse off, while the initial old agent in period 0 is made strictly better off.

This example can be generalized in a way that makes every agent better off. We start by defining transfers

$$(z_t^t, z_t^{t+1}) \in \mathbb{R}^2$$

where the first transfer is taken from (or given to) the young agent born in period t in that period, and the second transfer is given to (or taken from) the same agent in their second period of life $t+1$.

Now, fix two numbers $\epsilon > 0$ and $0 < r < 1$ and define the transfers as follows:

$$z_t^t = -\epsilon \sum_{j=1}^{t-1} r^j$$

and

$$z_t^{t+1} = \epsilon \sum_{j=1}^t r^j$$

Now, given any initial competitive equilibrium allocation $[x_t^t, x_t^{t+1}]_{t=-\infty}^{t=+\infty}$ we define a new allocation $[\hat{x}_t^t, \hat{x}_t^{t+1}]_{t=-\infty}^{t=+\infty}$ by

$$\hat{x}_t^t = x_t^t + z_t^t \hat{x}_t^{t+1} = x_t^{t+1} + z_t^{t+1}.$$

Given the linear utility specification, since

$$z_t^t + z_t^{t+1} = -\epsilon \sum_{j=1}^{t-1} r^j + \epsilon \sum_{j=1}^t r^j = r^t > 0$$

it follows that the hat allocation Pareto dominates the initial allocation. To show that the hat allocation is feasible, we note that

$$\hat{x}_t^t + \hat{x}_{t-1}^t = x_t^t + x_{t-1}^t - \epsilon \sum_{j=1}^{t-1} r^j + \epsilon \sum_{j=1}^{t-1} r^j = x_t^t + x_{t-1}^t.$$

So, since the initial allocation was feasible, the new allocation is also feasible. The last thing we need to do is make sure that we never have to violate the total resource constraint in making the transfers, i.e., that $|z_t^t| \leq 1$ (the endowment of each young agent.) Given r , this can be shown to require that $\epsilon < \frac{1}{r} - 1$. The reader will, of course, recognize this as a standard Ponzi scheme. Shell later shows (in the 1980's), in a series of papers written with Yves Balasko, that a lower bound on the curvature of indifference hypersurfaces through an allocation will rule out these kinds of transfer schemes and guarantee that the allocation is Pareto optimal. Since it is the curvature of indifference surfaces that determines an agent's time preference, this example reintroduces Fisher's ideas about the role of impatience in the determination of interest rates.

From this initial example of competitive inefficiency, Shell then makes a series of observations. First, and perhaps most importantly, he notes that the aggregate value of the competitive allocation when the interest rate is zero is infinite, which immediately implies that the competitive equilibrium prices are not in the dual space to the commodity space, which is l_∞ , in contrast to the results in Debreu's 1954 paper showing that with finitely many infinite-lived agents who discount the future appropriately, dual support prices always exist and, via the supporting hyperplane theorems, the corresponding allocations are Pareto optimal. While Shell doesn't say this in the paper, the infinite valuation of the competitive equilibrium also makes it impossible to apply the standard general equilibrium argument for proving the first welfare theorem.

In the next section of the paper (section 7), Shell then views the economy in the same way that Cass and Yaari did, from the general equilibrium perspective of a collection of infinite-lived individuals who only value consumption over two periods. Unlike Cass-Yaari, though, Shell views the inefficiency result as a matter of there being infinitely many agents, setting up a scenario akin to the so-called Hilbert hotel⁵, in which a hotel with infinitely many beds can always accommodate a new guest by having every existing guest move down one bed. In the economic case, Shell substitutes chocolates for beds, but the same idea is at work. This is also the reason that, in a finite truncation of the economy, the chain of improving transfers breaks down, and the original competitive allocation is then Pareto optimal⁶. Finally, Shell's observation on the infinite valuation of the zero-interest rate allocation is also relevant here, since truncation of the economy at a finite time renders the valuations finite, making the usual general equilibrium argument for the first welfare theorem operative. Both of these observations are formally codified in the Balasko-Shell papers.

To complete the discussion of Shell's implicit take on Cass-Yaari, he states, in discussing the potential fragility of Samuelson's social contrivance of money that the repudiation of intergenerational promises is not without empirical precedents:

In the actual dynamic world, "monetary reform" merely leads to frustration of the $(k-1)$ th generation's expectations. The most notable instance of such frustration is when the peasants' sons leave home for the city and fail to provide adequately for their parents' old-age. (*Ibid*, p.1007)

⁵ Shell attributes the example to George Gamov from his book *One, Two, Three...Infinity: Facts and Speculations of Science*. Gamov apparently neglected to cite Hilbert.

⁶ The same reasoning is at work in counter examples to the Brouwer fixed point theorem's applicability in infinite dimensional spaces; the ability to always shift over a dimension leads to a breakdown in the usual notions of compactness required by the Brouwer theorem.

In Cass-Yaari, an analogous situation would arise from either their inefficient storage equilibrium or the efficient monetary equilibrium the first time a young agent invented a productive technology and offered to sell ownership shares in the new technology to his peers.

We also note that Shell's double infinity interpretation is not inconsistent with the Cass-Yaari reasoning that restricted participation is the source of the Samuelsonian inefficiency, since Cass-Yaari maintain the possibility of on-going resource shifts by (in math terms) adding a point at infinity, which makes the time structure circular.

Shell concludes his paper by reconsidering the growth model's infinite-lived agent approach in the one-sector model and showing that a competitive inefficiency can arise there if agents don't discount the future. He shows this by converting the standard model into one equivalent to what Ramsey studied, with the long-run satiation point given by the golden rule steady state equilibrium. He then notes that with competitive agents holding stocks of capital sufficient to maintain the golden rule allocation, the zero interest rate at the golden rule equilibrium will lead these agents to liquidate all of their capital in order to increase consumption, leading to a deviation away from the optimal golden rule steady state.

Section 2.6 Starrett and Gale

Working at roughly the same time that Shell was, Starrett (1972) clarified a number of issues relating to the interaction of production and distribution in OLG models and refined the notion of efficiency in the model.

Starrett sets up a general OLG model with population growth, but one of the first important things he establishes is that one need not limit the notion of growth to simply that of population. If there is exogenous growth in total factor productivity (as well as in population), then this will have the same effect in terms of distribution possibilities as does simple population growth.

The second key observation that Starrett makes is that of distinguishing between models in which time is bi-infinite (running from $-\infty$ to $+\infty$), and those in which there is a beginning at $t=1$, with an initial “generation” of agents consisting of everyone born prior to the first period. The beginning in such a model can be considered either an artificial truncation of the bi-infinite timeline serving the purpose of the analyst, or as an actual beginning in terms of a new policy action or shock process impinging on the economy. Under either interpretation, Starrett shows that for the purposes of defining optimality, the two models are different.

For the bi-infinite model, the Samuelsonian or golden rule equilibrium is always optimal, since, as Starrett puts it

[T]he type of transfers we have been discussing require that everyone participate, and this means everyone in the indefinite past as well as the indefinite future. If the economy starts planning at some base period (so that the indefinite past is irrelevant), these transfers are not the relevant ones. (*Ibid* p.281)

Starrett then notes that if the economy has a beginning, the pre-existing old generation will never willingly engage in intergenerational trades for the simple reason that they can never be compensated for doing so. Again, quoting Starrett

Thus, in a model with beginning, there is a cost to consuming early; it is the cost of disenfranchising those alive at the beginning. Interest rates above the growth rate may be efficient after all, since they induce people to postpone consumption and hence help out the [initial] generations. (*Ibid* p.282)

This observation is then the basis for a key result extending the Diamond, Cass-Yaari, and Shell observations on efficiency showing that competitive equilibria in which interest rates exceed the economy's growth rate are efficient, while those with interest rates below the growth rate are inefficient.

The last paper we discuss in this chapter is Gale (1973). Gale was not part of the new crop of economists coming out of graduate school in the '60's with their interest piqued by Samuelson's new model. Rather, he was part of the old guard (having earned his Princeton doctorate in 1949), but as a mathematician with interests in economic dynamics, his interest in the OLG model was natural. We include his paper because of the overlap of its findings with those of Starrett. The coincidence of the two papers is also an example of the universal phenomenon of ideas that are "in the air" leading to near simultaneous publication, but where the senior author gets more recognition for the work than does the junior author. In this particular case, Google Scholar shows 67 citations of Starrett's paper (despite its publication in the top-tier *Journal of Political Economy*), while Gale's *JET* paper gets 690 cites.

Gale's approach to the OLG model was similar to Starrett's in that he first set up a general model that encompassed both the two- and three-period models considered by Samuelson and the later writers, though, like the others, Gale restricts his attention to the two-period setting for the explication of his results.

Gale identifies two distinct steady states for the general model, which he labels "balanced" and "golden rule" with corresponding equilibria which he calls (respectively)

“classical” and “Samuelsonian.” Gale’s use of the term “balanced” refers to the accounting sense of a balance of payments, rather than the notion of a balanced growth path. He characterizes the classical case of a balanced steady state equilibrium as one in which aggregate social savings (and hence debt) is zero. The Samuelsonian case has as its equilibrium the golden rule steady state.

Like Starrett, Gale defines optimality in terms of whether the transition from intergenerational autarky to the golden rule steady state is Pareto improving. In the context of the two-period model with only a single good, the autarky equilibrium is simply one of no-trade. Gale’s theorem 1 then states that a steady state must be either be the golden rule, or autarky, a result he characterizes as “amount[ing] to little more than a bookkeeping identity.” This identity had already been derived by Samuelson in mathematical form, which we expressed earlier as the identity $(1 - R)m = 0$, with (again) R being the gross rate of interest, and m the valued quantity of money held by agents in the economy.

The next part of Gale’s paper considers the stability properties of non-steady state equilibria, notably using the kind of offer-curve analysis that contemporary explications of the model rely on. Gale shows in this analysis that in the forward dynamics of the model, the optimal equilibria are always unstable. Specifically, he shows that in the classical case (no-trade, positive interest rates), any increase in saving will ultimately lead to an infeasible outcome, while any increase in debt will lead to movement away from autarky toward the golden rule steady state (though this must happen at the expense of the initial old generation). Since we cannot draw the graphic analysis of this situation any better than Gale did, we reproduce his diagrams below in Figures 6 and 7.

The first diagram represents the classical case, where endowments of the old are larger than those of the young. The curve labeled PQ is the offer curve, and, as the diagram indicates, the intersection of the offer curve with the downward sloping 45-degree line – which indicates the resource constraint that what one generation gives up, the other receives – at the point e is the optimal autarky equilibrium. The non-stationary equilibrium trajectories are indicated by the zigzagging lines between the offer curve and the resource constraint. One interesting thing to note from Gale's diagram (which he fleshed out later in the paper with an example, p.27) is the possibility that the equilibrium dynamics could cycle. This anticipates later work by, among others, Jean Michel Grandmont a decade later, though Gale considers it simply “an amusing example of a ‘business cycle’.”

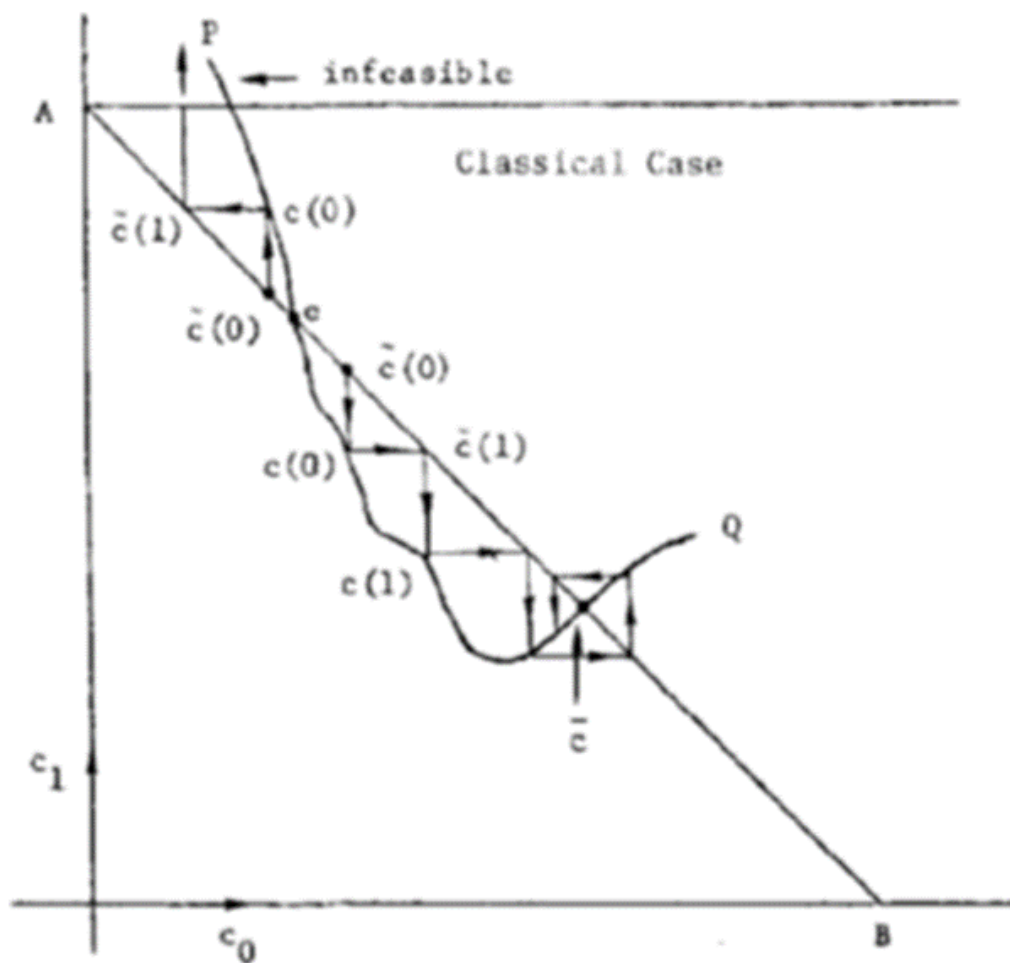


Figure 6

Gale's offer curve analysis of classical case

Copied from Gale (1973)

The next diagram illustrates the Samuelsonian case, where the endowments of the young exceed those of the old.

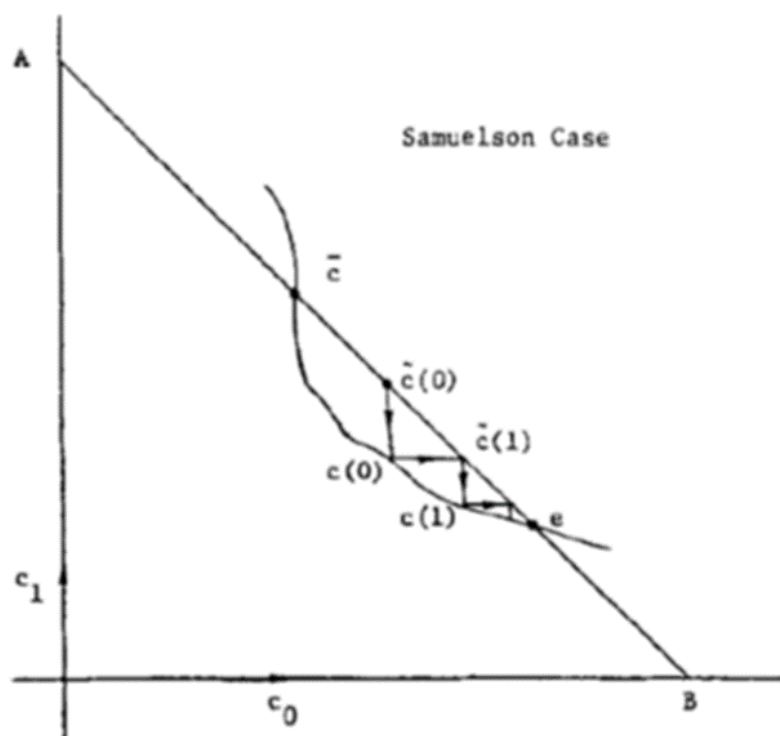


Figure 7

Gale's offer curve analysis for Samuelsonian case

Copied from Gale (1973)

In this diagram e is the endowment point, and $\bar{c}$ is the equilibrium (optimal) consumption point. The curve again represents the representative agent's offer curve, and the zigzag lines between the offer curve and the 45-degree line indicate the dynamics of the model starting from a small perturbation away from the equilibrium point on the right.

We note that Gale's focus on stability was similar to that of Meckling in his critique of Samuelson's analysis. Gale interprets these stability results in the paper, saying

[in the classical case]. Certainly, rapidly growing economies are objects of actual experience. Our findings suggest that the impetus to growth is given by starting conditions in which the [aggregate saving or debt] is negative. If [aggregate saving or debt] was initially zero it would take some kind of interference with the competitive mechanism to get things going (we discuss how this might be done in a later section) but the point is that such intervention would in our simplified world only be required once. Thereafter the invisible hand of prices could be allowed to take control and the economy would grow toward the golden rule optimum. It seems to me not inconceivable that things of this sort may be going on.

In the same spirit, what can be said about the perversely nonoptimal behavior in the Samuelson case? Here I expect we will look in vain for real life examples. While the Samuelson world is perfectly conceivable from a logical point of view it is probably not the one we live in -- exactly because of the empirical facts adduced by Fisher relating to impatience and investment opportunities. (*Ibid* p.17-18)

In the later section Gale refers to, he discusses how a transition from the competitive outcomes to the golden rule might come about, noting that in the classical case, a small altruistic transfer from the initial old to the young would then start the market moving down toward the golden rule, as in Figure 6 above.

For the Samuelsonian case, Gale introduces an intermediary (a la Cass-Yaari) which he calls a clearinghouse and suggests that the clearinghouse can issue some small initial credit to the existing old generation at the time (which he likens to Samuelson's social contrivance of money), allowing them to increase their second-period consumption by transferring the credit to the contemporaneous young agents. But Gale then notes that unless the clearinghouse can somehow coordinate the initial transfer so as to implement the golden rule equilibrium immediately, competition will drive the economy back to the inefficient autarky.

The odd thing about this discussion is its apparent unawareness of the determinacy issue initially raised by Samuelson in his response to Meckling, and particularly the possibility that rational forward-looking agents would find a way to make the jump to the appropriate equilibrium despite the simple forward-looking dynamics. What makes this even odder is that

Gale was perfectly aware of dynamic models of growth where the steady state equilibrium of the model is only saddle-path stable (see, for example Brock and Gale [1969].)

One additional aspect of Gale's paper merits attention. In the introduction, Gale states that

Samuelson's paper may be thought of as analyzing two special examples of the Fisher model using the population point of view. Regarded in this way, however, it presents one striking and perhaps curious feature. As Samuelson himself observes, his examples have characteristics exactly opposite to those considered to be typical by Fisher and Bohm-Bawerk. Instead of impatient people whose income is delayed, he considers people who receive income in the early periods of their lives but none at the end. This leads him to situations involving negative rather than positive interest rates. Thus, in introducing his new approach to the interest rate problem Samuelson has at the same time chosen to analyze an economic world which is in a sense the reverse of that of the classical writers. (*Ibid* p.14)

This statement would seem on its face to deny the reality of what Samuelson was trying to capture – that (at least in the modern world) people have to save for retirement. Samuelson was explicit about this in setting up the three-period model initially, where he notes that it is best able to capture the hump-shaped distribution of income by age. Much of the rest of Gale's paper then takes the two-period version of the model perhaps more seriously than it ought to as a way of explaining the empirical fact of positive interest rates by assuming that older people always have more money.

In this sense, the distinction between classical and Samuelsonian cases looks like an example of the old parable of the blind men and the elephant. Nevertheless, the distinction persists in the literature for the simple reason that even in general, multi-period exchange models, the golden rule steady state can coexist with a classical steady state in which interest rates are greater than the population growth rate. But, as Diamond and Starrett showed, once we introduce production, interest rates will always be positive (as long as capital is productive), but

competitive equilibrium may still be inefficient in a Samuelsonian sense, or efficient but not at the golden rule allocation.

The papers of Gale and Starrett, taken together, along with Samuelson's sections on the two-period model seemed to have had the effect of cementing the idea that two-period OLG models were somehow sufficient for general analytic purposes. While this would later be shown to be true by Balasko, Cass and Shell (1980), the habit of working in the simplest possible framework seems to have started here.

What is interesting to recall here is that Gale's 1973 approach had appeared in an earlier University of California Berkeley Industrial Engineering and Operations Research Department Working Paper entitled "Price Equilibrium for Infinite Horizon Economic Models" (Gale [1969]). The research was supported by the Office of Naval Research and done while Gale held a Senior Postdoctoral Fellowship from the National Science Foundation at the University of Copenhagen, Denmark. According to Gale's abstract:

The paper analyzes one-sector models of general equilibrium over an infinite time horizon in which there are an infinite number of agents, these being the members of successive generations. It is argued that such models are more realistic than the customary equilibrium models with finitely many agents. Three aspects of these models are then examined including: (1) non-optimalities in the form of productive inefficiency (as noted by Samuelson and others) and consumption non-optimality due to "inflationary" equilibria, (2) the role of "credit" and its effect on the equilibrium is discussed and (3) it is shown that for simple exchange models steady state equilibria are possible with a permanent imbalance of trade in which one country constantly exports to the other, a phenomenon which cannot occur in the classical model.

As we will see in the last chapter of the book when we examine the contemporary debate over the viability of the New Keynesian DSGE model, the issues Gale identifies here are precisely the ones that critics of contemporary macroeconomics feel should be addressed.

Chapter 3: Expectations and the Neutrality of Money c. 1972: From OLG to SOLG

While a purely logical approach to our topic would dictate looking next at the development of OLG models as models of money, the fact that history doesn't always proceed in neat logical steps dictates that we first deal with two important breaks in the logical flow. One, which will come in the following section, harkens back to the growth model and examines how the growth model's central planner became a dynastic family. The other, which we deal with here, is Robert Lucas' leapfrogging of the profession with his Nobel-prize winning 1972 paper (Lucas [1972]).

When we say Lucas leapfrogged the profession, we mean that while OLG theorists such as Barro, Grandmont and Laroque, Shell, Starrett and Wallace were working to address the criticisms raised by Lerner, Meckling and others of OLG as a model of money, Lucas simply accepted that (in the words of his Nobel prize lecture, Lucas [1996]):

[Samuelson's model] introduced a deceptively simple example of an economy in which money with no direct use in either consumption or production nonetheless plays an essential role in economic life. (p. 672)

He used the OLG framework to study the empirical puzzle in the relationship between money, prices and real economic activity. The puzzle, which Lucas first encountered in David Hume's 1752 essays *Of Money and Interest*, stems from the fact that in the long-run, money and prices are perfectly correlated, as one would be led to conclude from a simple reading of the quantity equation

$$MV = PY$$

holding velocity V and real output Y constant at their long-run (detrended) average values. Lucas illustrates this with a graph taken from McCandless and Weber (1995), which we reproduce below as Figure 8.

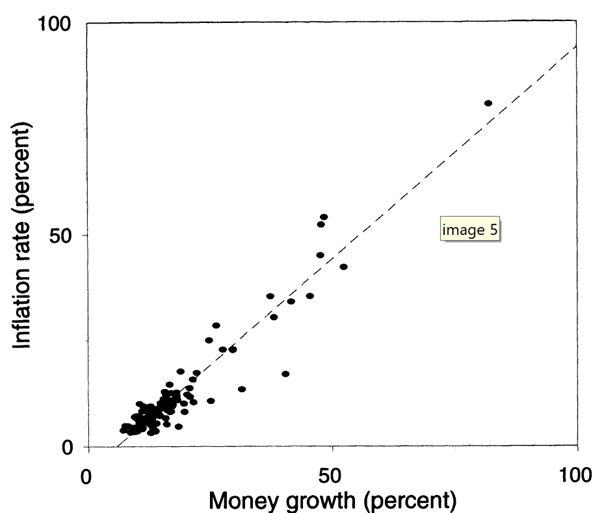


Figure 8

Long-run neutrality of money

The monetary policy puzzle is that over short periods of time, changes in the money supply do have real effects. Lucas illustrates this with a series of graphs showing scatterplots of the unemployment rates and inflation rates for the U.S. over a number of different, relatively short periods of time. This data is taken from Stockman (1996). We reproduce these below as Figure 9.

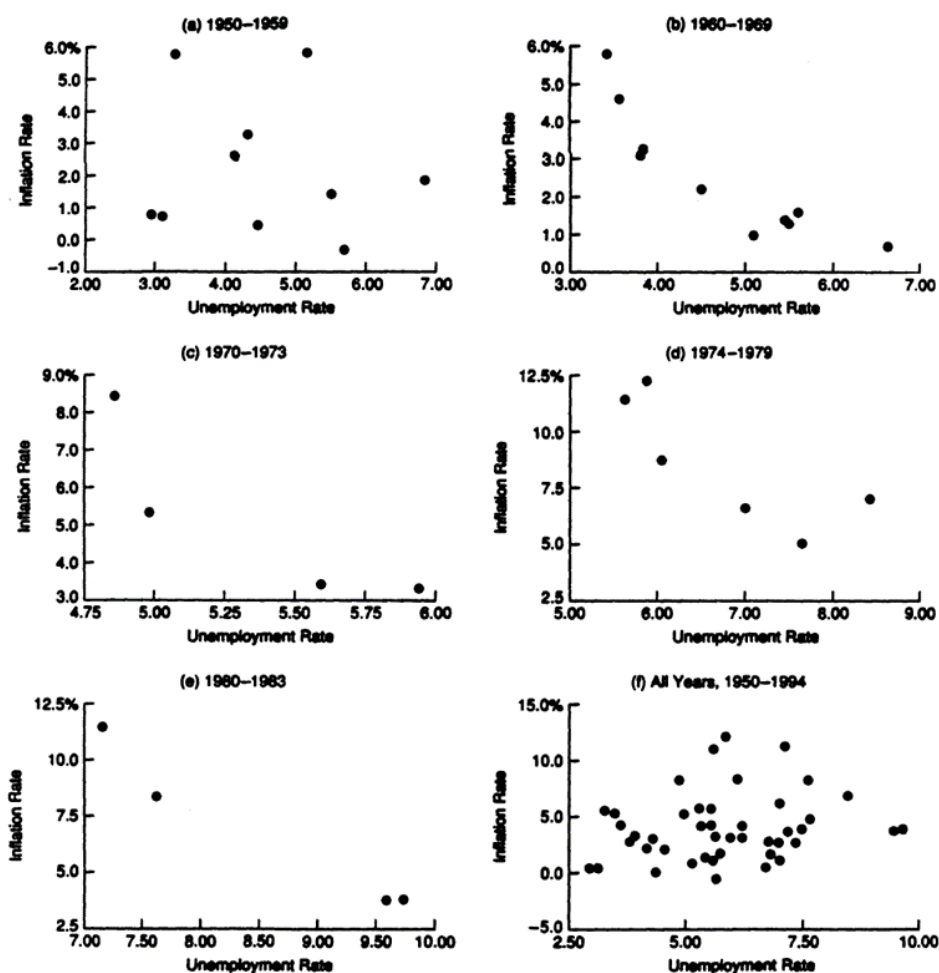


Figure 9

Short-run non-neutrality of money

The inverse relationship between these two measures seen in all of the graphs except the 1950-59 one (and the last one, which covers the long-run period from 1950-1994) is known as the Phillips curve (after William Phillips, who first noted the statistical regularity). It was this short-run relationship, together with the fact that there is no long-run relationship, that Lucas sought to explain theoretically using Samuelson's model.

In tackling this problem, Lucas made history. His training in macroeconomics, together with his rejection of the existing Keynesian approach of modeling aggregates directly, led him to think about how individual firms in the economy dealt with co-incident changes in both money and demand. This disaggregated approach was typical of the thinking of people Lucas was working with at the time at Carnegie Mellon University's Graduate School of Industrial Administration (GSIA). These people included Dave Cass, Marty Geisel, Leonard Rapping, and Thomas Sargent (per Lucas' *Professional Memoir*, Lucas [2001]), all of whom approached macroeconomics as a general equilibrium phenomenon in which equilibrium is determined by the collective actions of forward-looking, optimizing agents who cannot be treated as if they were molecules in a balloon.

Lucas's early foray into understanding the Phillips curve relationship involved work with Rapping, in which they tried to explain the Phillips curve relationship in terms of agents' money illusion and adaptive expectations (see Lucas and Rapping [1969]). Lucas discusses this paper and its reception by the profession in detail in his memoir, concluding with the following:

Before our paper was finished, Milton Friedman had used his Presidential Address to the American Economic Association to argue that in the long run, the unemployment rate would be independent of the inflation rate: There would be no Phillips-like trade-off between inflation and unemployment. Friedman's argument was theoretical, but his premises all seemed to Leonard and me to hold for our model. Yet our model did imply a long-run trade-off. Later, we came to see that this difference was due to our use of adaptive, rather than rational, expectations, but at the time we simply accepted it as an unresolved puzzle. (*Memoir*, p.18.)

The problem of adaptive expectations, of course, was that it generated a standing inconsistency between what agents in the economy were forecasting and what was actually happening. Earlier in his memoir, Lucas comments on this:

Dick Schramm's GSIA thesis showed how to make a multiple capital good model operational econometrically. His model involved a lagged adjustment of price expectations to movements in actual prices. I remember a seminar presentation of Schramm's at which Jack Muth asked Dick why he had not assumed rational expectations. Of course, we all knew and admired Jack's paper "Rational Expectations and the Theory of Price Movements," but none of us knew how to exploit this idea econometrically. At that time, I thought of rational expectations as an elegant rationalization of the kind of lagged adjustment of expectations that Schramm assumed in his thesis, not as an alternate model. This was a half-truth that kept me from seeing the force of Jack's objections. (*Memoir*, p.12.)

In his Nobel lecture, Lucas elaborates on the consistency issue:

As intertemporal elements and expectations came to play an increasingly explicit and important role, this modeling inconsistency became more and more glaring. John Muth's (1961) "Rational Expectations and the Theory of Price Movements" focused on this inconsistency and showed how it could be removed by taking into account the influences of prices, including future prices, on quantities and simultaneously the effects of quantities on equilibrium prices. The principle of rational expectations he proposed thus forces the modeler toward a market equilibrium point of view, although it took some time before a style of thinking that recognized this fact had a major effect on macroeconomic modeling. (*Nobel Lecture*, p.671)

It was Lucas himself who forced the profession to confront the inconsistencies of the *ad hoc* modeling of expectation formation, and he did so by pulling together the tools of stochastic general equilibrium theory he'd been using in the early work with Rapping, Samuelson's model of overlapping generations, and a set of newly learned results from functional analysis in the 1972 *JET* paper. We note that Lucas' choice of Samuelson's model was a natural one not only because it admitted a role for money, but also because at the time Lucas was working, his choice of dynamic models would have been the RCK model or the OLG model. The RCK model back then, as we've argued, was viewed as a planning model. Lucas was interested in examining how actual firms would respond to changes in the money supply or real demand in an environment where price signals confounded the two effects. This interest itself dictated that he use a model populated by actual agents, rather than one in which individual actions were all squeezed into a

planner's social welfare function. As we will see in the next chapter, Robert Barro's reinterpretation of the RBC framework as one of a finite number of dynastic families would end up nudging OLG aside as the model with the most realistic treatment of firms and families.

The model in the '72 *JET* paper is a two-period, stochastic overlapping generations model in which there are two sources of uncertainty. One source is the government's changes in the money supply, which Lucas models as a stochastic shock to the quantity of money. The other is a real, stochastic change in demand. The underlying idea behind this approach to explaining the short-run Phillips curve relation is that of *signal extraction*.

In the model, firms observe only the current price for their output, and this price can be affected by both the monetary and real shocks. Firms want to respond to the real shock, but not to the monetary shock. Under the assumption that firms form their expectations rationally, an individual firm's forecast of future prices will be the same as the economy's pricing function, which maps the stochastic state of the world (i.e., the monetary and real shocks combined) into a market-clearing equilibrium price. In Lucas' model, this mapping is monotone, so that firms, by observing the equilibrium price, can infer the underlying stochastic state of the world. This implies, for example, that if only one shock were operative, firms would know the state of demand or the money supply and make the optimal adjustment. In particular, if only the monetary shock is operative, firms will know this and take no action since nothing real has changed. In the opposite direction, if only the real shock is operative, firms will also infer the direction of change in demand from the price movements, and again respond optimally.

When both shocks are active simultaneously, firms must try to use their observation of price changes to extract the signal – real changes in demand – from the noise – changes in the money supply. Lucas shows in his model how the rational expectations assumption

operationalizes this process, and demonstrates that over the short-run, firms will make real adjustments after observed increases or decreases in the rate of inflation, but that over the long-run, the persistence of the money supply shocks relative to the real shocks means that real output fluctuates around a long-run average, independent of the rate of inflation.

We turn next to the technical accomplishments of Lucas' paper, which had a more profound effect on macroeconomics than did Lucas' actual explanation of the Phillips curve puzzle.

To develop these ideas, we first present a simplified version of Lucas' model. We simplify the model by focusing on pure exchange, and by imposing the simplest possible stochastic structure on the model: the endowments of the young fluctuate randomly with realizations being determined by the flip of a coin. Other than this, the model is the same. Agents live two periods, trade a single good, are risk-averse, and use fiat money to transfer wealth between youth and old age. As in Lucas' model, we assume that agents are in a Samuelsonian economy, so that their endowment when young (in either state of nature) exceeds their endowment when old.

Following Lucas, we focus on the stochastic equivalent of a steady state equilibrium. In this version of the model, the simplest such equivalent has the prices depending only in the exogenous endowment realizations, so we can write the prices as $p_t \in \{p^H, p^T\}$ (i.e., without any other time dependence). The rational expectations assumption in this case is captured by the assumption that agents know, conditional on the state of nature, that these prices will be the equilibrium prices.

With this specification, then, we can also write the lifetime allocation of the agent as $(x_1^s, x_2^{ss'})$, with $(s, s') \in \{H, T\} \times \{H, T\} = \{H, T\}^2$. Note that in the OLG framework, young agents are born into a realized state of nature, so that the only uncertainty a typical young agent faces is generated by the stochastic fluctuations in next period's price. The dependence of the second period consumption on the first period's state realization is through the fact that money holdings are determined in the first period of life and can, in principle, depend on that period's state realization. With this simplification, we can write a typical young agent's intertemporal optimization problem as

$$\pi^H u(x_1^s, x_2^{sH}) + \pi^L u(x_1^s, x_2^{sL})$$

subject to

$$p^s(x_1^s - \omega_1^s) + m^s = 0$$

$$p^{s'}(x_2^{ss'} - \omega_2) - m^s = 0$$

for all $(s, s') \in \{H, T\}^2$. We note that by adding the two budget constraints, we can convert the optimization problem into one with a single budget constraint. We write the constraints this way to emphasize the role that money plays in transferring wealth between periods.

Under standard assumptions on the utility function, the demand functions generated from this optimization will be differentiable everywhere. From the assumption that the economy is Samuelsonian, we know there will be a monetary equilibrium for the model.

To show the existence of an equilibrium for this model, write the excess demand functions of each agent as

$$z_1^s[p^H, p^T, \omega_1^s] = x_1^s[p^H, p^T, \omega_1^s] - \omega_1^s$$

$$z_2^{ss'}[p^H, p^T, \omega_1^s] = x_2^{ss'}[p^H, p^T, \omega_1^s] - \omega_2.$$

for $(s, s') \in \{H, T\}^2$. The equilibrium conditions for the model are then given by

$$z_1^s[p^H, p^T, \omega_1^s] + z_2^{s's}[p^H, p^T, \omega_1^s] = 0$$

for $(s, s') \in \{H, T\}^2$.

Strictly speaking this is a system of 4 equations in two unknowns. However, because of Walras' law, not all of the equations are independent. Note first that from the budget constraints, we have

$$p^s(z_1^s + z_2^{ss}) = 0$$

which implies that $z_1^s + z_2^{ss} = 0$ as long as $p^s > 0$. This reduces the market-clearing conditions to

$$Z^H = z_1^H + z_2^{TH} = 0$$

$$Z^T = z_1^T + z_2^{HT} = 0.$$

We now show the following: $Z^H = 0 \Leftrightarrow Z^T = 0$. To prove this, note that

$$\begin{aligned} p^H Z^H + p^T Z^T &= p^H z_1^H + p^H z_2^{TH} + p^T z_1^T + p^T z_2^{HT} \\ &= (p^H z_1^H + p^T z_2^{HT}) + (p^T z_1^T + p^H z_2^{TH}) = 0 \end{aligned}$$

since each of the expressions in parentheses is a budget constraint. The result then follows.

So, we are now left with a single equation in two variables. The two variables are not independent, however, since it is clear from the cross-state budget constraints that scaling both state prices by the same amount leaves the budget constraint unchanged. Hence, we normalize relative prices as $\rho = \frac{p^H}{p^T}$. To complete the existence proof, we have the following:

Proposition: Z^H is an odd function (i.e., as $\rho \rightarrow 0$, $Z^H > 0$ and as $\rho \rightarrow \infty$, $Z^H < 0$).

Proof: Write the cross-state budget constraints as

$$\begin{aligned}\rho z_1^H + z_2^{HT} &= 0 \\ \frac{1}{\rho} z_1^T + z_2^{TH} &= 0\end{aligned}$$

As $\rho \rightarrow 0$ non-satiation and the fact that endowments are positive implies that $z_1^H \rightarrow \infty$.

Since $z_2^{ss'}$ is bounded below by $-\omega_2$, it follows that $Z^H = z_1^H + z_2^{TH} \rightarrow \infty$. Next, as $\rho \rightarrow \infty$, $z_1^T \rightarrow \infty$ by the same argument. From the second budget constraint above, then, for large ρ , $\frac{1}{\rho} z_1^T > 0$, so that $z_2^{HT} < 0$. From the first budget constraint, we can write

$$z_1^H + \frac{1}{\rho} z_2^{HT} = 0.$$

Again, by desirability, as $\rho \rightarrow \infty$, $z_2^{HT} \rightarrow \infty$ and for large ρ we have $z_2^{TH} > 0$ and hence $z_1^H < 0$. Combining the results, we have for large ρ that $Z^H < 0$. By continuity then, there must exist a $\hat{\rho}$ for which $Z^H = 0$, and hence, an equilibrium exists.

For general stochastic structures, this kind of argument generally isn't available. What makes the example here transparent is the fact that the finitely supported stochastic structure generates a model in which the endogenous prices are finite-dimensional vectors, and the resulting model can be solved using standard finite-dimensional techniques.

In models with continuous stochastic structures, such as Lucas', we can view the prices as infinite-dimensional vectors (or equivalently, as functions), but the problem of solving for the zeroes of the aggregate excess demand functions (of which we now have infinitely-many) becomes (infinitely?) more complicated. In fact, since it is well known that finding the zeros for

systems of mappings is equivalent to solving a fixed-point problem, we can illustrate the complications that arise in general models via this kind of analysis.

So, suppose we have a continuous random state variable σ – for example we can let the first period endowment have a continuous rather than Bernoulli distribution. In order to solve their consumption optimization problems, young agents will need to forecast how they think the equilibrium price is going to depend on the stochastic state variable. Suppose the young agent in period t believes that $p_{t+1} = f_0(\sigma_{t+1})$.

If we take this forecast as fixed and solve for the market-clearing price given the forecast, we obtain what is known as the *temporary equilibrium*⁷ price function $p_t = f_1(\sigma_t)$. Now, in general, the forecast and temporary equilibrium price functions will be different. At the rational expectations equilibrium, however, we will have $f_1(\sigma) = f_0(\sigma)$, which is to say the rational expectations equilibrium price function will be a fixed point of the mapping from forecasts to temporary equilibrium price functions. Lucas himself notes this in his memoir:

Calculating a perfect foresight equilibrium for an industry, say, sounds as though it should involve solving a fixed point problem in a space of price paths: the actual path implies a forecast path, which in turn implies an actual path. (*Ibid*, p.13)

Lucas solved this problem in the '72 paper via an application of the Banach contraction mapping theorem. While graduate students today learn this result in the context of forward dynamic programming applications in both micro and macro, at the time Lucas used the result, it was not well-known and would have been encountered only by practitioners working with the then-new tools of dynamic programming. Lucas discusses his introduction to these tools in his memoir:

⁷ Temporary equilibrium models were introduced in the late 1970's by Jean Michel Grandmont. We examine his work in this area – which was based primarily on overlapping generations models – later in the book.

In my first years at GSIA I worked furiously to pick up the mathematics of optimization over time -- calculus of variations, the maximum principle of Pontryagin, Bellman's dynamic programming -- and of the differential equations systems these optimization problems produced. I worked on applications of these methods to problems in economics and operations research. Theoretically-minded economists of my cohort were doing the same thing all over in the world. I was studying the dynamics of investment at the firm and industry level, as were many others. Still others were studying optimal growth of an economy, optimal accumulation of human capital, optimal advertising and R and D investment, and so on. (*Ibid*, p.14)

Lucas goes on to say that it was Ed Prescott who first introduced him to dynamic programming. When we asked Lucas where he first encountered the contraction mapping theorem, he replied:

I was going over the CMU library copy of Bellman's book *Dynamic Programming*, and someone had written in pencil 'he means this is a contraction' at just the right place. I followed up this tip. I'm not sure Banach was mentioned there but somehow I got to Banach's book in French. I got a French Canadian student to help me to translate -- which was easy because it was mostly math symbols! Later on Prescott told me about David Blackwell's great paper. (Private email communication, September 11, 2014.)

In follow-up communication, Lucas added

My paper appeared in print in 1972 but was written in the late 60s. If had given the reference (as I should have!) that would send you to the right place, but I don't think I did. Anyway, this application was not to a Bellman equation, though the idea of a contraction is certainly there. This [forward dynamic programming] is all later than my early work with Prescott, who had introduced me to Blackwell's work.

In terms of my education, the connection of this use of Banach is distinct, I think, from the "contraction" someone wrote in Bellman's book.

The student would probably have been Marcel Boyer, who is now emeritus at Montreal. Marcel wrote an outstanding thesis that used the idea of a Banach space in a central way. Another possibility is Leon Courville, who went on to a successful career in banking. (Private email communication, September 11, 2014.)

In addition to opening up a new research trajectory for him, Lucas' introduction to Bellman's and Blackwell's work also led to a new approach to teaching graduate

macroeconomics. In the early 1980's, Lucas' lecture notes on dynamic programming had been typed up by someone unknown to us. These notes were later expanded on and published as Stokey and Lucas (1989), which is now one of the most widely used texts in graduate economics education.

We will return to discuss this book in more detail later, but we note here that the only material in the book on overlapping generations models is in section 17.1 which essentially reproduces Lucas (1972).

Chapter 4: Infinite Lived Agents: Dynasty, 1970-1980

As we indicated at the beginning of the previous Chapter, we need to make a second digression in our story to examine how the other main workhorse model in dynamic economics – that of the infinite lived representative agent – came to be understood as a model of finitely many dynastic families. This interpretation is in marked contrast to the structure of the OLG model, which consists of infinitely many unconnected individual families.

As we noted in Chapter 1, the original interpretation of the optimizing agent in the early development of the neoclassical growth model was that of a central planner. The planner's objective function could be simply aggregate consumption, an undiscounted social welfare defined over aggregate consumption, or a discounted social welfare function. The infinite "life" of the central planner was an approximation of the fact that planning institutions, or more generally, governments, typically outlive the individual people who populate the society over which the government reigns. Treating the government or planner as being infinitely lived (particularly when the objective function is discounted) is then simply a mathematical shortcut rather than any kind of profound statement about the organization of society

This interpretation was not universal, as we noted above. Debreu's 1954 extension of the finite-dimensional Arrow-Debreu-McKenzie model to allow for infinite-dimensional commodity spaces simply assumes that there are finitely-many consumers whose preferences are defined over the commodity space. At this level of abstraction, there was no reason for Debreu to ponder the nature of his consumer household, so there was no corresponding discussion of planners versus families. Debreu's result, however, has been widely used in applications of the growth

model, since it implies that one can always find a competitive equilibrium by solving for a Pareto optimum, i.e., by solving the central planner's problem.

Cass (1965) foreshadowed the reinterpretation of the planner as a dynastic family by showing how to fully solve the growth model with the planner's objective function being the discounted utility of consumption. Cass himself never interpreted the growth model in anything but normative terms. While we don't have direct evidence of this in Cass's writing, his long-time friend and co-author Karl Shell shared his remembrance of Cass's view in an email:

[Dave] was irritated by the application of "Ramsey-Cass-Koopmans" to non-planning problems. Dave and I agreed (possibly with somewhat different emphases) that optimal control problem is a central planning problem. (Email communication, August 21, 2021.)

When the RCK model was translated into discrete time and solved using dynamic programming techniques, the interpretation of the objective function as belonging to a long-lived private-sector entity became natural.

The first explicit interpretation of the infinite-lived agent as a family appears in Arrow and Kurz (1969). Their analysis focuses explicitly on the optimization problem faced by a dynastic family which they define as consisting of a parent household which is concerned with "the welfare of all its present and future members." (p.68) They also choose not to work with bequests, noting:

The reason for our interest in this problem is the inadequacy of the common formulation of the problem as a consumer optimization over a finite horizon with a bequest motive. The point is that the bequest function itself is part of the problem; that is, the individual's bequest motive is founded on his concern with the welfare of future members of the family, and thus the infinite horizon formulation makes the bequest function part of the problem to be solved in connection with a continuous process of intergenerational transfers. The process of bequest itself is not as discontinuous as would be implied by the finite horizon formulation since the most important form of bequest is education. (p.69)

This is in contrast to Yaari (1964) which considers a finite lived family's optimization problem given an exogenously specified bequest function. It is the arbitrariness of the bequest function in Yaari's paper that Arrow-Kurz object to and eliminate from their analysis.

Arrow and Kurz go on to set up a model in which bequests are implicitly defined based on the solution to the control problem they solve to characterize the dynastic family's optimal allocation of consumption and wealth.

Gale also considered bequests in Gale (1973) (discussed above) by way of generalizing his key findings for the two-period model to a general multi-period setting. Gale's treatment of bequests is not one of trying to define an optimal bequest, or (as in Yaari [1964]) to determine optimal or equilibrium consumptions in the presence of bequests. Rather, he allows the bequest process to be arbitrary, and considers the aggregate bequests that are possible in the steady state of the model. Following Gale, we let A denote aggregate assets (including bequests), ρ the equilibrium gross interest factor, and γ the gross rate of population growth. Gale's main finding (Theorem 5) is that

$$(\rho - \gamma)A = 0$$

which implies that either the economy is at the golden rule equilibrium steady state, or the asset accumulation is zero, meaning that there are no intergenerational transfers occurring. Gale does not link the possibility of bequests in the OLG setting with the neoclassical growth model, nor does he consider the question of family structure in this analysis.

The final step in completing the reinterpretation of the central planner as a dynastic family comes in Barro (1974). Barro starts with an overlapping generations model but adds the possibility that parents can leave bequests to their children. To simplify the model, Barro

assumes a representative lineal family structure. He deals with the bequest function problem identified by Arrow and Kurz by assuming that the parent generation cares about the next generation's welfare (i.e., their utility), so that that preferences in the model are non-paternalistically altruistic. Of course, this assumption ignores the caveat in Arrow and Kurz that for most families, the major bequest from parents to children is education, whether in the form of taxes paid to support public education or direct expenditures on private education.

With this assumption on altruism, Barro observes that this gives rise to intertemporal preferences of the form

$$V_i = U(c_{iy}, c_{io}, V_{i+1})$$

where the function U is an aggregator over the consumptions of generation i and the future preferences of generations $i+1$. Barro then shows that given this specification of the family structure and preferences, the OLG model is equivalent to a model with infinite lived agents, as long as the bequests between generations are strictly positive (or, in Barro's terminology, operative).

While Barro did not make use of the new tools of forward dynamic programming in his paper, the model he proposed meshes perfectly with the mathematical methods of forward dynamic programming, and we can use this framework to indicate the equivalence of Barro's model with the RCK model. We note first, though, that these techniques were formally introduced in the context of economic growth by Radner (1967), although there were indications throughout Bellman's writings about the applicability of dynamic programming to problems in economics.

To illustrate this applicability, let us consider a simple one-sector capital model with a single good that can be used either for consumption or as capital in production. Since this example is the one that Stokey *et al.* use to motivate the dynamic programming tools in their *Recursive Methods* text, our digression here will also show why the forward dynamic programming approach has come to dominate the optimal control methods previously used to analyze problems like these.

We assume that labor supply is fixed at one unit and inelastically supplied by households in each period. Production is given by a standard production function $f(k)$ which is strictly increasing, strictly concave, and satisfies $f'(0) = \infty$. The infinite lived household in each period must allocate current output $y_t = f(k_t)$ between current consumption and future capital (or future capital plus undepreciated current capital, which for simplicity we assume is zero). The household's preference over current consumption is given by a standard concave utility function $u(c)$ which is also strictly increasing and satisfies an Inada condition at $c=0$.

With this specification, the reduced form period t utility of the household will be denoted

$$V(k_t, k_{t+1}) = u(f[k_t] - k_{t+1}).$$

As is standard in the capital theory literature, we refer to the function V as the *return function*.

The household's intertemporal optimization problem is then one of choosing a sequence of capital stocks $\mathbf{k} = [k_0, k_1, \dots, k_t, \dots]$ to maximize intertemporal utility, which is given by

$$W(\mathbf{k}) = \sum_{t=0}^{\infty} \beta^t V(k_t, k_{t+1})$$

where $0 < \beta < 1$ is the discount factor.

If we look at the one-sector optimization problem as we've defined it above, we note that since the return functions V are strictly concave (given the assumptions on utility and production functions), the household's objective function is concave. Hence, if standard concave programming theory holds for this class of problems, the first-order conditions should be necessary and sufficient to characterize the solution. The first-order conditions (also known as the Euler equations) for the problem here are

$$V_2(k_{t-1}, k_t) + \beta V_1(k_t, k_{t+1}) = 0, \text{ for } t = 1, 2, \dots$$

Here, the subscripts on the V function denote partial derivatives with respect to the first and second arguments. Since $V_{12}(k_t, k_{t+1}) = -u'(f[k_t] - k_{t+1}) \neq 0$ the implicit function theorem tells us that we can solve for k_{t+1} as a function of the other variables,

$$k_{t+1} = g[k_{t-1}, k_t]$$

which is a second-order difference equation. We will show that in fact, the correct optimal solution to the problem is given by a first-order difference equation $k_{t+1} = h(k_t)$. We will discuss why the current approach doesn't give a full solution after this.

For the dynamic programming approach to this problem, we define an arbitrarily given *continuation* or *value* function for the problem $W_0(k_{t+1})$, where the 0 subscript denotes an initial choice of value function (implying the possibility that the function can change over time), and

we assume that the value function tells us the optimized value of our capital accumulation process from time $t+1$ forward, given that we start with capital k_{t+1} at time $t+1$. With this definition, we can define the actual value of the accumulation program at time t with capital stock k_t by

$$W_1(k_t) = \max_{k_{t+1} \in T(k_t)} \{V(k_t, k_{t+1}) + \beta W_0(k_{t+1})\}.$$

Here $T(k_t)$ is the constraint correspondence that restricts the choice of future capital to lie in the set $\{0 \leq k_{t+1} \leq f(k_t)\}$.

This functional equation – functional because it takes the function W_0 and produces a second function W_1 – defines an operator T on the space of functions in which the value function lives. If we can find a fixed point $\hat{W} = T(\hat{W})$, we would then have the value of the optimization program defined recursively in terms of itself. Let $\hat{k}(k)$ be the solution to the maximization above when $k_t = k$. By iterating the recursion at the fixed point, we get

$$\begin{aligned} \hat{W}(k) &= V(k, \hat{k}[k]) + \beta \hat{W}[\hat{k}(k)] \\ &= V(k, \hat{k}[k]) + \beta V(\hat{k}[k], \hat{k}^2[k]) + \beta^2 \hat{W}[\hat{k}[k], \hat{k}^2[k]] \\ &\vdots \\ &= \sum_{t=0}^{\infty} \beta^t V(\hat{k}^t[k], \hat{k}^{t+1}[k]). \end{aligned}$$

Since we are maximizing the value of the return function at each period of time, we are clearly maximizing the intertemporal discounted sum of return functions, so this is the solution to the optimization. The functional equation at the fixed point is, of course, the Bellman equation

$$\widehat{W}(k_t) = \{V(k_t, k_{t+1}) + \beta \widehat{W}(k_{t+1})\}$$

and the key thing to note about this is that (assuming differentiability of $\widehat{W}$), the first-order condition for the maximization is

$$V_2(k_t, k_{t+1}) + \beta \widehat{W}'(k_{t+1}) = 0.$$

If the value function is twice-differentiable⁸ we will generally be able to solve (at least implicitly) for k_{t+1} as a function $k_{t+1} = h(k_t)$. This is clearly a first-order difference equation, as opposed to the second-order equation that arises from the Euler equations for the direct approach to solving the model.

The efficacy of the forward DP comes from the fact that it converts the infinite-dimensional direct optimization problem into a recursive finite-dimensional one. Instead of having to solve an infinite-dimensional optimization, we can solve a standard finite-dimensional one. As long as the various functions involved are concave – and it can be shown that as long as the return function V is concave, $\widehat{W}$ will be – then standard concave programming results apply, and the first-order conditions will be necessary and sufficient to characterize the solution. Of course, the algorithm does not (and cannot) banish the underlying infinite nature of the problem. Rather, it replaces one infinity – the discounted sum of utilities – with another – the value function, which is itself an infinite-dimensional object.

⁸ Conditions under which this will obtain were established independently in Araujo (1991) and Santos (1991).

The potential difficulty of working with this new infinite-dimensional object should be clear from looking at the functional equation that gives rise to the Bellman equation. The operator T defined by the equation maps functions to functions. When we impose the typical assumption that the functions are continuous and bounded, we specify the domain of T as an infinite-dimensional normed linear vector space. It is well-known that finding fixed points of mappings on such spaces can be problematic. While the Brouwer theorem (for example) remains true, getting sets to be compact in these spaces involves significant restrictions on the subspaces one can consider, which limits the generality of the results obtained.

Fortunately for the dynamic programming framework, there is one fixed point theorem that applies very generally and works for the forward dynamic programming algorithm. This is none other than the contraction mapping theorem that Lucas saw penciled in the margin of Bellman's book. Contraction mappings don't require stringent assumptions on the underlying spaces they operate on, they have unique fixed points, and have the property that one can pick a starting value for the mapping and then iterate the map until it converges on the fixed point. This last property is particularly useful for computational applications.

In the discounted dynamic programming case, Bellman's student David Blackwell provided a pair of easily verified sufficient conditions for a mapping to be a contraction: that it be pointwise monotone and contracting on constant functions. These conditions have seen widespread application throughout dynamic economic analysis.

From this discussion, we can also see why the first-order conditions to the direct problem are only necessary. Since resources are bounded, the underlying commodity space that our infinite lived agent operates in is l_∞ . The return function and discounting map sequences in this space into a sequence of summable utility payouts, which lie in l_1 . Since the resource inputs into

the return function are bounded from below, we can (possibly with translation) view the sequence of discounted utility payouts as living in l_1^+ , the space of summable sequences with non-negative elements. This, in turn, means that the intertemporal value function can be viewed as the sum of the discounted return function payouts, which will have its graph in the space $\mathbb{R} \times l_1^+$. Because l_1^+ has empty interior, so will $\mathbb{R} \times l_1^+$. Hence, when we try to set up the supporting hyperplane argument used to show that the first-order conditions are also sufficient, the application of Minkowski's theorem fails, since this requires that at least one of the two convex being separated has a non-empty interior⁹.

For the direct problem, then, one needs to look at optimal control techniques for the necessary and sufficient conditions, and, as Cass showed in his 1965 paper using methods from Pontryagin's *Maximum Principle* text (Pontryagin [1960]), the additional required condition is the transversality condition.

The analysis here applies directly in Barro's model (based as it is on Diamond's OLG model with production) once one specifies a linear aggregator function U and assumes that parent households discount their offspring's utility.

The combination of Barro's results and the ease of application of the forward dynamic programming algorithm proved to be a heady mixture for macroeconomics, and it led to an explosion of work based on the neoclassical growth model with the optimizing agent interpreted as a dynastic family. Foremost among the new applications were the extension of the model to

⁹ John Tukey published the first counterexample to the proposition that the interiority condition in Minkowski's theorem could be dropped. Interestingly, he published his paper "Some notes on the separation of convex sets" in the journal *Portugaliae mathematica* in 1942 in order to ensure that the German mathematicians would be able to access the paper despite the wartime embargoes on trade in place at the time.

stochastic environments by Brock and Mirman, and the application of this model to the study of business cycles by Kydland and Prescott.

This new interpretation of the growth model was not without its criticisms, and we will examine these later. But in the meantime, chronological time dictates that we first look at what was happening in the OLG world in terms of its role as a model of money, led primarily by Neil Wallace in the late 1970s, and the work of general equilibrium theorists in this period on understanding the dynamics of Samuelson's new model.

Chapter 5: OLG and Money, 1970-1980

Lucas' 1972 "Expectations and the Neutrality of Money" paper opened the door for monetary theorists to explore the effects of money and monetary policy in the overlapping generations model, in spite of the objections raised by Lerner and Meckling in their commentary on Samuelson's paper about the lack of robustness of the monetary equilibrium.

Section 5.1 Neil Wallace

The pioneer in applying Samuelson's new model to monetary economics was undoubtedly Neil Wallace. Wallace was a member of the 1960's cohort of graduate students whose training in macroeconomics was based on the Keynes-Samuelson notion that macroeconomics could be separated from individual optimizing behavior (what we now call the microfoundations). From Wallace's *Macroeconomic Dynamics* interview (Altig and Nosal [2017]), we know he wasn't satisfied with the contemporary approach to macro and monetary economics:

Tom Sargent and I were trying to build models which were elaborations of IS-LM. I won't speak for Tom, but I was stuck in a static mode. We had the view that things depend on people's expectations of inflation, and we wanted to endogenize the expectations. Our approach was to be clear about stocks and flows in IS-LM. But when you start thinking about that, you're naturally led to think about some kind of dynamics. At the time, I was also working with John Kareken and Tom Muench. Muench was a good operations researcher, so to him the idea that an equilibrium condition is a difference or differential equation didn't seem strange. But to me, that was just way out there. So although we were thinking about dynamics, it was in a very static way. Then I saw this paper by Lucas; I don't know why I read it, or tried to read it. In part, I did because Bob had been a classmate at Chicago and I had a high opinion of him. I picked up the paper, and it's talking about people, two-period lived people. What's that? There are no people in macroeconomics! I pretty much saw that the Phillip's curve ideas that Sargent and I had been working on were a dead end. That paper (Lucas '72) raised the standard, and there was no turning back. (*Ibid*, p.1794)

Wallace's first paper using the OLG framework to study monetary policy issues was "On Simplifying the Theory of Fiat Money" (Wallace [1977a]). This paper was a largely non-technical piece arguing that monetary economics was about explaining patterns of exchange, and why fiat money has value, with the paper's focus on the second question. In the report, Wallace sets the stage by stating:

My goal in this paper is to try to convince you to take seriously as models of fiat money various versions of the Samuelson/Cass-Yaari overlapping-generations consumption-loans model. (*Ibid*, p.1793)

Wallace's key argument is that the OLG demographic structure and the need for people to save for retirement creates a natural demand for assets. He then sets up two models, one as in Lucas '72 where production using only labor generates output, and the second including a durable asset (land) which may be productive. The two models generate differences in the demand for money in the sense that if land is productive, holding equity shares in land will dominate holding money as a means for transferring wealth between periods. Wallace then notes that while this poses a conundrum for a model purporting to explain why fiat money can have value, this conundrum is one that all models of money would share:

The case is made by summarizing and interpreting what these models have to say about fiat money and by arguing that these properties are robust in the sense that they can be expected to hold in any model of fiat money. (*Ibid*, p.1792)

As Wallace states later in Kareken and Wallace (1981) on foreign exchange, "For a model of a fiat money economy to be acceptable, it must then be possible for the money (or if there are several, any one of them) to be dominated" and it is this feature that Wallace holds to be robust across any viable model of money. One implication of this requirement is that money not appear as an argument in the utility or production functions of the model. Rather, its appearance is restricted to the budget constraints of the individual or firm. This edict was at odds

with the view of money as a medium of exchange that cut through the inefficiencies associated with barter. We will discuss this controversy later in this chapter.

Wallace also addresses the tenuousness of the monetary equilibrium generated by the fact that every new generation has an incentive to repudiate the previous generation's money and reissue new money (also known as the seniorage problem). Wallace reiterates his contention here that any other form of money would also create this perverse incentive, and that only government restrictions can eliminate this possibility (*Ibid*, p.7). This observation raises an important feature of assets in the OLG setting. Wallace's discussion of the possibility of rate of return dominance between assets implicitly assumes that it is possible for agents in the model to share ownership of the productive technology. In practice, this requires the original inventor/owner of the technology to create a system for sharing ownership rights in return for payment from new shareholders. This system takes the form of the stock market, and what is traded there are pieces of paper stating that ownership rights, in a given proportion, have been transferred to the holder of the piece of paper. But this paper can be just as easily repudiated as a given generation's money issuance. Absent some enforcement mechanism, a firm could simply claim that stock certificates are fraudulent and refuse to honor requests for dividends. It is civil society and the rule of law behind these financial institutions that remove the tenuousness not only of fiat money, but of all paper assets. This argument also extends to real assets like land, which, absent a mechanism for enforcing ownership claims, can be seized as easily as any more portable asset.

Wallace (1977b) followed shortly on the heels of report no. 22 and provided the technical details that were missing in the first report. This report used a discrete-time version of the Cass-Yaari model, in the sense of providing a storage technology with the possibility for storage to

yield strictly positive returns (wine, for example) as well as an active, deterministic monetary policy. The key result of the paper is an affirmative technical demonstration of the assertions in the previous report, that when the real rate of return to storage exceeds that of money, there can be no monetary equilibrium in the model. The rate of return dominance question was one that ultimately led Wallace to reject the OLG framework as being the best model of money. From the interview:

For a while there, I thought the overlapping-generations model was going to be the model of money. My attitude was to try to put that model to work in a number of applications. A classic problem in monetary economics, identified by Hicks back in 1935, is why money exists when there are higher return assets around. I had this idea that if we legally inhibit intermediation, then rate of return dominance would emerge. I wrote a couple of papers about that using an overlapping-generations model. In retrospect, I think the profession was right to reject that view of money. (*op. cit.* p.1795)

Despite the pessimism this result generated for Wallace, the paper itself made a key observation about the determinacy issue that Samuelson first raised. This observation was based on the possibility of unstable forward dynamics of non-stationary perfect foresight equilibria, and Wallace made the point that unless the economy were literally starting *de novo*, then the choice of an initial price in period 1 would be constrained by the expectations of the young in period -1 in order to satisfy the perfect foresight assumption.

A third OLG-related paper to come out of this period was the well-known Kareken and Wallace (1981) paper. This paper was highly influential in the debate over whether foreign exchange rates should be fixed or allowed to float freely. Kareken-Wallace use an OLG model to show that under the *laissez-faire* free-floating exchange rate regime, the monetary equilibrium in the OLG model is indeterminate in the sense that the money demands for agents across the two countries can't be pinned down. This paper, together with the less well-known Kareken-Wallace (1977) paper established the OLG model as a workhorse for international trade.

Section 5.2 The 1978 Minneapolis Fed Conference

In 1978, Kareken and Wallace organized a conference held at the Minneapolis Fed on “Models of Monetary Economies.” As their introduction to the conference volume states:

The Federal Reserve Bank of Minneapolis sponsored the “Models of Monetary Economies” conference precisely because some of its research staff were so doubtful about macroeconomic models, or about the monetary and fiscal policy implications of those models. The hope was that the Bank might, by financing the conference help in the development of more satisfactory models. (Kareken and Wallace [1980], p.1)

This conference attracted many of the major scholars working on OLG models and in monetary economics, and it generated a deep and impactful debate about how economists should think about money. The various papers presented, together with the comments of assigned discussants, and several written replies submitted after the conference, were put together in a conference volume edited by Kareken and Wallace, entitled “Models of Monetary Economies” issued by the Minneapolis Fed in 1980. Since many of the papers and commentary dealt with OLG models, it is worth reviewing this material.

The papers presented in the conference were:

1. “The Role of Money in Supporting the Pareto Optimality of Competitive Equilibrium in Consumption Loan Type Models” (David Cass, Masahiro Okuno, and Itzhak Zilcha)
2. “The Overlapping Generations Model of Fiat Money” (Neil Wallace)
 - a. Discussion by James Tobin
 - b. Discussion by Jose Scheinkman

3. “The Capital Stock Modified Competitive Equilibrium” (Martin Shubik)

4. “Equilibrium in a Pure Currency Economy” (Robert Lucas)

a. Discussion by Leonid Hurwicz

b. Discussion by Milton Harris

c. Discussion by Frank Hahn

Post-Conference contributions were:

5. “The Optimum Quantity of Money” (Truman Bewley)

6. “Some Remarks on Monetary Policy in an Overlapping Generations Model” (William Brock and Jose Scheinkman)

7. “Transaction Demand and Moral Hazard” (John Bryant)

8. “Money in Consumption Loan Type Models: An Addendum” (David Cass)

9. “In Defense of a Basic Approach” (David Cass and Karl Shell)

10. “General Equilibrium Approaches to the Study of Monetary Economies: Comments on Recent Developments” (Ross Starr)

11. “Models of Money with Spatially Separated Agents” (Rob Townsend)

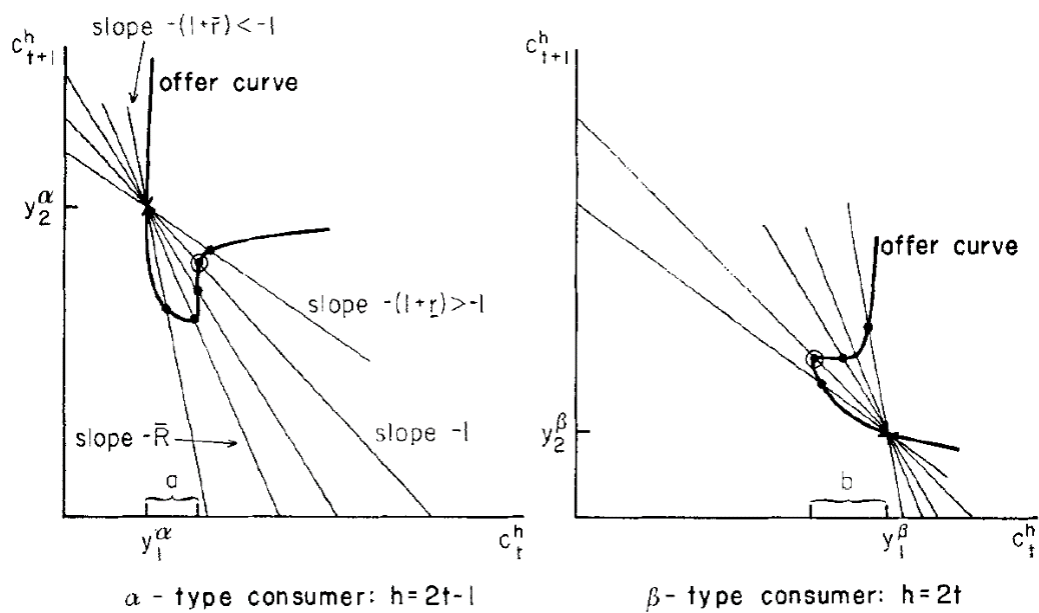
The Cass-Okuno-Zilcha paper is something of a constructive contrarian exercise aimed at showing that two of the most commonly asserted features of the two-period lived OLG model don’t hold. The paper characterizes these features in terms of two propositions about existence

and optimality of equilibrium. The existence proposition states that there is a monetary equilibrium if and only if there is no barter equilibrium which is Pareto optimal. The optimality proposition states that if there is a monetary equilibrium, then there is also a monetary equilibrium which is Pareto optimal. Via a series of counter examples, the paper shows that:

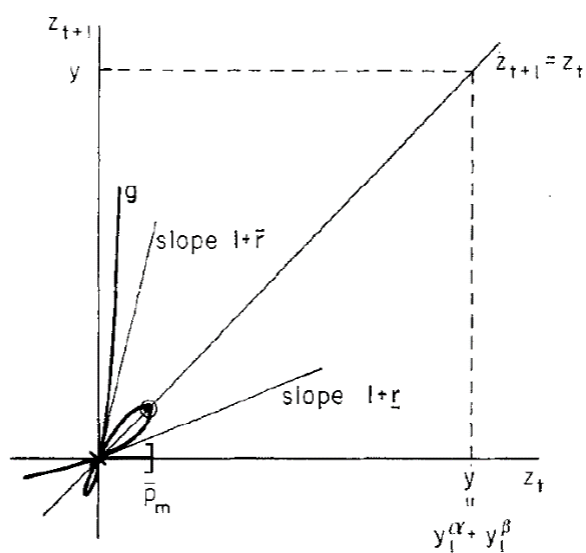
1. There can be both monetary and non-monetary equilibria in the model, both of which are Pareto optimal.
2. There can be both monetary and non-monetary equilibria in the model, neither of which is Pareto optimal.

The notable thing about this paper was its early application of the community excess demand decomposition results of Debreu, Sonnenschein, Mantel and others to construct the offer curves for a single good, two-period lived OLG economy with at least two agents.

We illustrate the constructive analysis of the paper using the diagrams from the paper itself in Figure 10 (from p.54 of the paper):



4a. Consumer Behavior



4b. Dynamical System

Figure 10

Cass-Okuno-Zilcha offer curve analysis

Copied from Cass *et. al.* (1979)

The diagram shows two offer curves for two different consumers in the top two panels. The third panel shows the result of adding these two offer curves to get the aggregate offer curve, which has been recentered at the endowment point and reflected to show savings rather than consumption. Equilibria occur where the aggregate offer curve crosses the 45-degree line, and as is apparent from the diagram, there are two equilibria: one with strictly positive aggregate saving (the monetary equilibrium), and one at autarky but with a positive interest rate. This corresponds to example (1) above.

The paper provides a similar construction (see p.55 of the paper) to show example (2). While these counter examples do demonstrate the possible contradictions that can arise in the model, the authors note that the examples are, in fact, special, in the sense that

these specific examples embody somewhat extreme behavior, since (partly) vertical or horizontal offer curves implicitly require a region of inferiority for the good whose consumption is unchanged. (*Ibid* p.56)

The key takeaway from the paper that the authors highlight is

[The] nonoptimality examples strongly suggest that -- in general -- something like continuous redistribution by means of creation (or destruction) of fiat money may be an indispensable lubricant for the efficient operation of an evolving market economy (*Ibid*, p.44)

While it isn't clear whether these results had any influence on the standard conduct of monetary policy, the use of the excess demand decomposition theorems was picked up four years later and fruitfully applied by Kehoe and Levine, as we will see in the next chapter. As we will also see later, Cass himself provided a general proof, for the case where agents have standard preferences in which all goods are normal, that there always exists a steady state equilibrium in which the interest rate is greater than or equal to zero.

Before proceeding, let us turn to the early development of the work of Okuno and Zilcha, before their collaboration with Cass. Okuno's 1974 Stanford Ph.D. was supervised by Kurz, with Shell, who was visiting Stanford, on his thesis committee. The thesis contained a chapter entitled "Equilibrium in an Infinite Horizon Model," which as Okuno recalled, while never published, was "a starting point of my joint research" with Zilcha (email April 4, 2020). Regarding how his collaboration with Zilcha came about, Okuno recalled (email April 4, 2020) that:

In my second year at Illinois, Itzhak came as an assistant professor and his office was next to mine. Because he knew a lot about math (especially infinite space topology, etc., which I knew very little). So I proposed to do joint research on OLG models, which he happily accepted.

When asked how Cass became a co-author of the Feb 1979 *JET* paper – and Cal Tech WP Dec. 1978 put out by Cass while visiting there – which was originally prepared for the Minneapolis Fed Conference, Okuno said:

We (I and Itzhak) wrote a small paper and sent to either him or Karl, who invited us to their conference. Dave pointed out some mistake about the paper, and we started to find some example which was going to play a key role in our joint paper. Dave was excited about it and wrote the entire paper by himself with our example in it.

Then, when asked why he stopped publishing papers about problems of OLG and money, Okuno replied:

My interest then was to find a sufficient condition for the existence of monetary equilibrium (price of fiat money being positive at the eq). I thought we had more or less solved this problem (approximate monetary eq., etc.) and also I started to think the issues of credit and trust are more important research topic than the fiat money. Moreover, when I came back to Illinois after a one year leave, those people I used to talk with (Itzhak, Andy Postlewaite, Rich Kihlstrom, David Schmeidler, etc.) were absent (most of them on leave), and I had to find a different research topic.

In an additional email dated April 5, 2020, Okuno wrote "I did not attend Minn conference and I don't recall participating in the work of writing the [Cal Tech] WP."

Regarding his work with Okuno and Cass, Zilcha, for his part, in an email dated April 4 2020 recalled:

First, we both were assistant professors at the Department of Economics, University of Illinois. (by the way, the theory group at this college of Business Administration included Len Mirman, Rich Khilstrom, Andy Postlewaite and Alvin Roth). I joined the department in 1975.

We...listened to a lecture by Neil Wallace in our seminar in which he claimed that in the Samuelson OLG model with fiat money there is a unique monetary competitive equilibrium (starting at date 0 and extending indefinitely). Okuno and I started working on this issue trying to contradict this claim. We had examples only, then we have written our *RES* paper which was published in July 1980 (See Okuno and Zilcha [1980]). This paper was submitted first to JET and rejected. Later in a conference David Cass told us that we did not write it well etc. but he was interested in the issue of existence of monetary equilibria and their Pareto optimality etc. We started collaborating with him and had nice results. In the meantime I moved to Cornell for one year and later to Tel Aviv University. I did not attend the Minnesota Fed conference.

Continuing with our discussion of the conference papers, Wallace's conference paper reprises the basic arguments in the two previous staff reports, though this paper puts more emphasis on the role of money as a store of value, and further amplifies the contention that money should never be an argument in an agent's utility function. On the store of value issue, he states:

One claim seems to be that the overlapping-generations friction accounts for the "store-of-value" function of money only. As a consequence, it is argued, models built on it are quite misleading. In particular, the tenuousness of equilibria in which fiat money has value in models built on the overlapping-generations friction is due, according to this claim, to the fact that this friction does not account for the "medium of exchange" role of money. But this claim fails to recognize that tenuousness is an implication of the two defining properties of fiat money, inconvertibility and intrinsic uselessness, and not of the overlapping-generations friction. (Kareken-Wallace [1980], p.50)

In contrast to his defense of the store of value role created by the OLG setting in the two staff reports, Wallace downplays the role of legal restrictions, simply stating that "one can impose legal restrictions that give fiat money value." (Kareken-Wallace [1980], p.50) This

reluctance to emphasize the role of government in the management of money leaves the door open for one of the discussants (Tobin) to claim that “there is no government commitment to the value of fiat money.” (*Ibid* p.85). It would remain for Cass and Shell, in their after-conference contribution, to argue that government debt – money or bonds – can serve no purpose if it doesn’t serve first as a store of value.

Wallace’s paper was followed by two discussions, one from James Tobin, and the other from Jose Scheinkman, so we turn next to these.

Scheinkman’s discussion turns on the technical issue of the stability of the monetary equilibrium, which he describes as the source of tenuousness in the value of fiat money. Based on this, he develops a very clever model of infinite lived agents who can trade on one of two markets. The first requires money for trade (implicitly imposing a Clower-type constraint), while the other market operates by barter. Trade in either market produces a desired composite good, and the trading mechanisms are such that neither market dominates the other. In this setup, then, Scheinkman shows that it is possible for a monetary equilibrium to become asymptotically demonetized, just as can occur in the OLG setting. He also identifies a condition on the agents’ common utility function which precludes this asymptotic result and interprets this as making money essential – in its medium of exchange role – for the existence of long-run monetary equilibria. This condition also appears in models in which real balances enter the utility function, for the same reason: money is so essential for trade that even if the government runs a large inflation tax, people will continue to hold large amounts of money.

The last part of Scheinkman’s discussion disputes Wallace’s contention that the monetary equilibrium in a standard OLG economy cannot survive if there are assets available that pay strictly positive returns. Scheinkman argues that if agents live three periods and have an asset

such as land or a tree available that can be purchased when young but only liquidated in their final period of life, then agents will wish to hold money in order to transfer wealth between their first and second periods of life.

We view this latter criticism of Wallace's contention as one of introducing a market incompleteness argument, since absent money, agents can only transfer wealth from period one to period three. In principle, this is no different than having a stochastic model with at least two states of nature, and one asset that generates positive returns. In this model, introducing a second asset – money, for example – will allow agents to improve risk-sharing and hence, money will have value.

While none of the conference papers or discussions address the stability issue in detail, we would note that, as in the earlier Samuelson-Meckling debate, questions about stability and determinacy are closely interrelated and depend strongly on whether or not rational agents' forecasts are subject to errors which can lead to the economy diverging from an optimal, monetary steady state equilibrium.

Tobin's discussion is less technical and, possibly because of this, both more contentious about the focus of the conference, but also prescient in anticipating the direction of future research in monetary economics.

Tobin makes six objections to the contention that OLG should be the model for monetary economics.

1. The Cass-Okuno-Zilcha results show that money will have value only when the non-monetary equilibrium interest rates are below the growth rate of population, but history shows us that money has always existed independently of interest rate conditions.

2. Historically, there have always been alternatives to money – land in particular – that serve as stores of value
3. Again, citing the Cass-Okuno-Zilch paper, money will only have value if society exhibits the life-cycle pattern of saving followed by dissaving, which has not always been the case historically, but in these societies, there was still a role for money.
4. Via Barro’s linkage of OLG and ILA, Tobin argues that even with such “constructive immortality” (which he acknowledges isn’t realistic) he would expect that money would have some value.
5. If one accepts the two-period set up of the OLG models examined in the conference, it implies that people hold money for long periods of time, on the order of 25 years, which has no empirical foundation.
6. Fiat money should be dominated by a formal government transfer arrangement such as Social Security, where young generations paying into the PAYGO fund would be more likely to trust a formal agency than fiat money. In Tobin’s words:

The answer is that explicit promise carries more conviction than the decentralized market expectation; there is no governmental fiat commitment to the value of money. (*Ibid* p.85)

These criticisms stimulated the Cass-Shell rebuttal, which we discuss in some detail below. The remaining two substantive sections of Tobin’s discussion make a number of important points which seem to predict the evolution of monetary theory in the hands of search theorists like Kiyotaki and Wright, and many others. Among Tobin’s observations in these sections are the following:

1. Money is a public good. Like language, it provides a way of communicating that will be understood universally within a society.
2. Money economizes on the information required to communicate relative prices, in the same way that it economizes on the multiple coincidence of wants needed for effective barter-mediated exchange.
3. Although contracts can substitute for money in mediating exchange, the need to write down contingency clauses in contracting can become quite costly, another example of the importance of search costs in exchange. In this context, money functions as an insurance policy against the possibility that rare unanticipated contingencies lead to the breakdown of trade itself.
4. When markets are incomplete, money can co-exist with other assets to facilitate risk-sharing.

Before we leave Tobin's discussion, we should note the irony in the fact that Tobin co-authored a paper in 1981 with Walter Dolde, (Dolde-Tobin [1981]) which used simulations of a deterministic life-cycle model to evaluate the welfare costs and benefits of a pay-as-you-go Social Security system. In these simulations, agents live for 6 periods, which provides a significantly finer grid on which to examine agent behavior relative to the simple two-period lives of the models Tobin was criticizing at the Minneapolis conference. This simulation work was a direct precursor to later work by Summers, and by Auerbach and Kotlikoff using simulations of multi-period deterministic OLG models to examine a variety of macroeconomic issues. The use of simulations for policy analysis was also a precursor to the practice now

known as calibration. While we will discuss this work later in the book, a good overview of Tobin's OLG work can be found in Dimand (2022).

The Shubik and Lucas contributions to the conference volume, and the discussions of these papers deal with non-OLG models, so we omit detailed discussion of them here. Lucas' contribution is noteworthy, though, because taken together with Lucas (1978) it represents Lucas' switch away from OLG in favor of ILA models. We know this from Lucas' Introduction to his *Collected Papers on Monetary Theory* (Gillman [2012]) where he states that

When I moved from Carnegie-Mellon to Chicago in 1975, I began to teach in the first-year sequence in macroeconomics, a practice I maintained for more than twenty-five years. For the core (as we called it) I thought I needed a coherent approach to the basics of monetary theory, which I thought of as the quantity theory of money and the evidence that supported that theory. I wanted to do this in a way that made maximum use of modern general equilibrium theory. I'm sure I began with Samuelson's 1958 overlapping generations model and the developments of that theory due to David Cass, Menahem Yaari, and others, but I became dissatisfied with the abstractness of that framework and sought a view in which money is an asset that we hold to pay bills with, a factor in a *payments system*. (Gillman [2012], p.xviii)

Examination of the "Pure Currency" paper from the Kareken-Wallace volume or the "Asset Prices" paper, makes clear that Lucas viewed the overall benefits of the infinite lived representative agent approach – its amenability to forward dynamic programming tools, its adaptability to time-series data of any frequency, and its straightforward extendibility into stochastic regimes – as sufficient reason to prefer a model that, at best, describes the economic workings of the upper 5% of the income distribution.

As we will see going forward, Lucas was not the only macroeconomist to abandon the OLG framework in favor of the representative agent model, so, as we noted in the Introduction, a key question for us is why this happened. At the time Lucas was working on these two papers, his decision was, perhaps, understandable. If one compares the steady state predictions of

Diamond's OLG model with those of the standard RCK model, the two look quite similar. As we will see later, one of the key OLG projects of the 1980's was Rao Aiyagari's work looking at whether OLG and ILA models were observationally equivalent. To the extent that the outputs that distinct models deliver look the same, then, using the model which is simpler to analyze and more easily adapted to the data available seems like an easy call.

In fairness to Lucas (and others not named), he noted explicitly that Clower constraint models and OLG were two different ways of modeling financial frictions in Lucas (1984)

The convention adopted in this paper that all traders alternate synchronously between centralized and decentralized markets is only one of many ways of utilizing the cash-in-advance constraint to study situations with incomplete markets. For example, Grossman and Weiss (1982) and Rotemberg (1982) examine models in which some agents are always engaged in securities trading but never all agents at the same time. Comparison of their results with those cited above makes it clear that the characteristics of the equilibrium depend critically on the nature of the assumed trading rules and timing conventions.

The intergenerational models introduced by Samuelson (1958) provide another context for analyzing monetary issues within the general equilibrium framework used in the theory of finance. See Wallace (1980) for a useful description of recent developments.

I do not see any way of judging which of these approaches will prove most useful for which questions that does not involve working out the implications of theories of both types. By pursuing the particular Clower-type approach used here. I do not mean to suggest that I view this question as closed at the present time. (Lucas [1984], footnote 7, p. 21).

We turn next to the after-conference contributions. The first paper, Truman Bewley's "The Optimum Quantity of Money" works with the infinite lived agents framework, so, as with the Shubik and Lucas contributions, we omit discussion of the paper here.

The next OLG paper in the volume is the Brock-Scheinkman paper. This paper uses a simplified version of the Lucas model to study the effects of random variations in the money supply. The simplifications are along the same lines as the "easy Lucas" model sketched above.

Specifically, in addition to the standard two-period lived agents assumption and the restriction to a single commodity, the Brock-Scheinkman models also focuses on pure exchange and discrete random shocks. As we noted above, these simplifications make analysis of the model considerably easier since the equilibrium objects in the model are now finite-dimensional Euclidean vectors rather than functions.

Brock and Scheinkman show the following results:

1. In a deterministic setting with perfect foresight, increasing the amount of money (i.e., generating inflation) leads to welfare losses (except to the initial generation).
2. With random money supply fluctuations, there is always a deterministic monetary policy which leads to the same welfare outcomes as the stochastic policy. In this case, however, because the relationship between the price level and the quantity of money is inverse (and hence, convex), increasing the variance of the random monetary policy can lead to welfare improvements.

John Bryant's conference paper would seem to have anticipated Tobin's criticisms, since it focuses on the fact that in an OLG setting in which money can serve as a store of value, the moral hazard issues associated with contracting can lead to a medium of exchange role for money as well. The observation that contracts – which can serve as an alternative to money for mediating exchange (and indeed, does in the conventional Arrow-Debreu model) – can be costly to negotiate or enforce goes back to Brunner and Meltzer (1971). While these informational issues are not specific to the OLG environment, they do figure in the later development of models of money based on search.

The Cass addendum, “Money in Consumption-Loan Type Models: An Addendum,” generalizes the Cass-Okuno-Zilcha example of the coexisting of non-optimal monetary and non-monetary equilibria to make the example robust to small perturbations of agents’ preferences and endowments. The construction still requires that agents can become satiated in second period consumption, so that while the example is robust to perturbations, it remains special in the sense of requiring the relaxation of common non-satiation assumptions.

The Cass and Shell after-conference contribution, “In Defense of a Basic Approach,” is a direct reaction to Tobin’s critique of OLG as a model of money. The main thrust of the critique is their statement that

If money and bonds do not serve as a value store, then they cannot serve any other useful function. (Kareken-Wallace [1980], p.251)

Cass and Shell emphasize the point that money cannot have value in models with finite time horizons via the usual backward induction (aka “hot potato”) argument, and they reject the argument that one can generate valued fiat money in a conventional Arrow-Debreu framework simply by including money in the utility function. Their rejection of money-in-the-utility function models is based on two observations. First, it doesn’t explain why intrinsically useless fiat money might serve as a store of value. Second, the usual argument in favor of these models – that they capture the medium of exchange value of money in cutting through Jevons’ “double coincidence of wants” – ignores the fact that money is itself a technology the management of which can generate positive or negative impacts. As Cass and Shell note, “such reduced forms are at best poor proxies for their structural counterparts.”

Fleshing out these ideas, Cass and Shell note that one can avoid the “hot potato” problem by working in an environment where the effective economic horizon is infinite, even though the

individual economic actors themselves live finite lives and consume finite resources. They distinguish the OLG model from what they call the “received doctrine” by noting two key features of the model:

First, it is genuinely dynamic. In this it significantly departs from the basically atemporal character of most received doctrine: There is explicit recognition of both the inherent mortality (as well as vitality) of the actors, together with the continual (as well as unceasing) evolution of their stage.

Second, it is fundamentally disaggregative. In this it is founded in perhaps the oldest theoretical tradition in economics: There is a clear distinction between economic agents' objectives and constraints – and hence the mainsprings of their individual behavior – and the economic system's coherent resolution of their joint interaction. (*Ibid* p.253)

They turn next to the question of whether there is any practical use in looking at the kinds of highly simplified and unrealistic models that were presented at the conference. The response to this question reads like a blueprint for the next thirty years of research on the OLG model.

They begin this discussion by noting that “the cardinal rule [is] that simple models can best be employed to produce counter examples,” and then point to the first important counter example, Samuelson’s demonstration that the model can have competitive equilibria which violate the first welfare theorem. The next counter examples they point to are the indeterminacy results first noted by Samuelson, based on the fact that the model can exhibit a continuum of non-stationary monetary equilibria (in the typical Samuelsonian setting), as well as cyclic equilibria which, as we saw above, were first noted by Gale.

Cass and Shell then venture into new and less developed territory by raising questions about the usual assumptions economists make about what constitute an appropriate set of state variables¹⁰. This leads to speculation about the possibility that purely extrinsic events – the

¹⁰ This is echoed in the Cass *Macroeconomic Dynamics* interview [Spear and Wright (1998)] when he talks about questioning Lucas about his assumptions that only the exogenous random shocks to money and population would affect the equilibrium prices in his '72 *JET* model.

occurrence or non-occurrence of sunspots, in their example – might have a self-fulfilling effect on economic equilibrium simply because agents in the model believed they would. As we'll see below, this speculation turned out to be correct, with papers by Cass and Shell, and by Costas Azariadis spawning a whole literature on extrinsic uncertainty and self-fulfilling prophecies in rational expectations equilibrium.

From this speculative discussion, Cass and Shell then note that

In fact, at this writing, there are virtually no well-established general theorems for the overlapping generations model available in the literature. The prime reason for this paucity of substantial results is clear: Very little is known about the most basic properties of this model except in its rudest embodiments. (*Ibid*, p. 255)

They then go on to provide a sketch of one possible generalization of the simplest two period, one commodity, representative agent model that also happens to address Tobin's comment about how unrealistic it was to assume people would hold money for 25 years. The generalization Cass and Shell outline – which appears formally in Balasko, Cass and Shell (1980) – involves allowing for (finitely) many agents who live arbitrary (but finite) lifetimes and consume any finite number of goods. They point out that by reinterpreting the length of the periods and appropriately adjusting the dimensions of the vectors of consumption and endowment commodities, the original model can be transformed into one where agents live for two periods with generations overlapping for one. As long as markets are complete, the two models are isomorphic.

The remainder of the paper enumerates a number of additional directions for research on OLG models. Cass and Shell note that whether one believes in OLG as a model of money *per se*, it is fundamentally a model in which agents need assets in order to finance consumption spending during their retirement years. As such, OLG provides a better microfoundation for both

capital theory and finance. Stochastic extensions of the model (following Lucas' pioneering paper) raise questions about the role of expectations (particularly given the later sunspot results conjectured here), socialization across demographic cohorts, and the need for models of economic learning.

The penultimate contribution to the conference volume is Ross Starr's "General Equilibrium Approaches to the Study of Monetary Economies: Comments on Recent Developments."

Starr makes two short points.

His first point notes that money can have value independently of its use as an asset for retirement saving if it is the only asset accepted by the government for the payment of taxes, or, even more simply, if agents in the model *expect* it to have value. The possibility that taxation gives money value is well-known (and was at the time of the conference), but Starr's comment on expectations reflected new developments in what became known as *temporary equilibrium* theory, pioneered by Jean-Michel Grandmont. We discuss this work in the next chapter.

Starr's second point is that standard sequence economies (whether with finite horizons or not) that don't have futures markets need an asset like money in order to implement equilibrium trades. This is akin to Arrow's results showing how to reduce the number of forward contingent claims markets required for an Arrow-Debreu equilibrium (with both time and uncertainty) by replacing the forward markets with assets that allow the transfer of wealth over time and across states of nature. What this argument lacks, however, is the motivation for saving to begin with. There is an obvious sense in which the OLG model builds in what would otherwise have to be assumed – i.e., that agents must save for retirement – but in the absence of individual human motivations for the creation of dynasties, this built-in retirement motive happens to be realistic.

The final contribution to the volume is Rob Townsend's "Models of Money with Spatially Separated Agents." Townsend looks at three distinct models.

1. The turnpike model, in which agents travel along an infinitely long road in one of two directions and are paired at each point of time with someone going the opposite direction. Under the assumption that the endowment distribution across agents is not Pareto optimal, Townsend shows that introducing money that is passed back and forth across the road (chickens?), agents can engage in welfare improving trades that are not possible without the monetary tokens, for reasons akin to those in the OLG model.
2. A modification of the turnpike model that allows agents to meet each other infinitely often, but still has the requisite mismatch of endowments between agents when they meet that makes Pareto improving trades possible. In this setting, agents could resolve bilateral debts, but Townsend sets the model up so that when creditors meet their respective debtors, they never have the real resources available to redeem their debts. This, then, creates a role for money both as a store of value and as a medium of exchange.
3. Townsend's third model is a variant of the Cass-Yaari circular model combined with the friction Lucas introduced in his conference paper to create a transactions demand for money.

While none of these models is realistic – even in densely populated cities, creditors generally know where their counterparties live – they do explicate the kinds of frictions that contemporary search theoretic models of money have rationalized. For a detailed accounting of

these developments, see the appropriately titled special *International Economic Review* issue *Models of Monetary Economies II: The Next Generation* (Wright [2005]). We will review the work combining OLG and search theory in Chapter 8.

Section 5.3 The Tenuousness of Money

The tenuousness of money issue, and the possibility that every new generation could generate seniorage gains by repudiating old money in favor of new money found formal theoretical expression in the literature exploring core allocations in OLG models. For the Samuelsonian case, the core of the economy is empty, for the simple reason that at any time t , current young will strictly prefer to issue new money, while all future agents are indifferent, so this coalition can block implementation of the standard monetary equilibrium. This first paper to show this was Hendricks, Judd, and Kovenock (1980).

While there were a number of papers looking at core-like issues in the OLG model – for example, Chae (1987) showed a short-run core equivalence result which is parallel to the concept of weak Pareto optimality – the core emptiness results for the full infinite horizon economy remains essentially a negative result. The original Hendricks, Judd and Kovenock paper gets only 26 Google Scholar citations, and this tops the list of papers based on a search for “core” and “overlapping generations” on Google Scholar. But the result itself is indicative of the essential tenuousness of fiat money, and of the overall process of intergenerational resource transfers.

The final definitive words on the relationship between the optimality of the competitive equilibrium and the existence of a monetary equilibrium were produced in two papers, one by

Cass and Larry Benveniste in 1986, and the second by Francois Duc and Christian Ghiglino in 1998.

Benveniste and Cass (1986) showed that when one removed the satiation assumptions that generated the Cass-Okuno-Zilcha counter examples, as long as agents can trade zero net supply bonds, a fixed money supply will support a Pareto optimal stationary equilibrium for the Samuelsonian case with many goods and a single two period lived representative consumer.

Curiously, the paper claims to provide an extension of the result to the case of many consumers, goods, and lifetimes under the condition that intergenerational trades from present to past generate Pareto improvements, in a follow-on unpublished working paper "On the Existence of Optimal Stationary Equilibria with a Fixed Supply of Fiat Money: II. The Case of Many Consumers with Arbitrary Lifetimes" (Cass [1982]). This paper does not seem to exist, and when we asked Benveniste about it, he replied (via private email correspondence dated May 29, 2022)

The paper with Dave in *JPE* was done while I visited Penn in 1981. Dave walked into my office with the question. Do stationary monetary equilibrium exist. I solved proved it and that really is my history. I visited for two more years but Dave was focused on sunspots and I was pursuing other questions. The multi consumer version is a mystery to me. Dave talked a little about it but I don't remember seeing a finished version. So any reference is possible. Dave may have finished it himself.

Burke (1999) also cites the Cass paper, and he was able at our request to unearth a copy of the hand-written notes, which as they are don't constitute a full paper, but are extensive, running to 41 pages. We include these notes online at:

<https://econ.tepper.cmu.edu/spear/cass92.pdf>

by way of completing the historical record on this work.

Cass apparently decided not to write the work up as a formal paper or try to publish it, but he did pass the notes on to Francois Duc and Christian Ghiglino for them to use in work they were pursuing on existence and optimality results for general OLG economies. The paper that came out of this was Duc and Ghiglino (1998), which cited the Benveniste-Cass work and provides a generalization of that work based on the theorem in Cass's extended notes that he provided Duc and Ghiglino. Ghiglino, in email correspondence (dated May 29, 2022) recalled that

Dave discussed with us his notes. These were his notes. I don't think there was ever a paper, only intentions and a non-polished (I think) [version] of the theorem. Dave was happy we used it and published it.

The theorem by Cass that Duc and Ghiglino use establishes the existence of steady state equilibria for the general OLG model with interest rates greater than or equal to zero. When the interest rate is zero, the steady state is nominal. When the interest rate is greater than zero, there are no intergenerational transfers of resources, and the equilibrium is non-monetary. Duc and Ghiglino show that these equilibria (which can involve trade between agents of different ages as well as types) are in fact Pareto optimal. Of course, when the model has only a single agent born per period and only one commodity, we are back in Gale's classical case world.

We complete our discussion of OLG as a model of money by noting that quite a few monetary economists agreed with Tobin's critique. Some of these additional critiques were aired at the 1983 Carnegie-Rochester Conference on Money, Monetary Policy, and Financial Institutions, organized by Karl Brunner and Allan Meltzer. One particular paper, by Bennett McCallum (1983), is worth reviewing in some detail.

McCallum raises objections to the use of OLG as a model of money, based on the dynamic tenuousness argument, and on an argument that purports to show that money cannot

serve as a medium of exchange in OLG economies. The tenuousness is the same as that posed by Tobin and others, which we have discussed above.

McCallum's argument that money cannot serve as a medium of exchange "builds upon the traditional presumption that an economy with a medium of exchange will be more productive than it would be if no such medium existed" (*Ibid*, p.24) He then suggests that in an OLG model with capital, because the same leisure and consumption opportunities are available to agents whether or not money is available indicates that money in the OLG model cannot serve as a medium of exchange.

An immediate objection to this argument is its assumption that the model is one with capital. As Wallace noted repeatedly, capital is an asset which, if productive, dominates money as a store of value, in which case the price of money is zero. For a model with capital, it would seem that the more relevant direct question would be whether capital in the model is accepted as a medium of exchange. Given the abstract level at which the model is presented (both in McCallum's paper and in the others we have been examining), this question is impossible to answer. It could well be that agents in the model actually break off chunks of a putty-like capital stock in order to trade (as long as it all ends up back at the factory in time for next period's production run). Or it could be, as McCallum suggests earlier in his paper, that the exchange process is unmodeled but exists in the background.

For a more relevant examination of the "traditional presumption," we would suggest looking at the Samuelsonian case in a pure exchange model, where money serves as a store of value. In this case, in the absence of money, the barter equilibrium cannot implement the intergenerational Pareto improving transfers that Samuelson first noted, so that the introduction of money does indeed make the economy more productive.

McCallum then examines the argument against money-in-the-utility function models. His development of these models takes the Cass-Shell criticism that these models are just reduced forms seriously, by positing a model in which the only arguments in an agent's utility function are consumption and leisure in each period of life. He then develops a sub-model based on Becker's home production paradigm which includes shopping time as an activity. This allows him to capture the inefficiency of barter relative to money as an explicit transaction cost that detracts from leisure. When this transaction technology is plugged back into the original utility function, the resulting indirect utility function looks like one in which money is an argument.

This particular approach, as we noted earlier, has been subsumed by more modern search theoretic models of money and transactions, but the basic idea of using an indirect utility representation remains a valid one.

To conclude this chapter on the early applications of OLG as a model of money, it seems fair to say that Wallace's ultimate rejection of OLG as THE model of money was correct. At the same time, though, there is a clear consensus in the papers and discussion that OLG creates a demand for assets – that is for a store of value – via the need for people to save for retirement, and the fact that storing commodities to meet this need is both unrealistic and inefficient. This demand-side consideration is absent in the alternative ILA-based models. This should be apparent from the fact that for money to have any value in these models, the use of money must be imposed exogenously via either cash-in-advance constraints, the condition that taxes must be paid in money, or the *ad hoc* inclusion of money as an argument in the utility function. We can also see this fact very simply by looking again at the simple one-sector capital model we examined in the section of dynastic models.

If we take the Lucas (1978) asset pricing model – where the technology is simply a Lucas tree that drops a given amount of output each period – and add the ability to store the output for one period, we can then ask whether our representative agent would ever avail herself of the storage option. The answer to this is simply no. As long as she discounts the future, there is no gain to saving any output in this case, since the tree will produce more output tomorrow. The only reason our infinite lived representative agent would save, then, is if saving generated a positive return, i.e., only if capital is productive. While most analyses of this kind of model focus on the fact that in the presence of productive capital, no agent will hold money because it is dominated in return, our point with this example is to show that the agent won't save – using any instrument – when the return to capital doesn't dominate other assets.

From this perspective, OLG generates a fundamental demand for assets of one kind or another, as a primitive of the model grounded in simple empirical observations of how people live. One might argue that the time-varying resource patterns that give rise to the demand for retirement saving could be built into ILA models, but for tractability, these would have to be cyclic, since it is difficult to see how retirement for eternity after working finitely many periods could be successfully financed. As Cass and Shell noted in the “Defense of a Basic Approach” post-conference contribution, it is the actual dynamism of new agents entering and old agents leaving that creates a world where finance isn't simply an afterthought of industry's need for machines.

Chapter 6: OLG and Theory, 1970-1980

While monetary economists were busy exploring the OLG model's store of value aspects, macroeconomists and general equilibrium theorists were also exploring the model, looking to establish the kinds of general theorems that Cass and Shell had suggested were needed in the "Basic Approach" paper. As we look at these developments, it will become apparent that the hub universities for theory work on the OLG model in the U.S. were Harvard, Minnesota, MIT and Penn, and scholars in France supported by CEPREMAP – which is an independent research agency under the aegis of the French Ministry of Research – and by the French Centre National de la Recherche Scientifique (CNRS). The French connection to OLG was channeled through Karl Shell, who spent a sabbatical year (1977-78) at CEPREMAP (supported by the Gugenheim Foundation and CNRS), where he met Yves Balasko, Jean-Michel Grandmont, Roger Guesnerie, and Jean-Jacques Laffont. Shell, together with Dave Cass, had NSF grants that made it possible for them to invite these French scholars to spend time at Penn during this period.

Section 6.1 Temporary Equilibrium

Proceeding chronologically, we begin with Jean-Michel Grandmont and his work on temporary equilibrium models. Like Lucas, Grandmont was motivated by monetary issues in macroeconomics in his theoretical interests as a graduate student. Grandmont was also solidly in the microfoundations camp, like many of his graduate contemporaries at the time. Summarizing from an extended email correspondence with us, over the period 1960-65, while studying engineering in Paris, Grandmont read a number of economic tomes including translations of Hicks' *Value and Capital*, and Samuelson's *Foundations*, which covered micro and general

equilibrium theory on the one hand, and mathematical forms of Keynesian macro on the other, noticing the absence of linkage between macro and micro approaches. The only glint of a connection he perceived was Hicks's temporary equilibrium formulation as manifest in *Value and Capital*, in which in a specific period rational decisions were made by agents given their expectations of the future environment, albeit in his view, there still existed a chasm between micro and macro.

Unlike Lucas, Grandmont was skeptical about the new rational expectations paradigm, believing instead that expectations should, of themselves, be taken as a psychological factor at work in economics (not unlike preferences), which was amenable to change through learning, but fundamentally more complex than the stochastic version of perfect foresight built into the rational expectations hypothesis. According to Grandmont, his publications that focused on temporary equilibrium were directed against the rational expectations and the Arrow-Debreu approaches, in as much as in his view they were both static and based on the rational expectations hypothesis that implied that all present and future decisions, including those of unborn agents, were taken at an initial point. In his view, Hicks's attempt to reconstruct dynamics in a realistic manner emanated from agents forecasting the future via a learning-based expectations function, dependent upon past information. This approach was advocated by theorists such as Arrow, Debreu and Hahn, allowing progress on temporary general equilibrium theory accordingly.

Grandmont's dive into temporary equilibrium theory started with a meeting with Debreu in his office at Berkeley, where he recalled that as a Ph.D. student he met Debreu and told him that he wanted to bring money into the general equilibrium framework. Debreu replied that Grandmont first had to complete his PhD coursework, and besides that, as he [Debreu] had never

addressed the topic, Grandmont would be on his own in dealing with the problem. Grandmont then said that luckily, one of the first courses he took in 1968 was an introductory Macro course given by Bent Hansen, based upon Patinkin's book, *Money, Interest, and Prices* which, at the time, was the leading monetary theory text. Grandmont then recalled that he came to understand the principle underlying the competitive nature of temporary equilibrium in the first term-paper he wrote, which consisted of reproducing Patinkin's approach with inside money in a two-period exchange economy with a single good. After a simple computation process taking a few minutes, he obtained two results. The first was that Patinkin's claim regarding temporary equilibrium with a positive price for money as existing via the real balance effect was actually not true in the case when expected inflation – fixed in the short run – was too large. The second was that existence could be reestablished in the case where the level of expected money price level was fixed; and this, by means of reintroducing intertemporal substitution. In other words, existence arguments based upon simplicity would illustrate the existence of monetary equilibria in a competitive sense, albeit in economies which were more sophisticated, that is with many agents and goods.

Grandmont completed his Berkeley Ph.D. thesis, one of the papers of which was the unpublished working paper Grandmont (1970), which gave rise to Grandmont (1973). This paper formally works with an infinite lived agents setting, but agents' preferences are structured so that they only care about consumption between "this week" and "next week," which allows Grandmont to capture the short-run equilibrium effects generated by different specifications of the expectations function in the model.

Grandmont's first use of the OLG model came on his return to Paris in late 1970. After turning to work at CEPREMAP, Grandmont, along with Younes, attempted to construct a coherent micro-foundational based approach to money and macro using a methodology based on

temporary equilibrium. The initial products of this project were two papers dealing with how money was integrated into models based upon agents who were infinitely lived and faced a cash-in-advance constraint. Grandmont – with Laroque – continued to work on the research program in the context of competitive OLG models initially with outside money. According to Grandmont, he and Laroque were the first to do this. They then continued to work with inside and outside money, and banking. According to Grandmont, these two frameworks, that is to say, the infinite horizon approach, with constraints based on cash in advance, and non-bequest OLG models, were characterized by financially based functions that could be analyzed by similar economic techniques. In further development of this work related to OLG-based on stochastic processes, Grandmont and Hildenbrand introduced stochastic shocks and a mathematical methodology that could be utilized irrespective of the length of agents' lives.

We can illustrate the basic setup of the OLG model in the temporary equilibrium framework using the simple two-period lived agents pure exchange model with a single commodity, as in the “easy Lucas” model outlined above.

The difference in the temporary equilibrium model is that agents are not assumed to have perfect foresight (in a deterministic setting) or rational expectations (in a stochastic setting). Instead, they must use currently available information (possibly including current prices in addition to lagged prices) to forecast the next period's price. Hence, a typical young agent will face the following sequential budget constraints

$$p_t \cdot x_t + m_t = p_t \cdot \omega_t$$

$$\psi(p_t) \cdot x_{t+1} - m_t = \psi(p_t) \cdot \omega_{t+1}$$

where p_t is the price at time t prevailing on the market, x_t is period t consumption, m_t is money holdings, ω_t is the agent's endowment, and the function ψ is the agent's point forecast of the future price. As Grandmont notes in his paper with Younes, one can use standard backward dynamic programming procedures to determine the time $t+1$ utility-maximizing demand for the agent, given his period t consumption and money holdings. Going back to period t , one then maximizes utility with respect to consumption and money holdings taking the period $t+1$ demand, and the forecast function as given. From here, having obtained the period t optimal demands relative to the given forecast, the temporary equilibrium is then simply the Arrow-Debreu equilibrium price for the static economy in period t . This equilibrium is referred to as "temporary" because with different information, agents will change their forecasts, generating a possibly different Arrow-Debreu equilibrium.

One of the features that Grandmont highlighted in this work was the fact that because the standard Arrow-Debreu model admits multiple equilibria, it was possible for a dynamic model to do this same. This became an issue for Grandmont while he was studying Lucas' 1972 paper.

Grandmont stated that in 1981 he started to read Lucas' 1972 paper with the intention of destroying its premise but was blocked in this effort because of issues arising from the signal extraction problem in the Lucas model. As Grandmont put it, Lucas assumed that observation of the market price revealed an otherwise unobservable variable. Lucas then searched for an equilibrium where prices were functionally related to that variable, showed that this equilibrium existed, and claimed that it was unique. According to Grandmont, such a procedure was valid only in the case where the revealed variable that was hidden was uniquely associated with an equilibrium price. which may not be the situation. Grandmont agreed with Lucas on existence, but asserted that as there may be multiple equilibria, the economic argument in Lucas 1972 was

invalid. Grandmont went on to say that he arrived at an understanding of what he saw as the basic error in Lucas 1972 via discussion with a colleague at CEPREMAP in 1981. Grandmont wrote to Lucas, with Lucas replying that there was indeed an error in the paper. Lucas wrote a corrigendum acknowledging the error but asserted that his argument was nonetheless still valid.

We will return to Grandmont's work in the next section, dealing with the theoretical work from the 1980s to '90's, but there is a slight irony here in that Grandmont's paper with Hildenbrand on stochastic processes of temporary equilibria has an error: the asserted existence of an invariant measure for the equilibrium process was proven on the assumption that the temporary equilibrium of the model depended continuously on the lagged data. Martin Helwig (Helwig [1980]) pointed out that this amounted to assuming that the temporary equilibrium of the model was always unique.

Nonetheless, Grandmont's temporary equilibrium work was important because it showed that under weak assumptions about expectations, all the structural characteristics of the standard static Arrow-Debreu model would hold for the temporary equilibrium model. This allowed the importation into dynamic general equilibrium theory of all of the basic results of standard general equilibrium theory, and in particular, the smooth economies tools first developed by Debreu and later elaborated on in the work of Yves Balasko, who explored the manifold properties of the equilibrium set (see, for example, Balasko [2009]).

These structural characterizations were later combined with measure theoretic techniques to extend the initial Grandmont-Hildenbrand characterization of temporary equilibrium stochastic processes to more general models (including the possibility of multiple equilibria) in a fully rational expectations setting. We will discuss these results further in the chapter on stochastic OLG models.

Section 6.2 Optimal Taxation and Public Finance

There was a constellation of papers on optimal taxation that used the OLG framework written during this period, with contributions coming from a member of the not-so-silent generation, Ned Phelps, and several more from the early Boomer generation who were completing their graduate work at this time.

The first modern optimal taxation paper was written in 1971 by James Mirrlees, who generalized Frank Ramsey's 1927 analysis of the relationship between demand elasticities and tax revenues by adding heterogeneous labor productivity (see Mirrlees [1971]). A natural extension of Mirrlees' work was to put the optimal tax problem into a dynamic context. This was important because taxes affect saving, which in turn affects capital accumulation and, in an OLG context, the ability of workers to finance their retirements.

Looking at this work chronologically (in terms of publication dates), Peter Diamond was one of the first to make such a dynamic extension, in Diamond (1973). Diamond's paper uses a two-period lived OLG model of growth to examine three possible ways a central planner would act to maximize a discounted sum of utilities over all generations. In the first scenario, the central planner can choose all quantities directly, and the solution to this is standard. In the second scenario, Diamond decentralizes the optimum obtained in the first scenario in the usual way, subject to the need for government debt to institute intergenerational transfers required of the optimum from the first scenario. (This need doesn't arise in the ILA setting because all intergenerational transfers in this model are internalized in the solution to the Pareto problem.) In the third scenario, Diamond considers the optimal taxation problem by letting his central planner choose prices, but not quantities, together with the assumption that there are now no

lump sum transfers possible. For this analysis, Diamond defines two notions of the intertemporal marginal rate of substitution. The first is the usual individual MRS (relative to a numeraire good), while the second is the societal intertemporal MRS (or SMRS) which reflects the intergenerational terms of trade between periods. Diamond's analysis of the optimal tax problem shows that these two rates need not be aligned at the optimum, though if they are, they will be equal to the marginal rate of transformation (MRT) determined by the technology of the economy. If all of the rates are equal, i.e., $MRS=SMRS=MRT$, then the allocation coincides with the fully optimal scenario one allocation. Diamond shows via an example of an economy with one consumption good plus a labor-leisure choice that the optimal tax equilibrium need not equal the full optimum.

A second early paper making the dynamic extension to the optimal taxation problem was Ordoover-Phelps (1975). This paper works directly with the Mirrlees model – in particular with Mirrlees' assumption that workers have different productivities – but adds the structure of Diamond's version of the OLG model in order to look at the effects of capital accumulation, population growth, and retirement consumption to the optimization mix. The social welfare criterion that Ordoover and Phelps adopt is not the usual utilitarian one, but rather a Rawlsian criterion based on maximizing the minimum utility obtainable across generations assuming the economy is evolving along a steady state balanced growth path (driven by population growth). The justification given for this is

The social welfare function we employ expresses the conception of economic justice championed by John Rawls: As a chain is no stronger than its weakest link, social welfare is only as great as the utility of the least well-off, the person or persons having minimum utility. Thus the redistributive optimum is "maximin." (Ordoover-Phelps [1975] p. 661)

The questions that the paper seeks to address are

Should profits be taxed? Should they be taxed more heavily than wages? Might it be optimal to levy a proportionate wealth tax at a rate exceeding the average rate of profit on capital? (*Ibid*, p. 661)

While these are interesting questions, to which the paper begins to provide (as the authors note) technical answers given the ethical choice of the maximin social welfare criterion, examining them fully is outside the focus of our study of the development of OLG models, so we leave it to the interested reader to pursue this in detail in the paper. We do note, however, that Phelps and Riley (1978) extended the analysis of Ordover and Phelps using the kinds of recursive dynamic programming techniques that later become popular via the Stokey-Lucas (1989) text, though in the context of Diamond's OLG model. This is possible because the central planner/government in the model is assumed to be an infinite lived institution, even though agents in the model are finite lived. There is one quote, however, from the Phelps-Riley paper that is worth noting:

[I]t must be a bit startling, on first encounter with the utilitarian doctrine, to see the saving by a multi-period household become an allegory of the proper accumulation by a multi-generation society. (Phelps-Riley [1978], p. 103)

Their meaning here is quite clear: households are, for the most part, finite lived families organized as overlapping generations. Utilitarian approaches that weight generations via discounting transform the "multi-generation society" into a simple (and unrealistic) "multi-period household," or, as Barro viewed it, a lineal dynastic family.

A second early application of the OLG model to public finance was Auerbach (1979), which used a version of the Diamond model to assess the effects of taxation on economic growth with capital heterogeneity. Auerbach also adopts the OLG dynamics, calling it "the neo-classical growth model introduced by Diamond." Auerbach was one of the first of the Boomer generation to adopt the OLG model in his work, and his nonchalance in simply referring to it as the

neoclassical growth model, suggests a degree of acceptance of the naturalness of the OLG framework, given Diamond's work in showing how standard neoclassical production fit into Samuelson's model. That said, when we asked Auerbach why he settled on the OLG model for this work, he indicated that it likely had to do with his thesis advisor Marty Feldstein's earlier related work being set in a static two-period model, with OLG being the natural dynamic framework in which to extend Feldstein's analysis.

Other early applications of the OLG model to issues of taxation include work by Laurence Kotlikoff (who was a year ahead of Auerbach at Harvard), and Lawrence Summers.

Kotlikoff and Summers (1979) also extends work by Feldstein on the topic of long-run versus short-run tax incidence using the two-period lived OLG framework of Diamond. Kotlikoff and Summers also include human capital accumulation in the model, although this is modeled as an individual agent's trading off leisure time versus training which leads to higher future income rather than as a parental investment in education.

As we noted above in our discussion of Tobin's use of multi-period simulated OLG models, all three of these Harvard-trained scholars would make important use of the OLG model in their later work and were together pioneers in the use of realistically multi-period OLG simulations for various types of policy analysis. We will return to discuss this work later in the book.

Section 6.3 The Balasko-Shell Analysis

We turn next to the work of Yves Balasko and Karl Shell in their definitive papers on the OLG model, along with a paper co-authored by Balasko, Cass and Shell during the same time period.

All four of these papers were influential in the development of the OLG model and were based on exploring the general equilibrium fundamentals of the model. These papers were

1. "The Overlapping-generations Model, I: The Case of Pure Exchange without money," *Journal of Economic Theory* 23.3, 1980, 281-306. (Balasko and Shell [1980])
2. "Existence of Competitive Equilibrium in a General Overlapping-generations Model," *Journal of Economic Theory*, 23.3, 1980, 307-322. (Balasko-Cass-Shell [1980])
3. "The Overlapping-generations Model. II. The Case of Pure Exchange with Money," *Journal of Economic Theory* 24.1, 1981, 112-142. (Balasko-Shell [1981a])
4. "The Overlapping-generations Model. III. The Case of Log-linear Utility Functions," *Journal of Economic Theory*, 24.1, 1981, 143-152. (Balasko-Shell [1981b])

As we noted above, Yves Balasko was a young scholar, recently graduated from the Ecole Normale Supérieure with a Ph.D. in mathematics when Karl Shell met him in Paris and recruited him to work on the project of providing a full and detailed mathematical characterization of the OLG model.

All four of the papers look at deterministic, pure exchange versions of the OLG model. Of the four papers, the Balasko-Cass-Shell is the most general, although it focuses only on the question of existence of competitive equilibrium the model. This paper includes a formal

demonstration of the claim made in the Cass and Shell “Basic Approach” contribution to the Kareken-Wallace (1980) volume that any OLG model can be converted into one in which agents live for two periods via a suitable reinterpretation of endowment and consumption vectors. The model examined here allows for heterogeneous agents within generations, general utility specifications – the only required properties of the utility function are continuity, non-satiation, and quasi-concavity – and the possibility that some endowments or consumptions are zero during some period(s) of an agent’s life.

According to Shell in private communication, the three Balasko-Shell papers were written prior to the Balasko-Cass-Shell paper. Shell notes that after the equilibrium existence proof given in BS-I, they talked about how to weaken the assumptions used in that proof to provide a general proof of existence (see the bottom of p. 289 and top of p. 290). These arguments were formalized in the three-author paper.

The three Balasko-Shell papers make two important simplifying assumptions:

1. There is a representative consumer born in each period who lives two periods
2. Preferences are smooth in the sense of the utility functions being continuously differentiable, with indifference curves bounded from below.

All three papers focus on pure exchange economies as well.

The BS-I paper looks at an economy in which there are forward markets on which agents can trade one period ahead to make or take delivery of commodities contracted over in the previous period. This assumption simplifies the basic analysis by ensuring that agents can freely transfer wealth between periods without the need for additional assets and asset markets. This leads, in particular to a typical young agent facing a single budget constraint.

Since it will be useful going forward, we introduce some basic notation that Balasko and Shell use in their analysis of the model. With a single consumer born in each period and agents living two periods, we follow Shell in referring to the young agent born in period t as Mr. t^{ll} . Mr. t 's lifetime consumption vector is $x^t = [x_t^t, x_t^{t+1}]$, where x_t^s is the vector of goods Mr. t consumes in period $s=t, t+1$. This notation – subscripts for birth dates, superscripts for time periods – has become conventional in the general equilibrium analysis of OLG models. Endowments are denoted similarly, by $\omega^t = [\omega_t^t, \omega_t^{t+1}]$. Preferences are specified via a (possibly time-varying) utility function

$$u: \mathbb{R}_+^{2l} \rightarrow \mathbb{R}$$

which is assumed to be differentiable monotonic, strictly quasi-concave, with the closure of all indifference curves strictly contained in the positive orthant. Here, $l < \infty$ is the number of goods consumed. These assumptions are sufficient to guarantee that demand functions, and hence excess demand functions, are differentiable. With this notation, the budget constraint for a typical young agent born in period t is given by

$$p^t \cdot x_t^t + p^t \cdot x_t^{t+1} = p^t \cdot \omega_t^t + p^t \cdot \omega_t^{t+1}.$$

The analysis is undertaken assuming that there is an initial period, though this can be interpreted as simply the point at which the analyst “breaks into” an otherwise on-going economy. For this, Balasko and Shell also define specific consumptions and preferences for the initial old generation, with their consumptions from youth taken as historically given.

¹¹ Penn graduate students in the 70's and 80's who took Shell's OLG course always got a chuckle out of the fact that, at the time, Shell was apparently unaware of the other Mr. T, a professional wrestler turned TV star.

Prices are introduced as infinite sequences $p = \{p^1, p^2, \dots, p^t, \dots\}$ with each $p^t \in \mathbb{R}_+^l$. Prices are normalized by taking the price of the first good in the first period, $p_1^1 = 1$. This specifies prices in present value terms, with good one as the numeraire. Balasko and Shell denote the resulting price space by S .

As we noted above, Balasko and Shell use the same notational shortcut as in Malinvaud (1953), by letting Mr. t 's consumption vector x^t also represent the infinite-dimensional vector

$$x^t = [0, \dots, 0, x_t^t, x_t^{t+1}, 0, \dots].$$

Via this shortcut, we can represent the consumption/endowment sets of each consumer as

$$X_t = (X_t^t, X_t^{t+1}) \subset \mathbb{R}_+^l \times \mathbb{R}_+^l \text{ or}$$

$$X_t = [0, \dots, 0, X_t^t, X_t^{t+1}, 0, \dots]$$

and the aggregate consumption/endowment set for the economy as

$$X = [X_0^1 + X_1^1, X_1^2 + X_2^2, \dots, X_{t-1}^t + X_t^t, \dots].$$

We make this point in somewhat more detail than do Balasko and Shell, because it highlights the fact that any economically relevant functions defined on the economy's aggregate consumption space would only need to satisfy nice properties – continuity or differentiability – component-wise, so that the only topology one needs to consider for the model is the product topology.

Once the paper defines basic commodity spaces, preferences, consumer optimization problems, and allocations, it formally defines the set of *weakly Pareto optimal* allocations as those that cannot be Pareto dominated (in the usual sense) by any finite reallocation of resources among agents.

The most important contribution of this paper to the development of the OLG model is its proof of the existence of competitive equilibrium. The technique Balasko and Shell developed in this paper has been applied quite broadly in a variety of contexts to showing existence of equilibrium in OLG models. So, it is worth spending a little time examining their proof.

The proof proceeds in three main steps.

For the first step, let $f_t^s(p^t, p^{t+1}, p^t \cdot \omega_t^t + p^{t+1} \cdot \omega_t^{t+1})$ be the demand function for Mr. t in period s , when prices are $[p^t, p^{t+1}]$ and Mr. t 's income or wealth is given by the value of his endowments $p^t \cdot \omega_t^t + p^{t+1} \cdot \omega_t^{t+1}$. Define Mr. t 's excess demand in period s as

$$z_t^s = f_t^s - \omega_t^s.$$

With this notation, Balasko and Shell define a *t-equilibrium* price sequence

$$p = [p^1, p^2, \dots, p^t, p^{t+1}, \dots] \in S$$

as any prices that solve the finite set of equations

$$z_0^1 + z_1^1 = 0$$

$$z_1^2 + z_2^2 = 0$$

$$\vdots$$

$$z_{t-1}^t + z_t^t = 0.$$

The next step in the proof shows that as long as resources in each period are bounded, one can find natural upper and lower bounds on the price vectors (see the paper for details on this). Imposing these bounds on the price vectors, one obtains a subspace of the price space $S^* \subset S$ such that each factor in S^* is compact. This then implies that S^* is itself compact in the product topology by Tychonoff's theorem.

The final step in the proof is to denote the set of t -equilibrium price sequences by $W(t)$, and to observe that by construction,

$$W(t) \subset W(t + 1) \text{ for all } t.$$

Since the equations defining each t -equilibrium sequence are those one would obtain for a standard, static Arrow-Debreu economy, the usual tools used to show existence of equilibrium in these economies shows that for all t , $W(t) \neq \emptyset$. Via the inclusion relationships, then, a generalized nested set theorem (Balasko and Shell refer the reader to a result in Bourbaki), the set

$$W(\infty) = \bigcap_{t=1}^{\infty} W(t) \neq \emptyset.$$

The remainder of the paper is devoted to a welfare analysis of the competitive equilibrium.

The first result shown is that the competitive equilibrium is always weakly Pareto optimal. This is the closest analogy to the First Welfare theorem that one can obtain in the OLG model. The analog of the Second Welfare theorem is that every weakly Pareto optimal allocation can be supported as a competitive equilibrium after a suitable reallocation of endowments. These two results are established in Proposition 4.4 of the paper. As we noted

earlier, Samuelson was the first to observe that the First Welfare theorem would hold for any finite truncation of the model; the Balasko-Shell result generalizes this observation.

The analysis of when allocations are fully Pareto optimal proceeds by defining sequences of transfers between agents which are feasible (in the sense of adding up to zero), then showing that if there is a Pareto improving transfer between (say) Mr. t and Mr. $t-1$, then there must be non-zero transfers for all following agents as well. We can think of this in terms of the simple one-good model in which a transfer from any young agent to an old agent must be compensated in the following period by a transfer from next period's young to the now old agent.

Balasko and Shell then show that under some technical assumptions on the boundedness of consumption sequences and supporting prices at any weakly Pareto optimal allocation, and given sufficient curvature of agents' indifference curves, then a version of the so-called Cass criterion

$$\sum_t \frac{1}{\|p^t\|} = \infty$$

is necessary and sufficient for the allocation to be Pareto optimal. In a one-good context, this result corresponds to having all interest rates being non-negative, or, equivalently, all intertemporal marginal rates of substitution being greater than or equal to one in absolute value. This result is established in Proposition 5.6 of the paper.

The second Balasko-Shell paper builds on the results of the first paper by introducing fiat money. In terms of notation, this amounts to adding definitions for Mr. t 's period s money accumulations $x_t^{s,m}$, money transfers from the government m_t^s , and a (present-value) price for

money $p^{t,m} \geq 0$. If the government pursues an active monetary policy, so that the $m_t^s \neq 0$, the total (nominal) money supply is given by

$$M^t = \sum_{s=1}^t \sum_{\tau=s-1}^s m_\tau^s.$$

Note that in this setting, Balasko and Shell allow the money transfers to be positive or negative, so that at this level of abstraction, there is no real distinction between fiscal and monetary policy.

The key change from the BS-I paper is in the specification of the budget constraints. For a typical young agent, they now face two budget constraints

$$\begin{aligned} p^t \cdot x_t^t + p^{m,t} \cdot x_t^{m,t} &= p^t \cdot \omega_t^t + p^{m,t} m_t^t \\ p^{t+1} \cdot x_t^{t+1} + p^{m,t+1} \cdot x_t^{m,t+1} &= p^{t+1} \cdot \omega_t^{t+1} + p^{m,t+1} m_t^{t+1}. \end{aligned}$$

These budget constraints can be combined into a single budget constraint by adding them, since markets will be complete as long as the price of money is strictly positive

$$\begin{aligned} p^t \cdot x_t^t + p^{t+1} \cdot x_t^{t+1} + p^{m,t} \cdot x_t^{m,t} + p^{m,t+1} \cdot x_t^{m,t+1} &= \\ &= p^t \cdot \omega_t^t + p^{t+1} \cdot \omega_t^{t+1} + p^{m,t} m_t^t + p^{m,t+1} m_t^{t+1} \end{aligned}$$

Competitive equilibrium in the model is then defined analogously to the definition in the non-monetary model, with the additional requirement that money markets clear, i.e., the desired money holdings of all agents add up to the total money supply as long as the price of money is positive. If the price of money is zero, then any money holdings less than or equal to the total supply is an equilibrium.

With these preliminaries, Balasko and Shell show a key result: along any monetary equilibrium price sequence, the present-value price of money must be a non-negative constant. The proof is straightforward. Suppose $p^{m,t} \neq p^{m,t+1}$ for some periods t and $t+1$. Have agent t set $x_t^{m,t} = -x_t^{m,t+1}$. From the budget constraints, we get

$$p^t \cdot x_t^t + p^{t+1} \cdot x_t^{t+1} + (p^{m,t} - p^{m,t+1}) \cdot x_t^{m,t} = RHS$$

(where RHS denotes the agents income). If the money price difference here is negative, then an arbitrarily large purchase of money in period t allows arbitrary large lifetime consumption. If the price difference is positive, then an arbitrary large sale of money (i.e., borrowing) again yields the possibility of arbitrarily large consumption. As long as agents' preferences are non-satiated, then, differences in the present value of money between periods leads to arbitrage opportunities which are inconsistent with equilibrium in the money market. Hence, in equilibrium, it must be that $p^{m,t} = p^m$ for all t . If $p^m = 0$, then we have a non-monetary equilibrium, while $p^m > 0$ gives a monetary equilibrium. As Balasko and Shell note, this condition is well-known from work on intertemporal capital accumulation, where it represents the no-arbitrage condition on durable capital.

Balasko and Shell use the monetary framework developed in the paper to develop a theoretical framework for analyzing monetary policies in the OLG model, though they note that at the level of abstraction in the paper, there isn't really a distinction between monetary and fiscal policy, since positive money transfer are also government spending (though without the imposition of a government budget constraint), while negative money transfers are taxes. To this end, the paper defines a *bonafide* monetary policy as a sequence of injections/withdrawals of fiat money which maintains a non-zero price of money.

Most of the middle part of the paper is devoted to examining properties of *bonafide* policies. These include:

1. Long-run neutrality of money: if all money transfers are scaled up or down in the same non-zero proportion, the equilibrium allocations remain unchanged (Proposition 4.7 of the paper).
2. If the allocation sequence is bounded above and $\|p^t\| \rightarrow 0$, so that interest rates are eventually positive, then $M^t = 0$. This is a general characterization of Gale's classical case (Proposition 5.8 of the paper).
3. Equilibrium money prices are indeterminate; different price levels for the economy can be consistent with the monetary equilibrium associated with a given monetary policy (discussion after Proposition 7.3 of the paper).

The last result is important because it implies that one cannot associate a unique equilibrium price of money with the set of equilibrium allocations and define this as the utility of having a store of value. This, in turn, means that any monetary theory based on money-in-the-

utility-function must ultimately be one of modeling the microfoundations of transactions and representing the result as an indirect utility of money.

The third paper in the series examines a specific instance of the OLG model in which all agents' utility functions are log-linear. For a typical young agent, the utility function takes the form

$$u(x_t^{t+1}, x_t^t) = \sum_{i=1}^l \alpha_t^{t,i} \ln \ln x_t^{t,i} + \sum_{i=1}^l \alpha_t^{t+1,i} \ln \ln x_t^{t+1,i}$$

where the coefficients satisfy

$$\sum_{i=1}^l \alpha_t^{t,i} + \sum_{i=1}^l \alpha_t^{t+1,i} = 1.$$

With this specification, the paper then examines the monetary equilibria associated with different monetary policies and shows that the set of *bonafide* monetary policies is convex, and that, for a given specification of endowments and monetary policy injections, the set of equilibrium money prices is an interval.

We note that much of the analysis in BS-II and BS-III makes heavy use of the price-income equilibrium analysis developed by Balasko and detailed in Balasko (2016). While an explication of this approach to equilibrium analysis is beyond the scope of this text, Balasko and Shell's application of these techniques illustrate how the practice of microfounding macroeconomic models makes possible the application of powerful tools from general equilibrium theory.

Section 6.4 Sunspots

The final topic we cover in this chapter is the development of so-called *sunspot* models, i.e., models in which agents' beliefs are based on random events that are extrinsic to the economy, but which are self-fulfilling in the resulting stochastic rational expectations equilibrium. Before we embark on our examination of this work, we refer the reader to the very thorough Cherrier and Saidi (2018) looking at the development and propagation of the sunspots idea in economics.

In looking at the genesis of the idea, there are three papers to consider. Chronologically, these are Shell (1977), Azariadis (1981), and Cass and Shell (1983).

According to Cass and Shell, the idea that there might be self-fulfilling rational expectations equilibria in an economy with no intrinsic uncertainty developed from conversations they began having after Cass moved from Carnegie Mellon to Penn in 1974. From the *Macroeconomic Dynamics* interview with Cass (Spear and Wright [1998]) (discussing the impact of Lucas' 1972 "Expectations and the Neutrality of Money" paper)

I wasn't so interested in the macro, but what struck me, and this is related to some of my later work, was the assumption that Bob made to solve for equilibrium, that the state variables were obvious (that is actually the first time that I thought about the sunspot idea). Bob and I had some long discussions, and I would say, "Well Bob, why is this the actual state space in this model?" That question came up – and now I am jumping ahead – after I came to Penn. At some point Karl and I started talking about that and we developed what we called the idea of sunspots. (p. 45)

Shell corroborates this in his *Macroeconomic Dynamics* interview (Spear and Wright [2001]):

I made a trial run on sunspots at the Penn theory workshop during the Spring of 1977. In May 1977, I gave the core of what became the Malinvaud lecture at the Dartmouth conference center on Squam Lake. It was at an MSSB [Mathematical Social Sciences Board] conference organized by Dave and me.

My lecture was in part a follow-up to my 1971 *JPE* OG Notes [*on the Economics of Infinity*]. Hence the focus was on the simple linear case. Immediately after the lecture,

while still at the conference, Dave and I did an analysis of the nonlinear case, using Gale-like phase diagrams. (p. 721)

As Cass and Shell were working on fleshing out this new idea, the Economics Department at Penn hired Costas Azariadis as an Associate Professor. Azariadis did his graduate work at GSIA (Carnegie Mellon's business school, now called the Tepper School of Business), where Ed Prescott was his thesis advisor, though Azariadis counted Bob Lucas as a mentor there as well. Azariadis' early work from his thesis and time at Brown University as an Assistant Professor was in contract theory and applications of that to macroeconomics. He later became interested in dynamic macroeconomics and (presumably based on his knowledge of Lucas' work) decided that the OLG model was the way forward. From his *Macroeconomic Dynamics* interview (Durlauf [2007])

The only problem was that few people, except Karl Shell and Dave Cass at Penn, cared much or knew much about OLG. When I suggested to Herschel Grossman that we had to bone up on that model if we wanted to do interesting work in macro, he asked how we should do that. To which I replied, only half in jest, that we had to move to Penn. And that is exactly what happened. I was offered a job at the Penn economics department in March 1977, moved to Philadelphia in August of the same year, and stayed there for the next 14 years. (*Ibid*, p. 255)

During Azariadis' first few years at Penn, he and Shell became friends, and they talked a good bit about the sunspots idea. From the Shell interview:

Dave and I are the co-inventors of the sunspot equilibrium concept. Our original invention was explicitly based on OG dynamics. We later introduced sunspots into other general equilibrium models. I returned from Paris in June full of my OG work with Yves and my sunspots work with Dave. Dave picked us up at the Philadelphia airport, but was out of circulation for the rest of the summer. I gave Costas a copy of the Malinvaud lecture, and filled him in on what Dave and I had done beyond this. Costas was, for quite some time, resistant to the ideas, but we had a lot of time together and eventually he was persuaded. Costas did something that Dave and I did not do: He found *stationary* (Markovian) sunspot effects (in the Lucas model). (*Op. cit.* p. 724-725)

Azariadis also recalls his skepticism about the sunspots idea in his interview:

[T]he issue of beliefs kept coming up in conversations between me and Karl Shell, mainly because he and I lived close to each other in West Philadelphia. We frequently walked home together from Penn, and used to debate about beliefs and other things. Karl had an early example of multiple equilibria in a 1977 French-language working paper. I found the idea interesting, but the example unconvincing. It was a static economy with flat indifference curves that became tangent to budget lines at many points. I told Karl his example was a fluke, something completely nongeneric. Rational expectations, I claimed, would surely rule out his “sunspots.”

I told Karl in 1979 that I was going to write down a robust example of an economy with indifference curves that were not flat. That, I hoped, would persuade him that “sunspots” could not exist. I had not counted on large income effects, which permit many equilibria in static economies, and many laws of motion in dynamic ones. When I finished my example, it turned out that I had made Karl’s case better than he had done. He was right, and I was wrong, for a large family of OLG environments. (*Op. cit.* p. 256)

For the record, we note that Azariadis is incorrect in his recollection that Shell’s “Monnaie et Allocation” paper (Shell [1977]) was in French; it was not. The title page was in French, but the body of the paper was in English. We should also note that the model in Shell (1977) was not static, but in fact use the OLG structure to construct sunspot equilibria around the deterministic steady state of the model.

The paper that resulted from this interaction was Azariadis’ “Self-fulfilling Prophecies” (referenced above.) From our perspective, this is a clear case of the kind of scientific discoveries based on intellectual cross-fertilization that have occurred in other cases in economics (see, for example, Young (1987) on the origins of IS-LM and Young (2014) on the origins of Real Business Cycle models).

However, at the time, Dave Cass didn’t view things that way. From the Cass interview, it is clear he didn’t share Shell’s enthusiasm for Azariadis’ paper:

My view is that Karl explained the idea to Costas a number of times, and Costas finally picked up on it and he wrote a paper about it. He realized, not from a utility approach, but by having a first-order Markov system of probabilities, that one can get sunspot equilibria. Steve’s thesis actually develops the general story, and he solved that problem long before, for example, Azariadis and Guesnerie did. But I have to credit Costas with

something. When Costas produced a working paper or maybe even before that we realized that if we were going to develop the idea, we'd better get to it. (*Op. cit.* p. 50)

The reference here to “Steve’s thesis” is to Spear (1984). That paper came about as a result of Cass’s resentment directed at both Shell and Azariadis for what he perceived as the unauthorized use of the sunspot idea. From Spear’s private recollection of the genesis of this paper:

Dave called me into his office one morning while I was finishing up some work on my first thesis paper (which was all about temporary equilibrium), and sketched out the idea behind sunspot equilibria, and about how to characterize these in a simple OLG model where the extrinsic uncertainty is just a coin flip. I found the idea intriguing and went to work on it, finally coming up with a set of sufficient conditions for the existence of these equilibria. I have to admit that at one point I was stuck, and asked Costas for some help, which he gave gladly, though not without a bit of a funny look my way. It was quite a few years later that I learned that Dave had given me the problem to work on because of his anger at Azariadis.

According to private communication from Shell, this incident was the beginning of what he has called the “Cass-Shell divorce” though there were earlier indications of friction over Shell’s openness to discussing ideas with colleagues and an apparent growing penchant toward secrecy on Cass’s part.

Since Cass sadly is no longer alive and thus cannot answer questions regarding this incident, we can only speculate as to what might have generated this penchant.

As a general matter, secrecy in scientific work can serve one of two purposes. It can help ensure that priority for an idea is correctly established, or it can ensure that results, when finally released, are correct and properly understood.

Looking at Cass’s career retrospectively, he never seems to have been particularly concerned about priority. In previous work (see Spear and Young [2014]) we have detailed the story of how a young Dave Cass, as a graduate student, worked out the solution to the

constrained optimal control problem for the one-sector continuous time growth model. A much more senior Tjalling Koopmans was also working on this problem, but his initial attempts to solve it were in error. After hearing from Edmond Malinvaud why his solution was wrong, Koopmans ended up getting help from Hirofumi Uzawa – who happened to be Dave Cass’s thesis advisor – in the form of a preview of the thesis chapter in which Cass had solved the problem.

After the publication of both Cass’s and Koopmans’ papers, the economics profession’s long-standing view of the work was that the two papers were developed independently and simultaneously, something we now know not to be true. Cass never let on that he knew this wasn’t the case and went to his death insisting that the work had been independent and simultaneous.

From the Cass interview, commenting on this episode:

Actually, Uzawa liked to one-up people. At some point he was talking to Tjalling about the problem, and Tjalling was describing what he was doing and Uzawa interrupted and said, “Well, I have a graduate student who did that problem.” Then Tjalling got very nervous about it, he was always very nervous about . . . , oh, authorship and who was first and that sort of thing, and we had some correspondence. (*Op. cit.* p. 38)

On the rigor aspect of secrecy, Cass always emphasized accuracy and could well have viewed his work with Shell as preliminary enough to warrant keeping it under wraps. Despite this, Cass was not particularly secretive about collaborative work with colleagues. From Spear’s private recollections:

It was quite common to see Dave, Karl, and Costas sitting in the CARESS [Center for Analytic Research in Economics and the Social Sciences] lounge working through their ideas about sunspot equilibria on the board. Dave liked the lounge because, after the University had banned smoking in offices, he had had a large exhaust fan installed in the lounge to vent the cigarette smoke. Most of us in the graduate program would see them

there when we went into the lounge to get coffee.

If we look carefully at the Cass-Shell “Do Sunspots Matter” paper in comparison with Azariadis’ “Self-fulfilling Prophecies” paper, one clear distinction becomes apparent. Azariadis was focused on the macroeconomic implications of sunspot equilibria as a potential driver of business cycle fluctuations – akin to Keynes’ notion of “animal spirits.” Cass and Shell, on the other hand, were focused on understanding and explaining the general equilibrium frictions that could give rise to self-fulfilling prophecies. In Azariadis’ paper, it is strong income effects that generate the possibility for sunspot equilibria. In the Cass-Shell paper, it is restrictions on trading opportunities and other frictions that lead to a failure of the welfare theorems that matter.

Since Cass and Shell had not yet worked these issues out in their research, it is entirely natural that Cass might have felt that Azariadis had been too cavalier with his results.

At the end of the day, we don’t have definitive evidence of what led to Cass’s resentment over the situation. It could be that the experience of being denied what should have been clear priority for his work on the growth model in his thesis might have left Cass less open to cross-communication of ideas during their early stages of development. This seems unlikely though, given that the neoclassical growth model is universally known today (and has been for many years) as the Ramsey-Cass-Koopmans model. It may be simply that Cass didn’t feel that income effects were sufficient justification for putting forward the sunspots concept absent a deeper explanation of how the phenomenon arose.

But Cass’s resentment did have repercussions. It led to the breakdown of a years long productive research relationship between Cass and Shell, and ultimately to Shell’s relocation to Cornell’s Economics Department in 1986. It left Azariadis, then a still junior faculty member, alienated from the senior faculty who had been instrumental in recruiting him to Penn. When we

asked Azariadis why he hadn't offered Cass and Shell co-authorships on the self-fulfilling prophecies paper, he replied

My, perhaps imprecise, recollection of the co-authorship issue you raise was that it never came up or was broached by anyone involved. One reason was that I started the prophecies paper in an attempt to show that Karl's 1977 example was a fluke. I thought that example could not be generic because rational expectations equilibria were locally unique in static models, as in the work of Radner, and saddles in dynamic models. It did not occur to me that Karl would co-author a paper that wanted to prove him wrong! (Email dated May 28, 2022)

While it seems unlikely that Cass and Shell would have accepted a co-authorship had it been offered, given the direction that their own work took, the whole situation would have been considerably less problematic if Azariadis had presented Cass and Shell with a mini-seminar on his discovery, and worked through the issues of how to present the discovery and the allocation of credit before the publication of the paper. To his credit, Shell did not let his own concerns with this situation bias him in his position as Editor of the *Journal of Economic Theory*, which processed the Azariadis paper through to publication.

Since these two papers have been influential – Azariadis' paper has 1143 citations on Google Scholar, and the Cass-Shell paper has 1460, as of this writing – and because it will help to explicate the differences between the Cass-Shell and Azariadis approaches to sunspots, it is worth spending some time looking at the results. We start with the Azariadis paper, since it was published first.

The key insight that Azariadis had in looking at the sunspot question was in recognizing that a very simple stochastic process – a Markov chain on two-point support – was, in the limit, a cycle. Indeed, if one writes the transition probability matrix for this as

$$P = \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix}$$

where π_{ij} is the probability of starting in state $i=1,2$ and making a transition to state $j=1,2$.

When $\pi_{ij} = 1, i \neq j$ then necessarily $\pi_{ii} = 0$ for $i=1,2$, and the transition matrix degenerates to

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

But this implies that the process alternates deterministically between the two states, which is to say, the process is simply a cycle of period 2. What Azariadis did was to show that when the simple OLG model has a cyclic equilibrium, it will then also have a sunspot equilibrium (i.e., one which is genuinely stochastic) because the equations that define the equilibrium – the first-order conditions and market-clearing conditions – depend continuously on the probabilities associated with the transition matrix. We can illustrate this using the simple version of the Lucas model that we developed above. Specifically, the equilibrium equation

$$z_1^H + \frac{1}{\rho} z_2^{HT} = 0$$

is the same as in the stochastic model, except now, none of the endowments entering into the definition of the two excess demand functions is actually stochastic. In this case, given our assumption that we are dealing with a Samuelsonian economy in which the endowments of the young exceed those of the old, we know that one equilibrium value of the relative price ρ will be one.

We illustrate the possibilities graphically in Figure 11

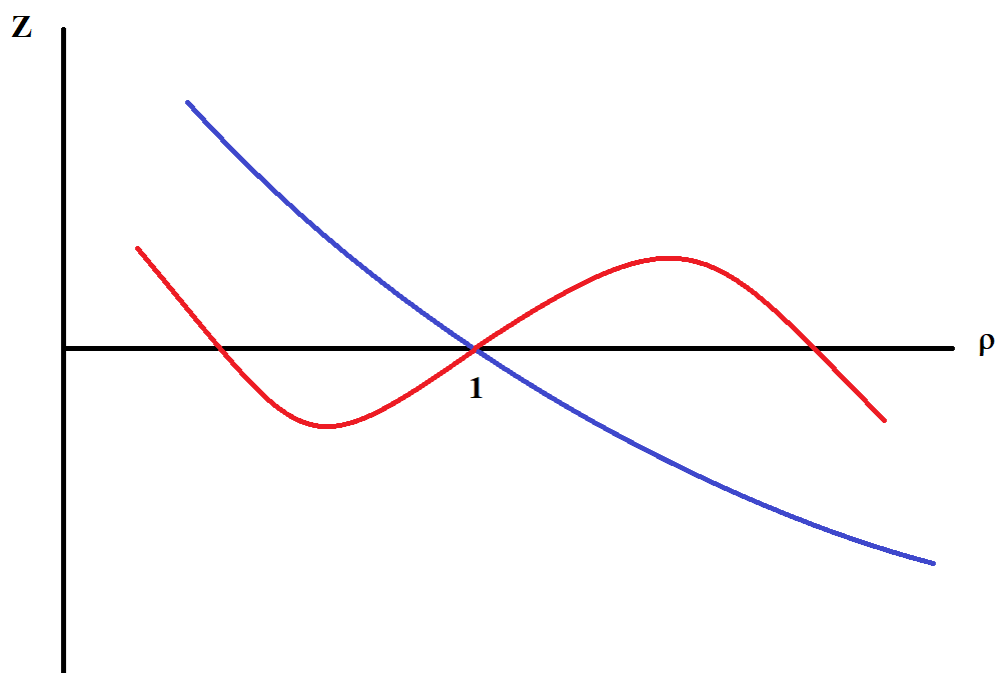


Figure 11

Excess demand functions for non-sunspot and sunspot equilibria

The diagram shows the graphs of two different deterministic excess demand functions. The blue graph corresponds to an economy in which there is a unique (monetary) equilibrium with relative price equal to one. The red graph corresponds to an economy with three equilibria, two of which are different from the one with $\rho = 1$. For the red economy, it is possible for the equilibrium to cycle between the higher and lower equilibrium prices. This possibility, as we noted above, was foreseen by Gale, and is illustrated in Figure 6 above, taken from Gale (1973). We replicate Gale's diagram in detail to show how the cyclic equilibrium can occur in Figure 12.

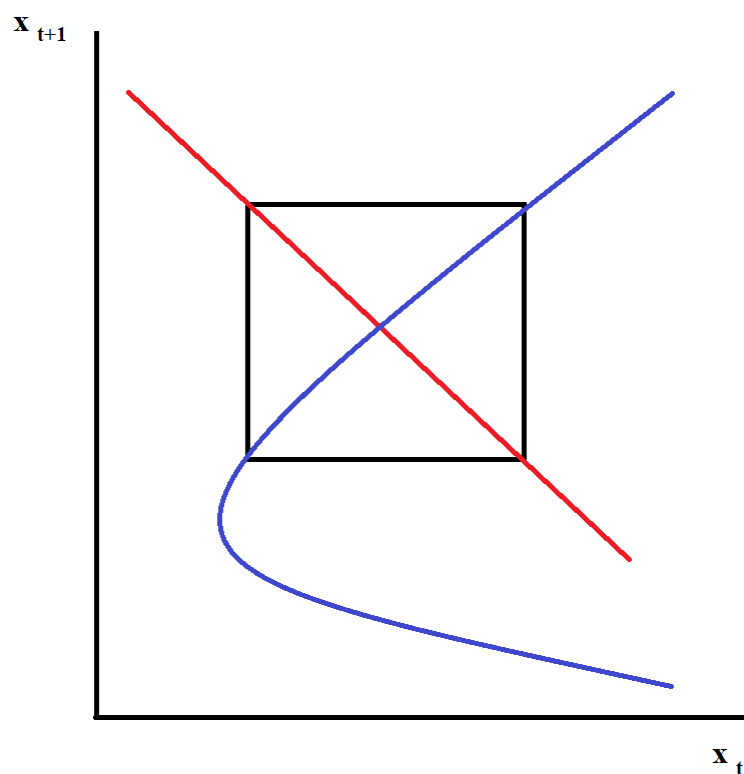


Figure 12

Backward-bending offer curve and cyclic equilibrium

In this diagram, the blue curve is the offer curve, the red line is the resource constraint (downward sloping 45-degree line) which, if continued down to where it intersects the offer curve would indicate the endowment. The black square, where it intersects the offer curve shows the two cyclic equilibria that the economy oscillates between. Note that this requires the offer curve to bend back on itself, which occurs when the income effects are such that young agents will wish to save less when interest rates rise and when they fall. In this Samuelsonian setting, interest rates would oscillate between being positive (at the upper equilibrium point) and negative at the lower point.

The Cass-Shell paper used a truncated version of the OLG model to simplify the analysis and focus on the general equilibrium aspects of the sunspot issue. As Cass and Shell note in a long footnote at the beginning of the paper:

These examples [from Azariadis' paper], however, are based on very complicated dynamic models which do not permit us to separate the respective roles of generational overlap, the infinite time horizon, and government debt. In the present paper, we focus on the implications of generational overlap in a very simple equilibrium model. (Cass-Shell [1983], p. 194)

The Cass-Shell paper focuses on the role of restricted participation on markets as the phenomenon that allows extrinsic uncertainty to matter, i.e., to have a real effect on consumers' allocations and utility. The restriction on participation in their model is implemented by positing a generation of consumers who live two modeling periods and have access to securities markets for inside money (zero net supply bonds) and Arrow securities for insuring themselves against the perceived (but extrinsic) uncertainty. A second generation of consumers is born after the realization of the sunspot variable, lives out the second period only, and can trade only on current period spot markets for commodities.

This restriction on participation is similar to that examined by Cass and Yaari (1966), but the limitation to two periods simplifies the model considerably.

Cass and Shell show a number of important results. First, if markets are complete and participation is not restricted, then sunspots can never matter. Second, they show that with restricted market participation, if the underlying certainty economy has multiple equilibria, then one can obtain sunspot equilibria by randomizing over the multiple certainty equilibria. This is a quick and easy way to show that sunspots can matter. A second important point the paper makes (in Proposition 7) is that if the welfare theorems hold in a model, then there can be no sunspot equilibria. Cass and Shell tried to show in the appendix to the paper that there could be non-

trivial sunspot equilibria in a model for which the certainty equilibrium was unique. According to private communication from Shell (email of September 5, 2022), two of Tom Muench's students had found a mistake in the appendix which couldn't be fixed. In later work (see e.g., Shell and Wright [1993]), Shell gave examples of models having unique certainty equilibrium and non-trivial sunspot equilibria.

While this third observation may seem trivial, its converse – that if the welfare theorems don't hold, the model can exhibit sunspot equilibria – raises a host of interesting possibilities for economic dynamics. Shell has dubbed this converse proposition the “Philadelphia Pholk” theorem because of the myriad results it has spawned as researchers looked into instances in which the welfare theorems failed to see if sunspots mattered.

The Philadelphia connection in Shell's quip is through Penn, via the number of economists who looked more deeply into the connection between market failures and sunspots. These economists include Cass, who spent much of the rest of his career studying incomplete markets, beginning with an example of how market incompleteness could give rise to sunspot equilibria (see, e.g., Cass [1989]). Shell and his student Jim Peck wrote several papers looking at the effects of imperfect competition as a source of sunspots, based on the Shapley-Shubik market game. They found an important connection between the notion of correlated equilibrium in game theory and sunspot equilibrium in the market game (see Peck and Shell [1991]). Spear showed, in the context of an otherwise conventional RBC model, that externalities in production could lead to the existence of sunspot equilibria (see Spear [1991]). A similar result was also shown by Kehoe, Levine and Romer in the context of taxation (see Kehoe, Levine, and Romer [1992]).

In macroeconomics, the sunspots idea saw applications as the driver of aggregate shocks in real business cycle models, as well as interpretations as Keynesian “animal spirits” in models that linked the psychology of expectations formation with dynamic economics. We will examine these issues in more detail at the end of the book.

Chapter 7: Boomers

As we noted in the Introduction, there was an explosion of new research done on the OLG model in the decade of the 80's-90's, much of it by baby boomers in or just finishing graduate school. As we noted at the beginning of the previous chapter, this work was concentrated predominantly at Minnesota, MIT and Penn, for the simple reason that these researchers were trained by the previous academic generation who did the original work on the model.

In examining this work, we will proceed roughly chronologically, but try also to organize consideration by focusing first on developments with the deterministic OLG model, and then on stochastic extensions of it.

Section 7.1 Kehoe and Levine

We begin by looking at the work of Tim Kehoe and David Levine. We consider this as coming from the MIT tradition, although Kehoe's Ph.D. was from Yale (working with Herb Scarf and Andreu Mas-Colell). While Kehoe's first job was at Wesleyan University (beginning in 1978), he moved to MIT in 1980. Levine's Ph.D. was from MIT in 1981, working with Peter Diamond and then-new assistant professor Tim Kehoe. The source of Levine's interest in OLG is obvious from the connection with Diamond. According to Levine, Kehoe's interest in OLG developed as they began looking at determinacy issues. From an email communication (dated July 14, 2022)

Tim arrived at MIT my last year, September 1980, I finished in September 1981, and we were working on [the OLG project] by March 1982. I did a bunch of determinacy stuff in my dissertation but on game theory. I'm pretty sure we started with the finite players infinite horizon determinacy using the Negishi method which Tim was fond of and taught me. This was a natural extension of the work Tim was already doing on the finite economy case. I think we quickly wondered what happened in the OG model, and my best guess is we started talking about this sometime between March 1981 while I was a graduate student, and before February 1982 when I was at UCLA.

Most of the Kehoe-Levine papers on OLG were produced during the early 80's, while Kehoe was at MIT and Levine was a new assistant professor at UCLA. As Levine noted in an email to us dated June 12, 2022, "we worked via telephone calls and fax back then."

Kehoe and Levine approached the OLG model using the tools of general equilibrium theory, applied within the framework of dynamic systems analysis. Dynamic systems analysis was not new to economics, having been widely applied to the study of the neoclassical capital model in continuous time. Continuous time formulations are generally easier to analyze since they are characterized by differential equations rather than difference equations, so it isn't particularly surprising that the early work of Ramsey, Malinvaud, Solow, Cass and others was first formulated in continuous time. Discrete time applications came about with the interpretation of economic time as planning time, where agents would formulate plans to be carried out over a discrete set of periods (weeks, for example). Basic dynamic systems theory applies equally well to both treatments of time, though, as we noted above, continuous time systems tend to be more tractable and amenable to analytic solutions. Thus, for example, in the stability analysis of the continuous time capital model, the optimal steady state is usually saddle-path stable. As we will see later, discrete time versions of the model can exhibit very different stability properties. The early 1970's was a period of rapid advancement in the understanding of discrete time dynamic systems in mathematics, so it isn't surprising that economists interested in

dynamic models would find a useful toolkit here. For details on these mathematical developments, see, e.g., Aubin and Dalmedico (2002).

As a preliminary matter, it will be useful to illustrate the basic structure of the discrete time approach to dynamic systems analysis.

A dynamic system is a specification of state variables and a function or mapping which defines how the state variables evolve over time. If we denote our state vector by $z \in \mathbb{R}^n$, then we can represent the dynamic system as a *forward difference equation*

$$z_{t+1} = f(z_t)$$

A fixed point of the mapping $\hat{z} = f(\hat{z})$ is called a steady state, or a stationary point, or a rest point of the dynamic system. A fixed point $\hat{z}$ is locally stable if there exists a neighborhood $N(\hat{z})$ such that for any $z' \in N(\hat{z})$

$$\lim_{t \rightarrow \infty} f^t(z') = \hat{z}$$

where $f^t = f \circ f \circ \dots \circ f$ ^{*t times*}.

The *invariant manifolds* associated with a particular steady state are hypersurfaces $M \subset \mathbb{R}^n$ such that $\hat{z} \in M$, and if $z' \in M$, then $f(z') \in M$. Every steady state has associated with it two particular invariant manifolds, the so-called stable and unstable manifolds. Every element of the stable manifold converges to the steady state under forward iterations of the mapping f , while every element of the unstable manifold converges to the steady state under backward iterations of f .

We can get information about the stability of a dynamic system in the neighborhood of a steady state by examining the Jacobian matrix of the mapping f at the steady state, $Df(\hat{z})$. In

particular, we will be interested in the eigenvalues of $Df(\hat{z})$. If all eigenvalues have modulus less than 1, then the steady state is stable. In general, some eigenvalues will be less than one while others are greater than one, in which case we call the steady state a saddle point. This is illustrated in the diagram below.

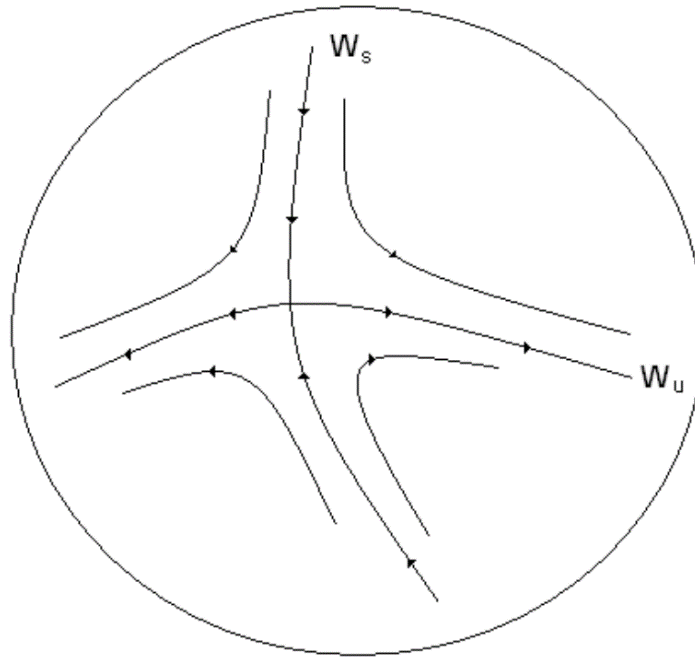


Figure 13

Local decomposition of invariant subspaces

A key result on dynamic systems is the *Grobman-Hartman Theorem*, which provides a characterization of local stability for so-called hyperbolic steady states of a dynamic system. A steady state $\hat{z}$ is hyperbolic if no eigenvalue of $Df(\hat{z})$ has modulus equal to one.

Theorem: If $\hat{z}$ is a hyperbolic steady state of the dynamic system

$$z_{t+1} = f(z_t)$$

then there exists a homeomorphic (generally non-linear) coordinate transformation of a neighborhood $N(\hat{z})$ with itself which, when applied to the mapping f , yields the linear dynamic system

$$w_{t+1} = \hat{z} + Df(\hat{z})[w_t - \hat{z}].$$

This result states that for hyperbolic dynamic systems, it is sufficient to study the stability properties of the linearization of the system around a fixed point. Note also that if we define new variables $x = w - \hat{z}$, then the dynamic system takes the simple form

$$x_{t+1} = Df(\hat{z})x_t$$

and the steady state now occurs at the origin. If we also make a linear coordinate change to put the matrix $Df(\hat{z})$ in canonical form, we get

$$y_{t+1} = \Lambda y_t.$$

Now, if all of the eigenvalues of $Df(\hat{z})$ are distinct, then Λ will be a diagonal matrix with the eigenvalues of $Df(\hat{z})$ down the main diagonal. In this form, the action of the dynamic system is particularly transparent. Along the eigenspaces corresponding to eigenvalues which are less than one in modulus, the dynamic system behaves like a contraction mapping. Along the eigenspaces corresponding to eigenvalues having modulus greater than one, the dynamic system is expanding. All trajectories not starting in one of the eigenspaces will be representable as linear combinations of the action of the system along the various canonical subspaces.

The first Kehoe-Levine paper to explore the OLG model as a dynamic system was Kehoe-Levine (1984). This paper started life as “Indeterminacy of Relative Prices in Overlapping Generations Models” (1983 MIT working paper) which was focused (as the title suggests) on the issue of indeterminacy, with the regularity and dynamic systems analysis developed there as tools. The 1984 “Regularity in Overlapping Generations Exchange Economies” paper split off the dynamic systems analysis (and broadened it considerably), postponing the determinacy analysis to a later paper.

Kehoe and Levine (K&L henceforth) simplified their analysis of the OLG environment by appealing to the same community excess demand results that Cass, Okuno and Zilcha did in their 1980 paper. These results state that in a general equilibrium economy with n goods, any continuous function $f: \mathbb{R}_+^n \rightarrow \mathbb{R}^n$ satisfying

1. f is homogeneous of degree zero (condition H)
2. f satisfies Walras Law: $p \cdot f(p) \equiv 0$ for all $p \in \mathbb{R}_+^n$ (condition W)
3. f is bounded from below (condition B)

will be the aggregate excess demand function for an economy consisting of at least n preference maximizing consumers. K&L add the condition that the aggregate excess demand functions they consider also be smooth (at least twice continuously differentiable), based on results from Debreu (1974) and McFaddin *et al.* (1974).

For the two-period lived agents model, K&L denote the aggregate excess demand functions of the old living at time t by $z(p_{t-1}, p_t)$, and that of the young agents living at time t by $y(p_t, p_{t+1})$, where we follow their notation and use subscripts to denote time periods. The heterogeneity of agents across different ages necessitates this level of decomposition. With n goods traded in each period, K&L need at least n agents of each type in order to invoke the community excess demand theorems. In terms of the two aggregate excess demand functions, the market clearing condition is

$$z(p_{t-1}, p_t) + y(p_t, p_{t+1}) = 0.$$

Following the literature, K&L assume the existence of a steady state equilibrium price p together with a scalar β such that

$$z(p, \beta p) + y(\beta p, \beta^2 p) = z(p, \beta p) + y(p, \beta p) = 0$$

where the second equality follows from the degree zero homogeneity assumption. The scalar β can be shown to be the intertemporal marginal rate of substitution for any individual consumer at the competitive equilibrium allocation. When $\beta=1$, the equilibrium is said to be nominal since it can be supported by having agents hold (fixed) money stocks in the amount $p \cdot z(p, p) = \mu > 0$. From this definition, it then follows that $(1 - \beta)\mu = 0$ at any steady state, a condition we've seen before, that indicates that if money has value and is held in any positive quantity, then the intertemporal MRS of all agents in the economy must be equal to 1, and the equilibrium is a monetary one. When $\mu=0$, then β need not equal 1. When $\beta < 1$, the associated competitive equilibrium will be inoptimal, while the opposite obtains if $\beta > 1$.

This observation leads to K&L's first regularity condition: the systems of equations

$$z(p, p) + y(p, p) = 0$$

$$-p \cdot y(p, p) = 0$$

has no solution. This condition rules out the case where $\beta = 1$ and $\mu = 0$ simultaneously. The paper shows how to perturb the chosen aggregate excess demand functions so that they continue to satisfy conditions (H), (W), and (B) to get these equations to not hold if it happens that one's choice of excess demands has led to the equations being satisfied. Via the perturbation analysis, K&L show that almost all excess demand functions satisfy this regularity condition, so that these economies (the regular economies) are generic.

With this assumption in place, K&L next go on to show that there will always be a steady state equilibrium for the model with $\beta = 1$, and a second steady state with $\beta \neq 1$. As we've seen before, if $\beta < 1$, this steady state is the autarky equilibrium. When $\beta > 1$, the equilibrium is Gale's classical case equilibrium. While K&L carry through their analysis for the case of both

real and nominal steady states, we will simplify the analysis here by focusing only on the nominal case.

K&L let $q_t = [p_t, p_{t+1}]$, and write the equilibrium market-clearing condition as

$$z(q_t) + y(q_{t+1}) = 0$$

with $q = [p, p]$ denoting the steady state equilibrium. They then linearize the equilibrium equations around the steady state in order to study the local dynamics on a neighborhood of the steady state. In order to do so, though, K&L need to invoke a second regularity condition, that

$$|D_{p_{t+1}}y(p_t, p_{t+1})| \neq 0,$$

in order to apply the implicit function theorem and infer the existence of a function $\hat{g}: \mathbb{R}_+^{2n} \rightarrow \mathbb{R}_+^n$ giving the equilibrium forward price p_{t+1} as a function of the prices at t and $t-1$. Since this formally defines a second-order difference equation system, applying standard dynamic systems theory to the model requires converting the second-order system into a first-order one, using the standard trick of writing

$$q_{t+1} = g(q_t) = \begin{bmatrix} \hat{g}(p_{t-1}, p_t) \\ p_{t-1} \end{bmatrix}.$$

Linearizing this first-order difference equation system around the steady state then yields

$$\begin{bmatrix} p_t \\ p_{t+1} \end{bmatrix} = q_{t+1} = D_{q_t}g(q)q_t = \begin{bmatrix} \hat{g}_1 & \hat{g}_2 \\ I & 0 \end{bmatrix} \begin{bmatrix} p_{t-1} \\ p_t \end{bmatrix} = Gq_t.$$

Here, $\hat{g}_i$ is the matrix of partial derivatives of the mapping $\hat{g}$ taken with respect to the i^{th} argument. These are determined in terms of the derivatives of the z and y functions using the standard implicit function theorem calculations. With these definitions, the paper then derives a set of restrictions on the derivatives of the excess demand functions implied by the economic conditions (H), (W), and (B). The boundedness condition doesn't restrict the derivatives, but homogeneity and Walras' Law imply, in turn, that β and I must be eigenvalues of the matrix G .

Of course, at a nominal equilibrium, the only restriction is that G has a unit root. Other than these restrictions, the eigenvalues of G can be anything, subject to the restriction that, because G is a real-valued matrix, complex eigenvalues must occur in conjugate pairs.

As we noted earlier, applying the Grobman-Hartman theorem requires that the matrix $G = D_{q_t}g(q)$ has no eigenvalue with modulus equal to one, which creates a problem. We can get around this problem, though, by restricting attention to the nominal case when the money supply $p_t \cdot z(p_{t-1}, p_t) = \mu$ is constant. For this case, K&L consider the equilibrium dynamics of the model are restricted to a $2n-1$ dimensional manifold which is invariant under the actions of the mapping $g_m = g|_{\mu=m}$ where m is the fixed money supply. In this case, the linearized local dynamics will be governed by the matrix $G_m = G|_{\mu=m}$. For this matrix, the eigenvector corresponding to the unit root in the original system will still be an eigenvector in the restricted system, but the corresponding eigenvalue is now zero, so that the restricted system will be hyperbolic. Per the Grobman-Hartman theorem, we know that there will be two linear subspaces E_s and E_u which are invariant under the actions of g_m and correspond to the stable and unstable invariant subspaces through the steady state price vector q . Because G_m is now hyperbolic, the dimensions of these subspaces will be complementary, i.e., $\dim E_s + \dim E_u = 2n - 1$.

K&L show how to analyze the case of a real steady state with $\beta \neq 1$ by noting that in this case, at the steady state, prices will be increasing or decreasing geometrically. By renormalizing prices by dividing them by a scalar function of p_t which is homogeneous of degree one, one obtains a stationary dynamic system similar to the one analyzed in the nominal case.

Kehoe-Levine (1984) is primarily methodological, though, as we noted above, it appears to have originated from a study of the question of determinacy of equilibrium in the OLG model. K&L return to this issue in Kehoe and Levine (1985). As we noted in our earlier discussion of

Samuelson's original paper (and his debate with Meckling), Samuelson viewed the equilibrium trajectories in the OLG model as being pinned down only once there was a suitable terminal condition specified for the system of difference equations that determined the economy's trajectory. By Samuelson's definition of determinacy, then, the non-stationary equilibrium in the two-period lived agents model with one good, log preferences and a monetary steady state equilibrium would be determinate, since given an initial condition close to, but not equal to the monetary equilibrium price, the resulting trajectory converged to autarky. In this example, the monetary equilibrium would have a terminal condition requiring that the equilibrium remain at the steady state, while in the non-stationary equilibrium, the terminal condition would be convergence to autarky. This notion of determinacy differed from the static general equilibrium notion which required that equilibrium prices were (at least local) isolated, so that comparative static analysis was possible based on perturbations of the underlying parameters of the economy.

In dynamic economic models where agents have perfect foresight (or rational expectations in stochastic settings), determinate terminal conditions can lead to non-isolated equilibrium trajectories which defy comparative static analysis. The simplest way to illustrate this is with the general Kehoe-Levine set-up outlined above, with an initial starting period. In this initial period $t=1$, we will have

$$z(a, p_1) + y(p_1, p_2) = 0.$$

Since agents have perfect foresight, they know how the equilibrium prices are determined, i.e., they know

$$p_{t+1} = g(p_{t-1}, p_t).$$

Young agents forecasting the future prices using the economy's law of motion will then generate an equilibrium set of market clearing equations of the form

$$z(a, p_1) + y(p_1, g[a, p_1]) = 0,$$

but in this situation, because the function g is obtained from the equilibrium equations via the implicit function theorem, the equation above is identically zero. Hence, any period one price will be an equilibrium price. In the particular case where the law of motion converges to the steady state for any starting price, comparative static analysis based on underlying perturbations of the parameters defining the economy (in this case the two excess demand functions) gives us no information on how the price in period one is determined. If the law of motion is divergent, then the only equilibrium that is viable is the steady state itself, which is locally unique¹².

This result has important behavioral implications. In static (i.e., simple two-period) models, the equilibrium price will be generically determinate by standard general equilibrium arguments, and all agents need to forecast is the equilibrium price. A standard argument in models like these is that agents are rational and know the structure of the economy and can therefore calculate what the equilibrium price will be. In the dynamic OLG setting, the argument presented here shows that how agents forecast future prices is important, and that the assumption that agents know the economy won't necessarily make the equilibrium determinate.

This was the problem that K&L set out to study in Kehoe-Levine (1985). They approach the problem by comparing the determinacy results for a standard ILA model with finitely many dynastic families and the two-period multi-good OLG model they analyzed in the "Regularity" paper.

For the ILA model, they use the Negishi (1960) approach together with techniques from differential topology pioneered by Debreu (1970) to show that the sequence of equilibrium prices

¹² One can argue that if the initial period is simply the point where an observer breaks into the bi-infinite timeline, then the only appropriate perfect foresight price in period one is the one the old in that period were expecting in the previous period. But this simply pushes the determinacy question back to $-\infty$. In the presence of an unanticipated policy shock, or (as we will see later) of an expectational shock, the indeterminacy argument is completely valid.

for this economy are uniquely determined by the Negishi welfare weights corresponding to the given initial endowments. We can view this result from a dynamic systems perspective by going back to our development of the one-sector capital model using the discounted dynamic programming methods. The steady state equilibrium for that economy is a saddle point. (This is a well-known result in capital theory.) The stable manifold associated with the steady state is simply the graph of the policy function that arises from the first-order difference equation delivered by the dynamic programming algorithm. The unstable manifold takes allocations asymptotically to zero or to the point where all resources are used to accumulate capital, leaving zero consumption. The optimal trajectory in this model is completely determinate, given the assumption that agents are rational and will know to follow the accumulation rule prescribed by the policy function. As K&L note, their determinacy result is a direct analog of the result developed by Debreu for static pure exchange economies.

For the OLG model, K&L note that the indeterminacy problem can be remedied if agents in the model use first-order forecasts, rather than the second-order forecast rule prescribed by the equilibrium law of motion for the economy. To show this, suppose agents forecast using the common forecast function $p_{t+1} = f(p_t)$. For this forecast function to be consistent with the equilibrium dynamics of the model requires that

$$\begin{bmatrix} f(p) \\ f \circ f(p) \end{bmatrix} = \begin{bmatrix} f(p) \\ g(p, f[p]) \end{bmatrix}$$

for all admissible prices p . K&L show how to construct such a forecast function based on the linearization of the system given by the Grobman-Hartman homeomorphism, in which the consistency condition requires that

$$\begin{bmatrix} F \\ F^2 \end{bmatrix} = \begin{bmatrix} 0 & I \\ G_1 & G_2 \end{bmatrix} \begin{bmatrix} I \\ F \end{bmatrix}.$$

If we now let $H^{-1}\Lambda_f H = F$ be the canonical decomposition of the matrix F , we have

$$\begin{bmatrix} H^{-1}\Lambda_f H \\ H^{-1}\Lambda_f^2 H \end{bmatrix} = G \begin{bmatrix} H^{-1}H \\ H^{-1}\Lambda_f H \end{bmatrix}$$

or

$$\begin{bmatrix} H^{-1} \\ H^{-1}\Lambda_f \end{bmatrix} \Lambda_f = G \begin{bmatrix} H^{-1} \\ H^{-1}\Lambda_f \end{bmatrix}.$$

This implies that the matrix $\begin{bmatrix} H^{-1} \\ H^{-1}\Lambda_f \end{bmatrix}$ consists of eigenvectors of the matrix G , with Λ_f a Jordan matrix with the eigenvalues associated with the matrix F on the main diagonal.

To determine what the eigenvalues of F should be, K&L impose the condition that the forecast prices lie in the stable manifold associated with the steady state equilibrium, so that given an initial price, the resulting sequence of equilibrium prices will converge to the steady state. This pins down the terminal condition for the model. In addition to this property, K&L also require that the forecast function f be homogeneous of degree 1, since the equilibrium price function g is, and that the function be smooth.

From the eigenvalue conditions above, one can construct the matrix F , and, by inverting the mapping between the linear and original non-linear systems, we find the desired forecast function f . Interested readers should consult the Kehoe-Levine papers for details. For our purposes, the key thing to note is that when equilibrium prices are restricted to the stable manifold of the steady state prices, it provides a natural decomposition of the price dynamics akin to the saddle-path stability that occurs in the capital model. In particular, if the eigenvalues for the model “split” – for a monetary equilibrium, this means having $n-1$ eigenvalues of G inside the unit circle, and $n-1$ outside the unit circle. Given the K&L results on the (lack of) restrictions on the eigenvalues of G , we can characterize equilibria from different models

depending on whether the associated matrix G has more stable roots than unstable roots, the same number, or more unstable roots. For the splitting case (equal numbers of roots), the equilibrium is determinate. For more stable roots than unstable, equilibria are indeterminate, and, as we will see later, sunspot equilibria can arise. The third case generates unstable equilibria which can only exist in steady state.

Kehoe-Levine (1985) invokes the Balasko-Cass-Shell result showing how to convert a multi-period model into a higher dimensional two-period model and given that they start out by working with aggregate excess demands and arbitrary (but finite) number of goods, their characterization results are quite general. We will see later, though, that because the Balasko-Cass-Shell transformation doesn't work for stochastic models, there is a need for constructing lower-order forecast functions in multi-period models. The extension of the Kehoe-Levine procedure outlined here was provided by Kim and Spear (2021a).

We note that a similar set of restrictions on forecasting functions in the context of linear rational expectations models in macroeconomics were derived and published in Blanchard and Kahn (1980). This paper also characterizes the various possible outcomes for this class of linear models in terms of the relationship between stable and unstable roots of the matrix that determines the equilibrium law of motion for the economy. Since this paper came before the Kehoe-Levine papers, these conditions are usually referred to as Blanchard-Kahn conditions.

Section 7.2 Chaos

While Kehoe and Levine were working out the local dynamics for general OLG models, Jean-Michel Grandmont was tackling a different question, that of characterizing the global dynamics of a very specific OLG model, the one commodity, two-period lived agents model of

pure exchange. We have noted earlier that Grandmont's interest in temporary equilibrium models grew at least in part out of his disbelief in the rational expectations hypothesis. His initial interest in the possibility of complex dynamics occurring in the OLG model was also motivated by his frustration with the fact that, after the publication of Lucas (1972), rational expectations rapidly became the dominant paradigm in dynamic macroeconomics, particularly with the development of real business cycle models in the early 1980's (which we will discuss at some length in chapter 9).

Despite Grandmont's determination to "kill off" rational expectations, the issue of multiplicity of equilibria which Grandmont focused on as a way of questioning the rational expectations hypothesis obviously didn't halt the widespread application of rational expectations in macroeconomics, though it did lead to serious questions about which equilibria might be selected, and how, when models gave rise to multiple equilibria. This issue was an important input in the then-developing literature on how agents might go about learning rational expectations. It was also an input into the development of measure theoretic techniques for characterizing rational expectations equilibrium in stochastic environments.

Grandmont's work did attract a lot of attention, though, for its demonstration that even very simple deterministic OLG models could generate equilibrium trajectories that were erratic, and which could be interpreted as business cycles. The paper which established this was Grandmont (1985).

Given its importance in our narrative so far, and the fact that, after Diamond and Samuelson, it is the most cited paper using the OLG approach, we present a brief history of the evolution of Grandmont's 1985 *Econometrica* article. The first version of the paper appeared in June 1983 as CEPREMAP working paper 8316, revised in September 1983 (Grandmont [1983]).

The paper was "prepared for the IMSSS summer workshop at Stanford University" with support from the NSF, among other institutions. In October 1983, a version of the paper was presented at the University of Minnesota Institute of Mathematics and its applications workshop on price adjustment, quantity adjustment and business cycles, organized by Hugo Sonnenschein, and discussed by Goroff, Scheinkman, Sims, Wallace and Woodford, whom Grandmont thanked for "their valuable comments and constructive criticisms" (Grandmont [1985], p. 995, note 2).

In 1984, the paper appeared as Technical Report Number 438 of the Center for Research on Organizational Efficiency, IMSSS, Stanford (Grandmont [1984]). It was supported by an Office of Naval Research global grant covering IMSSS research at Stanford. What is interesting to note here is that the 1983 and 1984 versions of what was to become Grandmont's seminal 1985 *Econometrica* paper were much longer than the version eventually published. The reason for this is apparent from examining the earlier manuscripts: Grandmont included an extensive development of his temporary equilibrium framework in section 2 of both the 1983 and 1984 working papers. In this development, he characterizes the periodic equilibria that can occur in the model as being subject to a learning process that eventually allows agents to correctly forecast future prices in these equilibria, a situation which he characterizes as one of agents having "rational" expectations. From this material, it is clear that Grandmont was still trying to provide a workable alternative to the rational expectations hypothesis. That this material doesn't appear in the published version of the paper is likely due to significant pushback from referees and/or an Associate Editor/Editor at *Econometrica*, itself indicative of how entrenched the rational expectations assumption was in general equilibrium and macro theory by that time.

Grandmont's 1985 paper on this topic was not the first to be published, as he noted in the paper. That distinction belongs to Benhabib and Day (1982). The difference between the two

papers in technical terms is marginal. In economic terms, though, the differences are significant. Benhabib and Day examined the dynamics around Gale's classical steady state, while Grandmont examined them around the Samuelsonian steady state. As we've noted earlier, the classical steady state in the one-good, two-period lived model is Pareto optimal, so that in the Benhabib-Day model, the only way to get the erratic dynamics started is by transferring resources from the initial old to the young, which can't be done in a way that compensates the old for their loss. The Benhabib-Day model can be interpreted as one of an on-going bi-infinite equilibrium, in which there is no beginning, though as we've noted previously in the context of determinacy of equilibrium, this just pushes the question of how this equilibrium was established back to minus infinity. A second drawback of the Benhabib-Day model is that it requires that there be an infinite-lived credit intermediary to enforce the repayment by the young of loans made by the previous generation's old agents, since they are no longer around to take repayment. Per private communication with us from Benhabib, they chose to work with the classical case because

[I]n the Samuelsonian case cycles and chaos dynamics were correspondences (or functions in backward dynamics), while in the classical case you got chaos with forward dynamics with a well-defined function. (Email from Jess Benhabib dated June 20, 2022)

Grandmont's model only requires Samuelson's "social contrivance of money" to facilitate the intergenerational trade required in the monetary equilibrium. In addition, as we've seen, in this case the monetary equilibrium is fully Pareto optimal. The drawback to working with the Samuelsonian case, as Benhabib noted is that the forward dynamics, under parameterizations that give rise to erratic dynamics, are given not by a function, but by a correspondence. In Grandmont's paper, he focuses, therefore, on the backward dynamics, where consumption at time t is a function of consumption at time $t+1$. We note, however, that for both

approaches, all of the equilibria obtained are stationary in the sense that their regular properties are invariant under shifts in the time parameter. Hence, a cycle of period n is the same going forward or backwards and will be a stationary equilibrium in the bi-infinite interpretation of the model. The differences in the two models then come down to whether the stationary equilibrium is stable going forwards or backwards, and the issue of how to interpret the initial starting period.

Both models generate their erratic dynamics results by exploiting the then newly developed mathematics of chaos theory, a subarea of dynamic systems analysis focused on non-linear systems whose stationary equilibria can undergo bifurcations from simple steady states to cycles to chaos. This mathematics was developed primarily during the 1960's and '70's by mathematicians like Feigenbaum, Hayashi, Lorenz, Li, Malgrange, Mandelbrot, May, Ruelle, Shaw, Smale and Yorke. It was also helped immensely by the simultaneous development of high-speed digital computers during this period.

While a full exposition of this math is obviously beyond the scope of this text, we can illustrate the fundamental ideas with a simple example, the well-known quadratic logistic map

$$x_{t+1} = ax_t(1 - x_t)$$

For $x_t \in [0,1]$ and $0 \leq a \leq 4$. The graph of the function that determines x_{t+1} given x_t is hump shaped, and is illustrated in Figure 14

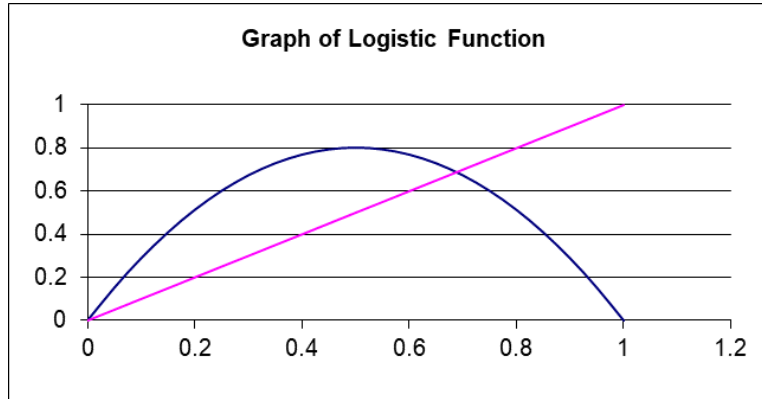


Figure 14

Graph of the quadratic logistic map

For different values of the parameter a , the time-series generated by iterating the logistic function will display different behaviors. For $a=2$, for example, it is easily verified that the fixed point of the simple dynamic system is $x=0.5$, and the derivative of the function at the fixed point is zero. This implies that the steady state at the fixed point is stable, and simple computer analysis of the dynamics for this case shows that, in fact, the mapping is globally convergent to the steady state.

For $a=3.2$, it is easy to verify that the derivative at the fixed point (which a simple calculation will show is equal to $\hat{x} = a - 1/a$) is negative, so that the graph of the function looks like that in Figure 14. Since the derivative of the function is given for general a by

$$f'(x) = a - 2ax,$$

evaluating this at the fixed point gives us

$$f'(\hat{x}) = 2 - a.$$

It then follows that the derivative at the steady state for the case of $a=3.2$ is negative and bigger than one in absolute value, so that the steady state equilibrium is no longer stable. Instead, in this case, we get a two-period limit cycle in the long run, illustrated in Figure 15 from an easy computer simulation.

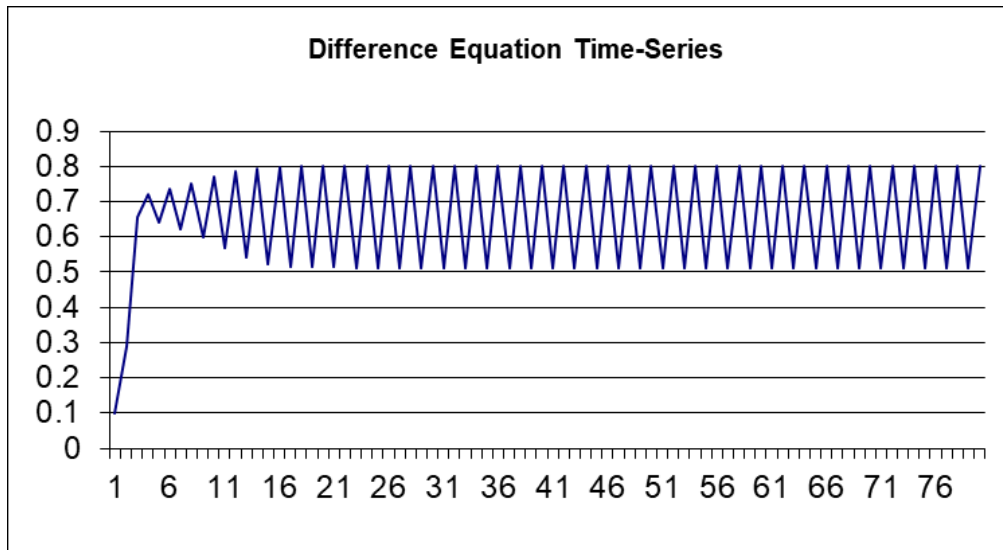


Figure 15

Time-series simulation for cyclic equilibrium

As the value of a gets larger, the two-period cycles will become unstable, bifurcating to a four-period cycle, then an eight-period cycle and so on with periods doubling at each bifurcation, until we obtain a cycle of “period” infinity, which Yorke dubbed “chaos.” We illustrate this in Figure 16 for the case of $a=3.8$.

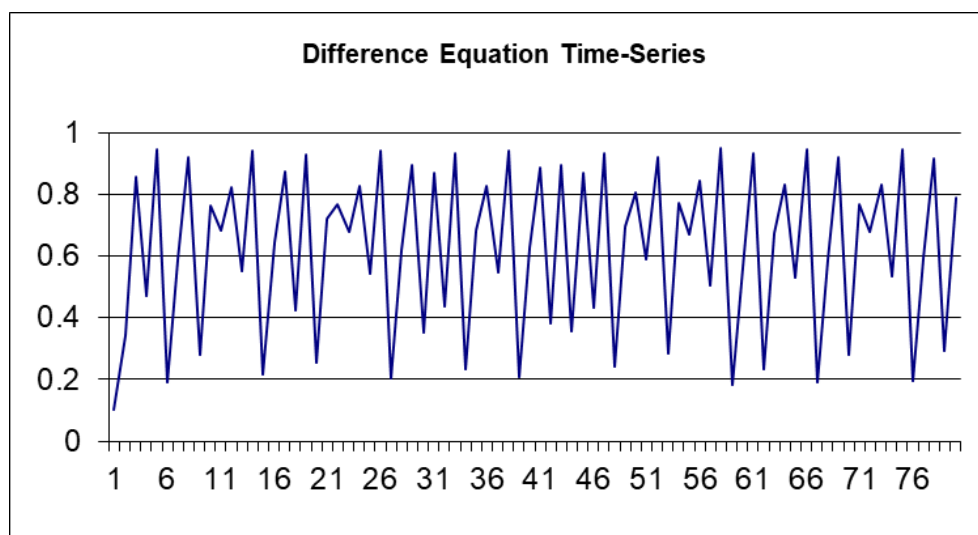


Figure 16

Time series simulation for chaotic equilibrium

The critical factor in getting a simple map like this to exhibit chaotic behavior is the hump-shape of the function, which needs to be steep on the end points, and flat through the middle and top. Iterating such a map will expand small neighborhoods near the extremes of the domain of the map, while contracting neighborhoods in the middle of the domain. This acts, in effect, the same way that a baker kneading bread dough does, constantly stretching and then folding the dough back in on itself. In fact, if the baker puts a drop of food coloring on the dough at the beginning of the process, the color will be smeared chaotically through the loaf by the time it is ready for baking. This is also the process by which soft plastics are colored, as many a fascinating YouTube video has demonstrated.

Since the non-linearity of the difference equation is the key to getting erratic dynamics, both Benhabib-Day and Grandmont need the OLG model to exhibit a hump-shaped offer curve, which imposes conditions on the curvature of the indifference curves of the young and/or old agents in the model. We can illustrate the workings of the two versions of the model using the offer curve diagrams introduced earlier. For the Benhabib-Day version of the model, the hump-shaped offer curve is illustrated in Figure 17.

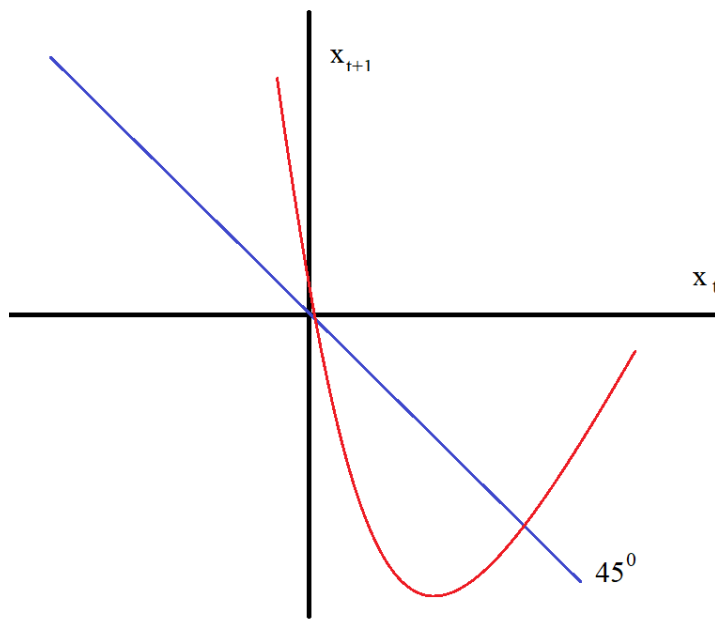


Figure 17

Benhabib-Day analysis

The corresponding diagram for the Samuelsonian case considered by Grandmont is illustrated in Figure 18.

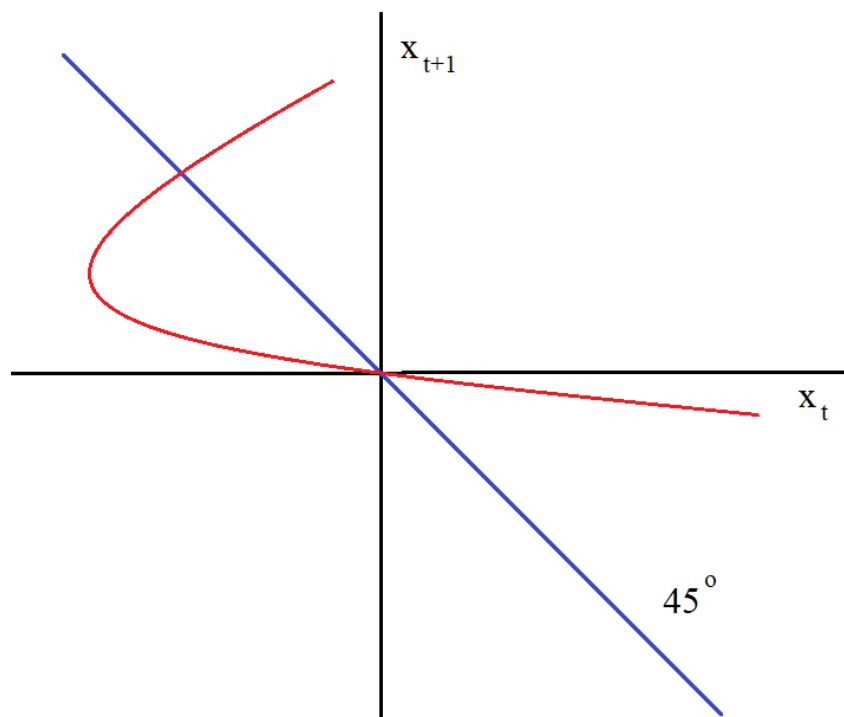


Figure 18

Grandmont analysis

The function vs. correspondence issue highlighted by Benhabib and Day is clearly illustrated in the two diagrams. In the second diagram, in particular, one can see that starting from any amount of saving by the young – i.e., a movement along the x -axis to the left from the endowment point at the origin – gives rise to two possible consistent equilibrium trades.

These two pioneering papers started something of a fad in macro and general equilibrium theory, since they raised an obvious question: was it possible to get complicated dynamics in the other workhorse model? The answer to this question was yes, at least for multi-sector growth models, and was demonstrated forcefully in the *Journal of Economic Theory* symposium volume Grandmont and Malgrange (1986). For the ILA models considered in the volume, there were two distinct approaches. One involved working directly with some version of the RCK model and showing conditions under which the policy function resulting from the representative agent's

intertemporal optimization problem would be sufficiently hump-shaped. The second approach involved imposing liquidity requirements or cash-in-advance constraints on the model to get it to perform more like an OLG model.

As Grandmont and Malgrange note in their introduction to the volume

These studies show by way of examples that complex deterministic dynamics may arise in models where agents optimize over an infinite horizon, and are not confined to, say, overlapping generations models with “short-lived” traders as in Benhabib and Day [6] and Grandmont [24]. The same point is made by Woodford in his contribution to this volume, as we shall see. These works suggest accordingly that endogenous fluctuations of the sort that is considered in this issue, may occur at frequencies that have nothing to do with human life, contrary to what some critics had argued previously. (*Ibid*, p. 5).

While a full accounting of the many important papers in this volume would take us too far afield of our current topic, we include this discussion because it clearly illustrates the concern that macroeconomists brought to the table over the need to be able to match time-series data of different frequencies to underlying theoretical models. We have touched previously on this issue with respect to Tobin’s and others’ use of computational multi-period OLG models and will return to discuss newer work in this vein later. But for now, we conclude our discussion of the chaos work by noting that all of this work has been criticized on the grounds of the unrealistic parameter value assumptions required to get the models to exhibit complex dynamics. For the OLG approaches of Benhabib-Day and Grandmont, the degree of risk-aversion required of one group of agents or the other is unrealistically high. Grandmont’s approach, for example, requires relative risk aversions for the old on the order of 9, while Benhabib-Day require those of the young to be around 5. Empirical estimates of these coefficients are routinely less than 2. These parameterizations generate the backward bending offer curve, which in turn implies that around the steady state of the model, increases in interest rates lead young agents to reduce their saving.

Many macroeconomists view this requirement as unrealistic and harbor doubts that real world economies generate the kinds of income effects necessary for this result to hold.

For the ILA models of chaos, the key condition for generating the appropriate curvature in the policy function is an unreasonably low discount factor (or correspondingly high rate of time-preference). Boldrin and Montrucchio's examples from the *JET* symposium required discount factors on the order of 0.02 before chaotic trajectories occurred.

This volume also included two additional important OLG papers, those of Reichlin (1986) and Farmer (1986). Reichlin's paper extended Grandmont's model to allow for more realistic production processes and showed that cyclic equilibria could exist under conditions where the savings elasticity with respect to interest rates is positive. Farmer's paper also shows the existence of cyclic equilibria in a model with production using techniques based Hopf bifurcations similar to those used by Reichlin. In Farmer's model, government debt policies that lead to the existence of equilibrium at the golden rule steady state also make possible the Hopf bifurcations in the presence of production. Farmer's results include two examples, one with CES production, and the second with Leontief production. More recent work by Bosi and Seegmuller (2005) has shown a basic equivalence between Reichlin's results and those of Woodford (1986). For a modern treatment of these results, see Benhabib (2021).

Section 7.3 Observational Equivalence

Working at roughly the same time as Benhabib, Day, Grandmont, Kehoe, and Levine, S. Rao Aiyagari began looking at the question of what distinguished the OLG and ILA models, below the surface of their obvious differences. Aiyagari was a graduate of the University of Minnesota's economics Ph.D. program, having worked under Neil Wallace's supervision there.

After graduating from Minnesota, he took a job at the University of Wisconsin's Madison economics department, though he also spent time as a visiting professor at New York University and Carnegie Mellon's Graduate School of Industrial Administration.

Aiyagari's observational equivalence (OE) research program was based on the observation that the Pareto optimal steady state equilibria, and much of the associated non-stationary dynamics around the steady state in versions of the Diamond model look the same statistically as those coming out of the RCK model using the same specifications of preferences, endowments and technology. From an econometric perspective, observational equivalence occurs when structural aspects of two different models can't be identified in the data. Aiyagari's analysis used functional analytic techniques to show how a given specification of either the OLG or RCK model could be mapped into the other model such that simulations of the two models would produce identical time-series data.

Aiyagari published three main papers on this topic. The first, Aiyagari (1985), looked at the simplest version of the Diamond model, with agents living two periods, discounting future consumption. Aiyagari shows that under the assumption that preferences and technology are such that the steady state of the OLG model yields interest rates in excess of the population growth rate, one can construct an ILA economy in which the representative infinite lived agent has the same preferences and technology as the OLG agents, which then gives rise to the same policy functions as in the OLG model. Hence, the two models are observationally equivalent. In the conclusion of the paper, Aiyagari notes that

It has been necessary to impose rather strong assumptions on the OLG model, assumptions which deliver a unique equilibrium, a unique and globally stable steady state, monotonicity, and differentiability of the function describing the evolution of the capital/labor ratio and more. With weaker assumptions, the range of dynamics that could be exhibited by OLG models is much larger, for instance, than that exhibited by

[Discounted Dynamic Programming] models and observational equivalence cannot possibly hold. (*Ibid*, p.220.)

This observation, together with the fact that the two-period lived agents assumption remained problematic in terms of how to fit data of different frequencies to the OLG model, led Aiyagari in his second OE paper to consider OLG environments with long-lived agents. Aiyagari (1988) looks at a sequence of OLG models in which agents live T period lifespans and examines the steady state equilibria of the sequence as T gets large. Unlike the previous paper, this paper doesn't undertake trying to show exact equivalence of the OLG and ILA models. Rather, Aiyagari made assumptions on the endowments of agents which guaranteed that the steady state equilibria along the sequence were classical, in Gale's sense. The paper then showed that as T gets large, the steady state equilibrium interest rate converges to the (common) rate of time-preference, the equilibrium steady state is Pareto optimal, and all agents' consumption profiles converge to the corresponding permanent income profile for the given discount rate. What this analysis suggests is that in multi-period OLG models which generate positive equilibrium interest rates, the qualitative properties of the equilibrium steady state and nearby non-stationary trajectories are the same as those generated by a comparable RCK model.

There is one additional important result presented in the paper, though it is buried in a short discussion section at the end of the paper. We reproduce Aiyagari's statement of this result here:

It is also possible to extend the results obtained here to include production and capital accumulation. The results parallel those in the classic one-sector growth model with a representative infinitely lived agent. A more difficult extension, attempted in Aiyagari [11], is to environments with uncertainty. For the special case of a logarithmic one-period utility function and random endowments, we were able to show that the one period ahead contingent claims prices converge (as T gets large) to the prices that would prevail in the analogous infinitely lived agent model. This was not possible for other utility functions because it turns out that so long as agents live three or more periods, consumption allocations and contingent claims prices depend on the entire infinite history of

endowment realizations. This is essentially the same problem that arises in Spear [14] even though the intertemporal preferences here are time separable. (*Ibid*, p. 114-115)

The reference here to Aiyagari [11] is to an unpublished (and apparently lost) working paper version of this paper. The reference to Spear [14] is to the paper "Rational Expectations in the Overlapping Generations Model" (*Journal of Economic Theory* 35.2, 1985, 251-275) which we will discuss below.

The third paper in the OE project is Aiyagari (1987). This paper is an extension of the previous paper in that it allows for population growth and shows that the old question of whether monetary equilibria exist turns only on the relationship between agents' commonly held discount factors and the rate of growth of the economy as long as lifetimes T are long enough.

There was related follow-on work on observational equivalence done by Blackorby and Russell (1992). Blackorby and Russell study the observational equivalence question from the perspective of aggregation, asking whether a given OLG model with multiple agents can be aggregated across agents alive contemporaneously to obtain a model that looks like an infinite lived representative agent model. Because a complete aggregation result *a la* Gorman (for static economies) is impossible in this context, Blackorby and Russell use Aiyagari's OE approach but generalize it to allow for robustness to perturbations in the technology available to agents, as long as there is latitude for reallocating endowments.

Aiyagari's OE work, and that of Blackorby and Russell was re-examined in Jones and Manuelli (1992). This paper showed that it is possible to get endogenous growth without increasing returns to scale in production in the OLG setting, as long as production exhibited constant returns to scale, and young agents were given sufficient income to drive the demand for growth. One of the consequences of this result, however, is that it contradicts the fact that in an infinite lived agents setting with the same preferences and technology, there can be no

endogenous growth. While the Blackorby and Russell extension of Aiyagari's results did not require the ILA model to have a steady state (and hence it was amenable to growth), their need for endowment reallocation in order to obtain their observational equivalence result precludes the possibility of endogenous growth.

The fact that the Blackorby and Russell work went unpublished is notable, since it speaks to the observed impact that this research program had on macroeconomics. From private communication with us, Russell comments on the struggle he and Blackorby went through trying to publish this paper:

[A]fter rejections by (as I recall) *Econometrica*, *QJE*, and *ReStud* (with very long periods of review at each journal), we killed the paper rather than go down market with a paper that we thought belonged in one of the premier theory journals. The negative referee reports found no errors but rejected the paper essentially on the grounds that growth and macro theorists had no interest in the question posed by the paper.

A few things to note: (1)[T]wo years after the rejection by *R.E.Stud*, we apparently had a change of heart about trying again and did submit the paper, without revision, to *Economic Theory* and then *J. Econ. Dev. Cultural Change*, with even more discouraging results. (2) It's interesting that the paper was taken most seriously by *Econometrica*, with a four-year review entailing two revise-and-resubmits. (3) By the time we got to the last two journal submissions, it seems that any initial interest had waned.

A second related paper in the OE literature was Huffman (1986) which focused on the asset pricing properties of SILA versus two period SOLG models and found that under some parameter restrictions, the two models were in fact observationally equivalent.

Looking back at Aiyagari's paper, the Google cites for the three papers were 36, 26, and 17 respectively, which would seem to indicate the same lack of impact that Blackorby and Russell got from the macro profession for their work. At the same time, it is hard to argue with Russell's contention that these results were important enough at the time to merit publication in

major outlets. Certainly, Aiyagari's work was visible, given *JET*'s visibility in the theory world. So why the low number of citations?

There seem to be two threads that one needs to follow to explain this. One is based on Aiyagari's ultimate findings (in the *JET* '88 paper) that stochastic RCK and OLG models were not observationally equivalent. We postpone this discussion until we complete our coverage of the development of the stochastic OLG framework. The second thread is the inherent negativity of the OE results for deterministic models. By "negativity" we mean the fact that if the two models were indeed equivalent, there would be no further reason for studying one or the other. We conjecture that as Aiyagari's results became well-known in macro, they helped to ultimately tip the market away from OLG toward ILA, and once this tipping occurred, there would be no further need to justify the use of the ILA framework by appealing to Aiyagari's equivalence results, leading in turn to the paucity of citations. The difficulty Blackorby and Russell had trying to publish their work, even shortly after the Aiyagari papers had appeared suggests that the tipping process was well underway by the early 1990's.

This observation leads naturally to the question of why the market would tip toward ILA and not OLG. We have already suggested that the development of relatively easy to use recursive dynamic programming methods played a key role here. To quote from Sargent (1996)

The recursive methods used by Lucas's *JET* paper, and also by Lucas and Prescott (1971) and Brock and Mirman (1972), have come to dominate macroeconomics. Virtually every macroeconomist today reads, writes, and thinks [in] terms of dynamic programming. Recursive reasoning unearthed the 'time-consistency' problem, and led to characterizing the types of government policies that could be implemented sequentially. (*Ibid*, p. 542-543)

A second reason for the market to tip toward ILA was the subsequent development of models based on the recursive ILA methodology which incorporated frictions that allowed the

resulting model to mimic behavior generated in the messier to analyze OLG setting, and it is to these models that we turn next.

Section 7.4 Incomplete Markets and Financial Frictions

When Aiyagari realized that his observational equivalence program had reached a dead end, he restructured his research to focus, for the most part, on ILA models with frictions that delivered equilibria resembling those that could arise in OLG environments. Indeed, a Google Scholar search for Aiyagari's work including the phrase "overlapping generations" returns 18 single- or co-authored papers listed in the search results. Without the "overlapping generations" phrase included the search returns 120 papers. (Aiyagari was a prolific scholar and would undoubtedly have produced many more top-quality papers if his life hadn't been cut short.)

Foremost among the new models that Aiyagari examined was an ILA model in which markets are incomplete. His most highly cited paper in this research program is Aiyagari (1994), with 3798 Google Scholar cites. This paper took an otherwise standard one-sector RCK model and modified it by assuming that agents in the model face idiosyncratic shocks to their own labor productivity and face a borrowing constraint. The need for the borrowing constraint arises from the possibility that equilibrium interest rates could be negative, in which case disequilibrium Ponzi schemes become possible in the absence of some limit on borrowing. The borrowing constraint also arises naturally for the case where interest rates are positive given that consumption must be non-negative. The market incompleteness arises because agents cannot insure themselves against the idiosyncratic labor shocks.

Aiyagari's work built on the theoretical contributions of Truman Bewley in two papers, Bewley (1977) and Bewley (1986). For this reason, these models are generally referred to as

Aiyagari-Bewley models, and later Aiyagari-Bewley-Huggett models in recognition of Mark Huggett's 1993 incomplete markets paper (Huggett [1993]). The market incompleteness generated by the idiosyncratic shocks leads agents to self-insure, i.e., to save for the proverbial "rainy day." The precautionary savings so generated is what gives these models their OLG flavor, while the assumption that shocks are idiosyncratic means that there are no aggregate shocks to complicate the analysis. This "trick" of restricting uncertainty to be idiosyncratic has been widely applied, not only in conventional Aiyagari-Bewley models, but in OLG models as well (see, e.g., Storesletten, Telmer and Yaron [2007]). The usefulness of the trick is that because there are no aggregate shocks, one can ignore the wealth effects generated by changes in the wealth distribution in the model because they get washed out by the law of large numbers.

The second class of ILA-based models that generated OLG-like dynamics are the cash-in-advance models first introduced by Clower (1967) and featured prominently in Grandmont's and Malgrange's special issue of *JET* discussed above. The seminal papers in this literature were Lucas and Stokey (1987) and Woodford (1986a). Woodford's paper is significant for a couple of reasons that are relevant to our examination of the OLG model.

First, according to Grandmont (in his communication with us), in Woodford's contribution, workers with infinite horizons are faced with constraints on their ability to borrow, with the result that they also solve problems emanating from a sequence of two-period overlapping maximization, such that when they are young they work and save money, which is spent on consumption when old, that is, one period later, similar to two-period models based on OLG, albeit with a resultant period length that is small.

Second, Woodford uses the fact that indeterminacy of equilibrium in the model is necessary and sufficient for the existence of sunspot equilibria in the two-period one good model – which was originally shown by Azariadis and Guesnerie (1986) – to introduce an old econometric modeling trick, known as “bootstrapping,” into macroeconomics. Bootstrapping involves setting up the first-order conditions (FOCs) for a dynamic optimization, but instead of setting the equations equal to zero, one sets them equal to a random variable whose mean is zero. Assuming that the FOCs can be solved for the forward endogenous state variable as a function of the lagged state variables and the bootstrap shock, one can show that this is in fact a rational expectations equilibrium, as long as the shocks are sufficiently small, and the steady state equilibrium is locally forward stable.

We illustrate this process in a simple two period, one good pure exchange OLG model. Let a typical young agent’s stationary utility function $u(x_t, x_{t+1})$ over consumption in period t and $t+1$ satisfy the usual properties – at least twice continuously differentiable, strictly concave, differentiable monotonic, and satisfying an Inada condition as consumption in either period goes to zero. Also, assume endowments (ω_z, ω_2) are also stationary and such that the steady state equilibrium is monetary with a fixed money supply $\bar{m} > 0$. The young agent’s budget constraints take the form

$$p_t x_t + m = p_t \omega_1$$

$$p_{t+1} x_{t+1} = p_{t+1} \omega_2 + m.$$

From these, we can solve for consumptions in each period in terms of prices and money holdings (together with the fixed endowments) and reduced the agent’s budget constrained utility maximization problem to

$$\max_m u \left[\omega_1 - \frac{m}{p_t}, \omega_2 + \frac{m}{p_{t+1}} \right].$$

The FOCs for this problem are

$$-\frac{m}{p_t}u_1 + \frac{m}{p_{t+1}}u_2 = 0$$

where u_i is the partial derivative of u with respect to the i^{th} argument. We assume that the utility function is such that this equation can be solved (implicitly) for p_{t+1} as a function of p_t . To construct the bootstrap sunspot equilibrium, let $\{\alpha_t\}$ be a sequence of i.i.d. random variables with mean zero and variance σ^2 . Set the FOCs above equal not to zero but to α_{t+1} ,

$$-\frac{\bar{m}}{p_t}u_1 + \frac{\bar{m}}{p_{t+1}}u_2 = \alpha_{t+1}$$

and evaluate the FOC at the fixed money supply $\bar{m}$. Solving this for the forward price gives us a first-order stochastic difference equation for the equilibrium (via the implicit function theorem)

$$p_{t+1} = g[p_t, \alpha_{t+1}].$$

If agents in the model now use this difference equation to forecast future prices, we will have

$$-\frac{\bar{m}}{p_t}u_1 + \frac{\bar{m}}{g[p_t, \alpha_{t+1}]}u_2 \equiv \alpha_{t+1}$$

since the function g was obtained from the implicit function theorem. If we now integrate on both sides with respect to the given density (or measure) for the α random variables, we get

$$\int \left\{ \frac{\bar{m}}{p_t}u_1 + \frac{\bar{m}}{g[p_t, \alpha_{t+1}]}u_2 \right\} d\mu(\alpha_{t+1}) \equiv 0.$$

But this is the first-order condition for a stochastic optimization problem. If the equilibrium stochastic process on prices remains on a sufficiently small neighborhood of the steady state equilibrium price for the model, the equilibrium process will then be ergodic. In this case, we have a *bona fide* rational expectations equilibrium for the model, since $p_t = g[p_{t-1}, \alpha_t]$ is a solution to the equilibrium law of motion above. To guarantee that the equilibrium stochastic

process doesn't wander away from a neighborhood of the steady state, it is sufficient that the steady state equilibrium be forward stable, and that the variance σ of the sunspot process $\{\alpha_t\}$ be sufficiently small that no shock realization moves the REE price outside of the basin of attraction for the steady state.

We note that Woodford's paper develops the stochastic side of the model as one of a stationary infinite sequence of shocks and then derives equilibrium prices as functions of these abstract shock variables plus lagged prices. The analysis we present here is based on the more general results contained in an unpublished 1986 working paper Woodford (1986b). As we will see later, this paper continues to be important.

This is the appropriate place to note, however, that Woodford's development of the infinite shocks formulation of the OLG and CIA models owed its impetus to a debate he conducted at the time with Grandmont. According to notes that Woodford sent to Spear and Sanjay Srivastava while they were discussing what to do with their Spear, Srivastava and Woodford (1990) paper, Grandmont had objected to Woodford's use of the bootstrapping technique in his contribution to the *JET* Symposium volume, on the grounds that the resulting equilibrium could not be "decentralized" because an "auctioneer" would have to select one of an arbitrary number of possible equilibrium forecasts for the model. This criticism was based on the fact that because the equilibrium law of motion is used as the forecast function, any initial time t price (or function of previous prices and exogenous variables) will generate an equilibrium. This criticism is difficult to follow because, if one used an arbitrary (temporary equilibrium) forecast function to specify $p_t = h(p_{t-1}, \alpha_{t+1})$, we would get market clearing, but the agent's forecast would be wrong. The equilibrium becomes rational only if the agent uses the actual law of motion for the economy as his forecast.

Woodford made this argument to Grandmont, but to no avail, so he then undertook to reformulate the model using the infinite history of exogenous (sunspot) shocks as the state variables. Using functional versions of the implicit function theorem together with results from the Kehoe-Levine regularity paper, he showed that for the given sunspot stochastic process, the resulting equilibrium was in fact determinate. Grandmont apparently accepted this argument because Woodford's paper was accepted for the symposium volume (and, as we've seen, praised by Grandmont), even though Woodford notes that he never received a formal reply to his letter to Grandmont explaining the new approach.

If one compares the equilibrium derived here with the one we found for the easy Lucas model in Chapter 3, the equilibrium stochastic process for that model was of the form $p_t = f(\alpha_t)$, i.e., there was no dependence of the equilibrium stochastic process on lagged endogenous state variables. This observation led a number of macroeconomists, Bennet McCallum notable among them, to propose the so-called "minimum state variable approach" to specifying equilibrium. McCallum (1999) justifies this approach by noting that it corresponds to the "bubble-free" solution of the model, reflecting only economic fundamentals. McCallum does acknowledge that alternative selection criteria based on learning are also useful, though he omits any mention of Woodford (1990) which shows how sunspot equilibria can be learned from empirical observation. We mention these controversies because they again illustrate the disagreements over what the right set of state variables is for OLG-type models.

The original Lucas-Stokey cash-in-advance paper gets 1173 Google Scholar cites, while the Woodford paper gets 470. All of the work on these types of OLG-like models is summarized in the Benhabib-Farmer (1999) *Handbook of Macroeconomics* chapter, which itself gets 681 citations. So, clearly these two substitutions of the basic ILA framework for an original OLG

foundation have had an impact on macroeconomics, reinforcing our contention from the previous section that the relative ease of the recursive techniques available in the ILA model, and the apparent observational equivalence of the original OLG setting with the ILA plus friction convinced many a young macroeconomist that these were the models to work with. The large number of citations that these papers get also explains the low numbers accruing to Aiyagari's original observational equivalence work: with positive new models to reference, practitioners didn't need to reference the observational equivalence results (right or wrong) in Aiyagari's effort to conjoin the two models.

Section 7.5 Pushback

Over the same period that Aiyagari was looking at observational equivalence and Grandmont, Lucas-Stokey, Woodford and others were looking at ways to generate OLG-like results in ILA environments, another group of young scholars was pushing back against the way ILA models were coming to dominate macroeconomics with the popularity of real business cycle models (see Young [2014] for this history) and the emergence of the New Neoclassical Synthesis of RBC and New Keynesian models¹³. To analyze this pushback, we focus on several papers by Alan Auerbach, Douglas Bernheim (and co-author Kyle Bagwell), Laurence Kotlikoff, and Lawrence Summers.

We begin by looking at Bernheim-Bagwell (1988). This paper pushed back against Barro's contention that ILA and OLG models were essentially the same as long as families in the OLG model left active (i.e., positive) bequests to their offspring. The Bernheim-Bagwell

¹³ If one accepts the contention laid out in Chatelain and Ralf (2018) that the New Neoclassical Synthesis will go the way of the old Neoclassical Synthesis because of its failure to explain involuntary unemployment and to identify the source of aggregate shocks, we should perhaps be referring to the Latest Neoclassical Synthesis here.

criticism focused on Barro's assumption that dynastic families are lineal, pointing out that real family structures form trees, given that it takes two people to procreate and the standard taboos against incest¹⁴.

The Bernheim-Bagwell analysis starts by taking the tree structure as the given family structure and invokes a result due to Erdos and Reny (1959)¹⁵ to justify the assumption that all of the nodes in the family tree are connected in the sense of allowing for positive transfers of wealth between families. Given this assumption, the analysis of the paper looks at the transfer game across the network of families and demonstrates, in an example, that government policy actions – for the specific example this is a tax on wages – have no effect because adjustments in transfers between families undo the distortion introduced by the taxes. The main analysis of the paper generalizes this and demonstrates that there can be no role for markets in such an economy: all economic interactions exhibit Ricardian neutrality.

These results obviously suggest that the likelihood that real economies are populated by large, interconnected dynastic families is quite low, which leads us back to the question of why macroeconomics became so dependent on such an unrealistic model. In an opening footnote to their paper, Bernheim and Bagwell address this issue:

No doubt its popularity in part reflects considerations of analytic convenience. Overlapping-generations models not only are generally less tractable but also often give rise to equilibria with undesirable properties. Specifically, equilibria in overlapping-generations models may fail to be either efficient or locally unique (see Balasko and Shell 1980; Kehoe and Levine 1985). Failure of local uniqueness is particularly troubling in any exercise involving comparative statics or dynamics. Thus Judd (1985) unabashedly attributes his adoption of the dynastic framework to analytic convenience. Unfortunately, this advantage may be illusory. In a recent paper, Gale (1985) has pointed out that, while Barro's dynastic solution is an equilibrium for the model he

¹⁴ Kotlikoff (1983) produced a similar analysis which was never published. In his *Journal of Economic Perspectives* paper (Kotlikoff [1988]) he reports result akin to those obtained by Bagwell-Bernheim.

¹⁵ We note in passing that this is one of the earliest applications of network theory in economics to model local game theoretic interactions between players.

considers, this model also generally gives rise to a continuum of subgame-perfect intergenerational equilibria (see Selten [1965, 1975]), many of which are inefficient. By adopting dynastic assumptions, one therefore does not necessarily succeed in avoiding the problems that arise in the standard overlapping-generations framework. (*Op. cit.* p. 309)

As we saw earlier in looking at the Arrow-Kurz analysis of optimal bequests, the issue of how important intergenerational transfers are to economic activities has been a matter of close study. This study has continued with the work of Bernheim and a large group of other authors looking at the types of altruistic behavior at work in the real world. From this analysis, it seems clear that one cannot fully explain the shape of the wealth distribution without some form of dynastic intergenerational wealth transfers. But the key question here is how this shapes the operation of the macro economy. In private communication with us (email of December 8, 2021), Bernheim addresses his own conclusions on the issue by stating that

Despite that paper ["Fiscal Policy with Impure Intergenerational Altruism," *Econometrica* 2001, with Andy Abel], my eventual conclusion was that non-paternalistic altruism, which is essential for the dynastic representation, just isn't a plausible model of intergenerational transfers. Sure, I wrote some papers showing that bequest motives can be important, and I believe that too. I just don't think the non-paternalistic altruism framework is the right one (or even close). Some of that traces back to Bernheim-Shleifer-Summers, but I also think there are factors like warm-glow giving (Andreoni), as well as various kinds of intrafamily signaling issues (Bernheim & Severinov, JPE 2003, and Severinov's separate paper applying the idea to Ricardian equivalence). With these alternative models, there are certainly connections between generations, but they behave more like the bequest is a type of consumption, so the properties are more OLG-like. (At least, more like OLG models with uncertain horizons where people accidentally leave bequests.)

Bernheim (1989) is follow-on paper, which looks at the Barro OLG formulation of the dynastic family model and demonstrates that the dynastic equilibrium could never arise as the solution to a standard social welfare optimization unless the social planner only cares about the current generation. In the usual case where we assume the planner (or government) puts positive welfare weight on all generations, Bernheim shows that the solution is one in which no parent

family leaves bequests to their offspring. The intuitive reason for this is straightforward. At any time t , if bequests are strictly positive, the parent generation at $t-1$ must be indifferent between its own consumption and that of generation t (as must all prior generations). But, because of the OLG demographic structure, the young in generation t will prefer to get a larger bequest (and future generations will be indifferent). Hence, as long as the planner puts positive weight on generation t , he will wish to increase the bequests of all generations, which is clearly infeasible. Thus, the dynastic allocation cannot be a welfare optimum, and any such welfare optimum must have all bequests set to zero. While the dynastic equilibrium will be optimal if the planner only cares about the current generation, the OLG demographics of the real world give democratic governments incentives to value members of all currently living generations, even if concern for the far future is minimal. In this case, the inoptimality argument Bernheim presents holds. Finally, as Bernheim notes, this argument also appears in earlier papers looking at gift giving equilibria (see, e.g., Bernheim, Shleifer, and Summers [1986] and Roberts [1984]).

There is a secondary macroeconomic issue worth mentioning in this context, which has a political dimension to it. The Barro (1974) paper implies that Ricardian equivalence holds in the ILA setting. Ricardian equivalence is the contention that rational agents will correctly anticipate that government spending must be paid for and will, therefore, increase saving if the government runs a deficit, anticipating future taxes. Similarly, they will reduce saving (and increase consumption) if taxes are levied contemporaneously since their future beneficiaries will benefit from the public goods (or transfers) they are paying for today. This question has been the focus of a great deal of study, with the modern consensus view being that the equivalence does not hold. (See, e.g., the Wikipedia article on this,

https://en.wikipedia.org/wiki/Ricardian_equivalence, for a summary of the empirical results and references to the major papers on the topic.)

The political economy dimension to this debate is obvious: opponents of government social spending use the Ricardian equivalence results to argue that such spending is ineffective because it gets undone by the actions of the private sector. On this dimension, one of the implications of the Bernheim-Bagwell result is that unless one believes that markets are unnecessary (because like government, they too are ineffective), Ricardian equivalence cannot hold.

More detailed studies of the motives for bequests have also bolstered the Arrow-Kurz contention that most intergenerational transfers take the form of human capital investment in children, with the promise of future payoffs from these “bequests” giving children the incentive to incur the disutility that can be associated with schooling.

Kotlikoff and Summers (1981) used data from the Census, the national income accounts, the IRS and the Social Security administration to estimate how much of U.S. annual saving for 1974 could be attributed to life-cycle retirement saving. From their calculations, they estimated that upwards of 80% of observed saving could not be explained by retirement motives. This estimate is in rough agreement with earlier studies cited by Kotlikoff and Summers, and in marked contrast to estimates obtained by Modigliani (1988), who estimated that only about 15% of observed saving was due to non-retirement motivated transfers. Modigliani also cites a number of other survey-based studies which agree with his estimates.

This raises the obvious question as to the motives for the excess saving (over retirement needs) observed by Kotlikoff-Summers and others. Is it for active bequest purposes (in the Barro sense) or precautionary, based on the uncertainties of health care costs combined with mortality

risk? There were a number of follow-on papers stimulated by the Kotlikoff-Summers findings that strongly suggest that the answer to this question is that the saving is precautionary.

Kurz (1984) uses data from the 1979 President's Commission on Pension Policy to test two hypotheses about savings motives, based on the fact that in U.S. society, giving and receiving transfers are generally viewed differently. As Kurz notes, outside of inheritances, there can be a stigma associated with receiving transfers given the typical American attitudes toward self-reliance. Giving transfers, on the other hand, is viewed positively as indicating compassion toward the needy. Kurz further posits that if the purpose of observed transfers was predominantly to support intra-family optimal wealth management, then estimates of the effects of positive or negative transfers in censored equations would be the same, since no direct utility effects would be generated. Based on this, he conducts two econometric tests. One looks at the effects of increases in Social Security transfers on private transfers both received and given. The second looks at the coefficients on the receiving versus giving regression equations and asks whether they are the same.

Kurz's first test finds that based on the receiving equation estimates, there is a significant dollar for dollar reduction in private transfers received for each dollar increase in Social Security support. On the giving side, however, he finds no relationship. This result weakly supports the Ricardian neutrality prediction of the dynastic family interpretation of the OLG model, though the lack of effect on giving suggests that what's actually happening is that children are reducing transfers to their relatively poorer parents as the government picks up the slack.

Kurz's econometric test of the second hypothesis – that the coefficients on the giving and receiving equations should be the same – forcefully rejects this hypothesis. So, both of Kurz's

test suggest that the dynastic bequest motive cannot explain the observed over saving relative to retirement needs.

While there is a whole literature on the question of whether the observed saving is driven by bequest motives or precautionary motives (indeed, more than we can possibly survey here), the fundamental issue in answering the question is one of econometric identification, since in data terms, both motives result in the same outcome. To date, the problem still hasn't been completely resolved, though Ameriks *et. al.* (2011) have made some significant headway by including survey questions in their econometric analysis. Their findings indicate that the precautionary motive for saving is quite strong, though they also find significant support for bequest motives among upper income families with children. These findings support the contention – based on the fat tails of the distribution of inheritances in the U.S. – that realistic macroeconomic models should be based on a hybrid of the OLG and dynastic family frameworks of the kind originally examined by Wilson (1980), Muller and Woodford (1988), and Wright (1988). That said, the jury is still out on this overall question. Benhabib and Bisin (2018) stress the importance of bequests in generating the observed skewness of the U.S. wealth distribution, but highlight a number of different models that can generate such bequests. Among these is the conventional Barro interpretation of the OLG/ILA model, but there are other non-dynastic ways in which the required bequests can arise. Gokhale *et. al.* (2001), for example, uses the Auerbach-Kotlikoff simulation model (which we discuss in detail below) with mortality risk, assortative mating, and differences in skills and time preference to show that purely accidental bequests due to imperfect annuitization can give rise to wealth distributions that match the observed data under realistic parameterizations of the model. Benhabib and Bisin also note that while fiscal policy has an obvious effect on the distribution of income and wealth, the converse

is also true, so that for a model to fully capture the forces at work in determining how wealth is allocated, we need to model political economy interactions as well. Finally, there are empirical studies of intergenerational wealth dynamics that indicate that wealth is mean reverting. This occurs because families grow, and in the process, dissipate the wealth that earlier generations have accumulated, though the overall process is slow. For more details on this, see Whal (2003).

The publication of Barro's 1974 paper also stimulated critical comment from Martin Feldstein and James Buchanan, published in the *JPE* in April of 1976. Both Feldstein and Buchanan based their criticisms not on the realism of the bequest mechanism at the heart of Barro's model, but on their contentions that Barro had overlooked (in Feldstein's comment) the possibility of economic growth, and (in Buchanan's comment) the real-world ways in which governments in a democracy decided on taxation or debt.

Feldstein's criticism looked back at the early development of the OLG model when a key concern was the growth of population (later amended to allow for endogenous growth), and the possibility that real interest rates might be below the growth rate of the economy as a whole. In this case, Feldstein pointed out that the government could maintain a level of debt in perpetuity, using growth in tax revenues to pay the interest on the debt, with the initial debt accumulation used for the benefit of the generation(s) alive at the time the debt is incurred.

Barro's reply to this concedes that in this situation, debt would increase net wealth, but argues that the stimulus afforded by the debt issue would have the effect of raising interest rates to the point that, in steady state, $r=g$. At the time of this debate, the U.S. economy had not been through any protracted periods where real interest rates were lower than the economy's growth

rate, though that changed in the decade following the financial crisis of 2008, prompting some economists to revisit the issue.

Buchanan's criticism, coming from a political economy perspective, was that most government provision of public goods or services was in fact financed by contemporaneous taxation, though he did hold up the Social Security system as an example where the means of payment of future benefits was not secured at the time the obligation to pay benefits was taken on. He cites Feldstein's work on the effects of Social Security on saving to bolster his point. Barro's response is basically one of disputing the validity of Feldstein's results.

Barro also directly addressed the Bernheim-Bagwell criticism in Barro (1989). Barro argues that if all of the members of a large economically interconnected family were all making the kinds of operative (i.e., positive) bequests that generate the excess neutrality results in Bernheim-Bagwell, then there would be incentives for members of the network to free ride on the contributions of others.

We find this argument suspect. In a conventional public goods model, agents care about their own consumption of the public good but would prefer not to pay for it if they think others will. Barro is interpreting a parent's altruistic care for their child as the public good, so that if that agent thinks her kid's spouse's parents will foot the bill for the down payment on their first house, they can free ride on that. While this likely happens sometimes in the real world, if the agents also care altruistically for their kid's spouse, and the altruistic preference relation between members of society is transitive, agents will care for their kid's spouse's parents as well and not want them to foot the whole bill. One could set up a model with differing degrees of altruism, depending on some measure of relationship distance, and then examine how neutral things are relative to connection strength, but as far as we are aware, no one has done this yet. Given the

empirical evidence against Ricardian neutrality (and presumably against everything neutrality), Barro's argument seems moot, at best. It is curious, however, that Barro chose this forum (i.e., his own book) to reply to the Bernheim-Bagwell critique rather than through the kind of communication in the *JPE* that Buchanan and Feldstein pursued after the publication of Barro's 1974 paper.

We note finally that an early version of Bernheim-Bagwell was cited by Becker and Barro (1986). This paper uses the Barro dynastic family formulation to study the effects of incentives, costs of childcare, and human capital investment costs on family fertility decisions. They cite Bernheim-Bagwell by way of acknowledging very briefly the problems associated with more realistic treatments of family formation but proceed to analyze a key component of what families do – procreation – using the tractable but unrealistic dynastic family framework. Barro and Becker wrote two additional papers, Barro and Becker (1988, 1989) examining family fertility decisions and the economic demand for children in the three years following their “Altruism and the Economic Theory of Fertility” paper. The second paper in the *QJE* cites Bernheim-Bagwell in the same throw-away manner as the “Altruism” paper. By the time the 1989 *Econometrica* paper is published, all mention of the Bernheim-Bagwell results is gone.

In 2018, we posed a series of questions to Robert Barro regarding his 1974 *JPE* paper. These related to: (i) origins; (ii) early reception; (iii) terminology; (iv) citations in the paper, and (v) the Feldstein and Buchanan comments on it. The first question (personal communication 10/12/2018) asked: “Can you tell us how you came to refer to the model in your 1974 paper specifically as an “overlapping generations model” as against using the term “consumption-loans” model, which, following Samuelson 1958, seems to have been the more frequent usage? He replied (10/12/2018):

I am surprised to see that the term overlapping-generations model is not in Samuelson 1958 or Diamond 1965. I used the term in my 1974 paper to describe what I thought was in Samuelson and Diamond. Malinvaud, 1987 JEL, argues that the idea of an overlapping-generations model is in Allais's 1947 book. However, I do not know whether the term overlapping generations was used in that book. And Allais's 1947 work seems to have had no influence on Samuelson 1958 or Diamond 1965 or surely on my 1974 paper (since I knew nothing of Allais's work). So, I really cannot say where I got the term overlapping-generations model for my 1974 paper.

We then asked about the following (10/13/2018):

“In your acknowledgements in JPE 1974, you wrote”:

“I have benefited from comments on earlier drafts by Gary Becker, Benjamin Eden, Milton Friedman, Merton Miller, Jose Scheinkman, Jeremy Siegel, and Charles Upton. The National Science Foundation has supported this research.”

1. Regarding the early drafts of the paper. Were any of the drafts given at seminars or meetings, and if so, can you recall where and when, and what were the reactions to the paper?
 2. Regarding the comments of those mentioned above. Do you recall what the comments were?
-

Barro replied (10/13 2018):

With respect to seminars, I remember it exactly. I presented the paper only twice, once in the money & banking workshop in Chicago Econ Dept. and once at a seminar in Chicago B. School. In the first, it is the only time I remember Friedman, Stigler, and Becker present together at a seminar. I recall an argument I had with Becker concerning the effects of inheritance taxes. I argued that these taxes made it less likely that an interior solution would apply for voluntary intergenerational transfers, but if the solution were still interior then the offsetting intergenerational transfers that produce Ricardian Equivalence would hold exactly. I remember Milton deep in thought on the matter for a couple minutes while no one said anything. Then he announced (apparently with great surprise) that Gary was wrong and I was right. I was also surprised in retrospect that Stigler never said anything about the term Ricardian Equivalence. That expression, not contained in my paper, became widespread only after James Buchanan's comment in the 1976 JPE.

In the B. School seminar, the only thing I remember is an interaction with Gene Fama. He kept complaining that a debt-tax swap should have no real effects. Then when I explained that his conjecture was consistent with my results, he decided that the results were interesting.

I also recall a lunch discussion that I had with Fischer Black. I took about 15 minutes to go through the argument for what is now called Ricardian Equivalence (based on offsetting intergenerational transfers). Fischer just listened carefully and said nothing. Then he finally said, “Sounds right to me.”

I don’t recall what the acknowledgments in my paper to other people were for.

Regarding the comments by Feldstein and Buchanan, we asked Barro (10/30/2018):

“Another minor puzzle, this time regarding your citations in your reply to Buchanan and Feldstein in *JPE* 1976. In their comments, they both used the term overlapping generations, and cited Samuelson as the original model. In your reply, *JPE* 1976, you cited Diamond and Cass-Yaari as OLG models, but didn't cite Samuelson at all...Do you recall why?”

Barro answered (10/30/2018):

As I recall, I thought Samuelson’s framework was not so interesting because it neglected capital and, hence, capital accumulation. So, one could not really think about the possibility of over- or under-accumulation of capital.

We note that this issue was addressed quite explicitly in Diamond’s OLG model with production and growth.

The second form of pushback against the dynastic family model came in the form of “bucking the trend” of using variations on the RCK model for policy analysis. This work involved applications of long-lived deterministic OLG models to study major public policy issues, continuing the work started by Tobin, as we’ve outlined previously. To examine these models, we focus on the contributions of Alan Auerbach, Laurence Kotlikoff, James Poterba and Lawrence Summers.

Alan Auerbach and Larry Kotlikoff were graduate students together at Harvard (Kotlikoff graduated in 1977, Auerbach the next year), both of them students of Marty Feldstein. Since Feldstein's work on taxation and social security was largely grounded in OLG, it is not surprising that Auerbach and Kotlikoff inherited his appreciation of the model. According to Auerbach, when we asked how he and Kotlikoff started working together on their OLG projects

I don't recall our discussing this while we were both in graduate school. We may have started when I was on the job market. Larry was in his first year of a three-year post-doc at UCLA and I stayed with him for a couple of days while I was interviewing there. We definitely did the initial research while he was still doing his post-doc, as our first RAs on the project were then UCLA graduate students (Maxim Engers and Jon Skinner). (Private email correspondence of July 5, 2022)

Their initial collaborations quickly produced two papers, Auerbach and Kotlikoff (1983a) and (1983b). Both of these papers use what has become known as the Auerbach-Kotlikoff simulation model, which is a 55-period deterministic Diamond version of the OLG model. This model is now generally referred to as the AK model. These simulations typically incorporated additively separable CES utility functions for households and Cobb-Douglas production functions along with realistic specifications of the government's policy instruments for taxation and transfers. As we noted earlier, this work was a precursor to Kydland and Prescott's (1982) introduction of calibrated simulation of general equilibrium models into macroeconomics. We characterize the work as a precursor because in addition to the calibrated simulations of their model, Kydland and Prescott also undertook a detailed decomposition of important macroeconomic time-series via the development and application of the Hodrick-Prescott filter. The results of this detrending of the data generated a set of statistical regularities that Kydland-Prescott and others then tried to match with their models. We also note that in this work, Kydland and Prescott were working with the stochastic growth model, so their analysis was more general than that of Auerbach and Kotlikoff. We will examine the problems associated with

simulating stochastic OLG models with long-lived agents in the next section, but, in terms of the basic idea of replacing bound-to-be-rejected econometric estimation with realistically calibrated simulations, the Auerbach-Kotlikoff model is every bit as innovative as the Kydland-Prescott model.

The first of the two AK papers cited above lays out the simulation model in detail and then applies it to examine four examples of policy issues which the model can address. These are (from the text of the paper)

1. The excess burden associated with the taxation of capital income provides some limited scope for improving the welfare of all current and future cohorts when lump-sum taxes and transfers are available. However, given that lump-sum taxes and transfers are not available policy tools, “tax reform” proposals are likely to significantly reduce the welfare of some cohorts and significantly raise the welfare of others unless annual tax rates and their associated deficit levels are chosen with extreme care.
2. The inter-cohort allocation of the tax burden of government expenditure is a significantly more important determinant of national savings than is the structure of taxation.
3. The long-run effect on the capital output ratio of switching from a progressive to a proportional income tax with no change in the stock of government debt is roughly 13%.
4. Short-run crowding out of private investment by balanced budget increases in government expenditure is on the order of 50 cents per dollar, while long-run crowding out is 20 cents per dollar of government expenditure. (*Ibid*, p. 460)

The second paper examines the effects of providing government incentives for saving versus investment via the tax structure. The starting point for this analysis is the observation that in conventional static public finance models, the impact of tax incentives for saving (the supply side) or investment (the demand side) don’t depend on who pays or receives the nominal tax or subsidy, just as with the conventional tax incidence analysis. In a dynamic, lifecycle context, though, the distinction affects different cohorts differently. Auerbach and Kotlikoff illustrate this using a simple two-period OLG model, showing that a subsidy to saving (new saving from

income plus existing asset holdings) makes the old agents in the model better off by raising the value of their assets in the market. A subsidy to investment, however, makes the old worse off since by encouraging new saving, it effectively acts as a tax on consumption, which hurts the old. One consequence of this is its implication that conventional approaches to providing investment incentives – reduction in corporate profit taxes or other property income – if financed by government debt, create long-run incentives for increased consumption and hence reduced saving and reduced long-run equilibrium capital formation. Conversely, the paper shows that there is a range of subsidies to investment that, by crowding out consumption by richer, older agents, crowds in capital formation and can, if rates are chosen correctly, make the subsidy self-financing.

These qualitative policy implications are confirmed by the quantitative analysis Auerbach and Kotlikoff do with their calibrated 55-period OLG model. This paper is also one of the earliest to adopt the structure of sketching qualitative results using “toy” models, and then confirming and sharpening these results using richer, more realistic quantitative models, a practice which has become the norm in contemporary macroeconomic analysis.

The work in these two papers was followed by an unpublished paper Auerbach-Kotlikoff (1985), and an *AER* paper Auerbach-Kotlikoff (1987). The unpublished Social Security paper makes the point that because of the opacity of the benefits formulas the Social Security Administration uses, workers are unable to see the linkage between their marginal tax payments and the benefits it provides them on retirement. This absence of linkage can generate sizeable efficiency losses. The paper reports on the results of simulations using the AK model on efficiency gains that can result from tighter linkage of marginal taxes and benefits. Specifically, gains on the order of 1% of GDP are possible, according to the model, suggesting that even

simple actions like providing workers with information about how their tax payments affect their benefits would be beneficial.

The *AER* paper looks at three major policy issues – consumption vs. income taxation, investment incentives, and the crowding out (or in) effects of deficit spending – and indicates the results of the simulation model's output in each of these policy experiments. While these results were all reported in earlier work, the publication of this paper in the *AER* (widely considered one of the premier general interest publication outlets in economics) was a strong signal to the profession of the value of large-scale simulations, and a strong endorsement of the core value of the overlapping generations model.

From the theory side of macro, some of the most adamant pushback against the conventional ILA approach to macro has come from Roger Farmer. Farmer received his Ph.D. from the University of Western Ontario in 1982. His main thesis paper, "A New Theory of Aggregate Supply" (Farmer [1984]) used the OLG structure in conjunction with information asymmetry – specifically the moral hazard problem that occurs if workers are completely insured against productivity shocks – to come up with a model of aggregate supply with implicit contracts and the possibility of firm bankruptcy. As Farmer notes in the introduction to the paper

In sharp contrast to either the new classical or the sticky price contract explanations of business cycle fluctuations, this theory is not a natural rate theory even in the long run. It predicts that the steady state employment level is an endogenous variable which varies systematically with fiscal policy. (*Ibid*, p. 920)

With a paper covering both OLG and implicit contracts – and one that would ultimately see publication in the *AER* – it isn't surprising that Farmer drew attention from the Penn faculty working in these areas and was recruited there as an assistant professor in 1983 (staying on until 1989 when he moved to UCLA, where he spent the rest of his career until retiring from there to

move to the UK at the University of Warwick). Farmer's tenure at Penn included a year Mike Woodford spent there as a visiting assistant professor right after he graduated from MIT. The coincidence of Farmer and Woodford's interests led them to produce Farmer and Woodford (1984). As the reference to the paper notes, this paper remained unpublished even as the original working paper version garnered over 100 Google Scholar citations. The paper was eventually published as a legacy paper in *Macroeconomic Dynamics*. The paper uses a simple linear-quadratic two period version of the Lucas '72 paper's OLG model, together with the bootstrapping procedure outlined above to examine how sunspot models could generate a number of macroeconomic results consistent with the empirical stylized facts. These included the pattern of interest payments on government spending and debt, the Phillips curve relationship, and the possibility of nominal wages being sticky even as real wages remain flexible. One innovation in this paper was the introduction of a fixed level of government spending financed by public debt. The effect of the government spending in the model is to then create two steady state monetary equilibria, one of which is forward unstable (and corresponds to the Samuelsonian monetary equilibrium when government spending and debt are zero) and the other forward stable. The bootstrapping technique then generates sunspot equilibria on a neighborhood of the stable steady state.

A second key paper from Farmer's Penn tenure is Farmer (1986). This paper studies a version of the Diamond model in which the government, at some point in the past, has incurred debt (positive or negative) but then reverts to a policy of balanced budget spending (so that the deficit going forward is zero). Under these conditions, Farmer derives conditions on the asset demands and wages as functions of the interest rate that lead to a Hopf bifurcation in the equilibrium second-order difference equation, determined by the government's debt position.

The paper shows that this dynamic system has two possible steady states, one where the debt is zero (Gale's balanced equilibrium) and one where the (gross) interest rate is one (the golden rule steady state). From here, a straightforward application of the implicit function theorem and related results from the Hopf bifurcation theorem allow Farmer to find conditions on the equilibrium difference equation such that the roots of the characteristic equation for the dynamic system are complex. When these roots are inside the unit circle, the limit behavior of the dynamic system is that of a continuous cycle. Farmer illustrates this with examples (both stable and unstable) based on plausible preferences and technologies.

We note that this paper was one of those included in the Grandmont-Malgrange *JET* symposium volume on complex dynamics. This established Farmer as one of the new generation of macro theorists following in the footsteps of Benhabib and Grandmont in exploring the implications of the OLG dynamic system. Much of Farmer's later work focused on the issue of indeterminacy more generally, in settings beyond the pure OLG model – in ILA settings, in the Blanchard-Yaari model (discussed below), and in dynamic games – and in this work he became one of the foremost young proponents of Shell's Philadelphia Pholk theorem. In pursuing this research, he has also become an important proponent of the need for macroeconomics to move beyond the New Neoclassical Synthesis and take seriously the theoretical results on complex dynamics, the role of expectations, and the possibility of indeterminacy in macroeconomic modeling.

Before we leave the subject of pushback against the ascendance of the ILA model, we should mention the development of what can be considered hybrid models, which combined features of the ILA and OLG model. There are four notable papers here. The first was Wilson

(1980), the second was Blanchard (1985), the third was Muller and Woodford (1988), and the fourth was Wright (1988).

Wilson's paper provides an abstract formulation of general equilibrium in infinite horizon dynamic exchange economies. The abstractness of his formulation allows him to nest OLG models (infinitely many finite lived agents) and ILA models (finitely many infinite lived agents) along with the possibility of models with infinitely many infinite lived agents. Wilson has two key results. First, if there are only finitely many infinite lived agents, or almost all of the wealth in the economy is held by a finite number of infinitely lived agents, then a competitive equilibrium exists. The second result establishes versions of the two welfare theorems when the economy consists of a finite number infinite lived agents, or (as in the existence proof) a finite number of infinite lived agents plus a negligible set of other agents. So, the positive results here pertain to the ILA model, though the paper does highlight what can go wrong (in terms of existence or optimality results) for OLG versions of the general model.

Blanchard's paper developed a continuous time model in which agents face a constant probability of death once they are born. Large groups of new agents are also born in each instant of time, so that the model has an overlapping generations structure. The fact that agents die stochastically presents a potential problem in the sense that if they are holding positive amounts of wealth, this needs to be bequeathed to descendants in some way. Blanchard solves this problem by using a result due to Yaari, introducing annuities in the model that, in a competitive environment in which the law of large numbers holds, returns all unconsumed wealth to the insurance company issuing the annuities as part of the optimal contract. The paper itself uses this framework to examine how lifecycle income (and associated finite horizons) and government fiscal policies interact to determine steady state equilibrium interest rates.

This paper stimulated a large literature exploring this new version of the OLG model, which we will examine later in the book. One aspect of the model which has been criticized for its lack of realism is the assumption that the probability of death is independent of agents' ages. This particular feature of the model makes working with it quite tractable, since it generates the kind of self-similar recursive structure found in the dynamic programming approach to the RCK model. The assumption isn't particularly realistic, since as people get older, their survival probabilities fall, but the tradeoff between tractability and realism isn't unreasonable.

In footnote 1 of the paper, Blanchard notes that Barro had suggested an alternative interpretation of the mortality probability, viewing it not as the probability of an individual agent dying, but rather as that of a family dying out, i.e., a family dies without producing offspring. The note states that this interpretation is a better justification for the assumption that the probability of dying is constant. But, somewhat confusingly, Blanchard points to Gompertz's law to illustrate the kinds of probabilities arising empirically. The confusion (at least to our minds) stems from the fact that Gompertz's law reflects individual mortality risk, not family risk. The relevant empirical framework for examining family mortality risk is the Galton-Watson process, which looks, variously, at the likelihood of a given family name disappearing, the probability that a maternal mitochondrial lineage disappears, or, more generally, at the extinction probabilities for two-couple families via the bisexual Galton-Watson process. Although the most realistic process – the bisexual GW process – is also the most difficult to analyze, there is a general result due to Thomas Bruss¹⁶ stating that as long as the mean reproductive rate per couple is bounded, and below one for sufficiently large populations, then the limiting family

¹⁶ See Bruss, F. Thomas. "A note on extinction criteria for bisexual Galton-Watson processes." *Journal of Applied Probability* 21.4 (1984): 915-919.

extinction probability is one. Based on the numbers from the GW process for name extinction, though, the probability that any given family name survives over 5 generations is in the 20-40% range. These numbers would seem to bolster the OLG interpretation of Blanchard's model, rather than Barro's interpretation of it as a realistic approximation of a model of dynastic families.

An early extension of Blanchard's model appears in Weil (1989). Weil modifies the Blanchard-Yaari model by allowing new generations of infinitely lived agents to be born at each point of time. This structure mimics the overlapping generations structure in the influx of new young agents but dispenses with the terminal condition of finite death dates in Blanchard's model. Weil shows that in this setting, most of the characteristics of conventional OLG models obtain. There are inefficient competitive equilibria, asset bubbles (or sunspot equilibria) can exist, and the Ricardian neutrality of the conventional representative agent ILA model doesn't hold. As Weil notes, these results show that infinite lives and representative agents need not be the same. As we will note later in our discussion of efficiency notions, Weil's results also suggest that the resource flexibility generated by having new generations entering without pre-existing constraints on their behavior imposed by their predecessor generations is what drives the differences between OLG and representative agent economies. While we also recognize Weil's key point that infinite lifetimes don't rule out OLG phenomena, it will be convenient for our purposes to continue referring to the dynastic family model as ILA.

The Muller-Woodford paper took the straightforward approach to dealing with the fact that while most parents don't leave large death bequests to their children, there are in fact families where large amounts of wealth are passed down through the generations. Their model

assumes that finite lived OLG agents (who don't leave bequests) coexist with infinite lived agents, interpreted as a finite number of (types of) dynastic families. Muller and Woodford first point out that the introduction of even a single dynastic family in an otherwise uniformly OLG environment can have a dramatic effect, since prices must be summable in this case to prevent the dynastic family's wealth from becoming infinite. This result, in turn, rules out the possibility of inefficient equilibria which could exist in the model in the absence of the dynastic family. There are, however, other OLG-specific equilibrium properties that persist in the model, specifically, the possibility that equilibrium can be indeterminate. The main result of the paper is its demonstration that "Indeterminacy continues to be possible even when Barro 'dynasties' own a large fraction of total wealth, as long as their consumption is not too great a part of total consumption" (*Ibid*, p. 257). These results have also been the basis for a great deal of follow-on work examining these hybrid models which we will be examining further later in the book.

The Wright (1988) paper also uses the hybrid OLG-ILA model (citing both Wilson's and Muller-Woodford's contributions) in order to analyze how the institution of contracting affects the observed data collected by government agencies and other entities if the underlying equilibrium is, in fact, at an Arrow-Debreu Pareto optimum. In Wright's paper, workers are OLG-type agents who live $N \geq 2$ periods. This allows the introduction of age-related heterogeneity – in terms of discounting, risk-aversion, and retirement horizon effects – that contributed to the observed differences in hours worked by age. Wright assumes that the employer in the model is an infinite lived household who owns an asset such as land or entrepreneurial skill that allows it to produce output by combining labor hired externally with the

household's unique asset.¹⁷ As in Wilson or Muller-Woodford, the presence of the infinite lived agent forces prices to be summable in order to guarantee that the aggregate value of the output stream is finite. Also, as in Wilson or Muller-Woodford, the resulting competitive equilibrium is Pareto optimal. Wright then introduces take-it-or-leave-it contracts in a way that has the resulting equilibrium allocation being equivalent to the Arrow-Debreu allocation. This allows him to compare the observed wages based on the data generated by the contracting equilibrium, and compare it with the actual wages determined in the Arrow-Debreu equilibrium. The observed individual wage based on contracting exhibits a bias which depends on the workers' interactions on the asset markets in the model, i.e., on workers retirement and precautionary saving. Wright shows that this bias does not average out over workers of different ages. The bias arises because the contracting outcome replaces both labor market and securities market interactions with the contingent outcomes specified in the contract.¹⁸ This paper, in a very real sense, takes Einstein's famous dictum that theory determines what we observe to heart. As Wright notes, he was not the first to make this observation – he credits Rosen (1985) with this – nor was he the last, since the paper is cited. But, the fact that the paper only gets 22 Google cites suggests that the profession was not (and is not currently) ready to confront the measurement distortions the paper delineates.

Section 7.6 Stochastic OLG

¹⁷ Strictly speaking, this should be a large number of small households who collectively own the means of the production, since the equilibrium concept used in the model is competitive.

¹⁸ This paper was written at a time when defined-benefit retirement contracts were giving way to defined-contribution contracts, so it would be interesting to see whether these changes in the contracting environment had an actual impact on the accuracy of the wage data collected.

Lucas (1972) together with the demonstration of sunspot rational expectations equilibria by Azariadis, and Cass and Shell, made the study of stochastic OLG (SOLG henceforth) models germane, particularly since they were not amenable to the kinds of recursive dynamic programming techniques codified in the Stokey and Lucas text.

A critical paper in this developing literature was Spear (1985) (from his University of Pennsylvania thesis, dated 1982). This paper looked at the existence of equilibrium question in a general, multi-good two period lived SOLG model, under the same assumption on the appropriate state variables that Lucas used in his '72 paper, i.e., that endogenous variables – price and allocations – depended only on the exogenous shocks to (in this case) endowments. Up to this point, all SOLG models (that we are aware of) worked with the assumption that only one good was traded, and agents lived two period lives. Spear's paper, together with an unpublished thesis paper by Rao Aiyagari appear to be the only papers from this period to consider more general models. Spear's results showed that generically in preferences, it was impossible to find a rational expectations equilibrium in the more general model where the prices and allocations depended only on the exogenous shocks, or even finitely many lags of the shocks.

From Spear's recollections of what got him started looking at this problem:

After I finished the work on the sunspot model that lead to the "Sufficient Conditions for the Existence of sunspot Equilibrium" paper that Cass had gotten me started on, it seemed natural, given the Balasko, Cass and Shell results showing that any OLG model could be transformed into a two period lived agents model as long as one allowed for multiple commodities and general preferences, to take a look at showing existence of the stochastic steady state equilibrium in a general multi-good model.

We can illustrate this result using the machinery developed earlier to look at the simple Lucas model. Recall for that model that we had two exogenous shocks, H and T , that affected the endowment of the young. The stochastic steady state market clearing condition for the model

with one good was simply that the aggregate excess demand in each state was zero. For the multi-good model, the market clearing condition is the same

$$z_1^H + z_2^{HH} = 0$$

$$z_1^H + z_2^{TH} = 0$$

$$z_1^T + z_2^{HT} = 0$$

$$z_1^T + z_2^{TT} = 0$$

where, as in the easy Lucas model, the excess demands are functions of the commodity prices p^H and p^T , though these are now vectors and the excess demand functions above are also vector-valued. Also as in the easy Lucas model, Walras' law restricts the equation system, since

$$p^s \cdot [z_1^s + z_2^{ss}] \equiv 0$$

for $s=H$ or T , and

$$p^s \cdot z_1^s + p^{s'} \cdot z_2^{ss'} \equiv 0$$

for $s \neq s'$, since these are budget constraints.

From the budget constraints, we also have two price normalizations available, since the first two constraints imply that we can normalize prices such that $\iota \cdot p^s = 1$ for $s=H, T$, where ι is a sum vector (i.e., a vector of 1's). With these preliminaries, we count equations and variables. The market clearing conditions constitute a system of 4ℓ equations, but because of the budget constraints (of which there are 4), we only have at most $4\ell - 4$ independent equations. Since prices are normalized, we have $2\ell - 2$ independent variables. In order to have at least as many variables as we have equations, then, we require that

$$4\ell - 4 \leq 2\ell - 2$$

which can easily be seen to imply that $\ell \leq 1$, i.e., we can only guarantee as many variables as independent equations in the one good model.

It is well known that in linear equation systems, having at least as many variables as independent equations is a necessary condition for finding a solution to the equation system. This need not be true for non-linear equation systems, which complicates the analysis for showing whether or not the simple equilibrium exists.

Spear's approach to showing that in fact there will not be a solution generally was based on differential topology methods, in particular a version of Rene Thom's transversality theorem, which states that for almost all mappings from one manifold to another, the image of the mapping in its range will intersect any submanifold in the range transversely (i.e., the tangent spaces of the image and the submanifold will intersect in general position). We illustrate this in figure 19.

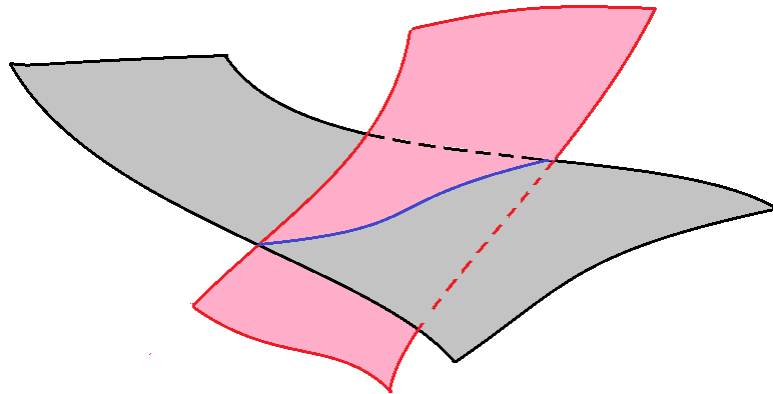


Figure 19

Transversal intersection

The pink and grey surfaces intersect along the blue line, and the intersection is such that small perturbations of either surface leaves the intersection qualitatively unchanged. This is the basic definition of transversal intersections. Non-transversal intersections are those involving tangencies. These are not in general position because a small perturbation of either surface leads

to either non-intersection or transversal intersections. We illustrate this in Figure 20 for one-dimensional surfaces (i.e., curves).

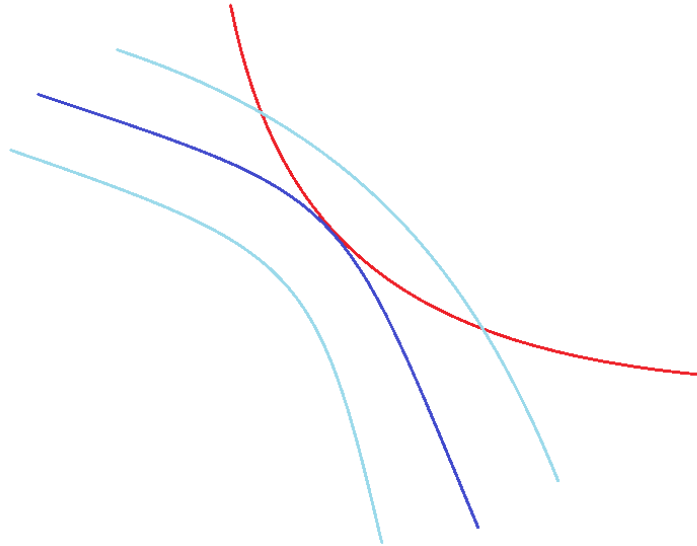


Figure 20

Non-transversal intersection

In the figure, the red and blue curves are tangent. A perturbation upward or downward of the blue curve destroys the tangency, resulting in either two transversal intersections or no intersection. From the definition of transversal intersection in terms of the general position notion, it should also be clear that non-intersections are transversal (vacuously).

this constitutes a system of linear equations, applying Thom's theorem boils down to counting equations and variables.

There were a couple of additional complications here. First, if the preferences in the model are additively separable over time, the simple equilibrium can be shown to exist.¹⁹ This is what necessitated looking for genericity results in the space of utility functions rather than simply with respect to endowments. Second, because preferences over uncertain outcomes are in expected utility form, a direct application of Thom's theorem to the space of utilities over random allocations was not possible, requiring instead consideration of the Mather (1970)'s multijet extension of Thom's theorem. This result was extended by Alex Citanna and Paolo Siconolfi to models with multiple heterogeneous agents in Citanna-Siconolfi (2007) using direct perturbation methods.

The significance of these results is simply that they imply that general stochastic overlapping generations models will have history dependent equilibria, i.e., equilibria which depend on the entire (infinite) history of shocks. This made working with general SOLG models difficult, since the state variable would be (generically) infinite dimensional.

We turn next to Aiyagari (1981) from his thesis. Chapter 3 of the thesis examines a two-sector model, which necessarily involves agents trading more than one good, so that Spear's result applies. From the thesis

We are here interested in showing the possibility of a stochastic steady state where average q may be less than one while aggregate investment is positive in the context of an overlapping generations model. It appears to be difficult to do the above with rather general specifications of technology and preferences. One alternative is to select specific functional forms and numerical values for the parameters of the model and to compute numerically the equilibrium steady state distributions for average q and aggregate investment. (*Ibid*, p.37)

¹⁹ According to Spear's recollections, Russell Cooper pointed this out during a Penn workshop on graduate student research. Spear and Cooper were classmates in Penn's Ph.D. program.

The functional form for the young agent's utility function over the two goods ends up being equivalent to an additively separable specification, so that for the example, at least, the stochastic steady state rational expectations equilibrium does exist. From this, it seems clear that Aiyagari was in fact aware that there could be problems with finding simple REE for general SOLG models. This fact is relevant for the next part of our story on the developments of the stochastic model.

Aiyagari spent the 1987 academic year visiting at GSIA. He had overlapped in Minnesota's Ph.D. program with Dan Peled, who was then an assistant professor of economics at GSIA and who arranged for Aiyagari to visit. While Aiyagari was at GSIA, he and Spear began talking about Spear's thesis results. From Spear's recollections:

Rao and I were both avid tennis players, and once we found out we had this mutual interest, we started playing once a week, usually in the late morning before lunch. It was natural, then, that we would spend time talking about research. Rao was quite interested in the result from my '85 *JET* paper.

One day, after we'd been talking about this, and I'd explained how I got the result, Rao disappeared for a week. When he came back he stopped by my office and said that he'd been looking at a three period stochastic model with additively separable preferences and a single good, and the same problem of more independent equations than variables arose in that model. As we talked about, we realized that what was happening was that the portfolio rebalancing actions the middle-aged agents took in response to the resolution of uncertainty in their second period of life made the equilibrium history dependent in the same way that the time non-separability of preferences did in the two-period multi-good model. I encouraged Rao to write the result up as part of a paper he was working on at the time.

Aiyagari did in fact write this up as part of what became his 1988 *JET* paper, discussed previously. When the paper came out, Spear noticed that the only aspect of Aiyagari's discovery that was in the published paper was the brief paragraph at the end indicating that because of the history dependence issue, SOLG and stochastic ILA models couldn't be observationally equivalent. Again, from Spear's recollections:

When I next ran into Rao – probably at a conference somewhere – I asked him what had happened to his write-up of the history dependence result, and he told me that he’d been asked by either a referee or *JET* Associate Editor (I don’t remember exactly which) to “get rid of the negative results.”

Unfortunately, the working paper versions of this paper appear to have been lost.

Aiyagari’s paper cites a University of Wisconsin working paper version from 1986, and via Spear’s recollection, he issued a GSIA working paper version with the history dependence results included in 1987. The GSIA working paper collection was disposed of in the early 1990’s when the school expanded into a new wing of its building, and a search by staff at Wisconsin’s economics department was unable to find a copy of the ’86 version.

Via Karl Shell, we were informed that the *JET* editorial office records from then were lost when the editorial office moved from Cornell to NYU. We also contacted the living members of the *JET* editorial board from that period to ask if any of them recalled handling the paper for the journal but got only negative replies from the 19 of the 26 board members who responded.

So, we have an example here of a key result that has no formal documentation of how the result was obtained or proved. Henriksen and Spear (2012) provide a formal proof of the non-existence of short-memory rational expectations equilibrium for the 3-period model with one good and additively time separable preferences (that same model Aiyagari worked with) and Citanna-Siconolfi (2007) indicate how their results can be extended to accommodate longer lifetimes, but by this time there had been many studies done using this framework – we discuss these below – with the understanding that the equilibria would necessarily be history dependent.

One of the earliest papers to work in the history-dependent SOLG framework was Woodford (1986) which we discussed above. In this paper, Woodford looks at a general, multi-good two period OLG economy and finds conditions under which a sunspot equilibrium will

exist. The sunspot variables in the model are infinite sequences of realizations of draws from a general stationary distribution, and it is the fact that the equilibrium Woodford derives depends on the full infinite history of shocks which makes the result compatible with Spear's result.

Woodford's technique in the paper involves a careful application of the implicit function theorem on Banach space, together with generalizations of the Kehoe-Levine results discussed above. One key result of this paper given its extension of the results presented in Woodford (1986a) to the multi-good environment is that it establishes that any degree of indeterminacy is necessary and sufficient for the existence of sunspot equilibrium. While we don't go into detail about the paper here, we do note that the result applies, with slight modifications, to a model in which endowments or other fundamentals are random.

Spear and Srivastava were the first to attempt to "tame" the history dependent nature of the REE in the SOLG model by observing that at the time agent's formed expectations of future prices and chose optimal consumption demands based on these expectations, the observables in the economy were in fact finite. They showed, in Spear and Srivastava (1986) that there were lagged endogenous variables – prices, consumptions, Lagrange multipliers, etc. – that were sufficient statistics for the infinite history of shocks up to that point in time. By including these as state variables, they showed how to demonstrate existence of equilibrium working in terms not of direct functions but of probability measures. This approach avoided the limitations of the approach used by Grandmont and Hildenbrand, also discussed earlier. It also reinvented a wheel originally introduced to the economics profession by Blume (1982).

While the Spear-Srivastava paper developed its results primarily in terms of marginal probability measures (and they called the resulting equilibrium a Fulfilled Expectations

Equilibrium), the appendix to the paper showed how to extend the results to allow for conditional measures and a fully rational expectations equilibrium.

We note that this paper was focused solely on the SOLG model. A significant broadening of these results was developed in Duffie *et al.* (1994) (also using Blume's techniques), which we will discuss in more detail below.

Peck (1988) showed how to use the non-stationary deterministic equilibrium dynamics of the model to generate sunspot equilibria using the "bootstrapping" technique which we examined above in the context of Woodford's work. Unlike Woodford's approach, which focused on stationary sunspot equilibria, Peck's work looked at the possibility of non-stationary sunspot equilibria based on the local non-stationary dynamics around the steady state equilibrium of the model. One reason for this is, as Peck notes, was:

Equilibria that appear to be nonstationary could in fact be stationary in a larger state space. In an economy with two or more commodities per period and random endowments, Spear shows that generically there do not exist (stationary) equilibria where the price is function of the current endowment. When the state space is expanded to include the current endowment and the previous price, Spear and Srivastava show that an equilibrium probability measure exists. (*Ibid*, p. 21)

The connection between the non-stationary dynamics and the possibility of stationary sunspot processes was implicit in Woodford's 1986 paper, though the interpretation was complicated by the infinite-dimensional state space. The connection would reappear later in Spear (1988), and then again in work by Victor Rios-Rull, and the Duffie *et al.* paper referenced earlier.

Since Peck's use of the technique appears in his 1985 thesis (see reference in the '88 paper cited above), his development of the technique was contemporaneous with Woodford's. Woodford has indicated (in an email dated July 10, 2022) that he was a visiting assistant professor at Penn in 1984, while Peck was a student there, and taught some classes on this topic.

Per our email conversation with Woodford, “the similarity in our work may not be entirely coincidental.”

When we asked Peck about his recollections, he told us (email correspondence dated July 11, 2022)

I am sure that we had conversations about sunspots, but I am not sure that there was any direct cross-fertilization. I always thought of my contribution as showing that sunspot equilibria existed quite generally, without strong assumptions like backward-bending offer curves. The technique was about supporting period t behavior with two outcomes for period $t+1$ rather than one outcome, and seeing that each of the period $t+1$ outcomes could be supported by continuing along the offer curve or branching into two outcomes again, and so on. The technique itself just seemed like a way of solving the problem and not necessarily something to take credit for beyond the task at hand. I first saw Mike's paper with Roger after I was well underway with my job market paper and never had the sense that they borrowed my technique or anything like that. So to the extent that we did something similar, I guess you could say it was simultaneous discovery or that it was just a natural way to solve a problem.

When we asked Woodford where he first encountered the bootstrapping technique, he pointed to Shiller (1978), which shows how to construct these kinds of equilibria in the context of linear rational expectations models. Peck advised us in a follow-up email that he first heard the term “bootstrapping” used by Karl Shell, though given the constellation of people around the Penn department at the time – Azariadis, Cass, Grandmont, Farmer and Woodford – the term was likely something in the air.

The next significant paper on SOLG chronologically was Spear (1988). Unlike the Spear and Srivastava paper, this paper imposes assumptions on the underlying model to guarantee that the rational expectations equilibrium prices are functions of state variables, rather than allowing for possible correspondences (due to multiple temporary equilibria). Furthermore, the paper shows that for an N agent, multi-good, two period lived economy, the lagged price is a sufficient statistic for the infinite history of shocks, as long as the shocks are sufficiently small. The basic methodology of the paper is similar to Woodford's unpublished '86 paper in using functional

forms of the implicit function theorem together with the Kehoe-Levine results to show that as long as the non-stochastic competitive equilibrium is determinate, the rational expectations equilibrium will be given by a stochastic difference equation of the form

$$p_{t+1} = f(p_t, \omega_t, \omega_{t+1}).$$

This paper provided the theoretical foundation for the follow-on paper Spear, Srivastava and Woodford (1990), which used the bootstrap approach to show the existence of sunspot equilibrium in a general multi-agent, multi-good two period SOLG model.

Again proceeding chronologically, the next big step in understanding the workings of general SOLG models was the Duffie *et al.* (1994) paper. This paper provided a general solution to showing how to obtain a Markovian equilibrium for general history dependent models with very general stochastic shocks, and provided examples of the construction of such equilibria for SOLG models, for ILA models with finitely-many heterogeneous agents and incomplete markets, and for stochastic games. This paper had an interesting history. The working paper version circulated, and was cited, for a number of years before the paper was finally published in 1994. Via a search of Google Scholar, the earliest citations of the paper (as a working paper) relate to the 1988 Stanford GSB Working Paper (Duffie, D., et al., 1988, "Stationary Markov Equilibria," working paper, Graduate School of Business, Stanford University).

There was some definite cross-fertilization between the work of Spear and Srivastava, and Spear, Srivastava and Woodford in the development of these results. Spear and Srivastava presented an early set of their results at the 1984 Summer meetings of the Econometric Society held at Stanford University in a session on Multiple Agent Models of Aggregates. The papers in the session were

1. "Reconciling Individual and Aggregate Differences in Labor Markets: An Adverse Selection Approach," Bruce Smith, Federal Reserve Bank of Minneapolis.
2. "Liquidity Preference, Stabilization and Welfare," Scott Freeman, Boston College.
3. "Equilibrium Dynamics in a Stochastic Overlapping Generations Model," Stephen Spear and Sanjay Srivastava, Carnegie-Mellon University.
4. "A Rational Expectations Complete Contingent Claims Model: Estimation Using Post War U.S. Data," Timothy J. Kehoe, M.I.T., and David K. Levine, U.C.L.A.

DISCUSSANTS: Martin Eichenbaum, David Levine, Ed Prescott, Robert Townsend

CHAIRPERSON: Charles Whiteman. (From the Program of the 1984 North American Summer Meeting of the Econometric Society, *Econometrica*, Vol. 52, No. 6, Nov., 1984, pp. 1562-1570.)

John Geanakoplos and Andrew Mas-Colell were both in attendance at this conference.

The Spear, Srivastava and Woodford paper was presented at both the 1986 Winter Meetings of the Econometric Society in New Orleans, in a session on General Equilibrium Theory, and a Mathematical Sciences Research Institute conference in March of 1986, with many of the same people in attendance. The papers included in the New Orleans session were

1. "Determinacy in Large Square Economies," Timothy J. Kehoe, Cambridge University; David K. Levine, University of California-Los Angeles; Andreu Mas-Colell, Harvard University; and William R. Zame, SUNY-Buffalo.
2. "On the Finiteness of the Equilibrium Price Set," Beth Allen, University of Pennsylvania.
3. "On the Existence of Implementable Rational Expectations Equilibria," Mark Feldman and Christian Gilles, University of California-Santa Barbara.
4. "On the Structure of Equilibrium in Stochastic Overlapping Generations Models," Stephen E. Spear and Sanjay Srivastava, Carnegie-Mellon University; and Michael Woodford, University of Chicago.

DISCUSSANTS: Darrell Duffie, Stanford University; V. V. Chari, Federal Reserve Bank of Minneapolis.

CHAIRPERSON: Larry Jones, Northwestern University. (From the Program of the 1986 North American Winter Meetings of the Econometric Society, *Econometrica*, May 1987, Vol. 55, No. 3, pp. 733-748.)

As was standard at such conferences, there was a lot of conversation outside of the presentations on on-going projects. Quoting from an email from Srivastava (dated July 14, 2022)

If I recall, they [Duffie, Geanakoplos, Mas-Colell, and McLennan] were already working on the same problem ... but it is interesting that several sets of people were working on the same double-infinity existence problem at the same time, not just in OLG models but other areas as well. In fact, around then, there was a lot of work going on with existence results with infinite numbers of agents and/or commodities, in competitive equilibrium models as well as others.

The main result of the Duffie *et al.* paper is the demonstration of the existence of what the authors call a “self-justifying set” of exogenous and endogenous state variables. The self-justifying aspect of this set is that if one starts from an element in the set and then determines the endogenous equilibrium values associated with this starting value, these will also be in the set. The method used to show the existence of this set is essentially the same one that Balasko and Shell used in the equilibrium existence proof for the non-stochastic model. One truncates the economy at a finite point of time $T < \infty$ and uses standard general equilibrium techniques to determine the possible equilibria. The basic existence results for the finite-horizon economies then generates a sequence of T -equilibrium sets which have the same kind of nesting property as in the Balasko-Shell model. The intersection of these sets then determines the self-justifying set, and the support for the equilibrium ergodic measure.

A second innovation in the paper is that of working with an *expectations correspondence*, which determines probability measures consistent with the temporary equilibrium determined for a given conditional probability measures that agents use to forecast future states. The need for

using a correspondence here, rather than a function, arises from the possibility that there may be multiple temporary equilibria. This, of course, was the limitation in the Grandmont-Hildenbrand result on existence of REE, as well as a built-in limitation in Spear (1988). When this correspondence is restricted to the self-justifying sets, it generates a *time homogeneous Markov equilibrium* for the model. Any fixed point of the correspondence so defined determines an invariant distribution for the Markov equilibrium process (using Blume's terminology here).

While we don't go into detail here on these results, one notable fact is that while the paper shows existence of Markovian equilibria for models with endogenous state variables, it doesn't identify which state variables should be included. The question of which endogenous state variables to include became a central concern with the work of the next paper in this sequence of work on the SOLG model, Rios-Rull (1996).

Rios-Rull's paper extended the modeling strategy of Auerbach and Kotlikoff to stochastic environments with aggregate shocks. Following Aurbach-Kotlikoff, the model in the paper has agents who live 55 periods, with multiplicative aggregate random shocks to production (as in conventional RBC models). In looking at the stationary equilibrium for the model, Rios-Rull assumes that the relevant endogenous state variables should be the wealth distribution, in the form of the asset holdings of each type of agent in the economy. Based on this assumption, Rios-Rull defines a *recursive equilibrium* as a mapping from the cross-product space of individual asset holdings and exogenous states to equilibrium good and asset prices which satisfy individual agent's optimization problems, the representative firm's profit maximization problem, and market clearing conditions. The term "recursive" here reflects the fact that in characterizing agents' individual optimizations, Rios-Rull uses a value function specification of the dynamic programming problem. This forward recursion is equivalent to the backward dynamic

programming solution to the finite lived agents' optimizations, although unlike in the ILA model, the value functions are generally time dependent. Nevertheless, the formulation gives a well-defined equilibrium concept, even if the terminology makes it easy to confuse with the equilibria that arise in ILA models via forward dynamic programming. Despite this confusion, the terminology has stuck.

Rios-Rull's choice of asset holdings as the relevant state variables is intuitive, since it is the portfolio rebalancing decisions of agents in the model that generate the history-dependence of the equilibrium. Assuming that the asset holdings (or wealth distribution) will be a sufficient statistic for the infinite history of shocks in the steady state is natural, although not formally justified based on the results of the Duffie *et. al.* paper. When we asked Rios-Rull whether he was aware of the Duffie *et. al.* paper, he replied (email date July 15, 2022)

It is hard to remember clearly, but probably I was aware of this paper from way before, although to be fair it was quite incomprehensible (measures on measures), at least for me.

That the state was the wealth distribution was sort of obvious. Even though aggregate wealth is sufficient for prices today when labor is not a choice, it is clear that when it is it depends on the wealth at each age. Moreover savings depend on who owns the wealth.

However, it turns out that this need not always be true, as a number of papers demonstrated in the early 2000's (see, e.g., Krebs [2004] and references there). A partial answer to this conundrum was provided 14 years later by Citanna and Siconolfi (2010) where they showed that set of OLG economies having recursive equilibria was dense in the space of all SOLG economies. In an echo of the earlier debates between Lucas, Cass and Shell over what constitutes the right set of state variables, this debate continues to this day. Spear's 1988 functional REE results have been applied by a number of authors working with general multi-period SOLG models, notably including Storesletten, Telmer and Yaron (2007), where the lagged bond prices serve as sufficient statistics for the shock history. One possible explanation

for why both approaches to summarizing the shock history work could be the fact that the equilibrium policy functions take the form of iterated function systems, which map the space of shock histories into the equilibrium realizations of the endogenous state variables. While the images of these mappings are generally quite complex, in simple cases where the aggregate shocks are finitely supported, they take the form of Cantor-like sets. Since every compact metric space is the image of the Cantor set under a continuous function (see, e.g., Aliprantis and Border [2006]), it may well be that there are homeomorphic mappings between the various possible state spaces that give rise to equivalent sufficient statistics for these models. We are at this time unaware of any formal results to this effect, however.

In addition to directly extending the Auerbach-Kotlikoff approach to stochastic OLG models with aggregate shocks, Rios-Rull also allowed for the possibility that agents in the model could die before reaching the maximum possible (finite) lifespan, by introducing age-specific mortality risk. His model thus nests and extends the Blanchard-Yaari model as well. As in the Blanchard-Yaari model, Rios-Rull assumes the existence of perfect annuities markets that allow agents to insure against the mortality risk.

Rios-Rull's simulation results are quite interesting. First, he shows that the time-series generated by the simulations are quite close to the same series generated by the conventional ILA models, on the production side. Second, the volatilities in aggregate consumption and investment in the calibrated model are in line with what is found in ILA models. Third, unlike ILA models, the SOLG model generates time-series of hours worked that match actual data fairly closely (though it underpredicts the observed working patterns for the 22-44 year old age group). Where the SOLG model differs significantly from the findings in conventional RBC models is in agents' deviations from the standard life cycle hypothesis of perfect consumption

smoothing. Rather, the observed age-consumption profile in the simulations is hump-shaped, in agreement with actual data. Subsequent studies have found that to fully match the observed age-consumption profiles in the data, however, one generally needs to include a borrowing constraint on the young. This requirement obviously suggests that asymmetric information about future prospects early in life is an important macroeconomic phenomenon. The fact that even without credit constraints the SOLG generates significant deviations from the predictions of the permanent income hypothesis has important welfare implications in its own right. In an earlier paper (Rios-Rull [1994]) based on the same simulations, Rios-Rull also showed that quantitatively, the allocations coming out of a model with sequentially complete markets relative to one with incomplete markets was negligible. Based on this, he concluded that it didn't matter for macroeconomic issues whether one worked with a model with a rich or sparse asset structure. This is something we will explore in more detail below.

Despite this important deviation on the distribution side from the predictions of RBC and related models, the macro profession's take away from Rios-Rull's work was that it was demonstrating some kind of observational equivalence between SOLG and SILA models, since it matched up well on the production side. As a result, the paper has not been widely cited (getting 425 Google Scholar cites as of this writing), even though this paper was arguably ahead of its time in terms of where macroeconomics ought to be going. Indeed, we consider the lack of attention to this paper one of the key indicators that the macro market was tipping away from OLG to ILA formulations. And again, the reason for this would seem to be that working with SOLG models was (despite Rios-Rull's choice of terminology) not recursive, borderline intractable, and irrelevant given the perceived observational equivalence of the two classes of models.

By way of documenting our claim that Rios-Rull's work has been widely ignored, we went through the first 26 pages of Google Scholar papers that cite Rios-Rull's 1996 paper (or earlier versions of it going back to 1990) and examined the papers to determine the reason for the citation to Rios-Rull. We stopped after 26 pages because although Google Scholar's ranking algorithm is not public, empirical studies of the results it returns – see, e.g., Beel and Gipp (2009) – indicate that the number of citations a paper gets generally moves it higher in the rankings and closer to the front of the displayed results. Based on this, we are looking at a sample of the most influential papers that cite Rios-Rull's 1996 paper.

We classified these reasons according to whether the work in the paper was genuine follow-on work (which we call “relevant”); work in other modeling frameworks or purely econometric which cited Rios-Rull as justification or contrast to the work presented (which we call “in passing” citations); summary citations (pedagogic, laudatory or handbook material); tangential citations where the underlying model was the same or similar to Rios-Rull's but without aggregate shocks; and vacuous citations (a cite in the paper's references but no discussion in the text or no actual citation). This sample of hits from Google Scholar constituted 218 papers. The table below summarizes the numbers in each category.

Category		%	Top 5
Relevant	31	14.22%	2 (6.5%)
In passing	63	28.90%	
Summary	42	19.27%	
Tangential	51	23.39%	10 (20%)
Vacuous	31	14.22%	
Total	218		

Rios-Rull Citation Summary

The “Top 5” heading indicates the number of papers in the “Relevant” and “Tangential” category published in one of the top 5 economics (or finance) journals. In the “Relevant” category, six of the papers (including one by Citanna and Siconolfi in *Econometrica*) are theoretical; of these five are by Citanna and Siconolfi in their series of papers examining the genericity of the recursive equilibrium, and one is by Kim and Spear characterizing the stochastic dynamic structure of recursive equilibrium in general SOLG models. The other top 5 publication in the “Relevant” category is Glover *et al.* (2020) which examines the intergenerational redistributions generated by the 2007-8 financial crisis and resulting recession.

Interestingly, in the “Tangential” category, 36 of the 51 papers (71%) deal with OLG models in which uncertainty is idiosyncratic, generally mortality risk or lifecycle income risk with variations on each of these. Of these papers, only one (Castaneda *et al.* [2003]) was published in a top 5 journal. Of the remaining papers on idiosyncratic risk, 14 were unpublished (39%) and the rest (58%) were published in top to middle tier field journals.

In the “Relevant” group, many of the papers make important points about the difference between SOLG and SILA models. For example, Gomme *et.al.* (2004) replicates Rios-Rull’s findings on hours worked by different cohorts, and notes that while the basic model can account for about 60% of observed variation, the model fits the data much more closely if one restricts attention to the middle-aged to retired cohorts. The model’s fit breaks down with younger cohorts. The paper draws a couple of important conclusions from this. First, the effect appears to arise from the differences in planning horizons between younger and older agents. This is consistent with the view that finite horizons matter, and that it is the double infinity of agents and periods that makes the OLG and ILA models different. Secondly, for young agents facing long time horizons for whom the aggregate shocks appear temporary, search frictions are likely to

matter in determining both contemporaneous hours worked and future ability to weather shocks that now seem more permanent. In a solo authored paper, Gomme (2008) compares the optimal inflation rates in comparably calibrated stochastic representative agent and OLG models and finds marked differences. In the ILA framework, Friedman's rule of deflating at the real rate of interest is optimal. Gomme finds in large scale SOLG models that significant inflation generated by lump sum money injections can generate Pareto improvements by transferring wealth from older, richer agents to poor, younger ones. Heer and Maussner (2012) work with an extension of Rios-Rull's model in which agents live 240 periods (to allow for calibration to quarterly data) to examine the effects of unanticipated inflation. Unlike Rios-Rull, their model is too complex to allow for computational simulations, but they are able to use Auerbach and Kotlikoff's techniques to solve for the model's steady state equilibrium and then use simulations to calculate impulse-response functions. Their findings in this paper supports the findings in Gomme (2008) in the sense that wealth inequality is reduced by the inflation shock via transfers from older agents to younger.

Rios-Rull (1994) makes an important follow-up contribution to his own work (follow-up despite the fact that the paper was published before the 1996 *REStud* paper; working paper versions of this work had been circulating since 1990). As we noted above, in this paper, he uses calibrated simulations of long-lived SOLG models with aggregate shocks to compare the outcomes under different degrees of sequential market completeness (i.e., by giving agents different numbers of assets with which to hedge the aggregate risk). He finds that the degree of market incompleteness across different asset regimes makes very little difference in the welfare outcomes of agents. This result is consistent with results derived later by Henriksen and Spear (2012) which we discuss below. Intuitively, though, when one includes the wealth distribution as

part of the description of the economy's state – which is to say the lagged asset holdings of agents are included as state variables – then in the presence of aggregate exogenous shocks, one can never have as many assets as there are states, so, in this sense, markets are always incomplete.

The Glover *et al.* paper (on which Rios-Rull is a coauthor, and the only non-theory paper in our “Relevant” category to make it into a top 5 journal) directly applies Rios-Rull's simulation framework to examine the effects of the recession following the 2007-8 financial crises (the Great Recession) across households of different ages. Their review of the empirical data clearly indicates that this recession differentially affected different age groups, so the choice of the SOLG framework is clearly justified. Their findings show that while younger households were hurt by declining real wages during the recession, older agents were also hurt by declines in asset prices. These asset price declines actually benefitted younger households since they were able to fund retirement asset purchases more cheaply.

In addition to these papers, there were a number of important papers (not all of which cite Rios-Rull [1996]) which applied the multi-period SOLG model to examine the effects of unfunded (pay-as-you-go or PAYGO) social security systems. We will discuss these papers in detail in the next chapter.

We are persuaded by the overall statistical analysis presented here together with the sample of follow-on results generated by Rios-Rull's model that this was indeed pioneering work that showed not only how the more realistic multi-period stochastic OLG model could be used in macroeconomic analysis, but why it ought to be used in preference to the dynastic family alternative. We include our full bibliography of papers that cite Rios-Rull (1996) in Appendix 2 of the book for anyone interested in doing their own follow-up research.

Section 7.7 Welfare Issues

While we have looked in some detail at the welfare issues associated with the deterministic OLG model, there are additional welfare issues that are specific to the SOLG environment, so we examine these here. To do so, however, we need to first examine a couple of technical contributions that showed formally how the double infinity of the OLG environment differentiated the model from the ILA setting.

For deterministic models, we have two competing claims for why the first welfare theorem can fail in the OLG model, the Cass-Yaari restricted participation friction, and the Shell double-infinity friction. While the restricted participation friction has been important in explaining the role of money as a medium of exchange (and in the development of search models of money), money in these environments allows for Pareto improvements by effectively removing the restrictions on market participation. This is true in the circular model that Cass and Yaari originally considered, and it remains true in the turnpike (and other) models introduced by Townsend. Because these models are essentially static, one can't obtain the kind of inoptimality that occurs in OLG models where, for example, the monetary equilibrium has the value of money going asymptotically to zero.

This argument suggests that Shell's contention that it is the double infinity of agents and time periods that leads to the possibility of inoptimal competitive equilibrium. The mechanism Shell pointed to was the intergenerational transfer of resources analogous to the shuffling of guests in the Hilbert hotel. This mechanism was put on a sound mathematical foundation in

Geanakoplos and Brown (1982). While this paper was never published, the basic arguments of the paper are outlined in Geanakoplos (1989).

Geanakoplos and Brown use non-standard analysis to analyze the behavior of deterministic OLG economies “at infinity” by showing that all the basic properties of the model that hold at finite times T will hold when T is an infinite integer (see Robinson, [2016] or Anderson [1991] for details on this mathematics). As Geanakoplos points out in his Palgrave chapter, in the standard Balasko-Shell proof of existence of equilibrium, in the finite truncated economy, one conventionally sets the excess demand of the last cohort of young agents *when they are old* to zero. Via the Balasko-Shell argument, then, this yields a particular competitive equilibrium. Geanakoplos-Brown show that by loosening this particular restriction “at infinity,” we obtain all the possible equilibria for the model because of the slack in the markets at infinity. The significance of the slack being at infinity is based on the fact that it is society, not the individuals who make up the society, that has the resource slack available. This becomes particularly clear in the Geanakoplos-Brown analysis of what happens when money is introduced. In the monetary model, the initial old are endowed with some positive amount of money, which they can use to purchase actual goods from the young in period 1. What gives money value, in the formal non-standard analysis, is the fact that money is backed by the addition of an endowment of goods in (transfinite) period $T+1$. What this result formalizes is the fact that money will have value as long as there is a social commitment or contract collateralizing the stock of money. In our previous discussion of the tenuousness of money, it was the possibility that the social contract might be breached that led to this tenuousness. The bottom line of the Geanakoplos-Brown analysis can be summarized by saying that when the future is flexible, many things can happen.

We can contrast the environment analyzed by Geanakoplos-Brown with the conventional ILA environment of the dynastic family. In these models, individuals (or families) themselves have the ability to transfer resources back and forth through time (including infinite time), so that the social slack at infinity in these models is internalized. Indeed, it is this property that gives rise to Ricardian equivalence in Barro's framework, or to the neutrality of everything in the Bernheim-Bagwell model. This internal versus external distinction between the two models is important in explaining the history dependence (and related Pareto optimality) issues in the SOLG setting.

The second result we wish to look at is Labadie (1986). Labadie studies a conventional two period lived stochastic OLG model in which the endogenous variables are functions of the exogenous stochastic process on a non-labor factor of production and shows in her main results that changes in the risk aversion of agents in the model have ambiguous effects on the endogenous variables. The reason for this is straightforward. In the two-period OLG environment, an increase (for example) in second period income will have an ambiguous effect on the intertemporal marginal rate of substitution because this depends on how rich or poor the agent is in their first period of life relative to their second. In a stochastic model, how rich an agent is when they are young is largely an accident of birth. How much real wealth they have in old age depends on the randomness of the prices of the assets they accumulate, and this accumulation is affected by the income effects generated by price changes from one period to the next. Because income and substitution effects for finite lived agents can move in different directions, the response of the intertemporal MRS is ambiguous.

In the ILA environment – in Labadie's case a version of the '78 Lucas asset pricing model – an increase in next period's income always generates a negative response in the

intertemporal marginal rate of substitution. Furthermore, any income effects get channeled through the equilibrium prices and payments to the non-labor factor of production in such a way that the return weighted expected intertemporal marginal rate of substitution is the same in every period. This result is possible in the ILA model because of the internalization of wealth flows between periods. In the presence of complete markets, this is also the reason that the resulting allocation of consumption is Pareto optimal.

We introduce this section with these results because this is what distinguishes the stochastic OLG from its ILA counterpart. While the distinction can be difficult to see in deterministic settings, it manifests starkly in stochastic environments. We see this most clearly in the setting Aiyagari used, where OLG and ILA agents have the same additively time separable utility functions, trade a single good, and have the same random endowments (or produce with the same random technology). In both models, the portfolio rebalancing that agents undertake in each period generates pecuniary externalities that feed forward to subsequent consumption and savings decisions. In the OLG model, the effects of these externalities eventually come due at the end life, and, since some agents die every period, the changes in the wealth distribution generated by these externalities affect the equilibrium and generate the observed history dependence. In the ILA model, by contrast, the effects of the externalities are pushed forward “to infinity.” As long as the shock process is stationary, infinite-lived individuals (or institutions) can offset the external effects by rebalancing the internal transfers at work between generations, *a la* Barro’s bequest mechanism. This internalization of the pecuniary externalities eliminates any history dependence not built into the shocks and allows for the correspondingly simpler characterization of equilibrium relative to that of the OLG model.

With these preliminaries, we can extend the analysis of optimality from the deterministic model to the stochastic one.

The earliest results characterizing optimality in SOLG models that we have found is Peled (1984). This paper examines the optimality of a stochastic steady state equilibrium in a two period, one good SOLG model (with finitely supported exogenous shocks) under the assumption that endowments are such that agents always wish to hold positive stocks of money. Aiyagari and Peled (1991) extended these results for general endowment distributions. Since these two papers use similar techniques, we illustrate them in the context of the '84 Peled paper. The two period lives and single good assumptions mean that the stochastic steady state equilibrium of the model will have all endogenously determined variables given as functions of the aggregate shocks only. Peled also imposed the assumption that markets are complete, i.e., agents can freely transfer wealth between periods and across states of nature. The optimality concept is that of conditional optimality (first introduced by Muench [1977]). Conditional optimality requires that the central planner work under the same constraints as the agents in the model. Specifically, this requires considering young agents born into different states of the world as different agents who then face uncertainty only over what will happen to them when they are old. In particular, this perspective rules out a fully *ex ante* consideration of intergenerational welfare. It does, however, correspond to the same constraints that the markets operate under, and as such, it gives us a framework for comparing market outcomes to optimal outcomes for the stochastic steady state.

Following Peled, we work in a pure exchange environment where the endowments are random. For simplicity, we also assume that preferences are represented by additively time separable C^2 utility function. So, for this simple model, we have exogenous aggregate shocks

$s \in S = \{1, 2, \dots, N\}$ with $N < \infty$. We assume that shocks are generated by randomness of endowments and assume that the stochastic process on the shocks follow a first-order Markov process with the probability of state s' conditional on being in state s given by $\pi^{ss'} > 0$. Taking prices p^s to be prices of money in terms of consumption, we can use the budget constraints to write, for states $(s, s') \in S^2$

$$\begin{aligned} x_1^s &= \omega_1^s - p^s x_1^{m,s} \\ x_2^{ss'} &= \omega_2^{s'} + p^{s'} x_1^{m,s}. \end{aligned}$$

Here, $x_1^{m,s}$ is the young agent's demand for money. Note that we allow second period consumption to depend on the shock from the first period because the money demand will generally depend on the young agent's birth state shock. As was the case with the easy version of the Lucas '72 model, this will turn out not to matter in equilibrium, given a fixed initial injection of fiat money into the economy.

For a young agent born in state s , her optimization problem is

$$\max_{x_1^{m,s}} u(\omega_1^s - p^s x_1^{m,s}) + \sum_{s' \in S} \pi^{ss'} u(\omega_2^{s'} + p^{s'} x_1^{m,s}).$$

The first-order conditions for this problem are

$$-p^s u'(x_1^s) + \sum_{s' \in S} \pi^{ss'} u'(x_2^{ss'}) p^{s'} = 0.$$

Since the marginal utility $u'(x_1^s) > 0$, we can rewrite this as

$$\sum_{s' \in S} \pi^{ss'} \frac{u'(x_2^{ss'})}{u'(x_1^s)} p^{s'} = p^s.$$

Now, let $M = [m^{ss'}]$ be the $N \times N$ matrix with elements $m^{ss'} = \pi^{ss'} \frac{u'(x_2^{ss'})}{u'(x_1^s)}$. In finance, M is called the pricing kernel or stochastic discount factor. Next, let $p = [p^s]$ be a vector of state-contingent money prices. We can then write the first-order condition as

$$Mp = p.$$

Evaluating the money holdings at the fixed money supply m , we then get the statement that at the monetary competitive equilibrium, the vector of state-contingent money prices must be an eigenvector of the matrix M with eigenvalue equal to one. Using the fact that in equilibrium, the second period consumption won't depend on the birth state, we can decompose the matrix M as

$$M = \hat{B}^{-1} \Pi \hat{C}$$

where

$$\hat{B} = \text{diag}_s u'(x_1^s)$$

$$\hat{C} = \text{diag}_{s'} u'(x_2^{s'})$$

$$\Pi = [\pi^{ss'}] = \text{transition matrix.}$$

Since $\hat{B}$ and $\hat{C}$ are both real and diagonal, while the transition matrix is symmetric, the decomposition shows that M is real and symmetric. It then follows that all of the eigenvalues of M will be real, and that the eigenvectors of M will be orthogonal. Finally, by the Perron-Frobenius theorem, M will have one strictly positive root with an associated positive eigenvector. Since the eigenvector p is strictly positive (and orthogonality of the basis of eigenvectors implies that all other eigenvectors must have some negative components) it follows that 1 is the Perron-Frobenius root.

To check optimality of the equilibrium, let

$$M_0 = \left[\pi^{ss'} \frac{u'(\omega_2^{s'})}{u'(\omega_1^s)} \right]$$

be the matrix with the elements as indicated. Since M_0 is also a positive matrix, the Perron-Frobenius theorem applies, so let $\lambda_f(M_0)$ denote the dominant root of M_0 . We will show that if this root is strictly bigger than one, then the endowment allocation is not Pareto optimal. To do this, note that via the Perron-Frobenius theorem, there must exist a strictly positive vector $dx = [dx^s]$ associated with the Perron-Frobenius root, so that under the assumption that $\lambda_f(M_0) > 1$

$$M_0 dx = \lambda_f(M_0) dx \gg dx.$$

From the definition of M_0 , this implies that

$$\sum_{s' \in S} \pi^{ss'} u'(\omega_2^{s'}) dx^{s'} > u'(\omega_1^s) dx^s \quad \text{for all } s \in S.$$

But this immediately implies that for each state s , we can find a positive state-contingent transfer from the young to the old that makes everyone better off. The main result of the Aiyagari-Peled paper in their Theorem 1 is that an interior competitive equilibrium allocation is Pareto optimal if and only if the dominant root of the pricing kernel is less than or equal to one.

In the Aiyagari and Peled paper, they note (*op. cit.* p. 71) that Rodolfo Manuelli also provided a characterization of optimality for two period lived single good SOLG exchange economies under the assumption that there is a single type of agent born each period, for more general stochastic processes of shocks. For details, see Manuelli (1990).

On the macroeconomic side, Able *et al.* (1989) used the Diamond model as the basis for examining dynamic efficiency when the economy is subject to aggregate shocks. As we noted earlier, in the deterministic Diamond model, the steady state equilibrium is efficient if the interest rate is greater than or equal to the economy's growth rate. In stochastic environments, particularly those with multiple assets subject to different degrees of risk, deciding which interest rate to use for comparison becomes difficult. Because the standard Diamond model has two

period lived agents, complete markets, and the assumption of a single good, the stochastic steady state equilibrium that Abel *et al.* analyze is not history dependent, so their analysis is not complicated by needing to determine the endogenous distribution of asset holdings in the economy. The welfare criterion they use is that of *ex ante* expected social welfare based on a weighted sum of utilities for all generations. They justify the *ex ante* focus by noting that their interest is in the question of whether capital is being over-accumulated rather than on issues of intergenerational risk-sharing.

Using this set-up, the paper ends up highlighting the ratio of a social dividend defined as aggregate profit net of new investment, and the value of the market portfolio. The key result of the paper is if this ratio is strictly positive for all time, then the economy is dynamically efficient. Conversely, if the ratio is strictly negative for all time, then the economy is inefficient. Intuitively, a positive ratio means that the economy is generating wealth in excess of that needed to maintain the capital stock, and this wealth is available for consumption, so the economy is not over accumulating capital. Based on this criterion, and accepting that we do not have the full data stream required to apply the criterion to actual economies, the paper uses the reasonably long data series on U.S. GNP (from 1929 to 1985) to estimate the dividend to market value ratios, and shows that these have been consistently positive (on the order of 2.4-4.2) for the U.S. economy over the post-WWII period. Based on this data, Abel *et.al.* conclude that the U.S. economy has not been over accumulating capital.

The welfare criterion Abel *et al.* derive can also be expressed in terms of rates of return to assets relative to the economy's growth rate, where the standard Diamond criterion obtains in the sense that if the rate of return exceeds the growth rate for all time, the economy is efficient, while if the rate of return is below the growth rate for all time, the economy is inefficient. For this

comparison, working with the rate of return on safe assets is, in principle, sufficient for making the efficiency determination. However, Abel *et al.* note that

Neither implication is very helpful in judging the dynamic efficiency of actual economies, where capital gains and losses cause the growth rate of the market value of the capital stock sometimes to exceed and other times to fall short of the safe interest rate. The result here is illuminating primarily in suggesting that comparisons of the safe interest rate with the average growth rate generally are not sufficient to resolve the issue of dynamic efficiency. (*Ibid*, p. 12)

We note that the kinds of capital gains and losses the authors discuss here are precisely the results of portfolio rebalancing that drive the history dependence of equilibrium in more general SOLG models. Since these effects will also change the equilibrium values of assets in the economy, it isn't clear whether the original welfare criterion will hold up in a more realistic SOLG setting.

The next major paper to examine optimality in general SOLG models was Chattopadhyay and Gottardi (1999). This paper uses an extension of the Balasko-Shell characterization of optimality in deterministic models based on the curvature of agents' indifference surfaces to derive necessary and sufficient conditions for *inoptimality*, based on an extension of the Cass criterion. Their main results provide a characterization for potentially non-stationary equilibria in the presence of sequentially complete markets (complete markets in the stationary model examined by Aiyagari and Peled). For stationary environments, they impose the assumption that there is only a single good being traded (to avoid the history dependence problem) and show that their optimality criterion is the same as that obtained by Aiyagari and Peled. Finally, they also provide a characterization of *ex post Pareto optimality* for economies with sequentially incomplete markets. *Ex post* optimality involves looking only at the realized allocations – as

opposed to the *ex ante* allocations of the conditional optimality criterion – obtained in competitive equilibrium. As in the sequentially complete markets case, allocations here are allowed to be non-stationary, and the authors develop a criterion for inoptimality similar to the Cass criterion developed for the general complete markets case.

The Chattopadhyay-Gottardi results were extended by Gabrielle Demange to OLG economies in which agents live arbitrary long but finite lives, in Demange (2002). This paper works with explicitly history-dependent, possibly non-stationary, competitive equilibria, and examines both interim and full *ex ante* optimality, under the assumption that the stochastic process of aggregate shocks is finitely supported. Interim optimality conditions on history up to some point (and is obviously related to the conditional optimality concept used by Aiyagari and Peled), while full optimality is independent of history. Demange defines a sequence of state-contingent support prices for the model, and uses these *valuation processes* to examine the different optimality criteria. Demange's analysis first establishes (in parallel with the Balasko-Shell results) that for economies obtained by truncating at some finite time period t , allocations are Pareto optimal if and only if they are supported by some valuation process. For the full infinite economy, Demange obtains a sufficient condition for interim/full optimality in terms of the summability of the expected prices. This is, of course, closely related to the Cass criterion results obtained by Chattopadhyay and Gottardi.

Like Chattopadhyay and Gottardi, Demange also deals with stationary economies and stochastic steady state equilibria and shows that her results also imply the dominant root characterization results of Peled, and Peled and Aiyagari. But notably, neither Demange nor Chattopadhyay-Gottardi made any attempt to address the optimality question in the context of models with endogenous state variables.

The only paper we are aware of that attempts this is Henriksen and Spear (2012). The Henriksen-Spear paper looks at a three period lived, one good SOLG economy with a Lucas tree asset that pays a stochastic dividend. As we noted earlier, this paper formally demonstrates the generic non-existence of short-memory REE (generic because of the possibility that endowments may already be Pareto optimal). Because of this non-existence result, the paper examines the recursive competitive equilibrium, following Rios-Rull, in which equilibrium prices depend on the exogenous state of nature and on the lagged asset holdings of the young and middle-aged agents in the economy. The equilibrium for the model is then given by a pair of stochastic difference equations

$$a_{t+1} = A(a_t, s_{t+1})$$

$$p_{t+1} = P(a_t, s_{t+1})$$

where a denotes the vector of asset holdings of the young – market clearing conditions can be used to eliminate the asset holdings of the middle-aged – and p is a vector of prices. The assets in the economy consist of zero net supply bonds and equity shares in the Lucas tree, so that the prices are those of the bond and equity, with consumption being the numeraire.

Under the assumption that the aggregate shock can take one of two values, the markets in this model are sequentially complete. The corrigendum to the paper shows that this implies the conditional optimality of competitive equilibrium, since young and middle-aged agents in the model are sharing risk optimally. But, under optimality criteria that take explicit account of the lagged endogenous state variables, there is a significant problem. As long as agents in the model are risk-averse – and this is a standard assumption in this setting – no benevolent central planner would ever introduce additional risk beyond that imposed by nature. So, as long as the central planner is allowed to make allocations dependent on all possible state variables – as opposed to

being constrained to take the lagged state variables as given – she will choose not to make allocations dependent on the endogenous states. This result is intuitive, though Henriksen-Spear provide a formal derivation. The key result, then, is that any Pareto optimal allocation must be strongly stationary, in the sense of depending only on the exogenous states of nature.

We note that there is some confusion introduced by this paper's argument that markets are endogenously incomplete. As we noted earlier, markets are sequentially complete as long as there are as many assets as exogenous states of nature. But markets are incomplete more generally in the sense that there are always more states (both exogenous and endogenous) than assets. Indeed, as long as asset holdings are included as state variables, markets can never be complete in this sense.

The Henriksen-Spear paper also demonstrates explicitly that it is the income effects associated with portfolio rebalancing that generates the history dependent equilibrium. The paper does this by considering a different version of the model in which there is no ownership of the tree asset. Rather, the dividend in each period is arbitrarily shared (in fixed proportions) between the living agents in each period. The asset market is restructured so that agents can buy or sell one period zero net supply bonds, and Arrow securities that are required to be self-financing in the sense that they transfer wealth only across states of nature and not between periods. In static general equilibrium models, this asset structure is equivalent to any other structure as long as markets are complete, so that it serves only to illustrate the possibility of separating insurance activities from intertemporal wealth transfers. In the SOLG environment, however, the separation is important because, as Henriksen-Spear show, the resulting

competitive equilibrium depends only on the exogenous shocks, and the risk sharing between agents is fully optimal²⁰.

These results all extend to a model with production, though as with the tree asset, separating credit and insurance activities would require redistribution of the returns to capital. Since this separation also removes an important source of investment funds (the market for equities), actually implementing this kind of Pareto improvement seems problematic, to say the least. Nevertheless, the inefficient risk sharing in the model is significant. Figure 21 reproduces the simulated consumption time-series from Henriksen-Spear, and the variance associated with the consumption of old agents is almost an order of magnitude higher than that of the middle-aged or young.

²⁰ Kollmann (2022) has proposed incorporating private transfers from older agents to young that implement perfect risk-sharing as a way of making multi-period SOLG models tractable. The paper shows that under this assumption, aggregation across ages is possible and the resulting equilibrium looks like that of a conventional representative agent model. Of course, given the Henriksen-Spear results, the assumption amounts to substituting markets for risk-sharing with the kinds of interfamily transfers examined in Bernheim-Bagwell (1988).

	c_y	c_m	c_r
Low	0.19719	0.30268	0.45013
(SD)	0.00018	0.00099	0.00117
High	0.21388	0.32490	0.51122
(SD)	0.00020	0.00112	0.00131

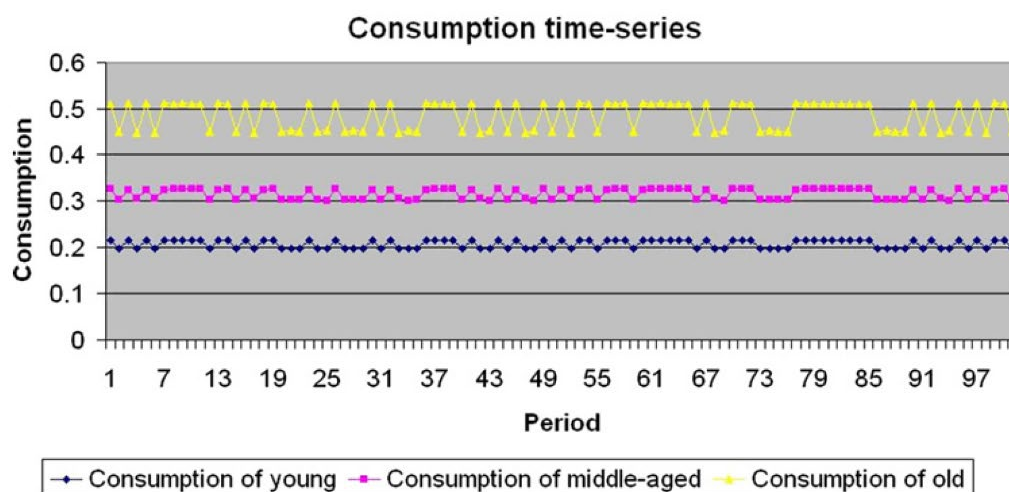


Figure 21

Simulations from Henriksen-Spear (2012)

In this diagram, if agents were sharing risk optimally, then in the case where agents didn't discount future utility, all three time-series lines would coincide, in keeping with the permanent income hypothesis. As we have noted previously, this inefficiency also shows up in Rios-Rull's simulations, where the age-consumption profile is hump-shaped (see Table 5, p. 483 of his paper).

These results also raise questions about the applicability of the Abel *et al.* results. On the one hand, the simulations in multi-period SOLG models do suggest that the history dependence of the equilibrium increases asset return volatility even though output moves only with aggregate shocks. Since, as Abel *et al.* have noted, real world GDP looks significantly smoother than the asset markets, it may well be that the history dependence in these more realistic SOLG models isn't that important for the actual performance of the economy. On the

other hand, the fact that financial crises historically have had severe impacts on the real economy suggests that asset and product markets are inextricably linked. And the fact that the last financial crisis occurred in 2008 suggests that we aren't done with them yet. Finally, the failure of risk-sharing in the history dependent equilibria of these models suggests that there is a potentially important role for government in the provision of social insurance. This issue, in the form of questions about Social Security, has been studied in a sequence of increasingly more realistic SOLG models in the past decade, and we will examine these studies in more detail in the next chapter.

Chapter Eight: Gen X and Beyond: 1995-2015

In this chapter, we focus on the research on OLG models produced in the period from 1995-2015. While much of this work was done by the now old guard Boomer scholars, the younger Gen X scholars coming out of graduate school during this period also made important contributions. If one searches in Google Scholar for the phrase “overlapping generations model” in the time period we are looking at here, it returns almost 20,000 hits. A similar query of the EconLit database for the same phrase and time-period returns 1250 hits. So, as we noted in the introduction, there was plenty of interest among economists in OLG, though it was dwarfed by the work being done using the ILA models of the new neoclassical synthesis. For our purposes, the plethora of OLG research forces us to narrow our focus and considered the work being done on stochastic overlapping generations models, since these offer the same tools for macroeconomic analysis of business cycles and government policy interventions as the dominant ILA models, though in a decidedly non-equivalent form.

We also note that among the Google Scholar hits for the SOLG search during this period were many policy applications based on stochastic two-period one good OLG models, despite the fact that it was well understood by this time that the results coming out of these models could differ significantly from those coming from more realistic models.²¹ We note this not by way of suggesting this work was without value, but rather to note just how difficult working with the more realistic model actually was. Subject to the caveat that the two period model might be

²¹ Fast forwarding to the present, this would include Olivier Blanchard’s 2019 Presidential address to the American Economics Association, published as Blanchard (2019).

unrealistic, economists working on issues where both lifecycle heterogeneity and uncertainty were important made what progress they could in the two period framework.

We will begin by examining the theoretical breakthroughs made during this period on the characterization of stochastic steady state equilibria, the efficiency properties of these equilibria, extensions of the model based on the Blanchard-Yaari perpetual youth model, and the detailed exploration of issues of indeterminacy that occurred during this time frame. We will then turn to look at a specific issue – Social Security – which received a great deal of attention during this period using long-lived SOLG models. We complete the chapter by looking at developments in computation technologies and algorithms for simulating complex SOLG models.

Section 8.1: Theory

The initial subject of our theory focus here is the work mentioned earlier of Alessandro Citanna and Paolo Siconolfi. Citanna and Siconolfi both did their Ph.D.s at Penn under the supervision of Dave Cass, with Siconolfi graduating in 1987 (putting him right on the cusp of Gen X), and Citanna graduating in 1995. By the time Citanna and Siconolfi were in graduate school, Cass had moved on from his OLG work to focus on models of market incompleteness. When we queried both Citanna and Siconolfi as to where they learned the OLG model, they both commented that the courses Cass taught were focused on his incomplete markets research, with their exposure to the OLG model coming from courses taught by Azariadis or Shell. Both Citanna and Siconolfi did their thesis work on incomplete markets, picking up OLG later in their careers.

Citanna and Siconolfi's first major SOLG paper was Citanna-Siconolfi (2007). This paper is an extension of Spear (1985) allowing for multiple types of agents in each generation, and, in principle, for longer than two period lifetimes. The paper defines short memory equilibria as those in which endogenous variables depend only on finitely many lags of the exogenous random shocks²². The proof that such equilibria do not exist generically is based on the same density of transversal intersections as in Spear's invocation of the multi-jet transversality theorems, but Citanna-Siconolfi use a clever trick to avoid assumptions on preferences needed to apply the multi-jet results. Specifically, the multi-jet results require that at each solution point to the system of equations under consideration, it must be possible to independently perturb the underlying parameters of the system. For the SOLG application, the parameters would be the endowments and utility functions. Spear (1985) assumed that all goods were normal and showed that this would guarantee applicability of the multi-jet theorem. Citanna-Siconolfi show how to delete equations and variables corresponding to instances where two equations are satisfied at the same point in the domain, and the fact that there will be at most finitely many of these cases, to then argue via the transversal density results that the set of economies for which short-memory equilibria exist must be nowhere dense.

Citanna-Siconolfi (2008) uses the perturbation techniques developed in their 2007 paper to show the generic existence of recursive equilibria (in the Rios-Rull sense) in the same class of models for which the various non-existence of equilibrium results are developed. The papers with these non-existence results are (in addition to Krebs [2004]) Kubler and Polemarchakis (2004) and Kubler and Schmedders (2005).

²² Spear (1985) calls these equilibria strongly stationary and shows in the appendix of the paper that the arguments used to establish non-existence for the i.i.d. shock case can be extended to apply to finitely many lags of the shocks.

Citanna-Siconolfi (2010) was the culmination of the techniques developed in the two previous papers, and its publication in *Econometrica* was well-deserved. The paper provides an alternate proof of the existence of what Citanna and Siconolfi call simple Markov equilibria, which are equivalent to what Duffie *et al.* called stationary Markov equilibria. The existence proof starts with the history dependent equilibrium that can be shown to exist using the same truncation method used by Balasko and Shell. From here, Citanna-Siconolfi show how to recursively build up equilibria in which histories are replaced by one-period lagged collections of exogenous shocks, individual wealth holdings, prices and individual Lagrange multipliers as the state variables. The sufficiency of these variables for describing the competitive equilibrium follows from the first-order conditions for individual agents and the market-clearing conditions. Using a process they call “grafting,” Citanna-Siconolfi show how to remove finite time sequences in which any initial Markov state returns to a previous state, and generate a consistent Markovian sequence of competitive equilibria.

Once the existence of the simple Markovian equilibria is established, Citanna-Siconolfi use the perturbation methods developed in their previous two papers to show that as long as there are sufficiently many households, the simple Markovian equilibrium can be obtained as a recursive equilibrium in which the state variables are the exogenous shocks and the wealth holdings of the agents. The need for heterogeneity across agents is to ensure that the relevant utility perturbations introduced in the previous papers can be applied.

In addition to the accomplishment of showing that for practical purposes one can work with the recursive equilibrium concept – and justifying Rios-Rull’s intuition along the way – this result also provides a resolution to the question originally raised by the Lucas (1972) paper’s assumption that the exogenous shocks alone were the right set of state variables. Citanna-

Siconolfi (2010) shows that the exogenous shocks and wealth distribution are the right variables if the equilibrium is determinate. If the equilibria of the model are indeterminate, then one can construct stochastic processes of sunspot equilibria associated with the stable manifold via the bootstrap process examined previously. The applicability of this construction is implied by the results in Woodford (1986b) but results on how to actually construct such equilibria in a Markovian form were first presented during this period in Bloise (2001) and Bloise (2004). These results leave open the possibility for psychological phenomena – Keynes’ animal spirits, contemporary notions of sentiments (Spear [1989], Angeletos and La’O [2013], Benhabib, Wang and Wen [2015]), independent belief formation (Farmer [2010]), or behavioral processes like mental accounting (Goenka [2003]) or hyperbolic discounting (Kim and Li [2020]) – to have real economic effects. We discuss these developments below.

We also note that this theoretical work is in contrast to that of Krusell and Smith (1998) who looked at the effects of changes in the wealth distribution on the equilibrium in an Aiyagari-Bewley incomplete markets representative agent model. The Krusell-Smith paper took the approach of including various moments of the wealth distribution as state variables and found that for the version of the Aiyagari-Bewley model they examined, the mean of the distribution alone was sufficient to summarize the history, at least to a very close approximation. The theoretical foundation for this approach is from Duffie *et al.*, together with Rios-Rull’s assumption that the wealth distribution was the correct endogenous state variable. In principle if one knows all the moments (or better, the characteristic function of the wealth distribution, from which all the moments can be obtained), then one knows the distribution itself and the recursive equilibrium can be computed. What is less clear is the robustness of their findings that only low order moments are important. Storesletten, Telmer and Yaron (2007) note in their computational

appendix (*op. cit.* p. 545) that their attempts to use finite collections of moments from the wealth distribution in their simulations did not give good results. Because of this, they instead adopted asset prices as the lagged endogenous state variable, which did work. It would seem from this observation that having a fully sufficient statistic for the shock history is important in general SOLG models in a way that isn't important in ILA models, even with market incompleteness. That said, however, Heer and Maussner (2009) claim to get good results using the Krusell-Smith algorithm for long lived SOLG models, which is based on simulating the wealth distribution at each step of the algorithm. As they note, though, the procedure is quite time consuming. We will examine computation issues in more detail at the end of this chapter.

A second area of theoretical results from this period looked at issues of efficiency of equilibrium, extending the earlier work of Chattopadhyay-Gottardi and of Demange. The key paper here is Bloise and Calciano (2008) which provides a general characterization of inefficiency in SOLG models.

The Bloise-Calciano approach uses the notion of *robust inefficiency* based on ideas introduced by Debreu (1951). Specifically, an allocation is robustly inefficient if it is Pareto dominated by some other allocation even after some resources in the dominating allocation are destroyed. Working in the same context of history-dependent equilibrium as Chattopadhyay-Gottardi and Demange and under the same assumption of sequential market completeness, Bloise-Calciano show that inefficient allocations can be characterized in terms of their supporting present-value state contingent prices by defining an operator on the space of potential transfers by

$$T(e)_t = \frac{1}{p_t} E_t p_{t+1} e_{t+1}.$$

Their condition for robust inefficiency is that there exists a contingent transfer sequence e such that $\rho T(e)_t \geq e_t$ for $0 < \rho < 1$. By iterating on this recursive relationship, one obtains Bloise-Calciano's Modified Cass Criterion on the convergence of the infinite sum of inverses of the prices. Bloise and Calciano also show that this criterion is independent of the number of periods agents in the model are assumed to live, by converting the model into one in which agents formally live only two periods, based on the original reduction to two periods given by Balasko, Cass and Shell (1980). Given this transformation, they show that if the allocation in the original economy was inefficient, it will also be inefficient in the transformed economy. One should note, however, that the transformation to two period lives does not produce an economy equivalent to the original one, since the young agents in the transformed economy are born into given states of the world that were as-yet unrealized in the original model. This means that where they were able to make contingent trades in the original model, given the assumption of sequential market completeness, these would not be available in the transformed model, so that sequential market completeness is lost. This doesn't matter for the efficiency characterization since it is the efficiency or inefficiency of the original allocation that is being examined, but it does show that once one recognizes that any improving sequence of transfers must continue forever, pairwise comparison of one period's prices with the expected values of future prices suffices to characterize efficiency. Finally, if one examines the proofs in the appendix of the paper, it is also apparent (as the authors note in their Remark 4) that the assumptions of a single good or a single agent per generation are easily extended to more general economies. One of the most interesting aspects of this characterization of robust inefficiency is the fact that it can be made in terms of the eigenvalues of the operator T . Specifically, T must have an eigenvalue

bigger than one corresponding to a sequence of Pareto improving transfers between agents. This is a direct extension of the Peled and Aiyagari-Peled results for the simpler two period one good model.

Much of the additional theory work during this period was based on Shell's Philadelphia Pholk theorem. Some of this work is relevant to the OLG story, and some of it is tangential. We include discussion of this work here for completeness.

As we noted earlier, Peck and Shell did their pioneering work on imperfect competition and existence of sunspot equilibria, and the relationship between sunspots and correlated equilibria, during this period. Cass, Geanakoplos and Mas-Colell, and Wilson started a research program looking at incomplete markets in the late 1980's, which spawned a literature over the next two decades aimed at fully characterizing the equilibrium and welfare properties of models with incomplete markets. While Cass's original incomplete markets paper was motivated by his intention to show the existence of sunspot equilibria when markets are incomplete, most of this work is tangential to our subject, so we refer the reader to the surveys in Magill and Shafer (1991) and Staum (2007) for details.

A second strand in the Pholk theorem thread was work done in the ILA setting with externalities. Externalities play an important role in the growth model when one introduces taxes, which are required for most public policy applications, and human capital accumulation, which operates through spillover effects and shared research impacts to generate external increases in the productivity of other inputs to production. The first papers in this program were Spear (1991) and Kehoe-Levine-Romer (1992) mentioned earlier. These papers established

indeterminacy results by showing that in the presence of externalities, the saddle path dynamics of the conventional RCK model would bifurcate to make the stable manifold around the steady state equilibrium higher in dimension than the number of predetermined variables. These results, in turn, imply the existence of sunspot equilibria via Woodford's results. Benhabib and Farmer (1994) examine related models in which production exhibits increasing returns to scale in economic environments which allow for competitive equilibrium. One of their models treats the factor that generates increasing returns (average human capital investment, for example) as external to the firms' profit maximization decisions, and hence they can apply the Spear or Kehoe-Levine-Romer results directly. The second model looks at monopolistic competition using the well-known Dixit-Stiglitz model, which generates external increasing returns through the provision of intermediate goods. Benhabib-Farmer then show directly for this model that the steady state equilibrium of the model is indeterminate.

Farmer and Guo (1994) use these results to construct two calibrated RBC-type models, one with monopolistic competition, and the second with external increasing returns to scale, in which fluctuations are generated by sunspots, or, in Keynes' terminology, animal spirits. They examine simulations of their models together with those of a standard RBC model (taken from Hansen [1985]). The simulations generated by the external IRS model with sunspot shocks generates time-series that match U.S. post-war data as well or better than the standard RBC model, and the sunspot model does a better job of matching impulse-response data than does the RBC model. These results present serious reasons for macroeconomists to consider models where demand-side shocks based on consumer sentiment, together with indeterminacies generated by OLG-type frictions, drive business cycle fluctuations, rather than the conventional modeling assumption that aggregate shocks are due primarily to random changes in productivity.

Coupled with the empirical evidence on the scarcity of aggregate productivity shocks, this paper should have received more attention than it has. We make this statement based on the 754 Google Scholar cites to Farmer-Guo versus 3319 cites to the Hansen (1985) paper.

The notion that sunspots – now redubbed animal spirits or consumer sentiment – might drive economic fluctuations experienced something of a revival during this period, independently of the work by Farmer and others on indeterminacy. The first of the new papers was Angeletos and La’O (2013) and Benhabib, Wang and Wen (2015). These papers generalize the earlier sunspot equilibrium results by allowing for heterogeneity in the sunspot variable, heterogeneity in the product markets, and for internalization of sunspot signals. Angeletos and La’O use a model of different islands that are paired bilaterally for trade in each period, while Benhabib-Wang-Wen use the Dixit-Stiglitz model of monopolistic competition, to generate the friction that lets sentiments matter. We note that the idea of internalizing the sunspot signal was not new to these papers. Spear (1989) used a model of two islands operating in separate “time zones” to show how each island’s equilibrium price could serve as the other island’s sunspot variable. Manuelli and Peck (1992) showed that the possible existence of sunspot equilibria in a model could serve to “amplify” the variation in equilibrium prices associated with small fluctuations in random fundamentals. Their model allowed for very small variance fluctuations in endowments in a model known to have a 2-cyclic sunspot equilibrium (as in Azariadis [1981]). They showed that in addition to a stochastic steady state with very small variance, there would be a second equilibrium with larger variance close to the sunspot equilibrium, so that the fundamentals of the model served, in effect, as the sunspot signals²³.

²³ The fact that it could be difficult to draw sharp distinctions between sunspot and non-sunspot equilibria was also made in Spear, Srivastava and Woodford (1990), though the paper doesn’t elaborate on this.

The papers on sentiments went significantly beyond the earlier general equilibrium demonstrations that fluctuations in beliefs could affect real economic activity, even in a rational expectations equilibrium, by embedding the sunspot mechanism in conventional and accepted macroeconomic frameworks. This represented a significant step away from the basic idea that shocks to production were responsible for business cycle fluctuations toward the more empirically plausible contention that demand fluctuations and indeterminacy of equilibrium were more likely to be the source of aggregate fluctuations.

In work related to sentiments but based on ideas coming out of behavioral economics, Goenka (2017) shows that a simple model of mental accounting (see Thaler [1999]) can give rise to sunspot equilibria. In Goenka's model, some consumers are boundedly rational and assign fixed expenditure shares to different subsets of commodities. These fixed shares associated with the mental accounting assumption manifest themselves in general equilibrium as if the boundedly rational consumers were credit constrained, and this friction then leads to the existence of the sunspot equilibrium.

These notions about demand-side variability were key drivers in much of the subsequent work on indeterminacy. Over the period 1991-2020, Roger Farmer wrote a number of articles denying the validity of two major elements in the New Classical Macroeconomic analytical toolkit: the Lucas Critique, and the Natural Rate Hypothesis (NRH). In Farmer (1991) he attacked the arguments and restrictions underlying the Lucas Critique argument as not only insufficient, but also presented a counterexample to the argument, and an alternative model selection criterion. Moreover, he identified general versions of OLG as being able to provide what he called a "Lucas-proof forecast rule" (Farmer [1991], p. 331). A decade later, in Farmer (2002), Farmer focused on the Favero-Hendry (1992) test of the Lucas Critique, and found that

the Lucas Critique, as he put it, “really does fail in practice” (Farmer [2002], p.112). A decade after this, in Farmer (2013) he presented an alternative to the NRH using a DSGE approach based on “beliefs” as a psychological parallel to “preferences” (Farmer [2013], p.244). Most recently, Farmer (2020) reviews the most recent advances in what he calls “the indeterminacy agenda” following on from Farmer (1993) and Benhabib and Farmer (1999).

Another related literature in this strand of ideas stems from William A. Barnett’s work on bifurcations in economic models. This work is summarized in Barnett and Chen (2015), although the best characterization of the importance of this work is found (in our opinion) in Barnett and Eriylmaz (2022) in their introduction:

Determinacy concerns the existence of a unique equilibrium path of a dynamic system. Equilibrium of a macroeconomic model is called determinate, as defined by McCallum (2009b), if it is locally unique and dynamically stable under relevant specifications of policy tools. Studying the determinacy issues in a wide range of models has been an important subject among macroeconomists. One of the reasons for high focus on the subject is the fact that uniqueness of the solution path plays an important role for policy makers and researchers. Determinacy is important to the monetary authority in determining policy to manage inflationary expectations and preventing self-fulfilling economic fluctuations. Indeterminacy permits existence of multiple solutions, which could be fundamental equilibria or nonfundamental sunspot equilibria, as suggested by McCallum (2003) and Bullard (2006), among other authors.

McCallum (2003) further divides the indeterminacy cases into two categories: failure of the model to determine the values of nominal variables (nominal indeterminacy) or of real variables (real indeterminacy). But Woodford (2003b) finds the distinction to be insignificant. He argues that both types of indeterminacy are quantitatively indifferent. Models with indeterminate solutions, as stressed by Bullard and Mitra (2002), are considered undesirable for macroeconomic analysis and policy design. In the presence of indeterminacy, as pointed out by McCallum (2003), Gauthier and Guesnerie (2005), Beyer and Farmer (2007) and others, non-fundamental shocks trigger extra variance and fluctuations in the economy. Then decision makers, unable to acquire full information, encounter unforeseen problems in monetary policy design. (*Ibid*, p. 2)

Much of Barnett’s work on indeterminacy has focused on the possibility that models used in policy analysis might exhibit bifurcations. Bifurcations are important because their possibility generally defines boundaries in the spaces of parameters that determine a model’s dynamics

which, when crossed, cause discontinuous changes in the dynamics around the model's steady state equilibrium. To the extent that calibrated simulations of a model are used to study the impact of a particular policy, the uncertainties associated with the estimation of parameters need to be evaluated for whether the confidence regions contain bifurcation boundaries. Barnett and Chen (2015) has examples of policy relevant models in which estimated parameter values are in fact close to bifurcation boundaries.

We close this discussion by noting that even in the absence of indeterminacy, general SOLG models generate the same kinds of “extra fluctuations” in the economy as Barnett noted above, due to the dependence of equilibrium on the distribution of wealth. The implications of such dependence were expressed nicely in Naryana Kocherlakota's Minneapolis Fed article (Kocherlakota [2010]):

[M]any modern macro models incorporate financial market frictions. However, these models generally allow households and firms to trade one or two financial assets in a single market. They do not capture an intermediate messy reality in which market participants can trade multiple assets in a wide array of somewhat segmented markets. As a consequence, the models do not reveal much about the benefits of the massive amount of daily or quarterly reallocations of wealth within financial markets. The models also say nothing about the relevant costs and benefits of resulting fluctuations in financial structure (across bank loans, corporate debt, and equity).

The extra fluctuations in general, multi-period SOLG models arise precisely from the portfolio rebalancing that occurs as agents realize trades and shocks and adjust their consumption and saving accordingly. Working with such models is, of course, difficult, as Victor Rios-Rull's work showed, but again quoting Kocherlakota from his Fed article:

Over the past 25 years, a great deal of work has used models that incorporate financial market frictions. Most of these models cannot be solved reliably using graphical techniques or pencil and paper. As a consequence, progress is closely tied to advances in computational speed.

Again, we see the importance of computation and will address this in more detail below.

There is an important postscript to this section, however. As we noted above, Roger Farmer's questioning of the empirical validity of the rational expectations hypothesis and the natural rate of unemployment hypothesis led him to a reconsideration of how markets (and labor markets in particular) come into equilibrium. It is well-known in economics that the standard assumption that prices move in opposition to excess demand is not generally valid (see Scarf [1960]). Since this mechanism for bringing markets into equilibrium can fail, and since wage rigidity is a well-known empirical fact in labor markets, Farmer opted to replace the usual wage equilibration mechanism with a search theoretic one where firms set supply to match demand in the product market, and then search for new employees if these are needed to meet this production target. Farmer (2010) lays out this model, and shows that in the absence of a price mechanism for bringing labor markets into equilibrium, firms' beliefs about future demand are what determines equilibrium wages and prices. Farmer then introduces a "belief function," considered as a fundamental psychological object in the same way that economists consider preferences as psychologically given, to close the model. This new formulation of equilibrium introduces a new kind of long-run indeterminacy in which it is possible for unemployment to persist for long periods of time. These ideas were the impetus for the recent declaration in Vine and Wills (2020) on the project to rebuild macroeconomics, in support of the indeterminacy agenda for macroeconomics. While the jury is obviously still out on whether this approach will be more successful than RBC or NK macro, it is interesting that the treatment of beliefs as a fundamental vindicates Grandmont's views on rational expectations versus temporary equilibrium expectations that we discussed above. It is also completely consistent with the large literature on learning rational expectations in economics and suggests that work on learning in search

moderated equilibrium frameworks will be an important agenda item for macroeconomists going forward.

Beyond the direct application of indeterminacy results in the analysis of business cycle phenomena, there were also direct applications of the sunspots idea in the analysis of banking and bank runs. The early modern work on banking theory is reflected in Diamond and Dybvig (1983)²⁴, which developed a simple model of the intermediation activities of banks in the uncertain circumstances where depositors can withdraw their funds prior to the maturation of projects the bank has invested in. The uncertainty associated with liquidity needs generates a coordination problem for the bank. If all depositors expect that most other depositors will leave their money in the bank until the investments mature and payoff, then the bank can in fact invest most of the funds deposited and generate a return for patient customers and a profit for the bank's owners. But, if depositors come to believe that their money isn't safe and choose to withdraw it independently of their liquidity needs, these expectations can cause a run on the bank. Absent some form of insurance against the run, the bank's failure becomes a self-fulfilling prophecy. Diamond and Dybvig themselves note that some form of sunspot dynamics could be used to provide a dynamic model of bank runs.

This possibility was shown to in fact hold in Peck and Shell (2003), where they construct a fully dynamic version of the Diamond-Dybvig model with a sunspot variable that coordinates agents' collective actions in the post-deposit game. Via this coordination mechanism, bank runs occur with positive probability in the model.

²⁴ This work was the basis for Diamond and Dybvig receiving the 2022 Nobel Prize in Economics.

A second notable follow-on contribution to this banking literature was Benhabib, Miao, and Wang (2016), who embed a simple model of banking akin to the Diamond-Dybvig model in a representative agent setting. As in the static Diamond-Dybvig or Peck-Shell models, self-fulfilling prophecies that manifest themselves as bank runs can arise, but because of the dynamic setting of the Benhabib-Miao-Wang model, the information asymmetries that drive self-fulfilling bank runs in Diamond-Dybvig generate a much richer set of possibilities. Because bankers in the dynamic model care about their reputation, and because depositors understand the moral hazard problem that arises because of the banker's superior information about investment opportunities, the Benhabib-Miao-Wang model has additional constraints (specifically the incentive compatibility constraints) relative to the Diamond-Dybvig or Peck-Shell models, and this generates non-linearities in the model that make complex dynamic outcomes possible. Under the assumptions of limited liability and enforcement of banking contracts, the paper shows the existence of multiple equilibria, including deterministic chaos, which can lead to coordination failures in the model, and hence bank runs, defaults and credit crises. The results suggest that banking regulations which include limits on leverage, central bank credit policies, and restrictions on bank size can be effective in stabilizing the banking equilibrium and preventing self-fulfilling crises.

Section 8.2 Social Security

The period between 1995 and 2015 also saw a resurgence of interest in Social Security. As we saw earlier, this was a topic of debate between Samuelson, Lerner and Meckling based on Samuelson's deterministic two- and three-period OLG models. By the late-1970's and early-1980's, the large size of the Baby Boom generation, coupled with the relatively smaller size of

Gen X began to generate policy concerns over where the retirement funds paid to Boomers would come from. This concern led to the creation of the Greenspan Commission to study the problem, and to the eventual passage of the 1983 Social Security Reform Act which raised the Social Security and Medicare taxes on working members of the Boomer generation, with the provision that the Social Security Administration invest the resulting surplus in U.S. government bonds.

Three notable early papers which attempted to study the effects of this tax change and other policy perturbations on Social Security were Huang *et al.* (1997), De Nardi *et al.* (1999) and Di Nardi *et al.* (2001). All three of these papers studied large-scaled OLG models with some form of idiosyncratic risk and used numerical simulations to examine the welfare effects of various policy proposals, ranging from pure privatization, to privatization with buyouts of existing claimants to retirement benefits, to continuation of the PAYGO system while allowing surpluses to be invested in non-governmental assets. The models used were extensions of the Auerbach-Kotlikoff modeling technique, with the addition of idiosyncratic risks to mortality or income. As we noted earlier in our discussion of the aftermath of Rios-Rull's 1996 paper, the incorporation of idiosyncratic shocks became popular during this period, particularly given that, thanks to the law of large numbers, they could be handled using the same computational tools available for deterministic models. In this setting without aggregate retirement risk, the main findings of these three papers were that the various reforms increased the welfare of future generations by increasing the size of the capital stock, but did so at the expense of current generations.

Much of the resurgence of interest in Social Security after 2000 was generated by policy proposals put forth by U.S. President George W. Bush's campaign in 2004 to allow working age

people to direct some of their Social Security payments to private investment accounts. During this debate, the Republicans stressed the fact that Social Security funds invested in U.S. Treasury bonds earned very little interest compared to what was available on the stock market. The Democratic side (under the direction of Senator John Kerry's presidential campaign) stressed the fact that the stock market exposed elderly investors to substantial risk, pointing back to the way many retirees in the 1930's lost their life savings in the stock market crash of 1929.

Many of the papers on the subject of Social Security written during this period were based on two-period lived SOLG models. Since the non-equivalence of the two-period and multi-period SOLG model was well-known by this time, the two-period assumption in this work was clearly made for tractability reasons and offered as an approximation of what more realistic models would produce.

One particular paper which avoided the two-period assumption was Conesa and Krueger (1999). This paper avoided the problems of history dependence in SOLG equilibria by working with a model of idiosyncratic shocks similar to the Storesletten-Telmer-Yaron papers. The question addressed in the paper was whether political economy considerations would impose an impediment to moving an economy running an unfunded pay-as-you-go (PAYGO) social security system to a fully funded system in which individual contributions would earn higher returns than those generated by the unfunded system. The voting friction introduced in the model has the effect of creating a bias in favor of the status quo since agents already invested in the unfunded system would effectively lose their contributions during the transition to the fully funded system. Conesa and Krueger's quantitative analysis further shows that the bias in favor of the unfunded status quo is stronger when the idiosyncratic shocks to income lead to significant wealth heterogeneity across agents of the same age.

A second response to the inadequacies of the two-period approach was the adoption of full-blown computational analyses of multi-period SOLG models to analyze the effects of social insurance, following in the footsteps of Riso-Rull's pioneering computational analysis. An early important contribution to this literature was Krueger and Kubler (2002). In this paper, Krueger and Kubler work with a three-period lived SOLG model and show that PAYGO social security programs can be viable only when asset markets are incomplete, as long as there is an asset that pays a positive dividend in every state of nature (such as land or a Lucas tree, for example). The reason for this is straightforward. With sequentially complete markets and a positive asset, returns to saving are always positive, so that private saving always dominates the pure transfers associated with unfunded social security systems. Krueger and Kubler work in a pure exchange environment subject to aggregate shocks and show that when the asset markets are incomplete, the introduction of a social security transfer system generates a Pareto improvement.

Krueger and Kubler (2006) follows up on their 2002 paper by adding production and capital accumulation to the model and increasing the number of generations from three to nine. We note that during the 4 years between the publication of these papers on social security, Krueger and Kubler also wrote an important paper on how to use projection methods to compute SOLG models, which made dealing with the computational complexity of the 9-period model workable. We discuss this paper in the section below on computation. In the 2006 paper, markets are again incomplete, for the same reasons as before, but now, the introduction of the social security program doesn't just reallocate savings as it did in the pure exchange model. It also has the effect of crowding out capital since the saving diverted to social security is no longer being used to fund firms' capital accumulation or replacement activities. Krueger and Kubler show that when only the risk-sharing aspects of the social security transfer are considered, the

program generates a Pareto improvement. But when the crowding out effect of capital is factored back in, the overall effect of the program is to reduce welfare. In their quantitative model, the effect of introducing social security is a 6% reduction in the capital stock, leading ultimately to a 2% reduction in average output.

While the Krueger-Kubler paper made significant strides in using the general SOLG framework, its need to limit the number of periods considered to 9 for computational tractability reasons left open the question of whether their results would change if agents lived longer lifespans. The first paper to tackle this question was Hasanhodzic and Kotlikoff (2013). Agents in this model live 80 periods, working for 45 of these. Production in the model is subject to standard AR(1) shocks to total factor productivity, as well as a shocks to depreciation. Agents in the model are borrowing constrained, in order to match the age-consumption profile generated by the model with the empirical profile.

Unlike Krueger-Kubler, Hasanhodzic-Kotlikoff focuses on assessing the degree of intergenerational risk generated by the model and assessing the extent to which markets or social insurance policies can mitigate this risk. Hence, while they find similar crowding out effects generated by the introduction of social security as in Krueger-Kubler, they control for these in examining the effects of social security on intergenerational risk. In terms of their results, they find for an initial calibration in which the shocks are set to match the variability of aggregate U.S. wealth, that the model generates little to no intergenerational risk. When they modify the calibration by increasing the size of the shocks to depreciation in order to match the variability of the returns to equity (following Krueger-Kubler on this), they find that the model generates significant intergenerational risk. This suggests that the conventional wisdom of looking at the risk the stock market imposes on retirees does have some validity. Finally, under both

calibrations, Hasanhodzic and Kotlikoff find that conventional asset markets or unfunded social security systems can effectively mitigate the intergenerational risk.

We note that earlier considerations of social security costs and benefits led Auerbach and Kotlikoff (with some additional coauthors) to propose the concept of “generational accounting” as an alternative way of measuring the burden of government debt on future generations.

Conventional measures of debt accumulation or decumulation look at a contemporaneous flow of funds away from or to consumers, but don’t take in account how the funds will be used – for the provision of public goods, public capital goods (infrastructure), or simply as transfers – with attendant costs and benefits that both accrue over time and confer differential benefits across agents of different ages. Generational accounting is a mechanism that attempts to breakdown the governments receipts and expenditures precisely in terms of cross-generation costs and benefits in a way that makes transparent the relationship between what a given cohort will pay in taxes over their lifetimes relative to what they will receive in benefits. This kind of accounting makes it possible to identify gainers and losers from given policy proposals, allowing a more straightforward application of the benefit principle to the intergenerational transfers such policies typically require.

While this accounting arrangement has been criticized on various grounds – see Cutler (1993) for a review of Kotlikoff (1993) – the concept does provide a way of thinking about the national income and product accounts in a dynamic context, organized around the graded age-set structure of the OLG model. Kotlikoff himself acknowledges the need (in Fehr and Kotlikoff [1995]) to consider dynamic accounting systems like the generational one in the context of fully dynamic general equilibrium models, and to a large degree, this idea comes to fruition in the Hasanhodzic-Kotlikoff paper.

The next paper in the series of work on social security systems in this period is Harenberg and Ludwig (2015). The quantitative model Harenberg-Ludwig use has agents active for 58 periods (implying a life expectancy of 78 years assuming agents enter the market at age 21), and agents face both aggregate and idiosyncratic shocks. The assumption that the aggregate TFP shocks follow a four state Markov process implies that the markets are not sequentially complete, so that the Krueger-Kubler necessary condition is satisfied. In its overall structure, then the Harenberg-Ludwig model is quite similar to that of Storesletten, Telmer and Yaron (2007), but the focus here is on the question of whether social security generates a welfare improvement. The answer to this question is yes. Like Krueger and Kubler, the introduction of a PAYGO social security system in the Harenberg-Ludwig models generates a significant degree of crowding out, on the order of a 12% reduction in the capital stock. Unlike the Krueger-Kubler results, however, Harenberg-Ludwig find that the combined effect of aggregate and idiosyncratic shocks is super additive, with the overall risk effects being significantly larger than the individual risk effects combined. In this setting, then, the social security system generates a significant improvement in intergenerational risk-sharing which more than offsets the losses due to crowding out.

The final paper we consider in this strand of literature on social security falls a little outside the time window we have set ourselves here, but we included it because of its close relationship to the other papers we've been considering. This paper is chapter one from Eungsik Kim's thesis (Kim [2019]), which considers an SOLG model with long-lived agents, calibrated along the same lines as in Hasanahodzic-Kotlikoff or Harenberg-Ludwig, but with the additional assumption of heterogeneity among agents within each age group. The heterogeneity is one of income differences, which Kim characterizes as arising from differences in education. He posits

three income classes, denoted as college educated, high-school educated, and drop-outs. The source of the income heterogeneity in Kim's model is increased productivity associated with greater education. Kim's calibration induces the required sequential market incompleteness per Krueger-Kubler, and he finds in his quantitative model that the introduction of a social security system has the same order of crowding out found by Krueger-Kubler or Haranberg-Ludwig (a 15% reduction in the capital stock and an associated 5% reduction in output). Unlike Haranberg-Ludwig, though, the introduction of social security affects different income classes differently.

First, because of the general equilibrium effects associated with crowding out, the introduction of social security in Kim's model raises the prices of the assets owned predominantly by the college educated types, making them better off even though they are paying taxes that support the PAYGO transfer system. The drop-outs in Kim's model also benefit because they end up receiving more in transfers than they lose in taxes. But the middle group of high-school graduates in Kim's model do not benefit, since they are rich enough to pay significant taxes toward the transfer system, but don't hold significant asset positions and hence miss out on the price improvements caused by crowding out. Kim's model demonstrates that in the face of heterogeneity across agents in the same cohort, some kind of political economy consideration is needed to determine whether or not a given society will wish to implement an unfunded social security system or not. In this sense, the paper is reminiscent of Conesa and Krueger (1999).

To complete our discussion of the social insurance applications of the SOLG model in recent years, we note that because most social security systems operate to transfer purchasing power from young agents to retirees, this mechanism misses the second inefficiency identified by Henriksen and Spear whereby young agents under consume in order to self-insure against asset

market risk when retired. This is also apparent in the empirical studies of precautionary over saving. As we noted earlier, there are a number of substantial U.S. government tax and spending provisions aimed at making life easier for the young, including the home mortgage interest deduction, deductibility of childcare and education expenses, public provision of K-12 education, public higher education institutions, and so on. These expenditures are on the same order of magnitude as those for social security, but there have been few attempts to study this aspect of intergenerational transfer in detail. Gale and Scholz (1994) provide a good overview on the importance of the issue. For a more recent study, see Yoo (2013), who shows using an applied three-period SOLG model with idiosyncratic income shocks that a reduction in the gift tax for *inter vivos* transfers generates a welfare improvement.

One strand of literature which deals with the nature of transfers between generations in an historical context, and which also arose during the period we are considering, was work on the so-called demographic transition. This is an empirical phenomenon associated with the onset of industrialization in which both mortality and fertility rates in the population fall in response to the increased standard of living afforded by industrial development. We examine this work in more detail in the next section.

Section 8.3 Demography and Wealth

Lifecycle models have long been used to examine the economic effects of demographic changes. One of the more profound demographic shifts that has ever occurred in human society is the observed demographic transition that occurs after the on-set of industrial development. This transition was observed historically in developed Western nations following the industrial

revolution which began in the late 1800's. The economic transition associated with industrialization is one of transforming the economy from an agrarian foundation to one based on manufacturing. The demographic transition associated with this economic shift is one in which families which were previously large and oriented toward farming become smaller and oriented toward factory work involving interactions with advanced technologies. Because these interactions require numeracy and literacy skills beyond those required of farmers, the transition sees parents investing in human capital across smaller cohorts of children, rather than in the large numbers of children that are useful in agrarian operations. The benefits of technology and scientific understanding of things like disease also lead to reductions in mortality, particularly among children, which also reduces the *ex ante* need for large families based on the expectation that some number of children would not survive.

While the literature on this topic is large, we will focus on the two most highly cited papers on this topic, Galor and Weil (2000) and Hansen and Prescott (2002). The Hansen-Prescott approach was based almost entirely on the changes in production that occurred over the course of the early industrial revolution, and, as they note

Although we found it convenient to study an overlapping-generations model in this paper, our results should carry over to an infinite-horizon context like that used in much of the growth literature (footnote 11, p. 1209)

The essential changes that took place during the industrial revolution were the mechanization of previously labor-intensive manufacturing activities, enhanced by the use of, first, wood-fired steam engines and later coal-fired engines. This process made labor more productive in the factories than on the farms and the consequent increases in wages drew workers away from the land to the factories. Hansen-Prescott model this process by postulating two distinct technologies, one based on agricultural production, and the other based on mechanized

manufacturing. The agricultural technology uses fixed land together with capital and labor to produce output. Hansen-Prescott refer to this as the “Malthusian” technology. The other technology, which Hansen-Prescott call “Solovian” technology, uses only capital and labor. The modern technology’s dispensation of the need for restrictive fixed land inputs captured the primary benefit of science in production by allowing for large amounts of output to be produced on small, compact parcels of land (the factory footprint) by multiplying labor productivity with steam-powered machinery. In addition to this specification of technologies, Hansen-Prescott posit an exogenous Malthusian mechanism whereby the population growth rate depends on the overall per capita output of the economy. They also assume that scientific contributions to the economy are exogenous and generate growth in total factor productivity at a constant rate (though they note in their calibration exercises that this need not be the case). The basis for the TFP growth assumption is Solow’s empirical finding that most of the observed growth in TFP over the 20th century cannot be explained by increases in labor or capital.

With these specifications of the model in place, Hansen-Prescott show analytically and via calibrated simulations that the Solow sector of their model, starting from initial values that held in England prior to 1800, eventually becomes dominant in production, with the agrarian Malthusian sector going to zero asymptotically. Via calibration of the population growth function that captures the demographic transition they are then able to replicate the observed patterns of the transition on investment in manufacturing relative to agriculture, the explosion in per capita income, and the decline in the value of land relative to output.

The one major criticism that can be levelled against the Hansen-Prescott approach is its inability to actually explain what was happening to households in terms of fertility and human capital investment decisions during the transition to industrialization.

The Galor-Weil paper takes the endogeneity of these decisions into account. They work with a three-period deterministic OLG model in which parents can trade-off numbers of children produced with the amount of human capital invested in each child. Households face a minimum consumption requirement, reflecting the Malthusian survival imperative. Unlike the production process in Hansen-Prescott, Galor-Weil posit that production of output is generated by an increasing returns-to-scale technology in which labor productivity is enhanced by human capital investment, which in turn affects the rate of technological progress (the TFP effect observed by Solow). Because the human capital investment decisions are made by households rather than firms, firms in the model take the human capital of workers as given in their production decisions and hence operate in a perfectly competitive environment. We do note, however, that like Hansen-Prescott, Galor-Weil also take the fundamental scientific progress that drives the growth in TFP as exogenous. The primary reason both papers make this assumption stems from the discoveries in Shell (1966) and Romer (1990) that full endogenization of TFP growth would require explicitly identifying how inventive activity (in Shell's parlance) or technological change (in Romer's) was organized and paid for. In Shell's models, research was either a public good, or one organized by a monopolist. In Romer's model, monopolistically competitive intermediate firms generated endogenous increasing returns-to-scale which, in turn, drove growth. Hansen-Prescott and Galor-Weil clearly recognized that the complexity of trying to introduce these features into their models would take the analysis too far afield of what they were trying to explain.

Given the dependence of scientific progress on human capital investments in the Galor-Weil model, together with simplifying assumptions on the nature of parental altruism that drives the quantity vs. quality trade-off in household fertility decision-making, Galor-Weil show three

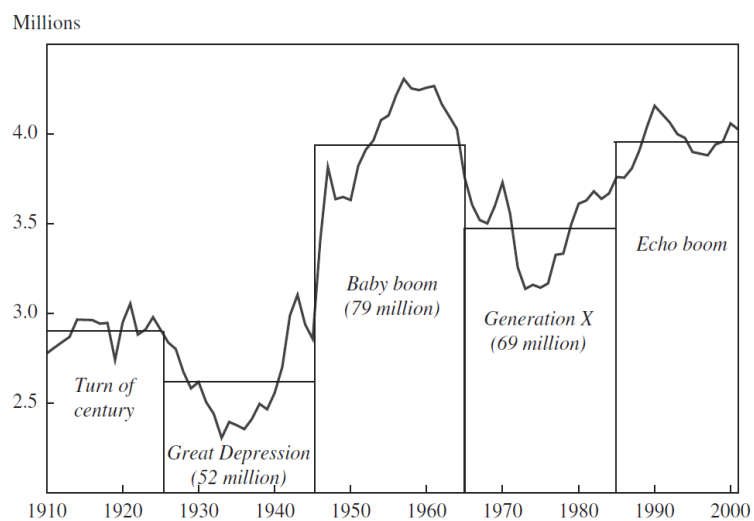
key results. First, if parental household income is such that the minimal consumption constraint is binding, then increases in parental income always lead households to produce more children. Second, once parental income is such that the consumption constraint is no longer binding, then increases in parental income lead to no changes in either the number or quality of children produced. Finally, if parents expect that technological progress will occur between their children's second and third periods of life, they will substitute away from numbers in favor of investing in quality.

Galor-Weil then show how the dynamic system generated by the model can explain the demographic transition observed to take place both historically after the industrial revolution in the UK, US and Europe in the late 1800's, and in developing economies during the latter part of the 20th century. The basic mechanism is the interaction between the way technological progress increases family income and investment in human capital, and the way that human capital investment then drives more technological progress. This identification of education as an important driver of growth in per capita income and standard of living had a lasting effect in the economics profession. Education now plays an important explanatory role not only as a driver of growth but in studies of income inequality, wealth inequality, discrimination and political leanings. It is hard not to wonder, though, whether the neglect of the issues raised by Shell and Romer – the fact that increasing returns technologies breed monopolies – might not play an equally important, though neglected, role in understanding these issues. That, however, is the topic for a different book.

Another important application of large-scale SOLG models during this period was to the study of the effects of demographic variation on economic activity. A number of studies from

the early 1990's (see, e.g. Mankiw and Weil [1989], or Poterba, Weil and Shiller [1990]) found that housing values didn't seem to follow the kinds of forward-looking pricing that other, more conventional assets did, but rather, were strongly (though not exclusively) affected by demographic variations. This observation led Geanakoplos, Magill and Quinzii (2004) (GMQ henceforth) to ask whether population movements might not affect the prices of assets that individuals accumulated and decumulated for retirement purposes. To analyze this, GMQ note that for the U.S. economy, live births over the course of the 20th century appear to follow a cyclic pattern, with a relatively large cohort born in the early 1900's, a much smaller cohort born during the depression years of the 1930s, followed by the baby boom cohort born in the years after WWII, followed by the relatively smaller Gen X cohort, and then the children of the baby boomers. We reproduce the graph from GMQ as Figure 22. From this pattern, GMQ decided to model the demographic fluctuation as if it were a simple 2-cycle, adopting a three-period OLG model as the simplest that would permit the existence of such demographic cycles. From this simplification, they then defined what they called the MY ratio, which was the ratio of the number of middle-aged agents living in a given period to the number of young agents. By comparing this ratio with the movement of the price-equity ratio for the Standard and Poor's

Live Births and the Five Major Cohorts, 1910–2001^a



Source: *Historical Statistics of the United States*, Series B1, and Bureau of the Census data.
 a. Numbers in parentheses are total births, and bar heights average number of births per year, during the indicated period.

index of stock prices (taken as a normalized measure of firms' value), they found a key piece of empirical evidence supporting the contention that demography influences asset values. We reproduce the diagram from GMQ below as Figure 23.

Figure 7. Average PE Ratio of Firms in Standard and Poor's Index and MY Ratio, 1910–2002

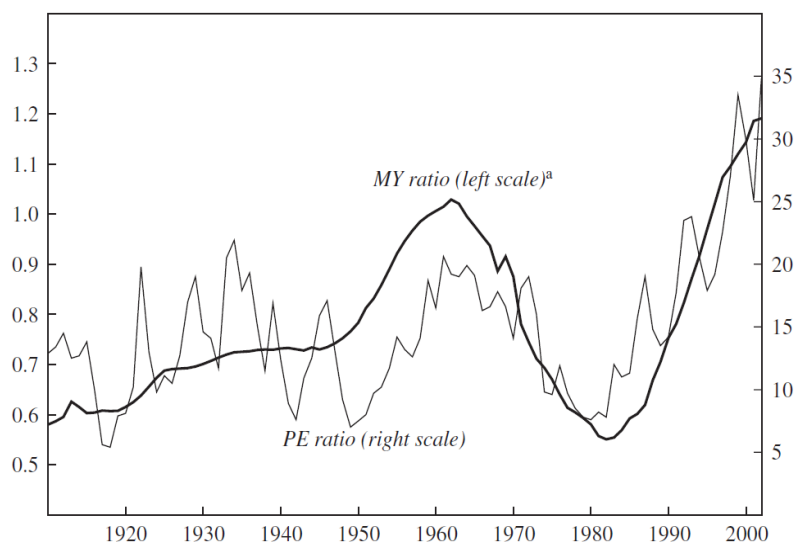


Figure 23

MY ratio and stock price correlation

Copied from Geanakoplos *et. al.* (2004)

To capture some of the randomness apparent in the asset prices, GMQ considered a stochastic version of the three-period OLG model having four possible aggregate shock states. The main findings of the paper are that the asset price variations in the model due to the endowment shocks are dwarfed by the fluctuations in generation size, to the point that asset prices become completely predictable in terms of population size. One of the predictions this model makes, then, is that the U.S. should expect to see a sustained bear market for stocks over the second and third decades of the 21st century as the Baby Boom generation retires and begins selling off assets to fund retirement. This prediction will also be something of a test of the older proposition about the importance of intergenerational transfers as a source of wealth relative to retirement accumulations, if wealthier Boomer generation parents don't consume their

accumulated wealth but rather pass it on to their children. For a different view of this hypothesis, see Poterba (2004).

In addition to their main findings, GMQ also identify what they call a “favored cohort” effect generated by the model. Specifically, smaller generations are strictly better off relative to larger ones. In the GMQ model, the favored cohort effect occurs because of competition between agents in a large generation for retirement assets. The collective bidding up of asset prices, coupled by increases in the real rate of interest to preclude arbitrage between the bond and equity markets, leads to reduced consumption and lower lifetime expected utility. Since the GMQ models does not include production, there is no wage channel for welfare improvements based on the number of workers in the labor market, but even in this setting, we would expect that increased competition for jobs would lead wages to move adversely against the larger generation.

Section 8.4 Search

As we noted in chapter 5, the fact that the OLG model embodied a natural motive for retirement saving which could give otherwise intrinsically useless asset (fiat money) value didn’t resolve the debate over whether the store of value function alone was a sufficient reason for the existence of money, or whether monetary theory would also need to include a medium of exchange role for money as well. The answer to this question came with the development of search models of money, the pioneering work in this area being Kiyotaki and Wright (1989). The Kiyotaki and Wright (KW henceforth) paper develops a simple model based on having

heterogeneous infinite lived agents who produce goods they do not wish to consume. The infinite lived agents assumption here does not involve the controversial interpretation of family structures, as it does in the conventional macroeconomic setting. Rather, it is a mathematical simplification for modeling sequences of frequent trading interactions without having to deal with the usual “hot potato” problem generated by a finite horizon. It shares this structure with repeated games and agency relationships.

In the KW model, agents meet randomly with other agents and can, depending on who they meet, trade for a good they will consume, trade for a good they will store in hopes of trading for something better later, or not trade. Storing goods is costly, with different goods having different storage costs. Under the assumption that the goods are not divisible and that each agent can only produce one unit of their production good, consume one unit of their consumption good, and store at most one unit of any good, the main result of the paper is a demonstration that commodity money equilibria arise quite naturally as a Nash equilibrium to the trading game in the model. The intuition behind the result is simple: using the cheapest-to-store commodity as a medium of exchange cuts through the double coincidence of wants inefficiency of barter. Kiyotake and Wright then introduce durable fiat money, which has no value in either production or consumption, and is cheaper to store than commodities. The model with fiat money then has an equilibrium in which fiat money is not valued, and a second one in which it is valued and plays the role of medium of exchange. In their discussion of this result, KW state that

This "tenuousness" of fiat currency is shared by the overlapping generations model, although not by the cash-in-advance model, and we think that it is a property that a good theory of money ought to have. The value of any medium of exchange, and especially fiat money, ultimately depends at least partially on faith. (*Ibid*, p. 943)

This observation points back to the debate at the Minnesota Fed conference of 1980, and forward to the on-going development of more general and more realistic search-based models of money. Lagos and Wright (2005) extended the basic idea from Kiyotaki-Wright to a general framework for monetary policy analysis by introducing a dual market structure in which agents could trade specialized goods (as in Kiyotaki-Wright) on a decentralized search market, and standard (Walrasian) goods on a centralized market using money.

While a complete discussion of the evolution of the literature on search and money is beyond the scope of this project, we do want to comment briefly on recent work explicitly integrating search theoretic and OLG models of money. Before turning to this, though, we need to highlight the Corbae, Temzelides and Wright (2003) paper (CTW henceforth) on matching markets, per a comment to us from Wright (email communication of November 4, 2022):

I have been pushing recently the line that simple search models and OLG models are both special cases of the general idea of a matching model. There is a set of agents and the agents interact over time in various ways. In Kiyotaki-Wright, e.g., each agent interacts each period (or at least in some periods) with a random draw from the population. In a standard simple OLG environment, let's say with one agent per generation for the sake of illustration, agent t interacts with agent $t-1$ at date t , and interacts with agent $t+1$ at date $t+1$, and that's it, although of course we can specify more than 2 period lived agents and make things more complicated. How the agents interact – bilaterally, in a large (Walrasian) market, etc. – is kind of a detail, which of course matters, but what is key is when and with whom they interact.

In the CTW model, agents have preferences over who they are matched with, as in the famous Gale and Shapley (1962) matching paper, but the incentive to match depends on the economic state of model: who wishes to produce, who to consume, and how to account for the trades that take place. The OLG aspect of this is clear once we recognize that saving for retirement creates resource slacks that in any efficient allocation must be used, which creates a need for the economy to match individuals who don't want to consume now (but do want to produce) with others who wish to consume but aren't currently producing. This, of course, is

just the intergenerational transfer process we associate with trade in OLG models. What the matching models show us is how closely tied the financial side of the economy is to this fundamental process.

On the subject of OLG and search, the first paper to formally integrate search and OLG was Zhu (2008), which combined the standard two period lived OLG model with the Lagos-Wright centralized and decentralized markets. Young agents in the model trade in two steps. In the first, they produce general goods and trade these with the co-existing old agents in exchange for money on the centralized Walrasian market. In the second step, they are randomly paired either as buyers or sellers with other young agents and can either produce or consume, depending on the matching, using money to facilitate these transactions. The surplus from these pairwise meetings is divided according to the Nash bargaining process. Since the pairing process is random, this introduces risk (as in Lagos-Wright), so that if old agents are risk-averse, changes in the money supply can induce real effects by altering the distribution of wealth. Zhu shows, for example, that if buyers in the matching markets have large bargaining weight in the decentralized market, the optimal rate of inflation will be positive.

More recently, Altermatt (2019) integrated OLG and Lagos-Wright in a model with active fiscal policy – government spending financed by taxes and government debt – as well as assets with random payoffs in aggregate, and showed (among other things) that there are significant general equilibrium effects in the determination of equilibrium portfolio holdings among money, assets and bonds, which effectively tie together the real and financial sides of the economy in the model. Altermatt, in private communication with us noted that (email of November 7, 2022)

[T]he OLG framework is particularly good at capturing the store of value role of money, while search is better at capturing the means of payment role of money. Thus, combining both allows to capture both roles of money, which is great in particular once one starts thinking about a world with multiple assets: Some might compete with money more as a means of payment, while others are inferior means of payments but better stores of value – and that may depend on policy, i.e., at the zero-lower bound, bonds and money serve equally well as a store of value.

Altrematt's model does include an infinite lived agent (or a continuum of them under the perfect competition assumption), which formally makes the model a hybrid. In this model, though, the infinite lived agent is not interpreted as a dynastic family, but rather as a firm, i.e., an on-going institution organized around production activities, managed by finite lived individuals when they are young. The firm is useful in the model as it can profit from the purchase of bonds whenever bond prices get too low. The infinite lived agent is also useful since it guarantees that real interest rates in the model will be non-negative.

Section 8.5 Computation

The ability to simulate large-scale SOLG models has grown alongside the computing power needed to solve these models, and with the development of algorithms for limiting the well-known problems that arise from the curse of dimensionality in all but the simplest simulations. The use of computers in economics goes back at least to the 1930s and attempts by Wassily Leontief to solve large systems of simultaneous equations, with a major boost from the work of John von Neumann in his forays into modeling economic growth and the development of game theory coupled with his role in the development of modern computing. (See Backhouse and Cherrier [2017] for a detailed history of the use of computers in economics.)

For OLG models, one of the earliest methods for solving for the deterministic steady state was introduced in Auerbach and Kotlikoff (1987) where they laid out the computational methods they used in their large-scale OLG simulation models. The solution techniques here involved solving for steady state equilibria of the models under different policy regimes, and then computing the transition paths between the old and new regimes. Auerbach and Kotlikoff solve their model for the initial and (when possible) final steady state equilibria using what they characterize as a Gauss-Seidel method. The approach involves guessing some plausible initial values for the aggregate variables – output, capital, labor supply and any policy aggregates – and then using these to compute the individual household and firm decision variables generated by the assumed aggregates in the model. From these, new aggregates are obtained by adding up the results of the household and firm decisions, and these new aggregates are adopted as updated guesses for an iteration of the procedure. Auerbach-Kotlikoff say the procedure generally converges to the steady state in 10-15 iterations. The same procedure can be used to compute the transition path between two steady state equilibria, though the computation itself is more complicated and time consuming.

Computation has also been an important analytic component for ILA models, going back to the work of Kydland and Prescott (1982) on real business cycle models. As these models have become more complex – with the introduction of imperfect competition in New Keynesian models, or the imposition of frictions like market incompleteness in Aiyagari-Bewley-Huggett models – computing power has also been a constraint. But unlike the work in SOLG models, the underlying computational algorithms for simulating ILA models haven't changed much since the pioneering work of Kydland and Prescott.

For ILA models, the recursive formulation we outlined in Chapter 4 is the key to solving models amenable to this specification. The fact that the recursive functional operator leading to the Bellman equation generates a contraction mapping is all one really needs to implement computational solutions for these models. Recall that the Banach contraction mapping theorem states that if an operator is a contraction, it has a unique fixed point, and this fixed point can be obtained by picking any starting value, and then applying the operator iteratively until it converges to the fixed point. In practice, since the contraction operator is typically applied to a function, the iterative result implies pointwise uniform convergence to the fixed point. For computational purposes, then, one only need discretize the underlying state space, pick an arbitrary starting value for the value function, and then iterate the operator on each point of the state space. For complicated applications, this procedure can be slow or computationally expensive, particularly for stochastic models. For these cases, various techniques have been developed to allow one to limit grid points to those most important for the computation. We can illustrate this with an example from Mitra, Montrucchio, and Privileggi (2003), who show that for the standard log-linear/Cobb-Douglas stochastic one sector growth model the support for the ergodic distribution for the model's equilibrium is a Cantor-like subset of the state space. Since Cantor-like sets are sparse, knowing that the equilibrium ergodic set has this property would allow one to restrict the grid in the state space for this model to an approximate guess about the likely realizations, and then dynamically adjust the grid as the model iterates toward a solution.

Coleman (1989) introduced a procedure known as policy function iteration which generalizes the value function approach and is applicable to models in which outcomes may not be Pareto optimal (such as in Aiyagari-Bewley-Huggett models). Both the value function and policy function iteration approaches take important advantage of the fact that the operator

updating initial solution guesses is monotone, and in many of the early simple stochastic ILA models, this property was satisfied. For models with multiple sectors, heterogeneous agents and real or financial frictions, though, a different approach was needed.

Working at roughly the same time as Coleman, Marcet (1988) introduced a distinctly different approach to computing stochastic general equilibrium models based on what he called the “parameterized expectations” approach (PEA). In this approach, one uses functional approximations of agents forecasts of future endogenous variables (expectations) and then either mathematically or numerically computes the temporary equilibrium corresponding to the given forecast function. The temporary equilibrium as a function of the state variables is then used to update the forecast function, generally by using projection methods to approximate the temporary equilibrium using the same functional form as the initial forecast. Iterating this procedure to convergence then yields the rational expectations equilibrium.

The parametrized expectations approach was more suited to applications involving SOLG models because it didn’t require that the first-order conditions or equilibrium conditions for the underlying model be a contraction. The approach was also logically grounded in the Stone-Weierstrass theorem’s results on the density of various computable functions in the space of all continuous functions. However, as noted in Marcet and Den Haan (1990), the projection methods used to update forecasts was not guaranteed to converge, though in applications, the process always did. Direct application of PEA has been supplanted over time by combining it with Coleman’s policy function iteration approach, since the relevant policy and implied forecast functions are the same at the fixed point of the temporary equilibrium operators (i.e., at the rational expectations equilibrium).

An answer to the question of whether (and why, in most applications) the PEA approach worked was given by Feng *et al.* (2014) where the approach was extended to allow for correspondences (when there are multiple temporary equilibria), which then required adding continuation utility values as state variables. The authors then show that the resulting temporary equilibrium operator is monotone and contracting, making the Banach theorem applicable. Because Feng *et al.*'s procedure uses a generalization of the Duffie *et al.* approach, it also recursively generates the self-justifying sets (in Duffie *et al.*'s terminology), with the implication that if one discretizes the state space by using the temporary equilibrium values of endogenous variables generated by the exogenous and pre-determined variables, this discretization will converge under iteration to the support of the equilibrium ergodic measure.

We note that there have also been fruitful applications of old techniques with new ones. Brumm and Kubler (2014), for example, adapts the Negishi approach used by Kehoe and Levine (1985) to provide an alternative way of finding the recursive equilibrium in SOLG models under assumptions on preferences and the structure of stochastic shocks that complement those typically used in simulations.

Contemporary work on computing economic models is embracing new discoveries in machine learning by leveraging the fact that neural networks are fundamentally function learning machines. These can be used directly with the iterative techniques given by Feng *et al.* to let the neural network learn the rational expectations equilibria of a given model. The use of neural networks for this purpose in economics first appeared in Kelly and Shorish (2000) based on work they did in the mid 1990's on learning rational expectations equilibrium in two-period SOLG models. Other early contributors to the use of machine learning techniques in economics were Marimon, McGratten and Sargent (1990) which used classifier systems to model the decision

behavior of boundedly rational agents in simulations of a Kiyotaki-Wright model of money. Holland and Miller (1991) introduced genetic algorithms as a mode of boundedly rational adaptation in agent-based models, effectively launching a new field of study in economics, though this is somewhat far afield of our current focus on OLG models.

Given the overall focus of this book, our discussion of computation has necessarily been short and incomplete, so we refer the reader interested in learning more about this topic and the many additional contributions beyond what we have included here to consult Judd (1998) or Backhouse and Cherrier (2017) for details.

Chapter Nine: The Ascendancy of ILA and the Future of Macro

Section 9.1 Why ILA?

Having now looked back at the development and applications of OLG models from their first appearance in the work of Allais and Samuelson through the recent past, we can now return to the question we initially raised in the Introduction: why isn't OLG the central workhorse model for modern macroeconomics?

We heard from a number of the protagonists in this story regarding their views as to why ILA became dominant. One reason cited was simply first-mover advantages: Ramsey's model was seen as the correct one for thinking about economic growth, and thus, much work in the early post World War II era went into refining it, via the Solow-Meade-Swan neoclassical approach, and optimal growth approach of Koopmans-Malinvand-Cass, as against the Keynesian-based Harrodian growth model (on this, see Young [1989] and Spear and Young [2014]). OLG models were still not well understood when Kydland and Prescott introduced rational expectations based RBC models into macro, and *inertia* kept the basic RCK framework in place even as RBC models were abandoned – because they couldn't explain the data – in favor of the new neoclassical synthesis approach. Karl Shell put it this way (private email dated September 5, 2022):

DSGE launched a generation of macro Ph.D.'s. There was room for some deep innovation, but there was also plenty of room for amending the basic DSGE to study the latest "puzzle" or the latest policy issue. Even the computer programs were ready to be used, originally written in Fortran. There is Thomas Kuhn history of science involved here. Young scholars and other scholars find safety in normal science. DSGE became normal science in macro.

How do we define "macroeconomics"? The denotation definition is "economics based on the large macro variables defined in the national income accounts." The connotation includes much more including banking and finance. In this broader view, beliefs about

beliefs of others play a very important role. Keynes, the Rheingold girl, animal spirits, bank runs, etc. In this way, macro is like game theory. Mostly DSGE has missed this. The typical DSGE exercise is to solve a representative agent optimization problem and take the dual prices as equilibrium prices, which they are for the single-agent economy. There are no bank runs in this economy; why would the single agent run on his own bank? There are exceptions: Roger Farmer, Jess Benhabib, and others have analyzed the role of interdependent beliefs within the superstructure of DSGE.

Shell hits on an important point here. The vicissitudes of getting promoted and tenured forced many a junior scholar interested in OLG models to work in other, more promotable, areas. The difficulties associated with general OLG models likely also persuaded more senior OLG protagonists to pursue other interests.

We have noted previously that even OLG's English-language inventor, Paul Samuelson, did very little further with his own model for the remainder of his career. Manny Yaari's work with Cass on OLG models was all he wrote on the subject, with other topics in growth and general equilibrium theory occupying him for the rest of his career. Dave Cass moved away from OLG to do pioneering work on models of incomplete markets, most of which were based on simple models with only two periods. As we also noted earlier, Karl Shell and his student Jim Peck moved away from OLG towards game theory, initially as follow-up to the conjectures of the Philadelphia Pholk theorem involving imperfect competition, but later in exploring bank runs and other sunspot-like phenomena in a variety of different models. Jim Peck is now a highly respected applied game theorist. Jean Michel Grandmont embraced versions of the ILA model that admitted complex dynamics in his co-edited special issue of *JET*. Neil Wallace stopped pushing OLG as the definitive model of money in favor of search theoretic approaches, and while some of his later work continued to use the OLG model, much of it did not. Many of the search theoretic tools that Wallace used were developed by Peter Diamond and his collaborators. And finally, as we noted earlier, Bob Lucas never worked with OLG models after his initial

foray with the Lucas (1972) paper. Alan Auerbach continued working with his coauthor, Larry Kotlikoff, on projects based around their numerical simulation model, but he also branched out with projects based on the stochastic RCK model, and with empirical macro-econometric analysis.

In the boomer generation, Tim Kehoe and David Levine continue to do some work using OLG models, but both have branched out to include work in more standard ILA-based DSGE models, game theory, and industrial organization. Rao Aiyagari turned away from OLG models to work with versions of the RCK model having incomplete markets. Spear and Srivastava are best known professionally for their work developing recursive techniques for analyzing infinitely repeated principal-agent problems. This research – together with that of Ed Green (see, e.g., Green [1987]) – became the basis for a large literature on dynamic public finance, virtually all of it based on the canonical ILA-based DSGE model. Mike Woodford wrote a well-known book, *Interest and Prices*, based on the combination of conventional ILA dynamics with monopolistically competitive intermediate goods and staggered wage/price adjustment. John Geanakoplos and Herakles Polemarchakis continued working in the general equilibrium paradigm, but moved away from OLG toward incomplete markets, game theory and finance applications. Costas Azariadis continued working with models that could generate self-fulfilling prophecies, but followed Woodford in working with more conventional versions of the growth model or new Keynesian variations on it. And, while Victor Rios-Rull never abandoned the use of the modeling approach he pioneered, he too wrote a number of papers based on the dominant ILA paradigm as well as a number of important empirical papers.

Among all the main contributors to this history of OLG, only Larry Kotlikoff continued what we would characterize as a steady pursuit of results based on OLG economies. This

included the on-going work he did with coauthor Alan Auerback using their AK simulation model, but it also included new pioneering work with Jasmina Hasanhodzic using Rios-Rull's stochastic extension to study issues around the U.S. Social Security system in a realistic SOLG setting. In light of Kotlikoff's dedication to OLG, it seems fitting that we quote him on the continuing problems he finds with the conventional ILA paradigm (private email dated August 31, 2022):

The ILA model has become the workhorse of macroeconomists despite its ludicrous assumptions and preposterous predictions - all of which are decisively rejected by the data. The explanation? Computational ease and analytical elegance. The result? Modern macroeconomics comprises beautiful models that are largely divorced from reality.

To put this in context, Kotlikoff is not referring here to the conventional DSGE model's ability to match moments or forecast financial crises or recessions. The data he refers to are the many theoretical and empirical studies of interfamily altruism and savings motives. As we've discussed previously, realistic models of family structure and non-paternalistic altruism yield too much neutrality. Empirical studies of interfamily gifts and bequests strongly reject the predictions of non-paternalistic altruism in dynastic families, and indicate instead that (as first noted by Arrow and Kurz [1969]) family altruism takes the form of investment in children's human capital or direct unrestricted gifts. Other studies, stimulated by Kotlikoff and Summers (1981) work on sources of wealth have shown that most households save for retirement and for precautionary purposes based on idiosyncratic health/mortality or income risk.

In light of this evidence against the assumption that the families that matter in a modern industrial economy are all organized as lineal dynasties, the only remaining plausible interpretation of the representative agent model is its original one: the optimizing agent is a central planner. And, in light of this, the only remaining option for any realistic model of how actual households in real economies interact is the overlapping generations model.

We share Kotlikoff's view on this and will detail why we share these views below. But before we undertake this analysis, we should point out that there have been other recent criticisms of the directions taken by contemporary macroeconomics, most notably those of Paul Romer in his 2015 American Economic Association panel discussion critique. In the discussion, Romer referred to the “mathiness”²⁵ of contemporary macro, meaning the use of sophisticated mathematics that contributed very little to the actual scientific understanding of the workings of the macro economy. In his working paper extension of this critique (Romer [2016]), Romer zeroes in first on the source of aggregate shocks in conventional RBC models, calling these “phlogiston” and then on extensions of these shocks (“forcing variables”).

While Romer doesn't mention OLG in this paper, the critique is clearly focused on the problems inherent in conventional RCK-based macro DSGE models. In an entry (Romer [2015]) on his blog entitled “What Went Wrong in Macro – Historical Details” Romer blames political differences between Keynesian macroeconomists – he singles out Robert Solow at MIT – and the new generation of macro theorists focused on rational expectations and microfoundations, based specifically on the policy questions of whether government intervention for business cycle stabilization was justified. Referring to the economists in the RE/microfoundations group as “the rebels,” Romer states that because of the policy differences,

[R]eal business cycle models, which were introduced by Prescott and Kydland in 1979, attracted support among the rebels, even though these models did an even worse job of matching the empirical evidence than the model that Lucas first proposed.

In particular, they assumed away the possibility that monetary policy and inflation could have any interaction with output and employment. In a telling shift, the rebels also decided that econometrics was getting in the way. Word came down from the top that they were abandoning the cutting edge econometric work that Sargent had specialized in and using calibration in its place. (*Ibid*)

²⁵ A play on Stephen Colbert's use of the word “truthiness” to refer to made-up facts.

Romer goes on to contend that there was a “window of opportunity” in 1978 when the policy/political disagreements could have been put aside in the interests of what Romer called “Feynman integrity” – a clear dedication to science – versus “Stigler conviction” – the belief in the correctness of one’s own views, saying

What’s sad is that the rebels could simply have declared victory. They had won the battle for mindshare among the next generation of macroeconomists. They could have taken credit for killing off the large multi-equation models that had previously dominated macro policy analysis.

From the MIT side, Lucas could have been embraced as a leading contributor to the larger intellectual program launched by Paul Samuelson, the founder of the department there. As Lucas solidified his position as the new intellectual leader at Chicago, he and Samuelson could then have healed the divide that had separated MIT and Chicago for decades. This divide had its roots in Stigler’s political agenda, but it played out in economics at the more damaging methodological level of Samuelson versus Marshall. At least for the young Robert Lucas, it was clear which side to root for in this fight. (*Ibid*)

The remainder of the blog post goes on to lay out a vision of how macro might have developed based on extending Lucas’ pioneering 1972 work on stochastic OLG models, combined with Dixit and Stiglitz’s (Dixit and Stiglitz [1977]) model of monopolistic competition, which Romer had used so effectively in his endogenous growth models.

While we are sympathetic to this view, we don’t believe it could ever have taken place given Paul Samuelson’s attitude toward the microfoundations approach. As we have documented earlier, Samuelson held steadfast beliefs that macroeconomics should be based on observable relationships between aggregate variables, on the pressure the gas in a balloon exerts to keep it inflated, not on the trajectories of individual molecules inside. As Romer also notes in his working paper (and as we have noted earlier), Lucas himself abandoned his own pioneering model in favor of real business cycle models, endorsing the Kydland-Prescott approach in his 2003 Presidential address to the American Economic Association. Based on this, it seems highly

unlikely that there could have been any reconciliation between what we now call the saltwater and freshwater approaches to macroeconomics.

The perspective outlined by Kotlikoff, that “computational ease and analytical elegance” were the main drivers of which macro models caught on, seem to us a more likely explanation of what went wrong with macro. We have referred several times to the seductive appeal of recursive dynamic programming which is at the heart of conventional DSGE macro models, so it is worth looking at how these techniques found their way into economics to begin with.

As we noted in our earlier discussion of the discounted dynamic programming framework, the mathematical techniques underpinning this were developed in the 1950’s and ‘60’s by Richard Bellman and his student David Blackwell. The first applications of the forward DP algorithm in economics appear in Arrow (1957) and then in Radner (1967) and Radner (1973) in a stochastic context. The techniques were also introduced into Operations Research by Shapiro and Wagner (1967). As our examination of Lucas’ introduction to the contraction mapping theorem showed, his discovery of Bellman’s work was independent of either Arrow’s or Radner’s use of the technique. Lucas did, however, become aware of the discounted dynamic programming framework based on Blackwell’s work during his time at Carnegie Mellon, since a typed set of lecture notes on the forward DP circulated among students at Carnegie Mellon and Chicago in the late 1970’s and early 1980’s. These lecture notes became the basis for the well-known text *Recursive Methods in Economics Dynamics* (referenced earlier) published in 1989. From an email from Nancy Stokey (dated May 20, 2022) she advised us that

Indeed there was a set of notes that Bob wrote at GSIA, and they were the starting point for *Recursive Methods*. Chapter 2, much of the material in Chapter 3 and parts of

Chapter 4 are direct descendants of those notes. There was also a treatment of systems with stochastic shocks, which (more loosely) was the starting point for Chapter 7.

The introduction and widespread use of this text cemented the dynamic programming technology into macroeconomic analysis, leading Sargent and Ljungqvist (2018) to title the first section of their book “Imperialism of Recursive Methods.” Curiously, Stokey and Lucas do not cite Radner’s 1967 or 1973 papers in their book, though they do cite Majumdar and Radner (1983) which references both of Radner’s earlier papers.

The forward dynamic programming algorithm had a major impact on areas of economics other than macro. In applied micro, it gave researchers a tool with which to model partial equilibrium interactions over individual lifetimes, by using the fact that when modeling periods are short and agents discount the future, the difference between a finite horizon and an infinite one is very small. Similarly, in industrial organization, the repeated game results of Abreu, Pearce and Stacchetti (1986) allowed researchers to work in well-defined dynamic partial equilibrium settings. In macro, recursion came in through two channels. One was the replacement of the Hamiltonian dynamic systems approach championed by Cass and Shell (and the need for transversality conditions) with Bellman’s recursive optimization approach. As we’ve noted earlier, the Banach contraction mapping theorem made working with Bellman’s functional equation easier (particularly for computational applications) than working directly with the state and costate variables in the Hamiltonian approach. The second channel was via the dynamic public finance applications that grew out of the work of Green, and Spear and Srivastava. Since the repeated agency results examined in these papers were based on a pair of infinitely lived agents, it was natural for macro applications of these results to be embedded in the conventional RCK framework, with the principal being the government and the agent being

the private sector. But, at the end of the day, it was the ease with which the recursive methods could be applied that made working with models amenable to the forward dynamic programming approach so popular. Perhaps the most telling indicator of how popular the approach has become (and reflecting Shell's notion of the safety of normal science) was the development and publication of the Dynare software platform ("macroeconomic modeling for all") in 1996 by a group of computational economists affiliated with CEPREMAP. This software package allows macroeconomic practitioners to implement the various recursive computational techniques for simulating large scale SILA models without having to write their own code. The platform will also solve some OLG problems, but these are limited to deterministic settings. To date, economists wanting to work with SOLG models must write their own code.

A second issue that separated OLG and ILA approaches to macro was one we've touched on repeatedly in the book: that of the correct specification of the state variables for macroeconomic models. Ljungqvist and Sargent, in the preface to their book, comment fairly extensively on the issue of the appropriate choice of state variables in macroeconomic applications, saying

This book reflects progress economists have made in refining the notion of state so that more and more problems can be formulated recursively. The art in applying recursive methods is to find a convenient definition of the state. It is often not obvious what the state is, or even whether a finite-dimensional state exists (e.g., maybe the entire infinite history of the system is needed to characterize its current position). Extending the range of problems susceptible to recursive methods has been one of the major accomplishments of macroeconomic theory. (*Ibid*, p. xxxiv)

While it isn't clear that "convenience" should be an important characteristic in specifying the state variables for an economic model – the possibility that the underlying structure of the real world might determine this choice also suggests itself – it does seem to have been one of the

guiding criteria in the development of macroeconomic models. As Ljungqvist and Sargent also note, it is possible for models to be history dependent – we’ve already seen how Woodford (1988) used the infinite history of shocks in a general SOLG model to demonstrate existence of equilibrium – in most applications there are relatively simple sufficient statistics for summarizing this history dependence. In the canonical deterministic RCK model, for example, iterating the policy function backwards traces out the economy’s dependence on the historical sequence of capital accumulations that lead to the current state of production. In Lucas (1972), the received convention of the two-period lived OLG model made it possible for Lucas to intuit that the exogenous shocks alone (given their i.i.d. specification) would be sufficient. In the early stochastic representative agent RBC models examined by Brock and Mirman, the aggregate capital stock is a sufficient statistic for the history of exogenous productivity shocks, since it together with the labor supply response of the representative agent completely determine the possible economic outcomes going forward. Hence, in these models, the capital stock and current exogenous shocks are the natural state variables. As one discovers on further reading in Ljungqvist and Sargent, various researchers have found clever ways of augmenting the states in more complex models to make seeming non-recursivities go away.

As we have seen, OLG models defy simple or convenient classification of the appropriate state variables, at least once one moves beyond the simplest specification of two-period lives. Even in these simple versions of the model, though, the development of the sunspots literature makes clear that intuitive choices for state variables may not encompass the full range of possibilities. Even the logical approach of McCallum’s minimal state variable criterion breaks down when the model is history dependent and requires the full infinite sequence of shocks to be understood. The discovery that general OLG models – two-period models with multiple goods

in Spear's thesis result, or multi-period models with a single good in Aiyagari's lost working paper – were generically history-dependent made the question of appropriate state variables even more critical for general SOLG models. Following on from Woodford's 1988 results on using infinite histories of shocks as the state, the Spear and Srivastava, and Duffie *et al.* papers started the hunt for sufficient statistics for summarizing the history of shocks in order to provide a recursive Markovian mechanism for solving and simulating these models. As we have documented, this search culminated with the work of Citanna and Siconolfi on the generic applicability of Rios-Rull's intuition that the wealth distribution would be sufficient for summarizing the history of shocks.

Rios-Rull's simulation work showed that while the recursions based on carrying along the wealth distribution as a state variable were doable, they were also quite difficult. In general situations, the wealth distribution could itself be an infinite-dimensional object, and obviously, swapping one infinite-dimensional state variable for another was less than satisfactory. Krusell and Smith (1998) used the equivalence of probability distributions and their characteristic functions to use the first few moments of the wealth distribution as proxies for the full distribution, and in the stochastic growth model application of their paper, this approximation worked quite well. As we've seen, though, in SOLG applications such as Storesletten, Telmer and Yaron (2007), the technique did not work.

There has been some progress in making SOLG simulations workable, however. The practice of working with a representative agent per generation can simplify the analysis somewhat, since each generation's asset holdings are the same. The fact that there is still heterogeneity across agents in different generations remains a complication, and as a result, to

date, most of the simulation work limits the number of assets to one or two (typically, a bond or money plus an equity asset). One can speculate that to the extent the two-fund theorem associated with the standard capital asset pricing model holds, consideration of two assets might be sufficient. This possibility has been studied in static general equilibrium settings with incomplete markets by Geanakoplos and Shubik (1990), with some follow-on work done in the context of two period lived OLG models, but none that we could find on more general SOLG models. But, even if this proves to be the case, the complexity of working with these simulations has been daunting enough that several academic generations of macroeconomists have either turned away from OLG in favor of ILA or worked in what amount to deterministic OLG settings by studying things like mortality or individual income risk which are plausibly idiosyncratic.

A third issue related both to the question of defining states and finding workable recursive formulations for solving a model is that of heterogeneity. In the SOLG setting, it is the heterogeneity of agents of different ages – the differences in their incomes, wealth and possibly preferences – that leads to the portfolio rebalancing activities and consequent external income effects in the model that force inclusion of the wealth distribution in the description of the state. The absence of heterogeneity in representative SILA models moots any need for tracking wealth, though, as Duffie *et al.* showed, when one introduces heterogeneity in these models, the need for augmenting the state space with lagged endogenous state variables also arises.

It is a great irony, though, that even as macroeconomics turned away from SOLG models in the 1990's and early 2000's, the limitations of the representative agent framework – in particular, the fact that in the absence of frictions the competitive equilibrium is always Pareto optimal, together with the lack of any role for money in the model – led cutting edge macro

people to start looking at models with frictions that generated heterogeneities among agents or types of agents in the model. As we noted above, most of these models assumed that shocks were idiosyncratic rather than aggregate, but the results derived from this work was fruitful enough to motivate follow-on work that included aggregate shocks. While the Krusell-Smith approach did work fairly well in models where the heterogeneities were idiosyncratic, the need for computational methods for handling more general heterogeneities spawned a new literature on new techniques (see Hommes [2018] for references). The irony, of course, is that if SOLG had become the dominant workhorse for macro, all of this work would have been started two decades ago and tailored to applications involving actual human life cycles and not the fantasy life of the upper 1% of the wealth distribution.

We close this section with a fitting quote from Larry Summers' paper "The Scientific Illusion in Empirical Macroeconomics" (Summers [1991]) where he notes in the conclusion "It is all too easy to confuse what is tractable with what is right." This sentiment was echoed by Robert Solow in his paper "Samuelsonian Economics in Twenty-first Century" (Solow [2006]), where he states:

The power of the idea was established clearly with Peter Diamond's "National Debt in a Neoclassical Growth Model" (1965). The basic model was equipped with a stock of real capital, and with a paper asset to serve as an alternative way of holding wealth. The OLG set-up became established as one of the workhorse models for macroeconomics. It provided, among other things, a way to escape the temptation induced by use of the Ramsey model to formulate any long-run equilibrium process in a way that guarantees nice properties without further thought.
(p. 39)

Section 9.2 Whither Macroeconomics?

In October 2016 the University of Oxford hosted a conference organized by David Vines and Samuel Wills around the theme of “rebuilding macroeconomic theory.” They invited a number of prominent economists, several of them Nobel prize winners, to present their thoughts on this topic. Vines and Wills viewed the financial crisis of 2007-08 and resulting recession as an economic event akin to the Great Depression or the stagflation of the 1970’s in terms of presenting major challenges to the reigning economic orthodoxy of the time. Their charge to the conference invitees was to suggest ways that the standard New Keynesian DSGE model could be improved in light of the events of a decade prior. Specifically, they asked the conference group to address the following questions (taken from the published conference proceedings, Vines and Wills [2018])

- (i) Is the benchmark DSGE model fit for purpose?
- (ii) What additions to this model are necessary to help us understand growth from now on?
- (iii) What are the important inter-temporal complications?
- (iv) What are the important intra-temporal complications?
- (v) Should the above questions be discussed with big or small models?
- (vi) How should our models relate to data?

The papers presented at this conference were:

1. The rebuilding macroeconomic theory project: an analytical assessment, *David Vines and Samuel Wills*
2. On the future of macroeconomic models, *Olivier Blanchard*
3. Ending the microfoundations hegemony, *Simon Wren-Lewis*
4. Where modern macroeconomics went wrong, *Joseph E. Stiglitz*
5. On the future of macroeconomics: a new monetarist perspective, *Randall Wright*
6. Is something really wrong with macroeconomics?, *Ricardo Reis*

7. Good enough for government work? Macroeconomics since the crisis, *Paul Krugman*
8. Stagnant productivity and low unemployment: stuck in a Keynesian equilibrium, *Wendy Carlin and David Soskice*
9. Macro needs micro, *Fabio Ghironi*
10. An interdisciplinary model for macroeconomics, *A. G. Haldane and A. E. Turrell*
11. The financial system and the natural real interest rate: towards a ‘new benchmark theory model,’ *David Vines and Samuel Wills*
12. DSGE models: still useful in policy analysis? *Jesper Lindé*
13. The future of macroeconomics: macro theory and models at the Bank of England, *David F. Hendry and John N. J. Muellbauer*
14. Modelling a complex world: improving macro-models, *Warwick J. McKibbin and Andrew Stoeckel*

While all of these papers are interesting and offer important insights into why the conventional NK-DSGE model hasn’t lived up to expectations, and while several of the authors – Vines and Wills, Blanchard, Stiglitz, and Reis – mention OLG models as being more realistic as foundational models, none of the papers here discuss the fundamental differences between the ILA and OLG structures, particularly in the context of stochastic modeling, that breaks the observationally equivalence claim about the two models. Indeed, none of the papers in this two issue conference proceedings volume even cites Rios-Rull (1996).

The best summary of the recommendations coming out of this conference are actually contained in footnote 2 of a later issue of the *Oxford Review of Economic Policy* (Vines and Wills [2020]) entitled “The rebuilding of macroeconomic theory project part II: multiple equilibria, toy models, and policy models in a new macroeconomic paradigm.” The summary of the previous project results states:

We recommended four main changes to what we called the ‘core model’: to emphasize financial frictions, to place a limit on the operation of rational expectations, to include

heterogeneous agents, and to devise more appropriate microfoundations. Achieving this would, we said, require changing all of the behavioural equations in the model governing consumption, investment, and price setting, and inserting a wedge between the interest rate set by policy-makers and that facing consumers and investors. (*Ibid*, p. 428)

Vines and Wills open the second volume with the question “Why do it all again?” They start to answer the question by stating:

Last time we were much more conservative (see Vines and Wills (2018)). We accepted the NK-DSGE modelling paradigm, and were optimistic about how it might be developed – providing a list of frictions to be added to the model. We even believed that a perfect Platonic form might exist: the *ideal* NK-DSGE model. (*Ibid*, p. 428)

They then state

We have changed our mind, basically because of the GFC and the COVID pandemic. Joe Stiglitz sets out the argument at the beginning of his article with Martin Guzman in this issue. (*Ibid*, p. 429)

The quote from Guzman and Stiglitz (2020) states that:

[t]he central problem of macroeconomics is to explain the large, deep, and often persistent downturns, accompanied by high levels of unemployment, that episodically afflict capitalist economies. Macroeconomic crises are extreme examples of economic fluctuations. But they are the most relevant. They are the events that teach the most about the stability properties of the economic system, in a way that small inventory cycles do not. And they are the events that matter the most for the lives of millions of people. (*Ibid*, p. 429)

Vines and Wills conclude their argument for the change of mind with two key points, stating that:

First, thinking about large downturns requires models with good outcomes and bad outcomes. This means models with multiple equilibria. It is not good enough to say that bad shocks cause bad outcomes; the risk of a lurch to a permanently bad outcome is always present, even with a temporary shock. Macroeconomic policy should be designed to guard against this risk. But the NK-DSGE modelling tradition does not do lurches. (*Ibid*, p. 429)

Second, to think about large downturns requires a detailed empirical analysis of what can go wrong, and why. (*Ibid*, p. 430)

They then conclude by saying that:

Because of these two inadequacies we no longer think that the NK-DSGE model can be thought of as a ‘core model’. Blanchard (2018) helpfully classifies macro models into five groups: foundational, toy, core, policy, and forecasting. In the first issue in this series we described the NK-DSGE model as a core model (Vines and Wills, 2018). According to Blanchard a core model should provide ‘a generally accepted theoretical framework for the profession’. However, the inadequacies described above mean that the NK-DSGE does not satisfy this test. (*Ibid*, p. 430)

The papers included in this volume were

1. Mapping types of macromodel to types of macro user
Simon Wren-Lewis
2. Implications of household-level evidence for policy models: the case of macro-financial linkages
John Muellbauer
3. Some important macro points
Ray C. Fair
4. Climate change and monetary policy: issues for policy design and modelling
Warwick J. McKibbin, Adele C. Morris, Peter J. Wilcoxon, and Augustus J. Panton
5. Why some places are left behind: urban adjustment to trade and policy shocks
Anthony J. Venables
6. Towards a dynamic disequilibrium theory with randomness
Martin Guzman and Joseph E. Stiglitz
7. The importance of beliefs in shaping macroeconomic outcomes
Roger E. A. Farmer
8. Clinical macroeconomics and differential diagnosis
Jeffrey D. Sachs

None of these papers, with the exceptions of Wren-Lewis and Guzman and Stiglitz, mentions OLG *in any context*. While most of Wren-Lewis’s paper focuses on the way different empirical needs shape the demand for different types of models, he does consider OLG a

foundational model. Guzman and Stiglitz, in footnote 2 of their paper make the point that indeterminacy of equilibrium in OLG models and the possibility of sunspot equilibria might be important for macroeconomics, stating that:

Hirano and Stiglitz (2019) show that in a rational expectations model with overlapping generations, there can be an infinity of trajectories. The large literature on sunspots (Cass and Shell, 1983; Benhabib and Farmer, 1999) focused on multiplicity of macroeconomic equilibria. Similarly, in debt markets there may be a low interest rate equilibrium, associated with and justified by low default rates; and another equilibrium with high interest rates, marked by high default. See Calvo (1988) and Greenwald and Stiglitz (2003a). For a general equilibrium discussion focusing on defaults in financial contracts, see Roukny et al. (2018). (Guzman and Stiglitz [2020], p. 653)

The most surprising omission of any mention of OLG in the volume is Farmer's, since much of the development of the ideas he presents in the volume about the importance of beliefs for macroeconomics were developed using the OLG framework. For example, in Farmer (2017) he states directly that

The representative household assumption is convenient because the mathematics of a model, populated by a representative household, is relatively simple. A single household facing a decision problem will always choose a unique action. But a household interacting with other households, most of whom have not yet been born, is a very different matter. Even if the physical world were unchanging, we would still face uncertainty as to the ways that our descendants will behave in the future. A model, populated by overlapping generations of finitely lived people, always has multiple equilibria.

I will repeat that statement because it is so important and so misunderstood. *A model, populated by overlapping generations of finitely lived people, always has multiple equilibria.* (*Ibid* p. 4)

A Google Scholar search with Farmer as author for the phrases “belief function” and “overlapping generations” returns 13 papers written since 1996. When we asked Farmer why he hadn't discussed the role of OLG in his *OxREP* paper, he said (email dated September 23, 2022)

The OxRep paper focuses on steady state indeterminacy and I wanted to use a familiar vehicle, the neoclassical RBC model, to showcase differences in labor market structures that I had originally modeled in my 2010 [*Economic Journal*] piece. The main idea is to generate demand-driven comovements in consumption and employment in a full blown

DSGE model with capital. The resulting model has a lot in common with the old-style Keynesian cross, including a continuum of steady states, but with intertemporal asset markets. The multiplicities and dynamic indeterminacies that arise in OLG models cannot explain the consumption-employment comovements we see in data.

In Farmer's *OxREP* paper, indeterminacy of equilibrium arises because markets for the allocation of labor inputs to firms are incomplete. This statement bears some further unpacking. In the conventional general equilibrium view, wages on markets for labor are the signal to firms about the value of hiring an additional worker, and for workers the value of offering an additional hour of labor. The key insight to come out of the conventional search theoretic Diamond-Mortensen-Pissarides (DMP) model is that there can be commodities for which formal, organized markets don't exist, due typically to informational frictions. In labor markets, formal markets are hampered by the need for firms and workers to collect information by which to evaluate the quality of potential employment matches. The informational frictions at work here for the firm arise from the fact that worker productivity can be hidden *ex ante* and hence, must be learned over time. On the worker's side, the working environment and support provided by the firm for enhancing productivity are also unknown *ex ante*. Overcoming these moral hazard and adverse selection problems requires effort, and this effort is captured in search models by assuming there is a matching technology that evaluates the quality of otherwise randomly occurring matches and determines the number of workers available to the firm. In conventional DMP models of the labor market, wages are then determined by a bargaining process – typically the Nash bargaining solution – and that closes the model, since workers are still assumed to be motivated to search by the level of wages available.

Farmer (2006) developed a model of what he called “Keynesian search theory,” following the lead of Howitt and McAfee (1987), in which the standard search process

determined employment matches, but firms hiring decisions were driven by the need to meet demand. In Farmer's model, firms and workers do not bargain to determine the wage. Rather, wages were determined competitively, in the sense that once the firm has determined how much labor to use in production, workers are paid their marginal product. This can be justified based solely on profit maximization or cost minimization principles²⁶. On the worker side, workers do not view the wage as a measure of the marginal value of their time, since they understand the problems associated with finding appropriate employment matches and allocate some fraction of their own or their family's time to job search.

These two key assumptions – that firms meet demand and workers don't respond to wage signals – lead to the indeterminacy of steady state equilibria in Farmer's model. To close his model, Farmer adds a belief function by which workers form expectations about their wealth or purchasing power, and this determines demand. One of the most interesting features of this kind of model is that as long as beliefs are changing in some way – Farmer uses an adaptive formation subject to random shocks – an initial belief condition will then pin down a unique trajectory for the economy. Because of the indeterminacy in the model, these *ad hoc* beliefs are self-fulfilling (just as they are in the conventional indeterminacy-based sunspot equilibria) and hence the equilibrium constitutes a rational expectations equilibrium. But, because of the indeterminacy of the steady state equilibrium, the model itself is silent on what happens to the economy if the shocks to beliefs are shut down.

As Farmer noted in his email to us, this kind of search friction can be developed in both ILA and OLG environments. Some very recent and preliminary work by Kim, Sood and Spear (2023) shows that the indeterminacy generated by Farmer's labor market incompleteness and

²⁶ We note that the wage could equally well be determined in a model of monopsony or other forms of imperfect competition such as the Shapley-Shubik market game.

Keynesian search microfoundation generate different results in the OLG model. Specifically, in general, multiperiod SOLG settings, product market clearing endogenously determines the wealth distribution (which, we recall, must be carried along as a state variable in any recursive equilibrium). This, in turn, means that agents' expectations can't be arbitrary, as they can in the ILA setting, if they are to be rational. Nevertheless, the indeterminacy of the steady state remains, and different specifications for stochastic processes of output will generate different sets of REE expectations consistent with the given output process. As with other indeterminacies, it becomes possible to introduce extrinsic randomization – sunspots or sentiment – into the model. What kinds of dynamic stochastic behavior are possible in this framework is a matter of further research.

Viewed through the lens of our OLG project, one can't help but notice the similarity between Farmer's belief functions and Grandmont's earlier expectations function approach to temporary equilibrium. Both Grandmont and Farmer view beliefs/expectations as primarily psychological constructs, akin to consumer preferences. While Farmer's model uses indeterminacy to generate self-fulfilling rational expectations equilibria, his work in Farmer (1991), (2002) and (2013) critiquing the Lucas critique and the natural rate of unemployment suggests that, like Grandmont, he has a skeptical view of the rational expectations hypothesis.

Farmer's approach and the fact that the Keynesian search model generates path dependent equilibrium trajectories has been controversial. One of the earlier criticisms of the temporary equilibrium approach was that one could explain just about anything in a temporary equilibrium model by positing an expectations function tailored to deliver the result. Indeed, this weakness and the *ad hoc* nature of adaptive expectations in the Keynesian models of the 1960s were an important motivating force in the development of the rational expectations paradigm.

The fact that one can make the same criticism about the choice of driving processes for models that exploit indeterminacy – and indeed it was Grandmont himself who raised this issue with Woodford’s 1986 *JET* paper (as we’ve documented above) – suggests that there is important interdisciplinary work to be done in understanding how human forward-looking psychology works. Some of this work is currently being done by economic theorists looking at learning (see, e.g., Evans and Honkapohja [2012]) and in the development of new theories of social learning. At the same time, empirical evidence strongly suggests that macroeconomic time-series are not stationary in the way most macroeconomic models assume, once the model is detrended to account for economic growth. The literature on learning in non-stationary environments goes counter to the kinds of results found by Evans and Honkapohja and suggests that rational expectations may not be the right way to model forward-looking behavior.

From a purely technical perspective, rational expectations – and its deterministic precursor, perfect foresight – serve to close dynamic economic models and make the equilibria they generate self-contained. These assumptions also serve (as we noted above) to discipline what results come out of a model by injecting a conservative bias that makes obtaining arbitrary results difficult. This is one of the key reasons why the findings of indeterminacy in OLG models, and the possibility of sunspot equilibria was so surprising to the economics profession. This is probably a likely reason why even economists like Farmer continue to present results based around indeterminacy and the existence of multiple equilibria as rational expectations equilibria. Until there is a compelling psychological theory of expectations formation, abandoning the rational expectations hypothesis in economics will be difficult. Indeed, this is akin to the way that economists continue to use expected utility formulations in their models despite the empirical evidence against it; until we know better how human beings process

uncertainty, we stick with the theory that can explain key economic responses to uncertainty like insurance and portfolio balance.

While there are other issues – such as how to bring path dependent models to data, or whether the apparent statistical regularities in economic data can be generated by path dependence – that suggest we still don’t fully understand macroeconomics, the indeterminacy school of macroeconomics that Farmer has pushed so consistently clearly deserves to be taken seriously (as evidenced by the Vines and Wills *OxREP* volume advocating for this approach). From our perspective, the overlapping generations model, with its realistic treatment of the human lifecycle, needs to be front and center in this effort.

As with any change in fundamental paradigms, the indeterminacy program won’t implement itself. Per the quote at the start of this chapter from Karl Shell, representative agent ILA-based DSGE models are now received science in macroeconomics. The economics profession has trained generations of practitioners to work with this model. It has directed the collection of data to support the analysis of the received model – per Einstein’s dictum that “it is the theory which decides what we can discover.” There is a deep computational infrastructure – Dynare – dedicated to the model.

The one major element in favor of the push for a new macroeconomics based on indeterminacy and multiple equilibria is the body of work that has gone into the development of general equilibrium theory and the overlapping generations model. Indeed, from our perspective, if OLG becomes the dominant workhorse model for macro (as it should have many years ago), the indeterminacy/multiple-equilibria program of Vines and Wills will follow naturally. Unfortunately, much of the basic knowledge needed to understand this approach isn’t taught in graduate economics programs anymore. An informal survey of the content of core

economics Ph.D. programs at top universities should convince the reader that general equilibrium theory either isn't taught or has been relegated to a couple weeks of coverage as part of the core first semester micro course. What used to be a semester of general equilibrium theory has been replaced by game theory and related content like contract theory and information economics. We don't view this displacement as a bad thing – indeed, modern game theoretic tools are critical (even in general equilibrium contexts) for understanding how mass production technologies generate large firms with significant market power, and for understanding how such large players interact.

When Dave Cass died in 2007, one of us (Spear) and Alex Citanna went through the papers in his office and at his home to try and preserve what we thought might be important for the historical record. One of the things we found was a set of hand-written notes from 1995 on “The future of general equilibrium,” which Cass had apparently jotted down in response to his negative reaction to the focus of a conference in honor of Gerard Debreu earlier that year. Among other things, Cass noted that conventional general equilibrium theory was foundational for both macroeconomics and finance, and that the natural “allies” for people doing general equilibrium theory were likely to be found in these subdisciplines. Our takeaway from this observation is that because general equilibrium theory is foundational for macroeconomics, advanced versions of it should be taught in graduate programs as part of the macro sequence. Today, in most programs, it is not, and the reason for this is that understanding the conventional ILA-based DSGE model doesn't require much more (beyond basic general equilibrium theory) than an understanding of forward dynamic programming. What current graduate students don't see any more is detailed modeling of the way income effects generate multiple equilibria in

conventional GE models, and the need for carrying along the wealth distribution as state variables in SOLG and heterogeneous agent SILA models.

This is changing, though, even in macro programs that are still focused on the conventional DSGE model, since the basic failures of the model are forcing even very conventional macroeconomists to consider models with heterogeneous agents, models with market incompleteness, and models of macro-finance with things like money and bonds and equity assets actively traded. As these models come to the fore – and as macro practitioners begin to realize that all of these possibilities are captured in the stochastic overlapping generations model – the need for teaching deeper general equilibrium theory as the foundation for macroeconomics and finance will become apparent. Looking beyond this, however, a full accounting for the operations of modern industrial (even post-industrial) macroeconomies will require modeling the oligopolies that populate the modern world, their interaction with government regulatory agencies, the political economy that dictates the environment in which these strategic rivals operate, and the financial lubricants that keep the whole system running.

For this work, we will need all of the tools of game theory, contact theory, and finance. And the modeling architecture in which this overarching theory is developed should be that of Samuelson's overlapping generations model.

Chapter 10: Morphology of OLG Models and Methods in Comparative Perspective

In the chapters above, we have presented a wide ranging and detailed account of the origins, evolution, development and dissemination of the OLG approach in economics. We have also dealt with the ILA approach in this context. This chapter attempts to bring together the diverse elements in our story by applying a general and systematic analytical method, that is to say, morphological analysis. Morphology deals with *both* structure and form and assists in determining evolutionary relationships (Zwicky [1969]). The main focus of morphological analysis has been in the physical and biological sciences. In these disciplines it is applied with reference to the analysis of the form of, and structural relationships between, various elements in what is being studied, and their formative processes. Prior to Zwicky's work, the morphological approach – as applied to social and managerial sciences – was not systematized, as in the physical and biological sciences. From the late 1940s onwards, Zwicky generalized the “morphological approach” – as applied to the social and managerial sciences – so as to deal with possible solutions to multidimensional non-quantifiable problems. As he wrote (*Ibid*, 1969, 34):

Attention has been called to the fact that the term morphology has long been used in many fields of science to designate research and structural interrelations – for instance in anatomy, geology, botany and biology ... I have proposed to *generalize* and *systematize* the concept of morphological research and include not only the study of the shapes of geometrical, geological, biological, and generally material structures, but also to study the more abstract structural interrelations among phenomenon, concepts, and ideas, whatever the character might be [our italics].

This is “General Morphological Analysis” (GMA) (Alvarez and Ritchey [2015]).

In economics, early morphological approaches were applied to the business cycle in its various forms, from the 17th century onwards. Indeed, Mercantilists, Physiocrats, Smith and the English Classical School, and others concerned with economic affairs mused over the nature of economic crises in terms of causes and characteristics. And, over the 19th and early 20th

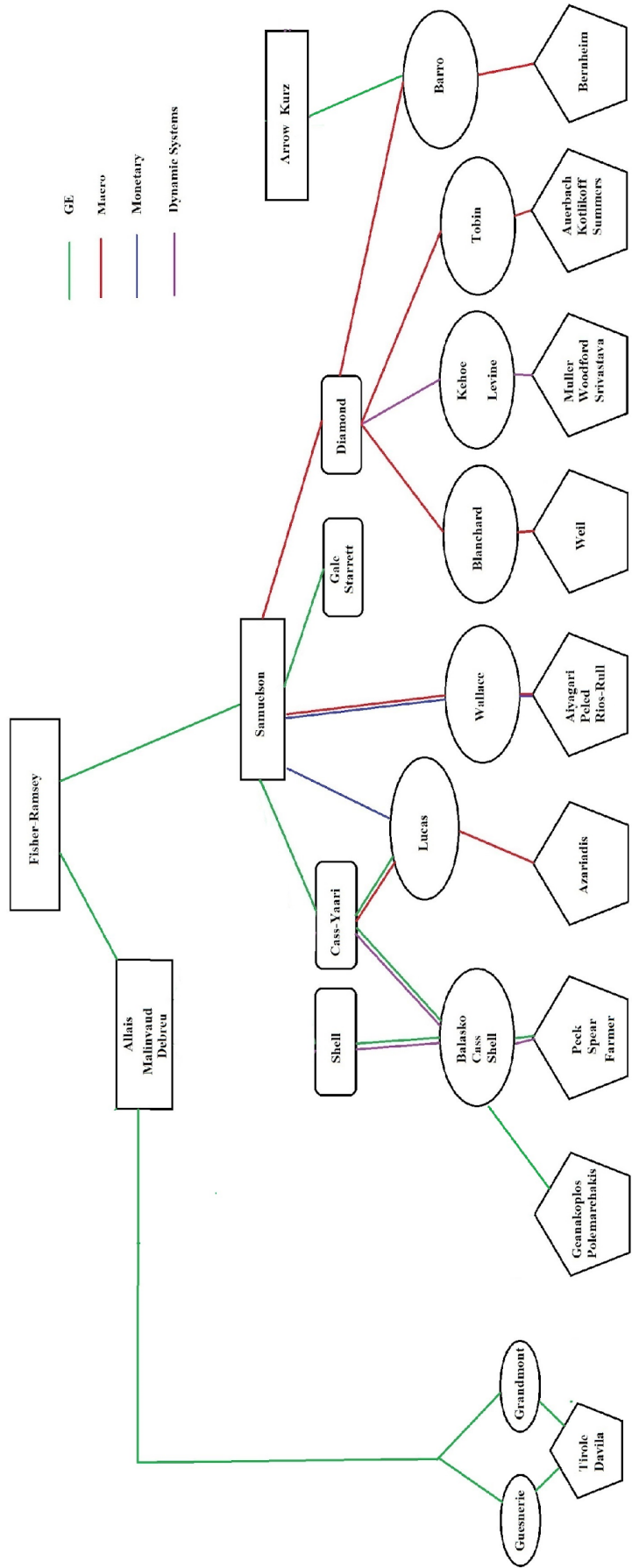
centuries, individuals using the tools of political economy, and later economic science, developed various morphological frameworks describing “credit,” “trade,” “industrial,” and “business” cycles respectively, analyzing their length, phases, and characteristics accordingly. Schumpeter (1939, 169-183) noted the distinction between three classes of cycles short, medium, and long-term. He identified the short-term cycle proposed by Kitchin (3-5 years), as against the medium-term cycle (7-11 years) of Juglar, and the long-term cycle of Kondratieff (45-60 years). But he went even further, noticing that the distinction between these cycles – in terms of what we now call “impulse” and “propagation,” were based mainly on economic factors and shocks. Regarding the Kitchin cycle, it was based on the inventory cycle; the Juglar cycle on fluctuations in fixed investment, while the Kondratieff cycle was the possible outcome of a number of factors, such as technological change, wars or revolutions, new countries being discovered, of fluctuations in gold production, the latter two mainly reflecting cycles from the 17th century onwards.

Moreover, besides Kitchin-Juglar-Kondratieff, there are a number of additional theories of cycles or “waves”: infrastructure investments, Kuznets (1930); aggregate demand and aggregate investment fluctuations, Kaldor-Kalecki (1935, 1940); cyclical growth based on wage-share unemployment dynamics, Goodwin (1967); financial instability and propensity to boom-bust, Minsky (1964,1986); and finally, that of Schumpeter himself, technological innovation and “creative destruction” processes, Schumpeter (1934,1939, 1942).

The GMA approach provides us with a systematic way to analyze the structures and evolution of OLG as against ILA models. It provides us with the following tools: (i) tree diagrams; (ii) category grids; (iii)morphology boxes; and (iv) morphology fields (Zwicky [1969], 115-120).

Another phenomenon described in the book that can be analyzed further using morphology is that of *symbiosis and cross-fertilization* of ideas between various cohorts at the main institutions where work on OLG was going on. This is addressed by use of a morphology tree as below. We can delineate three forms of cross-fertilization: generic (“in the air”), concurrent, and sequential. Cohort cross-fertilization would encompass faculty-faculty, faculty-student and student-student cases. The main institutions involved, as noted in the book, are Stanford, Yale (Cowles), Penn, MIT and Minnesota.

In the OLG morphology tree diagram below, we first identify four main branches of economic analysis which played dominant roles in shaping the OLG approach: General Equilibrium, Macroeconomics, Monetary Economics, and Dynamic Systems. These branches were extended – singularly or in combination – via the development of OLG approaches dealing with theoretical and applied issues. We can identify a group of *founding fathers*, and three *successive cohorts* of economists who advanced the overlapping generations approach. These cohorts are identified via the geometric figures in which they appear: *founding fathers* (rectangular); *successive cohorts* (curved sided rectangular, elliptical and pentagonal, respectively). The contributions of these individuals are described in the book. We note that the connections in the diagram aren’t meant to indicate direct influences (though in some cases they do), but rather the way the development and application of ideas about the OLG model played out in practice. Thus, for example, we link Auerbach, Kotlikoff and Summers to Tobin because of Tobin’s early pioneering work in using OLG models with long lived agents for policy simulation experiments.



OLG Morphology

Among the founding fathers' group, Samuelson would have had a direct influence on his MIT colleagues like Diamond and Shell (prior to Shell's moving to Penn), as well as on the next generation of MIT students that we document in the tables below, including Blanchard, Burke, Muller, and Woodford. The influence of Stanford's faculty working on general equilibrium and growth – Arrow, Kurz, and Uzawa – clearly prepared the next generation of their students – Cass, Shell and Yaari – for their later collaborations on the OLG model. In Spear and Young (2018), we documented the role that the University of Chicago, then home to the Cowles Foundation, played in bringing together faculty and graduate students with like-minded interests in macroeconomics and growth theory. Both Edmond Malinvaud and Tjalling Koopmans spent time at Cowles in the late 1950's and early '60's. Hirofumi Uzawa also spent time there and organized several conferences that involved the younger generation of theorists, including Cass, Lucas, and Shell. Neil Wallace was a Ph.D. student at Chicago at the time and, as we've noted earlier, got interested in Lucas' 1972 paper in part because he and Lucas had been grad students together. Harvard and Yale also contributed to the mix of scholars who ended up working on OLG issues, including Jim Tobin (whose interest, as we've seen, came later in his career), Tim Kehoe, and Robert Barro.

On the faculty-faculty collaboration nexus, the Cass-Shell partnership is legendary, as is the Kehoe-Levine connection (Kehoe and Peter Diamond were Levine's thesis supervisors). Prior to Cass and Shell collocating at Penn, Cass and Lucas traded ideas over lunches at Carnegie Mellon University's Graduate School of Industrial Administration, where Lucas did the work leading to his breakthrough 1972 paper. Karl Shell's French connections brought Yves Balasko into the OLG arena, and saw Jean-Michel Grandmont as a frequent visitor at Penn. Cass and Shell hired Costas Azariadis and set the stage for the contentious sunspots collaboration.

Among the large group of students trained by the Silent Generation scholars, we have detailed their interactions in the book. But, by way of indicating the interconnections among these scholars and the institutions they trained at, we present two grids based upon purposive samples. In the first, we delineate – at three institutions – OLG related doctoral dissertations along temporal, advisory and publication vectors, over the period 1981-2000. We observe that the decade of the 1980s reflects the accelerated OLG research program at these institutions accordingly, with the major publication outlet being *JET*.

Institution	Student	Title of Thesis	Year	Advisor	Journal Outlet
Penn	Spear	Three essays on the role of expectations in dynamic economic models	1982	Cass	JET (1984, 1985)
	Chae	Three essays in economic theory	1985	Cass	JET (1987, 1988)
	Peck	Essays in intertemporal economic theory	1985	Shell	JET (1987, 1988)
	Bertocchi	Essays on the financial structure of a dynamic economy	1988	Azariadis	EdLet (1991), JEDC (1995)
	Chang	International macroeconomic policy in optimizing models	1989	Azariadis	JME (1990)
	Heathcote	Frictions and incomplete markets: three essays in macroeconomics	1998	Atkeson	WP U. Penn
	Nishiyama	Altruism, uncertainty and intergenerational transfers	2000	Rios-Rull	RED (2002)

OLG Dissertations, 1980-1990 University of Pennsylvania

Source: compiled by authors

Institution	Student	Title of Thesis	Year	Advisor	Journal Outlet
Minnesota	Aiyagari	Putty-clay durable capital in stochastic overlapping generation models	1981	Wallace	JET (1985)
	Arango	On the uniqueness of the equilibrium in overlapping generation models	1981	Wallace	UCMadrid WP (1983)
	Millan	On the existence of optimal competitive equilibria in the overlapping generations model	1981	Wallace	UABarcelona WP (1982)
	Huffman	An overlapping generations model with asset pricing	1983	Wallace	JnlFin (1985), CanJec(1986), IER(1986)
	Leone	Inflation and the relative price of durable and non-durable goods in overlapping generations models	1983	Brownlee	Cen.Bank.Arg. WP (1983)
	Batima	An overlapping generations model with endogenous fertility	1985	Rosenzweig	Jnl.PubEc.(1986,1987)
	Loewy	An overlapping generations model with endogenous policy	1986	Wallace	JME (1988) EcLet(1988)
	Wright	Labor markets and labor contracts in an overlapping generation model with infinite lived employers	1986	Wallace	JOLE(1988)
	Manuelli	Topics in intertemporal economics	1986	Prescott	JET (1990)
	Maeda	Pairwise trading in overlapping generation models	1990	Wallace	JET (1991)

OLG Dissertations 1980-1990, University of Minnesota

Source: compiled by authors

Institution	Student	Title of Thesis	Year	Advisor	Journal Outlet
MIT	Woodford	Essays in intertemporal economics	1983	Kehoe	JET (1988)
	Mueller	Stationary overlapping generations economies with production and infinitely lived consumers	1984	Kehoe	JET (1988)
	Burke	Essays on equilibria in dynamic economics	1985	Kehoe	Jnl.Math Econ (1987), JET (1988, 1989)

OLG Dissertations, 1980-1990, Massachusetts Institute of Technology

Source: Compiled by authors

The second category grid shows a purposive sample of OLG related NSF grants over the period 1974-2004, along temporal, institutional and publication vectors. We observe that one institution, Penn, led the field in terms of grants, followed by Carnegie Mellon, where former Penn and Minnesota Ph.D. students now served on the faculty. Furthermore, almost half of the

papers that emanated from these OLG related NSF grants were published in one journal, again, *JET*.

OLG related NSF Awards, Purposive Sample, 1974-2004					
Investigators	Institution	Title of Award	Period	Amount (US \$)	Publications
Wallace (Aiyagari)	Minn	Theoretical Analysis of Alternative Exchange Rate Regimes (7722743)	1978-1980	69,300	JET (1985) Aiyagari
Cass, Shell	Penn	The Positive and Normative Roles of Fiat Money in Intertemporal Allocation (7806157)	1978-1980	175,171	JET (1980 a,b) with Balasko
Cass, Shell, Balasko	Penn	General Equilibrium Foundations of Intertemporal Decisions- The Overlapping-Generations Model (8007012)	1980-1983	377,398	JPE (1983)
Cass (Benveniste)	Penn	The Role of Financial Instruments in Facilitating Exchange (8309049)	1983-1986	101,400	JPE (1986)
Karni, Zilcha	Johns Hopkins	An Economic Analysis of Life Insurance, Annuities and Social	1984-1986	91,781	JPubEc(1986,1989)

		Security in an Overlapping Generations Model with Uncertain Lifetime (8408670)			
Srivastava, Spear	Carnegie-Mellon	Equilibrium in Stochastic Overlapping Generations Models (8420486)	1985-1986	73,013	RES (1987) JPubEc(1988) JET (1988,1990)

Smolensky, (Aiyagari)	Wisconsin	Overlapping Agents and the Representative Agent (8520440)	1986-1987	27,213	JET (1988,1989) QJE (1989) Aiyagari
Woodford	Chicago	Sunspot Equilibria in Infinite Horizon Competitive Economics (8710219)	1987-1990	52,538	JET (1990)
Wright	Penn	Foundations of Monetary Theory: Money as a Medium of Exchange (8821225)	1989-1991	46,749	EcTh(1993) with Shell
Rios-Rull	Carnegie-Mellon	Heterogenous Agents Economies with Capital: Model Development and Applications (9110911)	1991-1994	63,195	RES (1996)
Krueger,Kubler	Penn/Stanford	Social Security and	2001-2004	265,338	AER (2002,2006)

		the Wealth Distribution with Overlapping Generations and Aggregate Uncertainty (0004376)			
Brock and Scheinkman	Chicago	Optimal paths generated by optimal control problems (7419692	1974-79	223,400	In Models of Monetary Economies (1980)

Source: Compiled by authors from NSF Awards 1974-2004

The following table summarizes NFS grant activity from 1960 to 2020 in terms of grant support for research projects on the two types of models. As one can see, support for ILA-based research during what we might call the RBC/new neo-classical synthesis decades of the 1980's and '90's outpaces support for OLG-based projects two-to-one. This evens out during the first two decades of the 2000's, though at markedly lower levels of overall support for both research programs.

NSF Grants by period (same keywords as in Dissertations table)		
Years/Model	OLG	ILA
1960-1979	3	4
1980-1999	17	30
2000-2020	10	11
Total	30	45

Source: Compiled by auth from NSF awards 1960-2020

In their paper “The Seven Properties of Good Models” Gabaix and Laibson (2008) delineate properties which can be applied in our morphological analysis of OLG as against ILA based models. We have used three of the properties from their approach – parsimony, tractability, and prediction – and have added one, *teachability*. We can construct a morphology grid as below. There are various ways to fill in the cells and different readers may wish to fill the grid out differently. We have filled our grid in based on the arguments we have presented in the book.

Comparative Morphology

Models					
Properties		OLG	SOLG	ILA	SILA
	<i>Parsimony</i>	Yes	Somewhat	Yes	Yes
	<i>Tractability</i>	Yes	No	Yes	Yes
	<i>Prediction</i>	Open	Yes	Open	Open
	<i>Teachability</i>	Yes	Yes	Yes	Yes

Source: Compiled by authors

On the parsimony dimension, both the OLG and ILA models abstract heavily from reality in the interests of providing workable analytic frameworks for models of dynamic economic equilibrium. The two stochastic extensions of these models are also parsimonious in terms of their abstraction, but they differ significantly in terms of the state variables needed for characterizing rational expectations equilibria. This difference is the basis for describing the SOLG model as “somewhat” parsimonious, since it requires an infinite dimensional state variable for the characterization of competitive (and imperfectly competitive – see Kim and Spear [2021b]) rational expectations equilibrium. The SILA model without heterogeneity needs only the aggregate stock of capital and exogenous shocks as state variables, giving it a

particularly simple structure. Of course, once one begins looking at heterogeneous agent models, SOLG and SILA both require carrying along lagged endogenous state variables, so that parsimony and tractability differences become negligible. Similar concerns with the state variables underlie our “No” designation for SOLG on the tractability dimension, and for our “Yes” indication on predictability, because the structure of the recursive stochastic equilibrium generates testable predictions on the fractal nature of equilibrium prices (see Brock *et. al.* [1991] and Kim and Spear [2021c]), and on the history dependence of rational expectations equilibrium. Finally, all versions of the two models are taught in the published graduate texts on general equilibrium theory and macroeconomics, as we document in the appendix. We note, however, that the ILA model gets the most emphasis in these texts (with a few exceptions), and the treatment of solution methods for SOLG is sparse at best.

In the morphology grid below, we present the relative utilization of various OLG models based on the dimensions of *time* and *stochasticity*. The data are from EconLit, covering the period 1986-2022.

Comparative Morphology

Time treatment		Models	
		OLG	SOLG
	<i>Discrete</i>	34	6
	<i>Continuous</i>	78	4

Source: Compiled by authors

On the morphology table above, continuous time assumptions are due to *tractability*; differential equations are easier to solve than difference equations. The discrete time assumption is based on realism: people don't spend every nanosecond optimizing. Rather, they make plans and then execute them over discrete periods of time. A fully realistic treatment would allow the

length of time periods to be variable, contingent on unforeseen events. But these kinds of models (mixed differential/difference equations) are difficult to work with. On the stochastic dimension, continuous time stochastic differential equations are harder to work with than their discrete time counterparts, though they are used extensively in finance. This could be a second reason the discrete time growth model became prominent at the time that Brock and Mirman were working on stochastic extensions. At the same time, economic activity isn't constantly buffeted by random shocks the way rockets in flight are, so the discrete time/discrete shocks assumption is also reality-based.

Chapter 11: Summary and Conclusion

As we have seen, the foundational OLG model (using the terminology of Blanchard [2018, p.52]) of Allais, Samuelson and Diamond has come a long way since its inception. While DSGE models now dominate macroeconomics, they are essentially based on a “common core, roughly an RBC structure with one main distortion, nominal rigidities” which, as Blanchard notes “seems too much at odds with reality” (*Ibid*, p.52).

OLG, for its part, has metamorphosized from the two and three generational approaches of Samuelson-Diamond vintage to multigenerational models such as those pioneered by Dolde and Tobin, and Auerbach and Kotlikoff. Moreover, it has been applied in widespread fields such as Public Finance and Taxation; Social Security; Labor and Demographic economics; Money and Monetary economics; International economics and trade; and Environmental and Natural Resource economics, among other areas of investigation over the years since its initial formalization. The table below provides some indication of the extent of its application in the areas listed above (based on *EconLit.*, including books, published articles, working papers etc.)

OLG Applications	
Area	Number of Publications
Labor and Demographic economics	1548
Public Finance and Taxation	853
Social Security	686
International economics and trade	638

Money and Monetary economics	581
Environmental and Natural Resource economics	437

Source: Compiled from EconLit database 1958-2020

As seen in the table, the most extensive application of OLG was in the area of labor and demographic economics, followed by public finance and taxation. Applications of OLG in the areas of Social Security and international economics and trade are of the same magnitude, followed by monetary economics. Regarding environmental and natural resource economics, the early applications of OLG did not focus on these areas, but with the increasing importance of environmental issues and the implications of climate change for future generations, OLG applications have bloomed.

This book has presented a wide ranging and detailed survey of the origins, evolution, development and dissemination of the OLG approach – one of the workhorse models of modern economics – in contrast to the ILA approach, with which it competes in terms of efficacy and application. And, since the book's focus has been on this relationship between the two approaches and their respective relative impact, the widespread application of the OLG approach noted above has not been dealt with in detail and remains a subject for further survey and exposition.

In order to present a coherent conclusion to the mass of material presented in the book, we utilized the morphological approach outlined in the previous chapter. We can identify two essential model types: foundational and augmented, in addition to four areas of model efficacy: theoretical, computational, policy, and pedagogical (see for example, Jordi [2018, p. 87]). Given this, we propose that the plasticity of the two respective approaches enabled their metamorphosis from foundational to augmented types.

Now, the formalized canonical foundational OLG model of Samuelson (1958) took almost a decade to have an impact, and this came via the contribution of Diamond (1965). In the same year as Diamond's classic AER paper, path-breaking works by Cass and Koopmans applying the Ramsey (1928) approach appeared; this constituted what can be termed the canonical foundational Ramsey-Cass-Koopmans (RCK) ILA approach (see Spear and Young [2014]). Both the OLG and ILA approaches took time to disseminate through the profession, although the planning version of the RCK model was used early on. By 1972, Lucas had produced a canonical foundational stochastic version of OLG. A decade later, Kydland and Prescott (1982) presented the canonical foundational stochastic ILA model, based on the Brock and Mirman (1972) stochastic extension of the RCK model, which was termed the Real Business Cycle (RBC) Model.

Canonical augmentations of the OLG approach were the outcome of the work of Auerbach and Kotlikoff from the mid-1980s onwards, and, for the ILA model, the parallel development of the macroeconomic DSGE approach, of both New Classical and New Keynesian vintage, with RBC at its core. This was further augmented to provide both Representative Agent and later Heterogenous Agent New Keynesian (RANK and HANK) models (Kaplan et. al [2018]).

As we have documented, the ILA approach became dominant because of its relative advantages along the dimensions of tractability and teachability, particularly for stochastic variants of the two models. These advantages stemmed from the fact that dynamic programming allowed for tractable model solutions, if not on paper, then in simulations, something not available in the stochastic OLG case. However, the addition of heterogenous agents to the ILA

and SILA frameworks makes them as difficult to solve/simulate as the standard SOLG model (with a representative agent or multiple heterogeneous agents, see Duffie et. al [1994]).

The grid below gives us some illustration of comparison between OLG and ILA approaches in the book.

Approach and Application				
Canonical Model Type	OLG	SOLG	ILA	SILA
<i>Foundational</i>	Samuelson-Diamond	Lucas	Ramsey-Cass-Koopmans	Kydland-Prescott
<i>Augmented</i>	Auerbach-Kotlikoff	Rios-Rull	Ubiquitous	King-Goodfriend (NKDSGE)
Augmented Model Type				
Augmented Model	Auerbach-Kotlikoff	Rios-Rull	RANK	HANK
Efficacy				
<i>Theoretical</i>	Yes	Yes	Yes	Yes
<i>Computational</i>	Yes	Limited	Yes	Limited
<i>Policy</i>	Yes	Yes	Yes	Yes
<i>Pedagogical</i>	Limited	Limited	Limited	Limited

Source: Authors compilation

From the mid-1990s onwards, critiques of the empirical foundations of ILA-based models were raised by (among others) Deaton (1992), Hansen and Heckman (1996), and Browning, Hansen and Heckman (1999). The focus of their criticism ranged from issues of heterogeneity in terms of preferences, information, agents' skills and attitudes towards risk and discounting, and aggregation issues, in addition to problems relating to the accumulation of capital and the distribution of wealth. To sum up, as Deaton put it (*Op. cit.*, p.ix):

Representative agents have two great failings; they know too much, and they live too long. An aggregate of individuals with finite lives and with heterogenous information is not likely to behave like the single individual of the textbook.

And indeed, this is the comparative advantage of the OLG approach in all its variants – it is simply more realistic.

Appendix 1: Graduate Textbook Coverage of OLG

In this appendix, we survey the prominent graduate level textbooks in macroeconomics and comment on their treatment of the OLG model and its role in macro. This survey should not be interpreted as a review or endorsement of any of the texts examined. Rather, our comments are meant to highlight the treatment of OLG models – deterministic and stochastic – in the pedagogy provided to graduate students in economics. Our intent here is to buttress our morphological claims about OLG along the teachability dimension. The textbooks are ordered chronologically by the date of first publication of each book.

Year	Author(s)	Title	Publisher
1989	Blanchard and Fischer	Lectures on Macroeconomics	MIT Press
This text provides a pretty good treatment of OLG. While it omits much of Balasko-Shell/BCS results, this is to be expected in a macro text. It includes discussion of Grandmont's chaos work and Azariadis' work on sunspots, as well as Day's work on chaos, but not Benhabib-Day, and no Cass-Shell on sunspots. It covers RBC with aggregate shocks in detail, and includes Lucas '72 as an example of OLG with shocks, but no mention of the problem of history dependence in more general models.			
1989	Stokey and Lucas	Recursive Methods in Economic Dynamics	Harvard U. Press
The only OLG consideration in the book is based on the Lucas '72 model, presented in the context of the model itself, and then in the sections on various fixed point theorems. There is no consideration of history dependence, no discussion of Rios-Rull, and one brief mention of DGMM and Blume results. The material in the last chapter on equilibrium with distortions looks at examples where various people have figured out tricks for solving models with frictions when there are lagged endogenous state variables. However, aside from the brief historical mention of DGMM, there is nothing in the book that indicates that for general SOLG models, the DGMM results must be used.			

1991	McCandless and Wallace	<i>Introduction to Dynamic Macroeconomic Theory</i>	Harvard U. Press
As one would expect of a book co-authored by Neil Wallace, this text is organized around the OLG model. The text is pitched as an advanced undergraduate text, but the overall development of the deterministic OLG model would be useful to any macro graduate student. No consideration of stochastic models, however.			
1993	Azariadis	<i>Intertemporal Macroeconomics</i>	Wiley-Blackwell
This text uses a dynamic systems approach to frame the study of intertemporal macro. Much of the analysis in the book is based on the deterministic OLG model, and the penultimate chapter on market psychology introduces the concept of sunspot equilibrium in the context of the two-period SOLG model. Beyond this, however, this is no further discussion of stochastic economies.			
2002	De la Croix and Michel	<i>A Theory of Economic Growth: Dynamics and Policy in Overlapping Generations</i>	Cambridge U. Press
This text provides a very good treatment of deterministic OLG models. There is no stochastic analysis, but one mention of Rios-Rull extending the Auerbach-Kotlikoff simulation framework to allow for aggregate shocks.			
2002	Heijdra and Ploeg	<i>Foundations of Modern Macroeconomics</i>	Oxford U. Press
REQUESTED COPY FROM AUTHORS			
2004	Ljungqvist and Sargent	<i>Recursive Macroeconomic Theory</i>	MIT Press
The text provides a good comprehensive development of deterministic OLG with coverage of models with idiosyncratic shocks, and some mention of aggregate shocks – Rios-Rull’s work and DGMM paper – but no discussion of the fact that general multi-period SOLG models have history dependent equilibria. Aiyagari’s 1977 <i>JET</i> paper on observational equivalence is in the references, but not cited anywhere in the text.			
2007	Bewley	<i>General Equilibrium, Overlapping Generations Models and Optimal Economic Growth Theory</i>	Harvard U. Press

There is mixed coverage of both ILA and OLG in the text, but stochastic analysis is limited to ILA and permanent income applications.			
2009	Acemoglu	<i>Introduction to Modern Economic Growth</i>	MIT Press
The text provides a rudimentary treatment of deterministic OLG, and very basic stochastic OLG (two-period, one good). There is no discussion of history dependence issues in general OLG.			
2009	Sargent	<i>Dynamic Macroeconomic Theory</i>	Harvard U. Press
The book covers OLG as a model of money, plus stochastic OLG with two-period lives and one good (basic Lucas '72 model). There is no discussion of general stochastic OLG.			
2010	Tvede	<i>Overlapping Generations Economies</i>	Palgrave-MacMillan
This text obviously focuses on OLG, taking a good general equilibrium approach, following the Balasko-Shell formulation, with simplified analysis of the optimality criterion in the Balasko-Shell II paper. However, it provides no stochastic consideration.			
2010	Benassy-Quere et. al.	<i>Economic Policy</i>	Oxford U. Press
This text takes a very different approach than the others in this list by focusing on policy applications of the two workhorse economic models. The focus is one that Blanchard, in his Forward to the book, states that he wishes he had taken in a follow-on to his own text. The book gives a reasonably complete treatment of deterministic OLG models in a variety of applications, including some from political economy. There is no treatment of SOLG models, however.			
2012	Romer	<i>Advanced Macroeconomics</i>	McGraw-Hill
The text provides a rudimentary treatment of the deterministic Diamond model, but no consideration of SOLG at all. DSGE is defined as the New Keynesian variation of the conventional RBC model.			

Appendix 2: Rios-Rull follow-on citations

This appendix presents the citations to the papers that cited Rios-Rull's 1996 *ReStud* paper. The citations are organized according to the criterion we used to classify the work as genuine follow-on work ("relevant"); work that cited the paper as being similar to or different from the work presented in the paper ("in passing"); work which using models similar to Rios-Rull's but without aggregate shocks ("tangential"); work that summarized Rios-Rull's work for pedagogical purposes ("summary"); and work that cited Rios-Rull for no discernable reason ("vacuous"). All of these tables were compiled by the authors using information from Google Scholar.

Relevant Citations: Total = 32

Citation	Published	Date	Top 5	Theory
Carvalho, Vasco, Alberto Martin, and Jaume Ventura. "Bubbly business cycles." (2011).	NA	2011		
Chun, Young Jun. "The redistributive effect of risky taxation." <i>International Tax and Public Finance</i> 8.4 (2001): 433-454. (Actually extends Muller-Woodford to aggregate shocks)	INTL.TAXFI N	2001		
Citanna, A., Siconol, P., Recursive equilibrium in stochastic OLG economies: Incomplete Markets, <i>Journal of Mathematical Economics</i> (2012)	J. MAT	2012		x
Citanna, Alessandro, and Paolo Siconolfi. "On the nonexistence of recursive equilibrium in stochastic OLG	ET	2008		x

economies." Economic Theory 37.3 (2008): 417-437.				
Citanna, Alessandro, and Paolo Siconolfi. "Recursive equilibrium in stochastic OLG economies: Incomplete markets." Journal of Mathematical Economics 48.5 (2012): 322-337.	J.MATH	2012		x
Citanna, Alessandro, and Paolo Siconolfi. "Recursive Equilibrium in Stochastic Overlapping-Generations Economies." Econometrica 78.1 (2010): 309-347.	ETRICA	2010	x	x
Citanna, Alessandro, and Paolo Siconolfi. "Short-memory equilibrium in stochastic overlapping generations economies." Journal of Economic Theory 134.1 (2007): 448-469.	JET	2007		x
Ferraro, Domenico, and Giuseppe Fiori. "The aging of the baby boomers: Demographics and propagation of tax shocks." American Economic Journal: Macroeconomics 12.2 (2020): 167-93.	AEJ:Macro	2020		
Glover, Andrew, et al. "Intergenerational redistribution in the great recession." Journal of Political Economy 128.10 (2020): 3730-3778.	JPE	2020	x	
Gomme, Paul, et al. "The business cycle and the life cycle." NBER macroeconomics annual 19 (2004): 415-461.	NBER AN	2004		
Gomme, Paul. Measuring the welfare costs of inflation in a life-cycle model. No. 08001. Concordia University, Department of Economics, 2008.	NA	2008		
Hansen, Gary D., and Selahattin İmrohoroglu. "Business cycle fluctuations and the life cycle: How important is on-the-job skill accumulation?." Journal of Economic Theory 144.6 (2009): 2293-2309.	JET	2009		

Hasanhodzic, Jasmina, and Laurence J. Kotlikoff. Generational risk-is it a big deal?: Simulating an 80-period olg model with aggregate shocks. No. w19179. National Bureau of Economic Research, 2013.	NBER WP	2013		
Heer, Burkhard, and Alfred Maussner. "Business cycle dynamics of a new Keynesian overlapping generations model with progressive income taxation." (2006). (Probably use Krusell-Smith for solution)	NA	2006		
Heer, Burkhard, and Alfred Maußner. "Distributional effects of monetary policies in a new neoclassical model with progressive income taxation." Computing in Economics and Finance. Vol. 12. 2005. (Believe they use Krusell-Smith)	COMP.ECFI N	2005		
Heer, Burkhard, and Alfred Maussner. "The burden of unanticipated inflation: Analysis of an overlapping generations model with progressive income taxation and staggered prices." Available at SSRN 977126 (2007). (Again, use Krusell-Smith approach)	NA	2007		
Heer, Burkhard, and Alfred Maußner. "The burden of unanticipated inflation: Analysis of an overlapping-generations model with progressive income taxation and staggered prices." Macroeconomic Dynamics 16.2 (2012): 278-308. (They use Krusell-Smith technique, per their book)	MD	2012		
Heer, Burkhard, Stefan Rohrbacher, and Christian Scharrer. "Aging, the great moderation, and business-cycle volatility in a life-cycle model." Macroeconomic	MD	2017		

Dynamics 21.2 (2017): 362-383. (Likely they use Krusell-Smith)				
Kim, Eungsik, and S. E. Spear. "Singular Invariant Markov Equilibrium in Stochastic Overlapping Generations Models." JEDC (2019).	JEDC	2019		x
Krebs, Tom. "Recursive equilibrium in endogenous growth models with incomplete markets." Economic Theory 29.3 (2006): 505-523. (First counter-example to recursive equilibrium paper)	ET	2006		
Krueger, Dirk, and Felix Kubler. "Computing equilibrium in OLG models with stochastic production." Journal of Economic Dynamics and Control 28.7 (2004): 1411-1436.	JEDC	2004		
Kubler, Felix, and Karl Schmedders. "Life-cycle portfolio choice, the wealth distribution and asset prices." Swiss Finance Institute Research Paper 10-21 (2015).	NA	2015		
Larin, Benjamin. "A Quantitative Model of Bubble-Driven Business Cycles." (2016).	NA	2016		
Larin, Benjamin. Bubble-driven business cycles. No. 143. Working Paper, 2016.	NA	2016		
Matovu, John. "Composition of government expenditure, human capital accumulation, and welfare." (2000).	NA	2000		
Mennuni, Alessandro. "Labor force composition and aggregate fluctuations." (2013).	NA	2013		
Nakajima, Makoto. "Rising earnings instability, portfolio choice, and housing prices." Portfolio Choice, and Housing Prices (July 3, 2005) (2005).	PORT.CH	2005		

Pries, Michael J. "Social security reform and intertemporal smoothing." <i>Journal of Economic Dynamics and Control</i> 31.1 (2007): 25-54.	JEDC	2007		
Ríos-Rull, José-Víctor. "On the quantitative importance of market completeness." <i>Journal of Monetary Economics</i> 34.3 (1994): 463-496.	J.MONETARY	1994		
Ríos-Rull, José-Víctor. "Population changes and capital accumulation: The aging of the baby boom." <i>The BE Journal of Macroeconomics</i> 1.1 (2001).	BELL Macro	2001		
Silos, Pedro. "Housing, portfolio choice and the macroeconomy." <i>Journal of Economic Dynamics and Control</i> 31.8 (2007): 2774-2801.	JEDC	2007		
Wong, Arlene. "Refinancing and the transmission of monetary policy to consumption." Unpublished manuscript 20 (2019).	NA	2019		

In passing citations: Total = 63

Note: We don't summarize publication outlets and top 5 or theory status here since these aren't relevant to our analysis. We provide the citations by way of documenting the extent of our analysis.

Alvarez-Albelo, Carmen D. "Endogenous versus exogenous efficiency units of labour for the quantitative study of Social Security: two examples." <i>Applied Economics Letters</i> 11.11 (2004): 693-697. (Deterministic model; RR parameterizations)
Angeletos, George-Marios, and Laurent-Emmanuel Calvet. "Idiosyncratic production risk, growth and the business cycle." <i>Journal of Monetary Economics</i> 53.6 (2006): 1095-1115. (ILA)
Bagchi, Shantanu. "Can overconfidence explain the consumption hump?." <i>Journal of Economics and Finance</i> 35.1 (2011): 41-70. (Continuous time; RR parameterizations)
Basso, Henrique S., and Omar Rachedi. "The young, the old, and the government: Demographics and fiscal multipliers." <i>American Economic Journal: Macroeconomics</i> 13.4 (2021): 110-41. (Econometrics)
Benzoni, Luca, and Olena Chyruk. "Human capital and long-run labor income risk." Available at SSRN 2361497 (2013). (Continuous time)
Böhm, Volker, Tomoo Kikuchi, and George Vachadze. "Asset pricing and productivity growth: the role of consumption scenarios." <i>Computational Economics</i> 32.1 (2008): 163-181. (Two period model)
Bohn, Henning. "Risk sharing in a stochastic overlapping generations economy." (1998). (Two-period OLG)
Bonneuil, Noël, and Patrick Saint-Pierre. "Beyond optimality: Managing children, assets, and consumption over the life cycle." <i>Journal of Mathematical Economics</i> 44.3-4 (2008): 227-241. (Continuous time, deterministic)
Bowlus, Audra, Haoming Liu, and Chris Robinson. "Business cycle models, aggregation, and real wage cyclicalities." <i>Journal of Labor Economics</i> 20.2 (2002): 308-335. (Econometrics, RR parameterization)
Budría, Santiago, and Javier Díaz-Giménez. "Economic inequality in Spain: the european community household panel dataset." <i>Spanish Economic Review</i> 9.1 (2007): 1-38. (Econometrics)
Budría, Santiago, and Javier Díaz-Giménez. "Economic inequality in Spain: the European Union Household panel dataset." (2004). (Econometrics)

Budria, Santiago, et al. "New facts on the distributions of earnings, income and wealth in the US". Federal Reserve Bank of Minneapolis Quarterly Review 26 (2002): 2-35. (Econometrics)
Bullard, James, and John Duffy. "Learning and excess volatility." Macroeconomic Dynamics 5.02 (2001): 272-302. (Deterministic model)
Bütler, Monika. "The political feasibility of pension reform options: the case of Switzerland." Journal of Public Economics 75.3 (2000): 389-416. (Deterministic model)
Castaneda, Ana, Javier Díaz-Giménez, and José-Victor Ríos-Rull. "Exploring the income distribution business cycle dynamics." Journal of Monetary economics 42.1 (1998): 93-130. (Aiyagari-Bewley model)
Cazzavillan, Guido, and Patrick A. Pintus. "Capital externalities in OLG economies." Journal of Economic Dynamics and control 30.7 (2006): 1215-1231. (Two period model)
Chalk, Nigel A. "The sustainability of bond-financed deficits: An overlapping generations approach." Journal of Monetary Economics 45.2 (2000): 293-328. (Tw-period OLG)
Chatterjee, Satyajit, et al. "A quantitative theory of unsecured consumer credit with risk of default." Econometrica 75.6 (2007): 1525-1589. (Perpetual youth)
Choi, Sekyu, Alexandre Janiak, and Benjamín Villena-Roldán. "Unemployment, Participation and Worker Flows Over the Life-Cycle." (2013). (Econometric; builds on results from RR model on labor hours, but no formal model presented)
Cournède, Boris, et al. "Enhancing economic flexibility: what is in it for workers?." (2016). (Econometrics; RR parameterizations)
d'ALBIS, Hippolyte, and Emmanuelle Augeraud-Véron. "Continuous-time overlapping generations models." Optimal Control of Age-structured Populations in Economy, Demography, and the Environment. Routledge, 2013. 63-87. (Continuous time)
d'Albis, Hippolyte, and Emmanuelle Augeraud-véron. "Competitive growth in a life-cycle model: Existence and dynamics." International Economic Review 50.2 (2009): 459-484. (Continuous time, RR parameterization)
Den Haan, Wouter J. "Solving dynamic models with aggregate shocks and heterogeneous agents." Macroeconomic dynamics 1.2 (1997): 355-386. (Aiyagari-Bewley solution technique)
Denk, Oliver, and Jean-Baptiste Michau. "Optimal Social Security with Imperfect Tagging." (2013). (Continuous time)
Denk, Oliver. "How do product market regulations affect workers?: Evidence from the network industries." (2016). (Econometrics)

Dubovyk, Tetyana, and CeRP-Collegio Carlo Alberto. "Macroeconomic Aspects of Italian Pension Reforms of 1990s." Center for Research on Pensions and Welfare Policies Working Paper 101 (2010): 40. (Deterministic OLG)
Farmer ¹ , Roger EA, and Pawel Zabczyk. "The fiscal theory of the price level in overlapping generations models." (2019).
Feigenbaum, James. "Can mortality risk explain the consumption hump?." Journal of Macroeconomics 30.3 (2008): 844-872. (Perpetual youth model)
Ferraro, Domenico, and Giuseppe Fiori. "The aging of the baby boomers: Demographics and propagation of tax shocks." American Economic Journal: Macroeconomics 12.2 (2020): 167-93. (Econometrics)
Ferraro, Domenico, and Giuseppe Fiori. "The Aging of the Baby Boomers: Demographics and Propagation of Tax Shocks." Available at SSRN 2848156 (2016). (Econometrics)
Gervais, Martin, et al. "What should I be when I grow up? Occupations and employment over the life cycle and business cycle." NBER Working Paper S 20628 (2014). (Matching model)
Gornemann, Nils, Keith Kuester, and Makoto Nakajima. "Doves for the rich, hawks for the poor? Distributional consequences of monetary policy." Distributional Consequences of Monetary Policy (April 2016) (2016). (ILA)
Gornemann, Nils, Keith Kuester, and Makoto Nakajima. "Monetary policy with heterogeneous agents." (2012). (ILA)
Hairault, Jean-Olivier, François Langot, and Thepthida Sopraseuth. "Quantifying the Laffer curve on the continued activity tax in a dynastic framework." International Economic Review 49.3 (2008): 755-797. (ILA, RR parameterizations)
Hairault, Jean-Olivier, François Langot, and Thepthida Sopraseuth. "The Pre-Retirement Job Search: A Basic Explanation of the Older Worker Employment." Journal of European Economic Association 8.5 (2010): 1034-1076. (Econometrics)
Heathcote, Jonathan, Kjetil Storesletten, and Giovanni L. Violante. "The macroeconomic implications of rising wage inequality in the United States." Journal of political economy 118.4 (2010): 681-722. (Perpetual youth)
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Heijdra, Ben J., and Jenny E. Ligthart. "Fiscal policy, monopolistic competition, and finite lives." Journal of Economic Dynamics and Control 31.1 (2007): 325-359. (Continuous time)

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Tangential citations: Total = 51

Note: Since these are relevant to our analysis, we include the publication, top 5 and theory classifications.

Citation	Published	Date	Top 5	Idiosyncratic
Altig, David, and Charles T. Carlstrom. "Inflation, personal taxes, and real output: a dynamic analysis." <i>Journal of Money, Credit and Banking</i> 23.3 (1991): 547-571. (Idiosyncratic risk)	JMCB	1991		x
Attanasio, Orazio, Sagiri Kitao, and Giovanni L. Violante. "Financing Medicare: A general equilibrium analysis." <i>Demography and the Economy</i> . University of Chicago Press, 2010. 333-366. (Idiosyncratic risk)	DEMOG. EC.	2010		x
Bonneuil, Noël. "Viability, optimality and sustainability in vintage models." <i>Optimal Control of Age-structured Populations in Economy, Demography, and the Environment</i> . Routledge, 2013. 126-143. (Idiosyncratic shocks)	OPT.CONT.	2013		x
Börsch-Supan, Axel, Alexander Ludwig, and Joachim Winter. "Aging, pension reform, and capital flows: a multi-country simulation model." <i>MEA Discussion Papers</i> 64 (2004). (Idiosyncratic risk)	NA	2004		x
Buccioli, Alessandro. "Social security, self-control problems and unknown preference parameters." (2009): 001. (Behavioral, idiosyncratic risk)	NA	2009		x
Bullard, James, and James Feigenbaum. "A leisurely reading of the life-cycle consumption data." <i>Journal of Monetary Economics</i> 54.8 (2007): 2305-2320. (Idiosyncratic risk)	J.MON.EC	2007		x
Bütler, Monika. "Anticipation effects of looming public-pension reforms." <i>Carnegie-Rochester Conference Series on Public Policy</i> . Vol. 50. North-Holland, 1999. (Idiosyncratic risk)	CARN.ROCH	1999		x
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Conesa, Juan C., and Dirk Krueger. "Social security reform with heterogeneous agents." Review of Economic dynamics 2.4 (1999): 757-795. (Idiosyncratic risk)	RED	1999		x
Cubeddu, Luis, and José-Víctor Ríos-Rull. "Families as shocks." Journal of the European Economic Association 1.2-3 (2003): 671-682. (Idiosyncratic risk)	JEEA	2003		x
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Den Haan, Wouter J. "The importance of the number of different agents in a heterogeneous asset-pricing model." Journal of Economic Dynamics and Control 25.5 (2001): 721-746. (Idiosyncratic shocks) (Econometric analysis, no formal model)	JEDC	2001		
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Feigenbaum, James, and Geng Li. "A Semiparametric Characterization of Income Uncertainty over the Lifecycle." Available at SSRN 1247451 (2010). (Idiosyncratic risk)	NA	2010		x
Feigenbaum, James. The Contributions of Borrowing Constraints and Uncertainty to Aggregate Saving. Working Paper, 2006. (Idiosyncratic risk)	NA	2006		x
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